



ISO/TC 184/SC 4 "Industrial data"
Secretariat: ANSI
Committee manager: Tripp Dana Ms



Cumulative Resolutions 1st thru 91st plenary

Document type	Related content	Document date	Expected action
Meeting / Other	Meeting: VIRTUAL 8 May 2026	2026-05-08	INFO

Replaces: N 4049 Cumulative Resolutions 1st through 90th Plenary



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

CUMULATIVE LIST OF RESOLUTIONS

Supersedes ISO/TC 184/SC 4 N4049

Attached is a list of all resolutions adopted by the SC 4 Committee at each of the following meetings:

July	84	<u>Washington DC, USA</u>
March	85	<u>Paris, France</u>
January	88	<u>Rotterdam, Netherlands</u>
December	88	<u>Tokyo, Japan</u>
January	90	<u>Paris, France</u>
June	90	<u>Gothenburg, Sweden</u>
July	91	<u>Sapporo, Japan</u>
February	92	<u>Oslo, Norway</u>
October	92	<u>Dallas, TX, USA</u>
February	93	<u>Turin, Italy</u>
October	93	<u>Berlin, Germany</u>
May	94	<u>Davos, Switzerland</u>
October	94	<u>Greenville, SC, USA</u>
March	95	<u>Sydney, Australia</u>
October	95	<u>Grenoble, France</u>
June	96	<u>Kobe, Japan</u>
October	96	<u>Toronto, Canada</u>
June	97	<u>San Diego, CA USA</u>
October	97	<u>Florence, Italy</u>
June	98	<u>Bad Aibling, Germany</u>
October	98	<u>Beijing, China</u>
January	99	<u>San Francisco, CA USA</u>
June	99	<u>Lillehammer, Norway</u>
November	99	<u>New Orleans, LA USA</u>
February	00	<u>Melbourne, Australia</u>
June	00	<u>Bordeaux, France</u>
October	00	<u>Charleston, SC USA</u>
February	01	<u>Funchal, Portugal</u>

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

 Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

June	01	San Francisco USA
October	01	Fukuoka, Japan
February	02	Myrtle Beach, SC
June	02	Stockholm, Sweden
November	02	Seoul, Korea
March	03	San Diego, CA, USA
June	03	Stuttgart, Germany
October	03	Poitiers, France
March	04	Ft Lauderdale, FL, USA
July	04	Bath, United Kingdom
October	04	Seattle, WA, USA
March	05	Lillehammer, Norway
June	05	Valencia, Spain
October	05	HangZhou, China
March	06	Vico Equense, Italy
June	06	Toulouse, France
October	06	Hershey, PA, USA
March	07	Funchal, Madeira Island, Portugal
July	07	Ibusuki, Japan
November	07	Irving (Dallas), TX, USA
March	08	Louisville, KY, USA
June	08	Sun City, South Africa
November	08	Busan, Korea
May	09	Parksville, British Columbia, Canada
November	09	Rotterdam, The Netherlands
June	10	Bethlehem, Pennsylvania USA
October	10	Vico Equense, Italy
May	11	Portland, Oregon, USA
Oct	11	Suzhou, China
June	12	Stockholm, Sweden
Nov	12	Florida, USA
June	13	St Denis, France
Nov	13	Gyeongju, Korea
May	14	Philadelphia, USA
Nov	14	Belfort, France

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

 Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

Apr	15	<u>Vico Equense, Italy</u>
Oct	15	<u>Baltimore, USA</u>
May	16	<u>Sapporo, Japan</u>
Oct	16	<u>Seattle, USA</u>
May	17	<u>Oslo/Sandvika, Norway</u>
Nov	17	<u>Jeju, Korea</u>
May	18	<u>Beijing, China</u>
Nov	18	<u>Chicago, USA</u>
May	19	<u>Toulouse, France</u>
Nov	19	<u>Marina del Rey, USA</u>
May	20	<u>Virtual 79th Plenary</u>
Nov	20	<u>Virtual 80th Plenary</u>
May	21	<u>Virtual 81st Plenary</u>
Nov	21	<u>Virtual 82nd Plenary</u>
May	22	<u>Virtual 83rd Plenary</u>
Nov	22	<u>Hamamatsu, Japan</u>
June	23	<u>Paris, France</u>
Oct	23	<u>Saratoga Springs, USA</u>
May	24	<u>Seoul, Korea</u>
Oct	24	<u>Stavanger, Norway</u>
May	25	<u>Little Rock, Arkansas, USA</u>
Oct	25	<u>Nagasaki, Japan</u>
May	26	<u>Virtual 91st Plenary</u>



N4118

2026 – 05 – 15

RESOLUTION 1: (Gaithersburg - July 1984)

SC 4 recognizes the need for a new standard for the external representation of product model data. This standard will be based upon existing data exchange initiatives including the US IGES and PDDI, the French SET, the German VDA/BDMA-FS, and the UK NEDO.

RESOLUTION 2: (Gaithersburg - July 1984)

The SC adopts the following goal and objectives.

Goal: The creation of a standard which enables the capture of information comprising a computerized product model in a neutral form without loss of completeness and integrity, throughout the life cycle of the product.

Design Objectives: The standard must feature:

- Flexibility to permit expansion without invalidating existing portions of the standard.
- Efficiency for processing, communication and storage.
- Rigorous and formal documentation.
- The minimum possible set of data elements.
- Separation of data content from physical format.
- A logical classification of data elements.
- Compatibility with other existing relevant standards.

Near Term Objectives: The initial objective is to produce a Version 1 of standard for discrete mechanical parts including the following:

- A product model core to include:
 - * Geometry
 - 3-D wireframe/surfaces
 - Boundary representation
 - Constructive solid geometry
 - * Tolerances
 - * Features
 - * Bill of materials including: material, process specification, part identification, notes.
- Application data requirements or application influence on product model core

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

from the following areas:

- * Drafting
- * Finite element modelling
- * Machining
- * Quality assurance.
- * Data management

- A mechanism for standard parts
- Parametric design features.
- Data syntax and file structure independent of the above contents.

Drafting is to include existing national and international standard requirements. Where possible, common semantics will be defined for use with a flag identifying desired drafting standard to produce specific display information

More stable geometric forms of curves and surfaces preferred. Implementation guidelines should include conversion methods. Constructive and evaluated (result) forms of geometric entities are needed.

A draft proposal for Version 1 is required for ballot by SC 4 by the end of 1985.

RESOLUTION 3: (Gaithersburg - July 1984)

SC 4 recommends that its member bodies involved in the technical development of the standard for the external representation of product model data should adopt an organizational structure within their national activities to address four discrete main topics:

- Definition of data content (semantics)
- Definition of data syntax and file structure
- Specification of information for applications
- Implementation and validation of the standard.

The structure will support international collaboration within each topic, as well as integration of activities across individual national efforts.

RESOLUTION 4: (Gaithersburg - July 1984)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 agrees to establish Working Group 1, Technical Coordination and Support, with the following terms of reference:

- To provide technical support and recommendations for SC 4
- To coordinate national technical developments
- To resolve technical differences
- To collate national technical contributions for inclusion in the draft ISO document.

Jerry Weiss is appointed Convener.

SC 4 recognizes a possible need to establish other working groups in the future to cover:

- Guidelines for implementation
- Procedures for validation
- Data for specific applications
- Developments due to new technology

RESOLUTION 5: (Gaithersburg - July 1984)

ISO/TC 184/SC 4 thanks Mr. Bradford Smith for providing his excellent work as chairman of this meeting and proposes to nominate Mr. Smith as Chairman of ISO/TC 184/SC 4 for a three-year term.

RESOLUTION 6: (Gaithersburg - July 1984)

ISO/TC 184/SC 4 expresses its appreciation to NBS and ANSI and the members of these delegations for their gracious hospitality and for the excellent services provided for this meeting.

RESOLUTION 7: (Paris 1 - March 1985)

The drafting committee is constituted to include:

- B. LUCET for the French language
- I. BODNAR for the English language

RESOLUTION 8: (Paris 2 - March 1985)

The agenda SC 4 N 13 for the March 85 meeting is adopted as amended by SC 4 N 15.

RESOLUTION 9: (Paris 3 - March 1985)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 reconfirms its goal as expressed in Resolution #2.

To develop as soon as possible a single international standard for the exchange of product definition data to be called the standard for the exchange of product model data (STEP).

SC 4 will not concern itself with an interim solution based on any existing national standards, although it recognizes that these standards will continue in parallel use until STEP becomes fully operational.

All efforts will be concentrated to achieve a first version of STEP so that its effective industrial use can start in 1990.

RESOLUTION 10: (Paris 4 - March 1985)

SC 4 agrees that, whenever existing national standards are enhanced, development should be undertaken in parallel with the corresponding STEP effort, using the same reference model to ensure that compatible concepts are used. This will cause the standards to converge towards STEP and will simplify the eventual migration from the current standards to STEP.

RESOLUTION 11: (Paris 5 - March 1985)

SC 4/WG 1 is thanked for its accomplishments at its first meeting and is directed to:

- Develop documentation requirements for future work
- Develop a specific list of documents needed for a first version of STEP
- Develop precise functional requirements for Version 1 of STEP
- Develop summary functional requirements for Version 2 of STEP
- Develop a work program and time schedule at its Paris meeting which will aim at
the completion of the technical specification of STEP Version 1.0 by the end
of
1986.

RESOLUTION 12: (Paris 6 - March 1985)

SC 4 recognizes the excellent work of Mesdames LEBADEZET and CHAUMONT, and thanks AFNOR for its hospitality in hosting this session.



N4118

2026 – 05 – 15

RESOLUTION 13: (Rotterdam - January 88)

To approve the agenda for the Rotterdam meeting as amended.

RESOLUTION 14: (Rotterdam - January 1988)

SC 4 agrees to meet at intervals of not more than one year, to allow national standards organizations and ISO management to formally monitor progress in STEP development.

RESOLUTION 15: (Rotterdam - January 1988)

SC 4 will define, by April 1988, an efficient organization and method of operation, based on working groups and an agreed project plan, which will report to SC 4 every quarter to clearly demonstrate progress against that plan.

RESOLUTION 16: (Rotterdam - January 1988)

SC 4 will define and endorse, by 1 April 1988, a realistic workplan for STEP development, showing major milestones and taking into account existing material, available man hours, contributors qualifications and ISO procedures and to have this available to SC 4 members.

RESOLUTION 17: (Rotterdam - January 1988)

SC 4 reaffirms the technical objectives of the STEP project as stated in resolutions 2 and 9 (of SC 4). The STEP standard and its implementation shall feature:

- wider scope than existing national standards
- improved performance over existing standards
- structured application-oriented sets of entities and topical models
- unambiguous definitions
- modular structure
- upward compatibility between versions

RESOLUTION 18: (Rotterdam - January 1988)

SC 4 asks its secretariat to circulate for comments to the national bodies the following reference documents of WG 1 prepared at the January 1988 meeting:

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Document 0.0 - Project Plan
Document 0.1 - Preliminary Design
Document 0.2 - Work Plan Status
Document 9.0 - Requirements for STEP

These documents will be distributed before the end of February and comments should be received within two months.

RESOLUTION 19: (Rotterdam - January 1988)

SC 4 asks its secretariat to circulate for comments of the national bodies the following parts of STEP Version 1.0 as a minimum set:

Document 4.1.1 - EXPRESS
Document 4.2.1 - Physical File
Format Document 4.2.2 - Mapping
Document 4.1.2.1 - Geometry and Topology

These documents will be distributed after the next meeting of WG 1 in March 1988 to serve as a basis for a draft proposal. Comments should be received within two months. After this period, these documents should be forwarded to the Central Secretariat for registration as a DP and for ballot by SC 4 at their next meeting. Other documents considered by WG 1 to be ready for review may be distributed at the same time.

RESOLUTION 20: (Rotterdam - January 1988)

SC 4 directs WG 1 to produce a framework for software implementation, including implementation subsets, such as in VDAIS and application subsets, together with testing and validation procedures in parallel with each component of the standard submitted for approval.

RESOLUTION 21: (Rotterdam - January 1988)

SC 4 agrees that implementation of the STEP standard shall provide a method for migrating data from existing national standards such as IGES, SET and VDA-FS to the new standard, and directs WG 1 to establish the guidelines for the necessary tools and procedures to receive this data.

RESOLUTION 22: (Rotterdam - January 1988)

SC 4 wishes to formally recognize and welcome the potential contribution that can be

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

made by the proposed PDES cooperative in accelerating the development and implementation of STEP.

SC 4 considers that it is essential to establish a working relationship with the cooperative when it is launched, similar to that which has already been achieved with the ESPRIT CAD*I, French SET and German VDA projects.

RESOLUTION 23: (Rotterdam - January 1988)

SC 4 thanks Mr. Bradford M. Smith for his work as chairman of SC 4 and proposes to nominate Mr. Smith for a second three-year term.

RESOLUTION 24: (Rotterdam - January 1988)

SC 4 directs WG 1 to concentrate its effort to complete Presentation and Drafting documents (according to the ISO drafting standards) so it will be available to be included in the Draft Proposal of STEP Version 1 for adoption at the SC 4 meeting scheduled for 2 December 1988 in Tokyo.

RESOLUTION 25: (Rotterdam - January 1988)

SC 4 confirms that the STEP standard will be defined using the EXPRESS language.

RESOLUTION 26: (Rotterdam - January 1988)

SC 4 directs WG 1 to maintain a list of "Open/Resolved Issues", distributed prior to each WG 1 meeting, as a basis of preparation and subsequent discussion by the experts.

RESOLUTION 27: (Rotterdam - January 1988)

As standards developments must be based on technical excellence and use of the limited world wide expertise available, SC 4 recommends that ISO assist in obtaining cooperation and coordination within related standards work such as in the electrical/electronic area with EDIF, STEP and VHDL.

RESOLUTION 28: (Rotterdam - January 1988)

SC 4 wishes to accept the offer of a liaison with SC 24.



N4118

2026 – 05 – 15

RESOLUTION 29: (Tokyo - December 1988)

SC 4 approves the registration of STEP as an ISO Draft Proposal (DP) in accordance with ISO Directives Section 6.4.1.1. The DP comprises the following WG 1 documents:

N279 (Physical File Structure)
N280 (Mapping from EXPRESS to Physical File Structure)
N283 (Introduction, Scope and Definitions)
N285 (Test
Methods) N287
(EXPRESS)
N284 (IPIM) with the exception of the user-defined-entity (Chapter 4.4.1.13)

SC 4 directs its secretariat to circulate the DP for letter ballot according to ISO Directives Section 6.4.1.3 and 6.4.1.4. The ballot shall include voting on each clause and each section of Clause 4 of the DP. This letter ballot shall be open for three months from the date of circulation (6.4.2.1).

In addition to the DP, circulation shall include for information and commenting:

N286 (Classification Methods)
N288 (Annex D)
N289 (Annex E)

SC 4 also directs WG 1 to produce:

- A list of essential open issues related to the DP
- Guidelines for the requested comments, including a separate vote on the document(s) as described above.

These are to be circulated with and referenced in the DP to assist the review by the appropriate bodies.

RESOLUTION 30: (Tokyo - December 1988)

SC 4 establishes an editing committee to finalize the text of the Draft Proposal for submission as a Draft International Standard and maintain configuration control so that all future changes be documented and approved.

Each P-member is invited to nominate one representative to the SC 4 secretariat

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

by January 31, 1989.

The Editing Committee reports directly to the SC 4. Its tasks comprise:

- collecting and registering issues on the STEP DP
- collecting of comments
- assessing of comments as technical, editorial or irrelevant
- passing technical issues and comments to WG 1 for technical solution
- incorporating updates produced by WG 1
- editorial improvement of the standard
- conformance of the standard to ISO Directives and Drafting Rules.

All changes introduced by the Editing Committee must be approved by WG 1.

RESOLUTION 31: (Tokyo - December 1988)

SC 4 appoints Curtis Parks of NIST, USA to serve in the capacity of Chairman of the Editing Committee until that committee has convened and established their own chairman. The Chairman shall convene the Editing Committee no later than the next WG 1 meeting.

RESOLUTION 32: (Tokyo - December 1988)

SC 4 directs WG 1 to consider a logical subdivision of the STEP standard into an open set of consistent standards.

STEP shall be based on one single, complete, implementation-independent Product Information Model, which shall be the Master Record of the integrated topical and application information models. SC 4 recognizes that multiple implementation technologies exist and directs WG 1 to identify and harmonize support for those technologies.

Additional work plans to develop implementation-specific extensions must be approved by SC 4.

RESOLUTION 34: (Tokyo - December 1988)

SC 4 resolves that application profiles of STEP shall form separate parts of the STEP standard, in order to provide a basis for conformance testing.



N4118

2026 – 05 – 15

RESOLUTION 35: (Tokyo - December 1988)

SC 4 directs WG 1 to obtain SC 4 approval for the initiation of activities regarding the development of new topical and application models for STEP.

RESOLUTION 36: (Tokyo - December 1988)

SC 4 directs WG 1 to develop a topical model that deals with aspects of motion and kinematics as is required for the representation of products for which these aspects are essential (mechanisms, industrial robots, and others) as well as for the representation of assembly and disassembly operations.

RESOLUTION 37: (Tokyo - December 1988)

SC 4 directs WG 1 to develop a topical model that deals with aspects of product life cycle support. This model shall be added to the STEP Information Model.



N4118

2026 – 05 – 15

RESOLUTION 38: (Tokyo - December 1988)

SC 4 directs WG 1 to develop a topical model that deals with aspects of material properties with relationship to design, testing, and analysis of products, which shall be added to the STEP Information Model.

RESOLUTION 39: (Tokyo - December 1988)

SC 4 directs WG 1 to undertake a study of all existing initiatives relating to the use of CAD Standard Parts and to recommend any action required for international harmonisation and/or standardization.

RESOLUTION 40: (Tokyo - December 1988)

SC 4 asks its member bodies and liaisons to encourage implementation and validation of the STEP Draft Proposal or parts of it as soon as majority approval has been gained. This encouragement is to be directed to industries, universities, research organisations and organisations or projects that are funded explicitly for work related to STEP such as the US PDES Inc., the US NIST PDES Testbed, the European project CADEX, the German KCIM project and others. The member bodies are encouraged to identify these activities to the secretariat for purposes of harmonisation.

RESOLUTION 41: (Tokyo - December 1988)

SC 4 directs the secretariat to establish stronger liaisons with other international bodies, especially the appropriate bodies in the areas of AEC, FEM and Electrical/Electronics. Member bodies shall identify to the secretariat potential liaisons.

Furthermore, SC 4 directs the secretariat to prepare a package of technical documents with cover letters and send them to liaison organisations to encourage more active communication. Relevant papers for this package are to be sent to the secretariat.

RESOLUTION 42: (Tokyo - December 1988)

SC 4 directs that WG 1 develop a plan for the future development of STEP. The working plan is to be approved by SC 4 before October 1989 by mail ballot.



N4118

2026 – 05 – 15

RESOLUTION 43: (Tokyo - December 1988)

SC 4 and WG 1 would like to record a vote of thanks to Japan as host nation for the excellent facilities and support for the November 1988 meeting.

RESOLUTION 44: (Paris - January 1990)

ISO/TC 184/SC 4 invites and solicits proposals from IEC organisations and others for cooperative development and participative approval of the appropriate parts of STEP. In particular, we invite the representatives of EIA, VHDL, IPC D350, and EDIF to take part in this work through the exchange of reference models. This will follow the example of inter-committee cooperation set by the ISO and the IEC in the MMS project and companion standards.

RESOLUTION 45: (Paris - January 1990)

SC 4 respectfully recommends to the ISO/TC 184 chair that the title of the SC 4 Committee be changed to "Industrial Data and Global Manufacturing Programming Languages."

RESOLUTION 46: (Paris - January 1990)

Following the assignment of the accepted new work item on a standard parts library, ISO/TC 184/SC 4 establishes a new working group with the following title and task:

TITLE: Standard for the Neutral Representation of Standard Parts

SCOPE: The development of an international standard for the implementation of standard parts which is independent of any computer-aided design system. This should take into account the mapping of the external standard representations into the internal representations of a CAD system.

SC 4 accepts the offer of the member body of Germany, F.R. to undertake the convenership of this working group. Each "P" member of SC 4 is asked to nominate one expert and one alternate for the Working Group. Names shall be forwarded to the SC 4 chair by 1 April 1990.

RESOLUTION 47: (Paris - January 1990)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 affirms the formation of a new project within WG 1 to study the requirements and structure of libraries within STEP. The title of the project will be "Library Structures." Mr. P. Harrow (UK) and Mr. P. Dienst (FRG) are appointed as joint project leaders to ensure coordination of the results between this project and the Standard Parts effort.

RESOLUTION 48: (Paris - January 1990)

An advisory group called "Strategic Planning" is to be created to advise the SC 4 chair on matters concerning strategic planning and coordination of SC 4 activities. SC 4 accepts the offer of the member body of France to undertake the convenorship of this advisory group.

Terms of reference for the Strategic Planning Advisory Group are:

- a) Recommending goals and objectives for future work.
- b) Coordinating activities across the different areas of scope of SC 4.
- c) Recommending the general methodologies required to undertake SC 4 work.
- d) Acting as a liaison with other relevant bodies.

Each "P" member of SC 4 is asked to nominate one expert and one alternate for the Strategic Planning Advisory Group. Names shall be forwarded to the SC 4 chair by 1 April 1990.

RESOLUTION 49: (Paris - January 1990)

An SC 4 Project Management Advisory Group shall be established to provide project management and technical strategy for product data work items.

Terms of reference of the Project Management Advisory Group are:

- a) Set functional goals for the Product Data Definition Working Groups.
- b) Set project planning, work breakdown structure, and scheduling.
- c) Coordinate activities between Product Data Definition Working Groups.
- d) Set criteria for success.
- e) Evaluate quality.

Each "P" member of SC 4 is asked to nominate one expert and one alternate for the Advisory Group. Names shall be forwarded to the SC 4 chair by 1 April 1990.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

The SC 4 Secretariat shall ask each "P" member body for their interest in the convenorship.

RESOLUTION 50: (Paris - January 1990)

SC 4 resolves to produce a series of Technical Reports which document the present state of STEP technology. The contents of the report are subject to change. In particular, this effort requires:

- a) Identification by WG 1 subgroups of all Parts which are technically stable.
- b) Editorial work on Parts to ensure conformance to ISO style and incorporation of all ballot comments agreed to by Part owners.
- c) Balloting by SC 4 to register each Part as a Committee Draft (CD).
- d) Balloting by SC 4 P-members on each CD.
- e) SC 4 decision upon review of ballot results to publish as a Technical Report.

RESOLUTION 51: (Paris - January 1990)

SC 4 recommends the creation of a small ad hoc working group of SC 4 to select topics on which to develop application protocols. These topics should be circulated to national bodies for comment and prioritisation by 31 March 1990.

RESOLUTION 52: (Paris - January 1990)

In response to the request from ISO/TC 184, and in order to comply with the new ISO/IEC rules, SC 4 proposes to reorganise its work on STEP into the following Working Groups:

Resource Models	(ex WG 1
SG1) Application Reference Models	(ex WG 1 SG2)
Implementation Methods	(ex WG 1
SG3) Conformance & Testing	(ex WG 1 SG4)
Framework & Methodology	(ex WG 1 SG5/7)
Information Integration	(ex WG 1 SG6)

Under each Working Group, any required subdivision of the work shall be achieved by the definition of specific projects, with project leaders appointed by SC 4.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Member bodies are requested to review and comment on the proposal to the Secretariat by 31 March 1990 and provide proposals for the convenerships of the WGs.

An ad hoc group of SC 4 shall meet at the next WG 1 meeting to resolve any comments and prepare a formal proposal for balloting at the June meeting of SC 4.

RESOLUTION 53: (Paris - January 1990)

In response to the changing of scope, SC 4 is ready to welcome new work items, namely

- a) Industrial manufacturing management data
- b) Quality data
- c) Maintenance data

Members of SC 4 are solicited to send to the secretariat proposals for new work item definitions.

RESOLUTION 54: (Paris - January 1990)

The term "Application Profile" identified in SC 4 Resolution 34 (Tokyo #6) is to be replaced by the term "Application Protocol."

RESOLUTION 55: (Paris - January 1990)

SC 4 adopts the general strategy for breaking the DP 10303 (SC 4 N38) into a number of smaller volumes, each to be processed separately through approval of a Committee Draft and then to be balloted upon as a set of qualified and integrated documents representing a Draft International Standard.

RESOLUTION 56: (Paris - January 1990)

It is resolved that SC 4 shall create new versions of STEP where a version is defined as a specific, dated collection of Committee Draft parts that are advanced simultaneously to DIS.

RESOLUTION 57: (Paris - January 1990)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

The current ISO/STEP parts division and numbering scheme shall be considered tentative until a final recommendation is received from WG 1. For timely distribution to SC 4, this document must be received by the SC 4 chair by 1 March 1990.

RESOLUTION 58: (Paris - January 1990)

The development and integration of information models for draughting is a critical item in the STEP development schedule. SC 4 encourages member bodies to consider increasing their participation in this work to match the stated needs of their industries.

RESOLUTION 59: (Paris - January 1990)

SC 4 wishes to thank France for hosting the meeting of the committee this week.

RESOLUTION 60: (Gothenburg - June 1990)

SC 4 charges the Strategic Planning Advisory Group with carrying out the intent of SC 4 Resolution 44 (Electrical Harmonization), particularly working toward a resolution of the developing situation with regard to the EIA petition to IEC (dated April 12, 1990). (USA)

RESOLUTION 61: (Gothenburg - June 1990)

SC 4 asks its electrical project leader to develop one or more electrical schematic application protocols based upon documented efforts such as the German VNS, the French SET electrical, the IEC/TC 3, and the U.S. IGES and EDIF electrical projects.

RESOLUTION 62: (Gothenburg - June 1990)

SC 4 adopts the following recommendation of the ad hoc Application Protocols group:

Following the WG 1 decision that STEP Version 1.0 must include at least one draughting related AP, the SC 4 ad hoc group on Apes recommends that STEP Version 1.0 should include the proposed AP #3,

"Exchange of 2D geometrically explicit CAD drawings with explicit annotation."



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**


Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026-05-15

Proposed AP #3 is the top priority for STEP Version 1.0. The proposed APs #4, 5, 6 & 18 should be reviewed by the end of October 1990 to assess their adequacy. These APs could be included in STEP Version 1.0 if they do not require extensive SC 4 resources for completion. None of these APs shall be allowed to delay the delivery of STEP Version 1.0.

NB.: PMAG must assign the responsibility for assessing the adequacy of the proposed APs to the new WG, "Qualification & Integration."

- | | |
|--------|---|
| AP # 4 | Exchange of CAD drawings with reference to 3D geometry model and with explicit annotation |
| AP # 5 | Exchange of configuration controlled 3D product definition data |
| AP # 6 | Exchange of sculptured surface models |
| AP #18 | Exchange of boundary representation models |



N4118

2026 – 05 – 15

RESOLUTION 63: (Gothenburg - June 1990)

SC 4 adopts the following recommendation of the ad hoc Application Protocols group:

In order to ensure the timely delivery of APs, a project leader must be assigned for each of the APs listed in Resolution # 62. The project leaders will report directly to the PMAG.

SC 4 appoints Bruno Schilli as project leader for AP #3.
SC 4 appoints Steve Ryan as project leader for AP #5.
SC 4 appoints Ray Goult as project leader for AP #18.

The PMAG will appoint project leaders for APs #4 and #6 as soon as possible by the PMAG, to be confirmed by SC 4.

RESOLUTION 64: (Gothenburg - June 1990)

As well as the need for normative APs, SC 4 recognizes that there exists an industrial requirement for additional APs to be defined outside the SC 4 organization at the industry, national, or regional level. SC 4 requests the PMAG to define requirements for guidelines to permit the development of APs both inside and outside SC 4. Provision should also be made for the registration of APs with SC 4 to ensure the consistency of related standards.

RESOLUTION 65: (Gothenburg - June 1990)

SC 4 adopts the reorganization structure as proposed by the ad hoc reorganization group in Reston. The current WG 1 is to be disbanded, effective at the end of the Gothenburg meeting, and its current work is to be allocated to the following new working groups:

WG 3 Product Modeling
WG 4 Qualification & Integration WG 5
STEP Development Methods
WG 6 Conformance Testing
Procedures WG 7 Implementation
Specifications



N4118

2026 – 05 – 15

RESOLUTION 66: (Gothenburg - June 1990)

SC 4 thanks the WG 1 convener, Mr Jerry Weiss, for his pioneering leadership and personal dedication to the STEP development over the last six years, and looks forward to his continuing support as Project Management Advisory Group convener in the future.

RESOLUTION 67: (Gothenburg - June 1990)

SC 4 appoints the following working group conveners:

WG 3 Product Modeling: Barbara Warthen (USA)
WG 4 Qualification & Integration: Yuhwei Yang (USA)
WG 5 STEP Development Methods: Bill Danner (USA)
WG 6 Conformance Testing Procedures: Jon Owen (UK)
WG 7 Implementation Specifications: Jan van Maanen (UK)

RESOLUTION 68: (Gothenburg - June 1990)

SC 4 resolves that STEP Version 1.0 comprises the following parts:

Overview	(Part 1)
Express	(Part 11)
Physical File	(Part 21)
Conformance Testing	(Part 31)
Generic Product Data Model	(Part 41)
Shape Representation	(Part 42)
Presentation	(Part 46)
Drafting	(Part 101)

and one or more Application Protocols as per Resolution # 62.

Additional parts may be considered for Version 1.0; however, no additional part will be included if its inclusion will result in a schedule slippage.

Moreover, if there is slippage in the adopted schedule of any part in the above list, then SC 4 may vote to defer that part to a future version of STEP.

SC 4 recognizes the importance of both the documentation architecture and the

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

schema framework, and directs the PMAG to ensure their inclusion in STEP Version 1.0.

RESOLUTION 69: (Gothenburg - June 1990)

SC 4 directs the project leaders who are responsible for the development of the following STEP parts to complete their documents both technically and editorially, for CD ballot by December 1990.

Overview	(Part 1)
Physical File	(Part 21)
Conformance Testing	(Part 31)
Shape Representation	(Part 42)

SC 4 directs the project leaders who are responsible for the development of the following STEP parts to complete their documents both technically and editorially, as early as possible in 1991 for CD ballot, completing them no later than July 1991.

Generic Product Data Model	(Part 41)
Presentation	(Part 46)
Drafting	(Part 101)
AP #3	(Exchange of 2D geometrically explicit CAD drawings with explicit annotation)

RESOLUTION 69A: (Gothenburg - June 1990)¹

The dates stated in Resolution #69 shall be reconfirmed by letter ballot by SC 4 by 1 August, 1990, upon reconsideration of a schedule and plan to be submitted by the PMAG.

RESOLUTION 70: (Gothenburg - June 1990)

SC 4 directs PMAG to organize the work of the Working Groups in order to achieve the goals in Resolutions # 68 and # 69.

¹ This resolution was originally passed and designated as Resolution 70: (Gothenburg #11, June 1990). During the transition of the Secretary from one leader to the next (1995), the list of resolutions inherited by the new secretary was missing this particular resolution. As of 1997-03-18, the current secretary was given an old hard copy of early resolutions and was able to identify the dropped resolution. However, because the current numbering of resolutions has proliferated into many documents, the secretary deemed it best to insert the original resolution



N4118

2026 – 05 – 15

#70 as #69a.

RESOLUTION 71: (Gothenburg - June 1990)

SC 4 welcomes Part 11 EXPRESS (WG 1 N496) as the first CD of the STEP work item.

RESOLUTION 72: (Gothenburg - June 1990)

Whereas the new ISO/IEC procedures in 2.4.1 allow the circulation of documents for comment or for voting by P members on their approval as Committee Drafts (CDs), CDs circulated within SC 4 shall only be circulated for ballot.

RESOLUTION 73: (Gothenburg - June 1990)

SC 4 moves to accept for ballot a new work item on industrial manufacturing management data (SC 4 N60, submitted by France).

RESOLUTION 74: (Gothenburg - June 1990)

Terms of reference item "c" for the PMAG that is defined in SC 4 Resolution # 49 shall be changed so that it shall read, "coordinate activities between Product Data Definition Working Groups and the Editing Committee."

RESOLUTION 75: (Gothenburg - June 1990)

SC 4 establishes an Editing Committee to finalise the texts of the Parts of the STEP standard for submission as a Draft International Standard and maintain configuration control so that all future changes be documented and approved.

Membership of the Editing Committee is by nomination; each "P" member may nominate one representative to the Editing Committee. Additional members may be co-opted in order to achieve the objectives.

The Editing Committee reports directly to SC 4. Its tasks comprise:

Collecting and registering issues resulting from ballots and reviews of STEP Parts

Collection of comments



N4118

2026 – 05 – 15

Assessing of comments as technical, editorial or irrelevant

Passing technical issues and comments to the SC 4 Working Groups for technical solution

Incorporating updates produced by the SC 4 Working Groups

Making editorial improvements to the standard

Ensuring technical coherence among Parts of the standard

Ensuring conformance of the standard to ISO Directives and Drafting Rules

This committee will supersede the Editing Committee established by Resolution # 18 of SC 4 at the Tokyo meeting.

RESOLUTION 76: (Gothenburg - June 1990)

SC 4 wishes to thank Sweden for hosting the SC 4 meeting in Gothenburg.

RESOLUTION 77: (Sapporo - July 1991)

SC 4 wishes to thank the conveners, project leaders and the many technical experts who have given extraordinarily of themselves during the past years in our common effort to develop STEP version one.

RESOLUTION 78: (Sapporo - July 1991)

SC 4 directs all SC 4 working groups to use the methodology and tools developed for STEP, unless proven not suitable.

RESOLUTION 79: (Sapporo - July 1991)

SC 4 instructs PMAG to plan the application of adequate resources for integration, qualification, data interpretation and development of conformance testing material through the lifecycle of each project, as part of the project approval process.

RESOLUTION 80: (Sapporo - July 1991)

In order to facilitate the STEP development process, SC 4 asks each P Member that is

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

involved in the development of STEP application protocols and/or resource models to designate at least one representative to receive training in and to participate actively in the following:

- integration
- qualification
- editing
- interpretation
- conformance testing procedures

SC 4 encourages externally funded programmes and projects to provide and/or sponsor these representatives. SC 4 requests its Secretariat to invite P Members to nominate representatives to WG 4 to participate in the qualification and integration of resource models and application protocols.

RESOLUTION 82: (Sapporo - July 1991)

SC 4 requests PMAG to ensure the completion of an integrated set of methods and procedures for integration, qualification, data interpretation and development of abstract test suites, together with associated documentation and training materials, as soon as possible.

RESOLUTION 83: (Sapporo - July 1991)

SC 4 requests its Secretariat to invite P Members to nominate representatives to WG 5 to participate in the continued development of ISO 10303 methods.

RESOLUTION 84: (Sapporo - July 1991)

SC 4 approves the principle of setting up formal procedures within its organisation. SC 4 welcomes its Chairman's proposals for the following procedures:

1. SC 4 procedures for the approval of STEP
2. responsibilities of a WG convener
3. responsibilities of a project leader
4. SC 4 documentation approval levels
5. STEP part titles

SC 4 directs SPAG to review these proposed documents and to analyse their consistency with respect to the SC 4 methodology before SC 4 approves them.



N4118

2026 – 05 – 15

SC 4 directs PMAG to assess the adequacy of the procedures against their operational requirements.

Meanwhile, and in order to fulfil the ongoing work, SC 4 decides to use these new documents to approve the STEP parts, whilst SPAG solicits comments regarding their adequacy.

RESOLUTION 85: (Sapporo - July 1991)

Following the assignment of the accepted new work item on Industrial Manufacturing Management Data (N60), ISO/TC 184/SC 4 establishes a new working group (WG 8) with the following title and scope:

Title: Industrial Manufacturing Management Data

Scope: Method and development to standardise data which express information exchanged inside industrial manufacturing plants, except for product definition data

SC 4 accepts the offer of the member body of France to undertake the convenership of this WG. SC 4 appoints Mr. Albert Colin as convener of WG 8.

RESOLUTION 86: (Sapporo - July 1991)

Following the creation of a WG 8, SC 4 recommends that the following three projects be part of the new work item on Industrial Manufacturing Management Data.

To develop an international standard dealing with model, form and attributes of data exchanged between an industrial manufacturing company and its environment of manufacturing management activities.

To develop an international standard dealing with model, form and attributes of data able to reside in an industrial manufacturing company's resources database.

To develop an international standard dealing with model, form and attributes of data controlling and monitoring the flow of materials within an industrial manufacturing company.

SC 4 charges the WG 8 convener to recommend, to the SC 4 secretariat, candidates

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

for project leaders. SC 4 encourages as many experts as needed to participate in these projects. Names shall be forwarded to the SC 4 secretariat by 1st September 1991.

RESOLUTION 87: (Sapporo - July 1991)

SC 4 welcomes the proposal from IEC/TC 3 to work jointly on the topic of electrical and electronic product data standards and establishes a joint working group (JWG 9) with the following title and scope:

Title: IEC/TC 3 - ISO/TC 184/SC 4 joint working group for electrical and electronic applications of ISO 10303.

Scope: Develop resource models and application protocols using STEP methodologies taking into account documented efforts such as EDIF, IGES, SET and VNS.

SC 4 offers the convenership of JWG 9 to IEC. The current activities of the ISO/TC 184/SC 4/WG 3 electrical project will be transferred to JWG 9.

Care must be taken to ensure that the work can continuously proceed without losing the results achieved so far.

SC 4 accepts as a new work item proposal the work on STEP Data Access Interface described in SC 4 N71.

RESOLUTION 89: (Sapporo - July 1991)

SC 4 modifies the terms of reference for the SPAG (SC 4 resolution 48) as

follows: replace: "acting as a liaison with relevant bodies"

with: "acting as a strategic liaison with the relevant bodies and coordinating other liaisons at the working group level"

RESOLUTION 90: (Sapporo - July 1991)

The Strategic Planning Advisory Group is given the additional responsibility for:

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Reviewing any new work item proposal to be presented to SC 4.

Examining how each new work item proposal will fit within the SC 4 strategic objectives and other standards work.

RESOLUTION 91: (Sapporo - July 1991)

SC 4 asks that the SPAG review existing materials with the intent to select or create an enterprise taxonomy. The deliverable from this task would provide an approach for SC 4 to use, as a tool, to account for and to manage the development of standards within the enterprise-data domain so that overlaps and gaps in SC 4 standard coverage are identified.

RESOLUTION 92: (Sapporo - July 1991)

SC 4 sends out for mail ballot the proposed strategic policy statement developed by the SPAG in document ISO/TC 184/SC 4/SPAG N27.

RESOLUTION 93: (Sapporo - July 1991)

SC 4 appoints Mr. Jean-Pierre Letouzey as chairman of the Strategic Planning Advisory Group.

RESOLUTION 94: (Sapporo - July 1991)

SC 4 appoints Mr. Jerry Weiss as chairman of the Project Management Advisory Group.

RESOLUTION 95: (Sapporo - July 1991)

SC 4 moves to accept the offer of a liaison with ISO/IEC JTC 1/SC 21/WG 3 and appoints Mr Augusto Nieva (Italy) to serve in this role.

RESOLUTION 96: (Sapporo - July 1991)

SC 4 encourages and welcomes liaison with any external body which meets the ISO criteria and is developing STEP application protocols.

RESOLUTION 97: (Sapporo - July 1991)

SC 4 asks all P Members to assess their national requirements for future editions of

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ISO 10303. An initial list is requested by 1 Oct 91 to be sent to SPAG.

RESOLUTION 98: (Sapporo - July 1991)

SC 4 directs its Secretariat to establish an Editing Committee for ISO 10303 in accordance with the ISO Directives. In addition to assistance in the preparation of texts, consistent between themselves and with the ISO directives, the Editing Committee will provide review for technical coherence across texts.

The membership of the Editing Committee will comprise a chairman and all project leaders or their representatives responsible for text preparation. Each SC 4 P Member may nominate additional members. Other members may be co-opted by the Chairman in order to achieve committee objectives. This resolution supersedes SC 4 Resolution #76.

SC 4 directs its Secretariat to appoint Mr. Nigel Shaw as chairman of the Editing Committee for ISO 10303.

RESOLUTION 99: (Sapporo - July 1991)

It is SC 4 policy that the review period for first Committee Drafts of STEP Parts should be not less than four months.

RESOLUTION 100: (Sapporo - July 1991)

SC 4 wishes to thank the Japan member body and NICOGRAPH for their hospitality and for the quality of their meeting arrangements during the SC 4 session in Sapporo.

RESOLUTION 100: (Oslo - February 1992) - SPAG Resolution S03

SC 4 recommends the use of the term "initial release of STEP" instead of "STEP version 1".

RESOLUTION 101: (Oslo - February 1992) - SPAG Resolution S08

To fulfil objective # b6 of the SC 4 Strategic Policy and to embed consistency in the SC 4 procedures proposed in SC 4 Resolution #84 as well as in the SC 4 policies, SC 4 adopts SPAG N75 as the information model of the main entities used within SC 4 according to the ISO directives.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Objective #b6 reads: Assure that SC 4 participants understand the basic information and methods concerning SC 4 work.

RESOLUTION 102: (Oslo - February 1992) - SPAG Resolution S09

To fulfil the objective # c5 of the SC 4 Strategic Policy and the SC 4 Resolution #90, SC 4 adopts the SPAG document N64 (version 1.6, as amended with respect to confirmation of project leaders) as the policy to manage and approve the content of the SC 4 programme of work.

Objective #c5 reads: Refine the definition of present and future SC 4 areas of scope, work items and projects. Identify their relationships.

RESOLUTION 103: (Oslo - February 1992) - SPAG Resolution S10

To fulfil the objective # c5 of the SC 4 Strategic Policy, SC 4 adopts SPAG N59 as the base line approved list of SC 4 areas of scope, work items and projects existing on 1992-02-07.

Objective #c5 reads: Refine the definition of present and future SC 4 areas of scope, work items and projects. Identify their relationships.

RESOLUTION 104: (Oslo - February 1992) - SPAG Resolution S11

Following SC 4 Resolution #84, SC 4 adopts policy #4 of SC 4 document N85 on "SC 4 documentation approval levels" as GP#02 to approve SC 4 parts documents as well as other SC 4 technical documents relevant to the SC 4 standards.

RESOLUTION 105: (Oslo - February 1992) - SPAG Resolution S12

Following SC 4 Resolution #84, SC 4 directs its secretariat to maintain a dated list of Parts to be developed or under development for each SC 4 standard and to provide these lists to members of SC 4 at each meeting.

RESOLUTION 106: (Oslo - February 1992) - SPAG Resolution S13

SC 4 directs its secretariat to maintain a dated list of adopted SC 4 projects and of the corresponding appointed project leaders and to provide this list its members at each meeting.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 107: (Oslo - February 1992) - SPAG Resolution S14

SC 4 directs its secretariat to review all existing liaisons and to ensure that they have been formally registered by the Central Secretariat. Any liaisons not so registered shall be established by the Chief Executive Officer.

RESOLUTION 108: (Oslo - February 1992) - SPAG Resolution S15

SC 4 directs WG 8 and WG 7 to review the available documents on DCG and work together to propose an expert to be appointed by SC 4 to serve in the role of liaison with the DCG group.

RESOLUTION 109: (Oslo - February 1992) - SPAG Resolution S16

SC 4 adopts the policy that it shall endeavor to share and coordinate resources and results with any standardization effort outside SC 4 that is applicable to the SC 4 scope.

RESOLUTION 110: (Oslo - February 1992) - SPAG Resolution S17

SC 4 directs WG 3 to propose an expert to be appointed by SC 4 to liaise with the ISO/TC 10/SC 5 in the areas of draughting and tolerances.

RESOLUTION 111: (Oslo - February 1992) - SPAG Resolution S18

SC 4 directs its secretariat to draw the attention of the ISO Technical Board to apparent overlaps between the scopes of SC 4 and other ISO/TCs and SCs, and to request confirmation that the scope of SC 4 covers all aspects of product data representation and exchange in digital form within its current work items.

RESOLUTION 112: (Oslo - February 1992) - SPAG Resolution S19

SC 4 considers that the NWI on Software Product Data falls in the SC 4 Area of Scope A on Product Data.

RESOLUTION 113: (Oslo - February 1992) (SPAG Resolution S04)

SC 4 adopts the following six projects:



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

 Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

(PWA/PWB)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

P41 Mechanical Design Using Wireframe, STEP Part 206 P42 Sheet
Metal Dies and Blocks, STEP Part 207
P43 Life Cycle Product Change Process, STEP Part 208 P44
Electrotechnical Plants
P45 Electronic Printed Circuit Assembly, Design and Manufacture
P46 Electronics Test, Diagnostics and Remanufacture (PCB/LRM)

and appoints respectively as Project Leaders for P41-P46: Jon Aas, Jack Skeels,
Rick Bsharah, R. Nerke, J. Corley, and W. Russell.

RESOLUTION 114: (Oslo - February 1992)

SC 4 adopts the project Design Through Analysis of Composite and Metallic
Structures and appoints Mr. Keith Huntten as project leader.

RESOLUTION 115: (Oslo - February 1992) - SPAG Resolution S01

SC 4 directs its secretariat to circulate SPAG document N58 and N75 for P-member
comment by mid-June 1992

RESOLUTION 116: (Oslo - February 1992) - SPAG Resolution S07

With respect to the upward compatibility of EXPRESS, SC 4 directs WG 5 to consider
the document SC 4/WG 2 N43 and to report to SC 4 at the next plenary meeting how
the WG 2 requirement for Part Libraries can be satisfied.

RESOLUTION 117: (Oslo - February 1992) - PMAG Resolution P01a

SC 4 accepts the plan and schedule proposed by PMAG for the registration of the
initial release of STEP as DISs by 1 January 1993.

RESOLUTION 118: (Oslo - February 1992) - PMAG Resolution P01b

SC 4 encourages extra national effort to support the plan for the registration of the
initial release of STEP as DISs by 1 January 1993. Extra effort includes extra
meetings for integration, qualification, editing and for key project persons. Extra effort
also includes early return of ballot comments as shown on the PMAG schedule.

RESOLUTION 119: (Oslo - February 1992) - PMAG Resolution P02

SC 4 requests position papers for consideration by PMAG on "Requirements for



N4118

2026 – 05 – 15

coordinating and qualifying APs developed externally to SC 4".

RESOLUTION 120: (Oslo - February 1992)

SC 4 requests P-members to consider their PMAG representative assignments to assure that nominated members and alternates are available to participate regularly;

RESOLUTION 121: (Oslo - February 1992)

A common understanding of what application protocol interoperability means has been developed in an ad hoc group between April 1991 and January 1992. Based on this information, SC 4 directs its WG 4, 5, 6 & 7 to consider the development of methods and techniques to achieve application protocol interoperability and their implementation as a key priority requirement for STEP beyond the initial release.

RESOLUTION 122: (Oslo - February 1992)

SC 4 instructs its Secretariat to transmit the revised final CD 10303-11 being approved for publication as ISO/DIS directly to the ISO Central Secretariat for registration and publication as a Draft International Standard without further balloting, subject to consensus on the resolution of outstanding issues. SC 4 recognizes that these actions are fully in accord with clause 2.4.6 of ISO Directives Part 1 for the technical work.

RESOLUTION 123: (Oslo - February 1992)

SC 4 instructs its Secretariat to transmit the revised final CD 10303-21, without further balloting subject to consensus on resolution of outstanding issues, being approved for publication as ISO/DIS directly to the ISO Central Secretariat for registration and publication as a Draft International Standard.

RESOLUTION 124: (Dallas - October 1992)

SC 4 notes the consensus position reached by PMAG and WG experts on the issues relating to Initial Release implementation. This represents a solution to a number of potential ballot comments which may arise during the current CD ballot. The following represent the proposed solution:

Part 21:

ANDOR Supertype/Subtype Structures

- Instances based on AND will use external mapping.
- Other instances will use internal mapping.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

 Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

Entity names within the external mapping structure will be ordered alphabetically.

Part 41:

The Measures Schema recommended by WG 4 will be adopted.

Parts 42 and 46:

Other Parts will conform with the new Measures Schema.

Part 203:

Annex D of the current CD release will be removed

A new informative annex will be added (Annex J)

consensus:



N4118

2026 – 05 – 15

Text for the new informative annex should express the following

"This annex provides an example instantiation data structure that may help implementors understand how the Application Interpreted Model of Annex A might be restructured for internal implementation purposes."

"The data structure contained in this annex is not intended to be used as the required schema for the clear text encoding as defined in ISO 10303-21."

AP Guidelines and Supplementary Directives:

The informative annex (Annex J) as previously defined will be described as an optional annex for all APs.

RESOLUTION 125: (Dallas - October 1992)

SC 4 confirms the CD ballot target dates for existing STEP documents. The resulting comments, will be addressed at the workshop scheduled for 14 - 18 December 1992 at NIST. SC 4 instructs its secretariat to circulate the integrated set of consistent documents to P-members by 15 January 1993. The final decision to register the documents as DIS will be taken by SC 4 at the Torino meeting.

RESOLUTION 126: (Dallas - October 1992)

The Initial Release of ISO 10303 shall allow referencing mechanisms to a part in a standard part library using names and attributes. Such mechanisms shall allow the exchange through a STEP file of that information which is necessary to generate the same part in the receiving system when both sending and receiving systems contain the same part library. (SPAG 9)

RESOLUTION 127: (Dallas - October 1992)

Because of the likelihood of changes required to maintain overall consistency, ISO 10303-21 will be circulated to P-members on or before 15 January 1993, so that it can be included in the final decision on the registration of the Initial Release of STEP.

RESOLUTION 128: (Dallas - October 1992) - SPAG Resolution S11

SC 4 requests its secretariat to clearly and explicitly describe the expectations for the

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Initial Release of STEP as the basis for national bodies to prepare their own press releases.

RESOLUTION 129: (Dallas - October 1992)

SC 4 directs its Working Groups WG 4 and WG 5 to complete all documents required for the use of the STEP methodologies, and to submit them for approval as ISO Type 3 Technical Reports.

RESOLUTION 130: (Dallas - October 1992)

SC 4 requests member bodies to identify and provide qualified people with adequate long-term commitment and funding to support the critical central functions of qualification, integration, editing, training material development, training and configuration management. In addition, SC 4 requests PMAG to identify to member bodies the requirements for support for these functions.

RESOLUTION 131: (Dallas - October 1992)

SC 4 instructs its secretariat to send out for mail ballot the proposed strategic policy statement, developed by SPAG and WG 2 in the document SPAG N104 V1.4.

RESOLUTION 132: (Dallas - October 1992)

ISO/TC 184/SC 4 asks the ISO/TC 184/SC 4 Secretariat to make sure that an ASCII source listing will be available, containing an explicit representation of all characters of the CD text to be balloted represented as ASCII characters, without any formatting information or figures.

A 3.5" MS-DOS disk formatted at 720 K containing a complete 7 bit ASCII text must then be an integral part of any package for which a CD ballot is requested.

RESOLUTION 133: (Dallas - October 1992)

For each STEP Part on the NIST SOLIS, ISO/TC 184/SC 4 asks that the following forms be available, consistent and up to date:

Master copy in LaTeX or WordPerfect.

One derived copy containing a complete ASCII text of the STEP Part, containing as much text as possible.

One derived copy containing a POSTSCRIPT copy of the STEP Part, including

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

the cover sheet.

RESOLUTION 134: (Dallas - October 1992)

SC 4 asks its secretariat to register the work of WG 2 with ISO as a new standards project, different from 10303.

RESOLUTION 135: (Dallas - October 1992)

SC 4 requests its secretariat to ensure that the schedule for the principal meetings of the Working and Advisory Groups of SC 4 shall be determined at least one year in advance of such meetings.

RESOLUTION 136: (Dallas - October 1992)

At least eight weeks prior to each meeting of ISO/TC 184/SC 4, a memorandum shall be sent to each "P" member country indicating all issues, documents, projects, etc. for which votes are expected at that meeting.

RESOLUTION 137: (Dallas - October 1992)

SC 4 encourages WG 3/T8 (Product Life Cycle Support) and WG 8 (Industrial Manufacturing Management) to cover the domain of maintenance data. WG 3/T8 will particularly address maintenance support data. WG 8 will particularly address maintenance management data. The specific case related to diagnostics and resource status has to be solved in common. (WG 3)

RESOLUTION 138: (Dallas - October 1992)

SC 4 directs its Secretariat to send to the Central Secretariats of ISO and IEC, to the ISO/CEO, and to the secretariat of IEC/TC 93, the memorandum SC 4 N166 (SPAG N106) as a proposal of shared guidelines to be used as the basis for common agreements between ISO and IEC to achieve the cooperation required by SCIA in their decision #83.

RESOLUTION 139: (Dallas - October 1992)

SC 4 directs its Secretariat to respond to the IEC/TC 3 questionnaire IEC 5/1992 by sending back the SC 4 terms of reference containing the updated list of projects and the programme of work of JWG 9.



N4118

2026 – 05 – 15

RESOLUTION 140: (Dallas - October 1992)

SC 4 recognises the creation of IEC/TC 93 and desires to fully support the activities of IEC/TC 93 in the area of STEP AP development for design automation. SC 4 appoints Mr. Fritz Reuter and Mr. Pat Harrow to represent SC 4 at the first meeting of IEC/TC 93 in Paris, on 29 - 30 October 1992.

RESOLUTION 141: (Dallas - October 1992)

To fulfil the objectives #c2 and c3 of the SC 4 strategic policy (SC 4 N128), SC 4 adopts the document SC 4 N165 as the policy to manage the evolution of SC 4 standards and their parts. This document has been balloted with a positive result, and comments have been reviewed and resolved.

RESOLUTION 142: (Dallas - October 1992)

SC 4 directs PMAG to define the requirements for the configuration control to manage the evolution of STEP as defined in SC 4 N165. SC 4 directs PMAG will also define proposals of alternative procedures for meeting those requirements to fulfil the SC 4 policy adopted in Resolution #141. SC 4 directs PMAG to report these to SC 4 at least by mid 1993 for final decision.

RESOLUTION 143: (Dallas - October 1992)

To fulfil the objective #b6 of the SC 4 Strategic Policy (SC 4 N128) and to embed consistency in the SC 4 procedures proposed in SC 4 Resolution #83, SC 4 adopts the document SC 4 N164 as the information model of the main elements used within SC 4 according to the ISO directives.

RESOLUTION 144: (Dallas - October 1992)

SC 4 directs its secretariat to circulate the working paper SPAG N93 on "Global Programming Language for Manufacturing Applications" to P-member bodies, requesting national comments, priorities, and recommendations on whether SC 4 should or should not keep the area of scope within its program of work.

RESOLUTION 145: (Dallas - October 1992)

To ensure that incoherencies, gaps, and overlaps are recognized and eliminated, SC 4 recognizes the need for liaison between the business information domain covered by EDIFACT and other related standards, and the technical information domain

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

covered by SC 4 standards. SC 4 directs WG 8 to establish liaisons between SC 4 and the following bodies:

United Nations Trade
Division ISO/IEC JTC
1/WG 3 ISO/IEC JTC
1/SC 14

RESOLUTION 146: (Dallas - October 1992)

SC 4 directs its secretariat and SPAG to examine the feasibility of organizing during 1993 appropriate mechanisms to identify and structure requirements for STEP capabilities that will be needed during the next five years.

RESOLUTION 147: (Dallas - October 1992)

To fulfil objective c1 of the SC 4 strategic policy (SC 4 N128), SC 4 requests that the P02 (EXPRESS Language) project leader:

present to SPAG in Torino, Italy (February 1993 meeting) the status and the content of its requirement analysis for EXPRESS.

elaborate a strategy for future EXPRESS development, and to present it to SPAG for validation by July 1993. This strategy should include the context, the background, the main objectives on a mid-term and long-term basis, assessment of any alternative options, and the analysis of major risks. (SPAG 4)

RESOLUTION 148: (Dallas - October 1992)

SC 4 requests the P35 project leader, Mr. Mark Palmer, along with the participation of WG 4, WG 5, Editing Committee, interested members of PMAG, and experts within SC 4, to present their point of view and to make a proposal to SPAG and PMAG at their

February 1993 meeting in Torino, Italy. The point of view and proposal will address the topics of a reference model of SC 4 standards, concepts of "interoperability" and "companion standards," and requirements for coordinating and qualifying APs developed outside SC 4.

RESOLUTION 149: (Dallas - October 1992)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 adopts project P54 "Automotive Mechanical Design AP," and appoints Mr. Juergen Mohrmann as the project leader.

RESOLUTION 150: (Dallas - October 1992)

Following the resolution SC 4 #108, SC 4 appoints Mr. Albert Colin as the liaison with the ISO/TAG 7 / Data Element Coordination Group.

RESOLUTION 151: (Dallas - October 1992)

SC 4 appoints Ms. Sheila Lewis (UK) as convener of WG 6, in succession to Dr. Jon Owen, with the appointment to be effective upon the resignation of Dr. Owen. SC 4 expresses its thanks to Dr. Owen for his efforts in developing the framework for conformance testing for STEP.

RESOLUTION 152: (Dallas - October 1992)

Following the ISO/TC 184 resolution #153, SC 4 appoints Mr. Rainer Glatz as liaison with ISO/TC 184/WG 5.

RESOLUTION 153: (Dallas - October 1992)

Following SC 4 Resolution #110 (Oslo), Mr. Martin Holland is proposed by WG 3 as the liaison with ISO/TC 10/SC 5. SC 4 appoints him as a liaison.

RESOLUTION 154: (Dallas - October 1992)

SC 4 appoints Mr. Cliff Borchert as leader of project P29 on External Exchanges.

RESOLUTION 155: (Dallas - October 1992)

SC 4 appoints Prof. Wilhelm Dangelmaier as leader of project P30 on Material Flow Control.

RESOLUTION 156: (Dallas - October 1992)

SC 4 appoints Mr. Massimo Actis Dato as liaison with JTC 1/SGFS

RESOLUTION 157: (Dallas - October 1992)



N4118

2026 – 05 – 15

SC 4 requests PMAG to appoint a liaison with ISO/TC 172/SC 1 on optical systems.

RESOLUTION 158: (Dallas - October 1992)

SC 4 appoints Tom Baker as liaison with ISO/IEC JTC 1/SC 7 on software engineering.

RESOLUTION 159: (Dallas - October 1992)

SC 4 commends the Initial Release team and all the international experts who have contributed to the completion of the Parts in accordance with the schedule which was set by SC 4.

RESOLUTION 160: (Dallas - October 1992)

SC 4 thanks the U.S. for hosting this meeting of SC 4.

RESOLUTION 161: (Turin - February 1993)

SC 4 requests its Secretariat to register Parts 1, 21, 41, 42, 43, 44, 101 and 203 of ISO 10303 as Draft International Standards, following the incorporation of outstanding changes agreed at the Turin meeting.

RESOLUTION 162: (Turin - February 1993)

SC 4 requests its Secretariat to register Part 46 of ISO 10303 as a Draft International Standard, subject to the following conditions being satisfied by the Project Leader:

- a. The resolution of all outstanding issues from the second CD ballot with regard to Integration, Qualification and Draughting are documented in the issues log and circulated to member bodies.
- b. The results of these resolutions are incorporated in Part 46
- c. The processes of Qualification and Editing are completed and the results incorporated in Part 46
- d. The groups responsible for Qualification, Integration and Editing and the Project Leader shall sign off the final version of the document.

Member bodies will confirm that Part 46 (WG 3 N218) successfully resolves all outstanding ballot issues by the end of March 1993. The SC 4 Secretariat is authorized to assume that lack of response may be interpreted as confirmation.



N4118

2026 – 05 – 15

An electronic form of the document shall accompany the circulation.

RESOLUTION 163: (Turin - February 1993)

SC 4 requests its Secretariat to register Part 201 of ISO 10303 as a Draft International Standard, subject to the following conditions being satisfied by the Project Leader:

- a. The resolution of all outstanding issues from the second CD ballot are documented in the issues log and circulated to member bodies.
- b. The results of these resolutions are incorporated in Part 201
- c. The processes of Qualification, Integration and Editing are completed and the results incorporated in Part 201
- d. The groups responsible for Qualification, Integration and Editing and the Project Leader shall sign off the final version of the document.

Member bodies will confirm that the final version of Part 201 successfully resolves all outstanding ballot issues within one month of circulation by the SC 4 Secretariat. The SC 4 Secretariat is authorized to assume that lack of response may be interpreted as confirmation.

An electronic form of the document shall accompany the circulation.

RESOLUTION 164: (Turin - February 1993)

SC 4 recognizes John Rumble and his Initial Release team for their extraordinary dedication and contribution to the success of STEP.

RESOLUTION 165: (Turin - February 1993)

SC 4 resolves that, for all Part documents, the Project Leader shall confirm that the result of Qualification/Editing/Integration still meets the requirements of the project.

RESOLUTION 166: (Turin - February 1993)

SC 4 approves the start of the following Application Protocol projects:

10303-215 Ship Arrangement

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

10303-216 Ship Moulded
Forms 10303-217 Ship Piping
Ship Structures
Dimensional Inspection Process Planning

RESOLUTION 167: (Turin - February 1993)

SC 4 thanks UNINFO and their staff for hosting and supporting the meeting.

RESOLUTION 168: (Berlin - October 1993)

For each STEP Application Protocol an Abstract Test Suite shall be developed to document the guidelines for testing implementations for conformance to the Application Protocol. Before an Application Protocol can be registered as a DIS, the corresponding Abstract Test Suite must at least have started its CD ballot.

RESOLUTION 169: (Berlin - October 1993)

New projects shall be established to develop Abstract Test Suites for each approved STEP Application protocol. PMAG shall name a project leader for each project and shall create a project schedule by January 1994.

RESOLUTION 170: (Berlin - October 1993)

SC 4 resolves that documents may be circulated to gather comments from all member countries using the CD procedure, but the procedure for approving a document at the CD level requires a formal ballot. This supersedes resolution #73 from Gothenburg of ISO/TC 184/SC 4 N63 dated 10 June 1990.

RESOLUTION 171: (Berlin - October 1993)

SC 4 adopts project P50 "Printed Circuit Assembly Manufacturing Planning" (AP220) as part of the Work item on product data and appoints Mr Jack Corley as the project leader.

RESOLUTION 172: (Berlin - October 1993)

SC 4 adopts project P49 "Functional Data and Schematic Representation for Process Plants" (AP 221) as part of the Work item on product data and appoints Mr David Leal

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

as the project leader.

RESOLUTION 173: (Berlin - October 1993)

SC 4 adopts project P51 "Exchange of Product Definition Data from Design Engineering to Manufacturing Engineering for Composite Structures" (AP 222) as part of the Work item on product data and appoints Mr Glen Ziolkowski as the project leader.

RESOLUTION 174: (Berlin - October 1993)

SC 4 adopts project P52 "Exchange of Design and Manufacturing Product Information for Cast Parts" (AP 223) as part of the Work item on product data and appoints Mr Constantine Orogo as the project leader.

RESOLUTION 175: (Berlin - October 1993)

SC 4 adopts project P53 "Mechanical Products Definition for Process Planning Using Form Features" (AP 224) as part of the Work item on product data and appoints Mr Len Slovensky as the project leader.

RESOLUTION 176: (Berlin - October 1993)

SC 4 adopts project P54 "Structural Building Elements using Explicit Shape Representation" (AP 225) as part of the Work item on product data and appoints Dr Wolfgang Haas as the project leader.

RESOLUTION 177: (Berlin - October 1993)

SC 4 approves the principle of separate publication of ISO 10303-22 language bindings, such as late FORTRAN, late C and early C++, and directs PMAG to allocate part numbers as required, on the advice of the Editing Committee.

RESOLUTION 178: (Berlin - October 1993)

SC 4 requests its Secretariat to circulate document N230 (SPAG N110 V 1.5) to its member bodies for comment.

RESOLUTION 179: (Berlin - October 1993)

SC 4 directs SPAG to establish and maintain an SC 4 Strategic Policy Reference Guide and welcomes the proposition of SPAG to make tutorial seminars on the SC 4 strategic policy.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 180: (Berlin - October 1993)

SC 4 adopts document N231 (SPAG N111 V 1.2) as its strategic policy for liaisons.

RESOLUTION 181: (Berlin - October 1993)

SC 4 appoints Mr Neal Laurance as Convener of PMAG, and records its appreciation to Mr Jerry Weiss for his efforts in the project since its launch.

RESOLUTION 182: (Berlin - October 1993)

SC 4 accepts in principle that all Parts of ISO 10303 shall be published as joint ISO/IEC standards, and directs its Secretariat to work with the ISO Central Secretariat and the SC 4 Editing Committee to confirm that this may take place without delay of more than two months in their publication as International Standards.

RESOLUTION 183: (Berlin - October 1993)

SC 4 directs Working Group 6 to develop Abstract Test Methods to be used in assessing conformance of implementations using ISO 10303-22.

RESOLUTION 184: (Berlin - October 1993)

SC 4 requests its Secretariat to provide meeting reports, including resolutions, for distribution to member bodies within 6 weeks of each meeting of SC 4.

RESOLUTION 185: (Berlin - October 1993)

SC 4 requests its Secretariat to take into account those Resolutions of ISO/TC 184 concerning the work of SC 4.

RESOLUTION 186: (Berlin - October 1993)

SC 4 endorses the creation of an ad hoc group at the SC 4 level to address how to proceed with parametric capabilities for all SC 4 areas of scope. SC 4 endorses P. Boutinaud and H. Scheder as co-chairs of this group. The ad hoc group is requested to provide its recommendations to the next meeting of SC 4.

RESOLUTION 187: (Berlin - October 1993)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 requests its P-members, and the conveners of Working Groups to nominate experts as participants in the ad hoc group on parametrics.

RESOLUTION 188: (Berlin - October 1993)

SC 4 appoints Mr Adam Polly as project leader of project P39 (WI 1-WG 4-p2.2) on STEP resource model integration and thanks Mrs Yuhwei Yang for her contribution to SC 4/P39.

RESOLUTION 189: (Berlin - October 1993)

SC 4 appoints Mrs Yuhwei Yang as project leader of project P40 (WI1-WG 4-p6.2) on STEP application protocol integration and thanks Mr Steve Ryan for his contribution to SC 4/P40.

RESOLUTION 190: (Berlin - October 1993)

SC 4 appoints Mr Bernd Ingenbleek as project leader of project P18 on Part 201 Application protocol: Explicit draughting, and thanks Mr Peter Kruse for his contribution to SC 4/P18.

SC 4 thanks Barbara Warthen for her efforts in initiating and supporting this work on parametrics.

RESOLUTION 192: (Berlin - October 1993)

SC 4 directs PMAG to ensure that, at that point in time when PMAG is required to vote on any Part promotion which is CD or DIS promotion, the information about sign-offs having occurred earlier is available to the PMAG national representatives in an explicit form. This information shall indicate:

1. The person having signed off.
2. The committee for which the person has signed off.
3. The identifying N number of the document having been actually signed.
4. The issue date of that document version.



N4118

2026 – 05 – 15

5. The date of signature.

SC 4 asks PMAG to answer the following questions regarding AICs:

1. What is the role of AICs in STEP?
2. Who are the users of AICs?
3. Is there a requirement for AICs to become standards?
4. If (3) is answered positively: what are the expected criteria for them to be approved or disapproved by national member bodies?
5. Is planning of AICs required as part of an AP Planning Project?

RESOLUTION 194: (Berlin - October 1993)

SC 4 thanks DIN and ProSTEP for hosting and supporting the meeting in Berlin.

RESOLUTION 195: (Davos - May 1994)

SC 4 directs its Secretariat to incorporate the corrections developed in response to ballots on Parts 1, 11, 21, 31, 41, 42, 43, 44, 46, 101, 201 and 203.

In these documents, only changes of the following nature may be included:

- spelling corrections;
- changes to correct English grammar;
- changes to match ISO wording requirements;
- changes to ensure overall consistency so long as they correspond to ballot comments and introduce no new functionality.

Modified documents will be made available on SOLIS when the updates have been completed, in order to allow confirmation that the agreed edits have successfully been incorporated; this is not a reballot process;

Upon completion of the above process, SC 4 further directs its Secretariat to send the resulting documents to Geneva, no later than 1st September 1994, for publication as International Standards.

RESOLUTION 196: (Davos - May 1994)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

If any of the documents have not been received by the SC 4 Secretariat by 1 September 1994, SC 4 asks its Secretariat to submit the complete Initial Release DIS version of ISO 10303 to the ISO Central Secretariat for publication.

RESOLUTION 197: (Davos - May 1994)

In reference to resolutions 78, 82, 129 and 130, SC 4 wishes to reiterate its concern for the documentation of methods and procedures. SC 4 does not authorise the use or development of methods and procedures other than those used to develop the STEP Initial Release, until the methods and procedures for the Initial Release are documented and accepted by PMAG and SC 4.

RESOLUTION 198: (Davos - May 1994)

SC 4 directs the WG 8 Convener and the WG 8 P1 Project Leader, to perform a feasibility study to demonstrate how a data modelling methodology can be used in the EDI domain. This study will use STEP and MANDATE approaches as far as possible. A final report is expected to be delivered during the next ISO/TC 184/SC 4 plenary meeting.

RESOLUTION 199: (Davos - May 1994)

SC 4 encourages the WG 8 Convener to research with the bodies developing EDIFACT message models, the suitable material for manufacturing management requirements. If it is not possible to find such material, ISO/TC 184/SC 4/WG 8 should propose a new work item for developing a standard, refining the needed messages, preferably in accordance with the EDIFACT organisation.

RESOLUTION 200: (Davos - May 1994)

SC 4 instructs its Secretariat to circulate a new work item ballot on parametrics to pursue the goals, activities and workplan presented in the document ISO/TC 184/SC 4/WG 3 N318.

NOTE: include recommendation of Roberto Greening as Project Leader in the NWI ballot papers.

RESOLUTION 201: (Davos - May 1994)

SC 4 directs PMAG to recommend a solution for the interoperability of application protocols and circulate it to P-members before the end of 1994.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 202: (Davos - May 1994)

SC 4 directs AP projects for Parts 201 and 203 to deliver their respective Abstract Test Suites as Committee Drafts as soon as possible and not later than March 1995.

RESOLUTION 203: (Davos - May 1994)

SC 4 requests PMAG to organise an expert review of the technical lessons learned from the Initial Release of STEP, particularly with respect to the current architecture, methods and practices. The group shall make recommendations to SC 4 on methods and frameworks used in the standard to support future development. The expert group shall include, but not be limited to, one representative from each of the SC 4 working groups.

RESOLUTION 204: (Davos - May 1994)

SC 4 requests its Secretariat to organise a review of organisational issues and to make recommendations to SC 4 at its next meeting.

RESOLUTION 205: (Davos - May 1994)

SC 4 affirms that the EXPRESS language (ISO 10303-11) should be flexible enough to support the specification of data models regardless of the application area.

Requirements for the extension of EXPRESS resulting from this objective can be brought forward from other International Standardization bodies, National Standards bodies, and other such public agencies that develop and promulgate standards such as industry associations.

While inputs from private organizations are welcome, no special priority is given to their requirements.

It is recognized that requirements for the EXPRESS language that go beyond those supporting SC 4 projects may require additional resources before resolution is possible.

RESOLUTION 206: (Davos - May 1994)

SC 4 approves the new AP project, Plant Spatial Configuration (ISO/TC 184/SC 4 PMAG N130) with the understanding that it will be closely coordinated with AP 221.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 appoints Jack Skeels as Project Leader.

NOTE: Six countries indicated their intent to actively participate in this project;
FR,
DE, JP, NL, UK & US.

RESOLUTION 207: (Davos - May 1994)

SC 4 appoints Dr James Fulton as liaison to ISO/IEC JTC 1/SC 7/WG 11.

RESOLUTION 208: (Davos - May 1994)

SC 4 approves the new AP proposal, ISO/TC 184/SC 4 PMAG N122, Ship Mechanical Systems Application Protocol, and names Jerry Goodwin as the Project Leader.

RESOLUTION 209: (Davos - May 1994)

SC 4 EXPRESSES its thanks to Mr Bill Danner for his efforts as the WG 5 Convener in developing the STEP Development Methods and especially for the development of the EXPRESS language.

RESOLUTION 210: (Davos - May 1994)

SC 4 wishes to thank the Swiss member body for its hospitality and the excellent facilities and support at the May 1994 SC 4 and Working Group meetings in Davos.

RESOLUTION 211: (Greenville - October 1994)

SC 4 cancellation of a project shall become effective four months after date of action unless a project leader and resources are brought forth to the Secretariat in conformance with ISO new work item procedures.

RESOLUTION 212: (Greenville - October 1994)

SC 4 agrees to cancel project 16, Integrated Generic Resources on Form Features 10303-48, due to lack of activity.

RESOLUTION 213: (Greenville - October 1994)

SC 4 agrees to cancel project 60, Application Protocol on Dimensional Inspection Process Planning for Coordinate Measuring Machines 10303-219, and its corresponding ATS, 10303-1219, due to lack of activity.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 214: (Greenville - October 1994)

SC 4 agrees to cancel projects 41 and 73, Application Protocol on Mechanical Design Using Wireframe Representation, 10303-206, and its corresponding ATS, 10303-1206, due to lack of activity.

RESOLUTION 215: (Greenville - October 1994)

SC 4 directs its Secretariat to add to its program of work a project titled "Revision to the EXPRESS Language Reference Manual ISO 10303-11". Further, SC 4 confirms Dr. Phil Spiby as Project Leader. PMAG notes that this project was approved by PMAG at the Atlanta meeting.

In order to gain broad awareness and participation in this project, SC 4 directs the Project Leader to prepare a formal scope statement for the project (similar to the one on the New Work Item form) and a full list of requirements for the revision, to be delivered to the Secretariat by January 1995 for circulation to all SC 4 Members and organizations in liaison with SC 4.

RESOLUTION 216: (Greenville - October 1994)

ISO/TC 184/SC 4 recognizes the need to promote the introduction of STEP into industry by providing an organizational framework that supports:

an information exchange platform for implementors of STEP regarding clarification of questions arising during implementation

evaluation of proposed solutions to problems taking into account the STEP standard user needs and implementation practicality

endorsement of a selected solution which is to be used in the implementation

SC 4 requests all member bodies to investigate possible remedies to this need. The workshop concept established in the OSI-MMS environment for this purpose shall be taken into account. Member bodies are expected to provide a written position paper to the SC 4 Chairman by February 15, 1995, and the collection of positions will be distributed to SC 4 members by the Secretariat.

RESOLUTION 217: (Greenville - October 1994)

SC 4 establishes a new working group to resolve the technical direction and related

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

technical issues of SC 4 so that the results are consistent with the SC 4 vision and acceptable to SC 4 as a whole.

The objective is to define and maintain a consistent set of architectures for the standards within SC 4.

The principal deliverable of the working group is a standard or set of standards that document the architectures and the corresponding framework for methods.

Membership in this working group shall be limited. National bodies are asked to nominate members to the WG based on the desire to limit membership, the ability to carry out these terms of reference, and the need to include representatives from all areas of work within SC 4. It is also considered important that the members of the WG shall also be members of other working groups.

The terms of reference for the group are to:

- provide technical leadership across the standards under SC 4;
- collect, evaluate and respond to issues raised against the architectures and the corresponding methods;
- review, at intervals not exceeding three years, the suitability of the architecture against current progress and future requirements;
- identify and implement necessary changes and enhancements to the architectures.

Consideration shall be given to:

- other appropriate standards architecture;
- existing work in the global research and development community;
- prior work within SC 4;
- requirements for input from all existing work groups.

SC 4 requests P Members to recommend experts and their candidate for Convener of this new WG by 24 November. In making their recommendations, P Members are

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

also encouraged to provide suggestions for membership from other nations.

RESOLUTION 218: (Greenville - October 1994)

SC 4 approves SC 4 N230, Requirements for Future Capabilities of SC 4 Standards, and requests that PMAG map it into an operational plan of operations.

RESOLUTION 219: (Greenville - October 1994)

SC 4 directs its Secretariat to set up a quality assurance system within the SC 4 organization according to ISO/TC 184/SC 4 document N230, Requirements for Future Capabilities of SC 4 standards, objective C14: "Establish and implement a quality system for SC 4 standards."

RESOLUTION 220: (Greenville - October 1994)

SC 4 approves SC 4 N242, WG 2 Strategic Policy Statement.

RESOLUTION 221: (Greenville - October 1994)

SC 4 approves the requirement for an SC 4 strategic policy on Application Protocols as set forth in the following SPAG document N124 "Requirements for a SC 4 Strategic Policy on ", Application Protocols" dated 10/20/94.

RESOLUTION 222: (Greenville - October 1994)

SC 4 directs the Architecture WG to propose realistic strategic policy alternative solutions to the requirements defined in SPAG document N124.

The SC 4 Secretariat shall distribute, at least six weeks prior to each SC 4 meeting, a report on the status of SC 4 projects. This report shall have the following attributes:

A summary sheet showing summary status of each project. This will include a short description; significant changes to a project; and, if a project is not progressing, an indication of the problems and/or barriers to progress.

Detailed project descriptions about the above problem projects. This will include participants' lists, and project status in ballot process.

The NIPDE Electronic Library project information on STEP parts and APs will be reviewed by P members to ascertain if its information content is proper to contain the

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

information needed.

RESOLUTION 224: (Greenville - October 1994)

SC 4 expresses its desire to establish a Category A liaison with the ProSTEP Association.

RESOLUTION 225: (Greenville - October 1994)

SC 4 directs its Secretariat to amend its approval procedures for advancing an application protocol to CD stage to include providing evidence of substantive international review of the ARM and the Scope of the document. This evidence is in addition to the present requirements of approval by Qualification and Integration.

RESOLUTION 226: (Greenville - October 1994)

SC 4 directs its Secretariat to amend its approval procedures for registering an application protocol that has successfully passed the CD enquiry to include providing evidence of validation through some form of implementation of the application protocol as amended following ballot resolution. This evidence is in addition to the present requirements of approval by Qualification, Integration and Editing.

SC 4 directs WG 6 to complete Parts 33 and 34 and to submit them, before January 1995, for QIE approvals prior to CD ballot. SC 4 directs the projects responsible for Abstract Test Suites ATS- 1201, ATS-1203 and ATS-1207, to complete their documents and to submit them, before February 1995, for QIE approvals prior to CD ballot.

SC 4 directs its Secretariat that Application Protocols submitted for DIS ballot prior to August 1995 shall submit the Abstract Test Suite for CD review no later than three months after the start of the DIS ballot of the Application Protocol.

RESOLUTION 228: (Greenville - October 1994)

SC 4 appoints Neal Laurance as a physical liaison to ISO/TC 184/SC 5.

RESOLUTION 229: (Greenville - October 1994)

SC 4 expresses its desire to establish a liaison with ISO/IEC JTC 1/SC 21/WG 3/CSMF.



N4118

2026 – 05 – 15

RESOLUTION 230: (Greenville - October 1994)

SC 4 expresses its sincere thanks to its Chairman, Bradford Smith, to the team at NIST, and to all those who contributed to the on-time delivery of the Initial Release of STEP to Geneva for publication.

RESOLUTION 231: (Sydney - March 1995)

SC 4 establishes a Policy and Planning Committee to become the single advisory group to assist the SC 4 Chairman, Committee, Conveners and Project Leaders:

To facilitate the smooth running of the organisation, including a leading role in conflict resolution between WGs, monitoring progress in critical issues and ensuring effective communication across the organisation.

To complete the development of the SC 4 Organisation Handbook by 15 July 1995.

To monitor the effectiveness of the organisation and to make recommendations to the SC 4 chairman on improvements to the organisation and its processes and procedures as defined in the SC 4 Organisation Handbook.

In promoting the work and results of SC 4.

In strategic and tactical planning, including resource planning

The PPC shall not only provide advice in response to formally submitted requests but also operate meetings to address specific issues.

The PPC members shall act as "Ombudsmen", providing a focus for issues relating to the organisation, process and procedure. PPC members shall endeavour to resolve such issues by communicating with those involved, referring the issue to SC 4 only when settlement is not possible.

The PPC should establish an electronic capability for virtual meetings that can be proactive and reactive to expressed needs within two weeks.

RESOLUTION 232: (Sydney - March 1995)

SC 4 appoints Mr. Julian Fowler as the convener of Working Group 10, Technical Architecture

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 233: (Sydney - March 1995)

SC 4 appoints Bernd Wenzel as deputy convener for Working Group 10 on architecture.

RESOLUTION 234: (Sydney - March 1995)

For the new work item for Amendment 1 to 10303-1, SC 4 appoints Howard Mason project leader.

RESOLUTION 235: (Sydney - March 1995)

For the new work item STEP Development Methods, SC 4 appoints Julian Fowler project leader and assigns this project to WG 10. The WG 5 Project 2 is transferred to WG 10 to support this project.

RESOLUTION 236: (Sydney - March 1995)

For the new work item STEP Building Construction Core Model, SC 4 appoints Jeff Wix project leader and assigns this project to WG 3.

RESOLUTION 237: (Sydney - March 1995)

For the new work item Building Services, Heating, Ventilation and AC, SC 4 appoints Patrice Poyet project leader and assigns this project to WG 3.

RESOLUTION 238: (Sydney - March 1995)

For the new work item Building Structural Frame, SC 4 appoints Alistair Watson project leader and assigns this project to WG 3.

RESOLUTION 239: (Sydney - March 1995)

SC 4 requests its parent committee to revise the title of ISO/TC 184/SC 4 from Industrial Data and Global Manufacturing Programming Languages to Industrial Data.

RESOLUTION 241: (Sydney - March 1995)

SC 4 expresses its desire to establish a Category A liaison with ISO/TC 211 on Geographic Information Systems and appoints Barbara Warthen as that liaison to

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

actively manage information flow between the two committees.

RESOLUTION 242: (Sydney - March 1995)

SC 4 expresses its desire to establish a Category A liaison with EPISTLE.

RESOLUTION 243: (Sydney - March 1995)

SC 4 wishes to give its unanimous thanks to John Mitchell, and to the following members of Standards Australia: Brigitte Catteau, David Bruce-Steer, Rhonda Putman, Rosie Sabadina, Betty Titcombe, Linda Young, Cheryl O'Neill and Katalin Veres, for the wonderful staff support and the excellent meeting facilities. SC 4 members both at the meeting and from various countries on the Internet also wish to thank Dick Wittenoom from AusSTEP Consultants for providing computer support and SC 4's first use of meeting Web pages.

RESOLUTION 244: (Grenoble - October 1995)

SC 4 endorses the Secretariat's nomination of Richard R. Wandmacher to fill the position as Chairman of SC 4 for a term of six years and requests the Secretariat to forward the nomination to AFNOR for approval by ISO/TC 184.

RESOLUTION 245: (Grenoble - October 1995)

SC 4 adopts the Technical Corrigendum to ISO 10303-21 dated 7 September 1995 and directs the SC 4 Secretariat to forward the document to ISO Geneva.

RESOLUTION 246: (Grenoble - October 1995)

SC 4 agrees to cancel project #24, 10303-103: Electrical/Electronics Connectivity, due to lack of activity.

RESOLUTION 247: (Grenoble - October 1995)

SC 4 agrees to cancel project # 4, 10303-25: FORTRAN Language Binding to the SDAI, due to lack of activity.

RESOLUTION 248: (Grenoble - October 1995)

SC 4 requests its Secretariat to ensure that if any AIC drafts are recommended for promotion to DIS under current ballot (N315-N332, Parts 501-518) the parts will

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

incorporate agreed changes in response to technical comments received with the ballot prior to DIS balloting.

RESOLUTION 249: (Grenoble - October 1995)

SC 4 disbands its Strategic Planning Advisory Group and thanks Jean-Pierre Letouzey for his service as Convener.

RESOLUTION 250: (Grenoble - October 1995)

SC 4 disbands its Project Management Advisory Group and thanks Neal Laurance for his service as Convener.

RESOLUTION 251: (Grenoble - October 1995)

SC 4 appoints Shaun Devlin as Convener of JWG 9.

RESOLUTION 252: (Grenoble - October 1995)

SC 4 accepts the report of its Policy and Planning Committee. SC 4 requests its Secretariat to circulate the attached resolution for letter ballot to SC 4 members, and to issue a simultaneous call for a Convener of the proposed new Working Group, experts for this Working Group, a chair of the proposed new Quality Committee and experts for this committee.

RESOLUTION 253: (Grenoble - October 1995)

SC 4 requests its Secretariat to register the current ISO 10303-22 document as a Draft International Standard.

RESOLUTION 254: (Grenoble - October 1995)

SC 4 expresses its desire to establish a Category A liaison with ISO/TC 8 on Ships and Marine Technology and appoints Zabi Bazari (Lloyd Register of Shipping) as that liaison.

RESOLUTION 255: (Grenoble - October 1995)

SC 4 thanks the French national Body for hosting the meetings of SC 4 and its Working Groups on this occasion and wishes particularly to thank the staff of AFNOR, Sylvie Arbouy, Veronique Auger, and Ives Dudognon for their efforts.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 256: (Grenoble - October 1995)

SC 4 wishes to warmly and enthusiastically thank Brad Smith for his twelve years of service to SC 4 and for his personal contribution to our continuing success.

RESOLUTION 257: (Kobe - June 1996)

SC 4 reconfirms resolution #168 but approves exempting 10303-210, 10303-224, and 10303-225 from the associated abstract test suite requirements of SC 4 Resolution

#168 for these APs' initial DIS registration and ballot cycle if before 31 July 1996:

- (a) the CD ballot comments are resolved;
- (b) the DIS draft is approved by integration, qualification, and editing; and
- (c) the issues resolution log and draft AP for DIS registration are successfully loaded on SOLIS, with a production-ready hard copy of both documents to the SC 4 Secretary.

Compliance to SC 4 Resolution #168 is expected if an AP project team is unable to meet (a) through (c) by 31 July 1996. The requirements of abstract test suites will be enforced before FDIS.

RESOLUTION 258: (Kobe - June 1996)

SC 4 notes the following technical SEDS reports as found in SC 4 N 441, for incorporating into the affected part of ISO 10303:

reports 1, 3, 4, and 6
10303-42 & 10303-43: Report 5
10303-42 & 10303-203: Report 8
10303-11: Reports 13, 14, 16, 29, 31, and 33.

SC 4 requests the SC 4 Secretariat ensure these reports are made publicly available on SOLIS, and incorporated into technical corrigenda, amendments, or new editions when appropriate.

RESOLUTION 259: (Kobe - June 1996)

SC 4 notes the SEDS Coordinator and SC 4 Secretariat's decision that the following SEDS reports found in SC 4 N 441 are editorial in nature and can be processed accordingly: SEDS Reports 2, 7, 9, 11, 38, 39, 40-43, 45, 48, 49, 67. SC 4 requests

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

the
SC 4 Secretariat ensure these reports are made publicly available on SOLIS, and incorporated into technical corrigenda, amendments, or new editions when appropriate.

RESOLUTION 260: (Kobe - June 1996)

SC 4 accepts with regret the resignation of Yuhwei Yang, Sheila Lewis, and Albert Colin as Convener of WG 4, WG 6, and WG 8 respectively. The Committee wishes to thank Yuhwei Yang, Sheila Lewis, Albert Colin, Jan van Maanen, and Nigel Shaw for their leadership and personal dedication in the SC 4 standards development effort over the last decade, and encourages their continuing support of the SC 4 work.

RESOLUTION 261: (Kobe - June 1996)

SC 4 approves the list of project leaders and endorses the list of editors for the SC 4 work effort as found in SC 4 N439.

RESOLUTION 262: (Kobe - June 1996)

SC 4 approves forwarding 10303-225 for Draft International Standard registration upon meeting all requirements established by the ISO Directives and other SC 4 resolutions.

RESOLUTION 263: (Kobe - June 1996)

SC 4 agrees to cancel project #27, 10303-102, “Integrated application resources: Ship structures” due to lack of activity.

RESOLUTION 264: (Kobe - June 1996)

SC 4 agrees to cancel project # 132, 10303-516, “Application interpreted construct: Mechanical design context” as recommended by NWI/CD ballot comments.

RESOLUTION 265: (Kobe - June 1996)

SC 4 approves the publication of 10303-12 as a Technical Report Type II and requests the SC 4 Secretariat take appropriate action for its publication and dissemination.

SC 4 recognizes and supports the efforts of the STEP implementors to collaborate on

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

implementation issues, and encourages the implementors to continue to hold meetings in parallel to SC 4 Working Group meetings.

RESOLUTION 267: (Kobe - June 1996)

SC 4 requires that, unless a P-member expressly requests in writing an exemption from the SC 4 Secretary, all ballot comments shall be provided in, at minimum, ASCII format along with a backup hard copy. Distribution by the SC 4 Secretariat of ballot comments and general SC 4 correspondence shall be by 3.5" disk, providing the original format, PostScript, and ASCII files for each SC 4 document.

RESOLUTION 268: (Kobe - June 1996)

SC 4 accepts ISO/TC 184/SC 4 N438 as the prescribed guidelines when assessing fees for SC 4-related meetings, and requests the SC 4 Secretary forward these guidelines to ISO for the Secretary-General's approval as required by ISO Technical Management Board resolution 39/1995.

RESOLUTION 269: (Kobe - June 1996)

SC 4 approves the changes to the SC 4 Organisation Handbook, SC 4 N445, and requests the SC 4 Secretariat take appropriate action to publish and disseminate the revised Handbook. Any further changes to the Handbook will be undertaken using the procedures in Section 5.11 of the Handbook.

RESOLUTION 270: (Kobe - June 1996)

SC 4 resolves to establish a Change Management Advisory function. SC 4 further requests the SC 4 Chairman to work with the PPC to present appropriate changes to the SC 4 handbook to reflect the task of the Change Management Advisory function for Handbook ratification at the next SC 4 meeting.

SC 4 methods documents from SC 4 working groups and committees will be published as SC 4 standing documents which will provide methods and procedures for SC 4 to carry out its standards development more effectively within SC 4.

Copies of the proposed Standing Document will be circulated for ballot to P-members for a three month ballot. Approval of the proposed document shall be based on the procedures for a Committee Draft Document.



N4118

2026 – 05 – 15

Any comments provided with a yes vote will be considered for making improvements to the document, but the vote will be considered as in favor of making the document operative as an SC 4 Standing Document.

Any no vote must be accompanied by specific comments.

Changes to standing documents will be recommended by the working group or committee responsible. Changes will be made by providing a new version of the page containing the changes to the SC 4 Secretary for approval at the next SC 4 meeting.

Each Standing Document shall contain an annex Revision History Table indicating the Section or page change and the date accepted by SC 4 and a list of those SC 4 parts expressly exempt from the revision as approved.

RESOLUTION 272: (Kobe - June 1996)

SC 4 resolves to establish a new Working Group 12, SC 4 Common Resources. Responsibility for the Application Interpreted Constructs is assigned to this working Group. The Secretariat shall act as the Convener of this Working Group until nominations for convener can be solicited from the P-members and a selection by the Chair and Secretariat ratified by SC 4.

The nominated convener will prepare a proposal for the next SC 4 meeting to identify which other existing SC 4 Projects should be transferred to the new working group along with the timetable for such transfer. This proposal should recognize that resources for many of the common resources are currently involved in developing specific application protocols. The proposal should identify resources that will be available for work on the common resources and identify needed resources so that a call for experts can be initiated.

The task of this Working Group will be:

- Provide for the extension of existing common resources
- Ensure dynamic consistency between resource types
- When authorized by SC 4, create or enhance specific resources based on the requirements of
 - (a) All SC 4 Standards
 - (b) Integration requirements
 - (c) Requirements established by WG 10
 - (d) Requirements established by the Quality Committee

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Propose new work items for completion in the area of common resources to SC 4
- Establish Liaison with other standards activities working on information models useful as common resources for SC 4.

RESOLUTION 273: (Kobe - June 1996)

SC 4 appoints Mr. Jean-Jacques Michel as convener of WG 8, Manufacturing Management Data.

RESOLUTION 274: (Kobe - June 1996)

SC 4 appoints Allison Barnard Feeney as Convener of the Quality Committee.

RESOLUTION 275: (Kobe - June 1996)

SC 4 appoints Christophe Viel as Convener of WG 11, “EXPRESS Language, Implementation Methods, and Conformance Methods”.

RESOLUTION 276: (Kobe - June 1996)

SC 4 appoints David Price as Deputy Convener of WG 11, “EXPRESS Language, Implementation Methods, and Conformance Methods”.

SC 4 directs WG 11 to create an ad-hoc group to investigate the technical changes to parts 10303-11, 12, 21, and any other 10303 parts to allow the use of multi-language character sets, and to prepare new work item proposals to address this requirement. SC 4 nominates Neal Laurance to lead that group.

RESOLUTION 278: (Kobe - June 1996)

The development of ATS is of the utmost importance in accelerating the availability of high quality commercial AP implementations. So that information and experience of ATS usage in practice may be gathered, SC 4 resolution #168 is superseded by requiring that all Abstract Test Suites be advanced as Type 2 Technical Reports. Before an Application Protocol can be registered as an FDIS, the corresponding ATS must have started its ballot as a Type 2 Technical Report. The SC 4 Secretary and Chair are asked to identify the responsible projects within SC 4 and task them to establish the metrics for determining when the ATS Technical Report is adequate.

RESOLUTION 279: (Kobe - June 1996)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 strongly recommends that all P-members and A-liaisons have internet capability to send and receive ISO/TC 184/SC 4 documents by 1997-6-1.

SC 4 appoints the JWG 9 Convener, Shaun Devlin, as a participating liaison to IEC/TC 93.

RESOLUTION 281: (Kobe - June 1996)

ISO/TC 184/SC 4 requests IEC/TC 93 to designate a liaison on electronic design automation to JWG 9 who can participate in the planning and technical meetings of JWG 9.

RESOLUTION 282: (Kobe - June 1996)

SC 4 accepts the request for an A-liaison from NATO and requests its Secretariat to assist the NATO CALS Office to establish the liaison through the normal procedures.

RESOLUTION 283: (Kobe - June 1996)

SC 4 recognises the importance of the planned workshop on standards for the use of models that define the data and processes of information, and nominates the following delegates:

Guy Pierra to represent WG 2

Julian Fowler to represent WG

10

Phil Spiby to represent WG 11, using the invitation to SC 4

RESOLUTION 284: (Kobe - June 1996)

SC 4:

- welcomes the efforts by the Advisory Committee on Electronics and Telecommunications (ACET) to coordinate relevant activities across IEC
- recognises the strong industrial demand for integration between standards relating to mechanical and electrotechnical product model data
- emphasises the need for a joint ISO/IEC approach to this issue, drawing attention to the previous SC 4 resolutions on the topic (Resolutions 138, 140 - October 1992)
- expresses its willingness to contribute to the proposed Harmonisation Group, and nominates Shaun Devlin (Convener of JWG 9) as the principal point of contact

Dana Tripp

Secretary, ISO/TC 184/SC 4

trippdsc4@gmail.com



N4118

2026 – 05 – 15

for SC 4

- encourages the ISO TMB to recognise the Design Automation Harmonisation Group (DAHG) activity as a joint ISO/IEC activity.

RESOLUTION 285: (Kobe - June 1996)

SC 4 requests its secretariat to circulate a letter to P-members inviting nominations for each of the following liaisons:

JTC 1/SC 21/WG 3, JTC 1/SC 21/WG 7, JTC 1/SC 21/WG 11, JTC 1/SC 24, ISO/TC 29, ISO/TC 176, ISO/TC 213, Joint ISO-IEC-UN/ECE Coordination Committee on EDI

SC 4 requests its Secretariat to establish the corresponding liaisons where they do not already exist.

RESOLUTION 286: (Kobe - June 1996)

SC 4 resolves that

- an AIC shall be published as a separate part (500-series document)
- APs shall reference AICs only as a normative reference
- AICs shall be at the same or advanced voting status as an AP in which it is being used at the time of the AP's DIS or FDIS ballot.

SC 4 authorises exemption for 10303-202 and 10303-207, recognising the current ballot status of these documents.

RESOLUTION 287: (Kobe - June 1996)

SC 4 directs WG 10 to investigate the use of fully attributed EXPRESS ARMs and report to SC 4 at its next meeting.

RESOLUTION 288: (Kobe - June 1996)

SC 4 authorises preparatory (Stage 0) work on an EXPRESS mapping language and assigns Martin Hardwick as project leader.

RESOLUTION 289: (Kobe - June 1996)

SC 4 wishes to thank the Japan STEP Promotion Center and Japan Information Processing Development Center for their gracious hospitality in hosting this week's SC

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

4 Working Group and SC 4 meetings. Particular recognition for their dedicated support goes to Chie Kouchi, Junko Ito, Mina Miake, and Michiyo Kuwahara.

SC 4 approves the following new project leaders and/or project editors for the associated SC 4 projects:

ISO Project #	Project Lead	Document Editor
10303-45	Joseph Carpenter	Robert Swindells
10303-47	Bill Anderson	Bill Anderson
10303-202	Linas Polikaitis	Kevin Freund
10303-205	Jon Owen	Jon Owen
10303-207	Charles Cederman	
10303-215	Peter Lazo	John Kendall
10303-221	Jeroen Kingma	David Leal
10303-222	Floyd Ganus	Floyd Ganus
10303-223		Ching-Yang Wang
10303-229		Ching-Yang Wang
10303-232	Glen Ziolk	
10303-305	Jon Owen	Jon Owen
10303-318		Yuanxie Janke-Zhao
10303-323	Gerald Radack	Ching-Yang Wang
10303-329		Ching-Wang Yang
10303-332	Glen Ziolk	
13584-102	Robert Schuler	
14959	Michael Pratt	Dick Wittenoom

RESOLUTION 291: (Toronto - October 1996)

SC 4 endorses the Petrotechnical Open Software Corporation (POSC) as an A-liaison to ISO/TC 184/SC 4.

RESOLUTION 292: (Toronto - October 1996)

SC 4 authorises a temporary action to allow operation of the Quality Committee as outlined in ISO/TC 184/ SC 4 N483 (Rev 1) until the new pages are adopted by letter ballot.

SC 4 approves replacing the following text in the SC 4 Organization Handbook, Clause 5.7.3.2:

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

"...Copies of Last-Day Resolution Papers must be handed to the meeting desk by 15.00 in order to be available for pick-up by delegation leaders at 17.00 on the day preceding the start of the SC 4 meeting."

with:

"Copies of Last-Day Resolution Papers must be handed to the SC 4 Secretary or meeting desk by 10.30 in order to be available for pick-up by delegation leaders at 15.30 on the day preceding the start of the SC 4 meeting. A camera-ready hard copy and digital ASCII file shall be provided by the originating national body or convener."

RESOLUTION 294: (Toronto - October 1996)

SC 4 approves inserting the following into the SC 4 Organization Handbook for Clause 5.9:

5.9 Balloting Procedures

5.9.1 International Standards and Type II Technical Reports

Balloting procedures for developing ISO standards or Type II Technical Reports can be found in the ISO/IEC Directives Part 1, chapter 2. Information on how to acquire a copy of these procedures can be found in Appendix A.

5.9.2 Standing Documents

SC 4 methods documents from SC 4 working groups and committees will be published as SC 4 standing documents which will provide methods and procedures for SC 4 to carry out its standards development more effectively within SC 4.

Copies of the proposed Standing Document will be circulated for ballot to P-members for a three month ballot. Approval of the proposed document shall be based on the procedures for a Committee Draft Document.

Any comments provided with a yes vote will be considered for making improvements to the document, but the vote will be considered as in favor of making the document operative as an SC 4 Standing Document.

Any no vote must be accompanied by specific comments.

Changes to standing documents will be recommended by the working group or committee responsible. Changes will be made by providing a new version of the page

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

containing the changes to the SC 4 Secretary for approval at the next SC 4 meeting.

Each Standing Document shall contain an annex Revision History Table indicating the Section or page change and the date accepted by SC 4 and a list of those SC 4 parts expressly exempt from the revision as approved. [SC 4 Resolution 271]

RESOLUTION 295: (Toronto - October 1996)

SC 4 notes that work under the Parametrics project (allocated the ISO number 14959) has identified areas which may require modifications to existing IRs, the preparation of a new IR or IRs, and the development of a document or documents under ISO 14959 covering the emerging area of General Case Parametrics.

SC 4 resolves that these activities will for the present be combined within the project 14959 which will be allocated to WG 12 with Mike Pratt as project leader. These arrangements will be subject to ongoing review as the structure of project deliverables is clarified.

SC 4 thanks Roberto Greening for his work as leader of the Parametrics project during the formative period.

RESOLUTION 296: (Toronto - October 1996)

SC 4 agrees to cancel project # 50, 10303-211, "Application protocol: Electronics test diagnostics and remanufacture" and project #78, 10303-311, "Abstract test suite: Electronics test diagnostics and remanufacture" due to current lack of activity.

RESOLUTION 297: (Toronto - October 1996)

SC 4 authorizes preliminary (stage 0) work on Systems Engineering Data Representation and appoints Sylvain BARBEAU as project leader.

RESOLUTION 298: (Toronto - October 1996)

SC 4 recommends that the SC 4 organization handbook be amended as follows: "SC 4 Conveners and deputy conveners shall be nominated by the SC 4 P-members and shall serve for a maximum period of three years or for such shorter period as may be designated when they are elected. Successive extensions each of a maximum of three years may be approved by a vote of SC 4."



N4118

2026 – 05 – 15

RESOLUTION 299: (Toronto - October 1996)

SC 4 accepts the revised schedules and justification, and reconfirms the continued standardization efforts for the following SC 4 projects:

10303-1DAM, 10303-34, 10303-106, 10303-215, 10303-216, 10303-217, 10303-218, 10303-221, 10303-222, 10303-226, 10303-227, 10303-228, 10303-230, 13584-101, 13584-102, and 14959.

RESOLUTION 300: (Toronto - October 1996)

SC 4 approves Phil Kennicott as the convener of WG 12, SC 4 Common Resources.

RESOLUTION 301: (Toronto - October 1996)

SC 4 authorizes preparatory (stage 0) work on Java programming language binding to the Standard Data Access Interface specification and appoints Tom Rando as project leader.

RESOLUTION 302: (Toronto - October 1996)

SC 4 agrees to cancel Project #113, ISO 10303-13 “Architecture and methodology reference manual”, and to redirect this work towards an SC 4 Standing Document and to the Part 1 amendment project.

RESOLUTION 303: (Toronto - October 1996)

SC 4 authorizes preliminary (stage 0) work on the extension of capabilities of ISO 10303- 21 and appoints Neal Laurence and David Price as joint project leaders, and requests a statement of scope of requirement be circulated to P-members.

RESOLUTION 304: (Toronto - October 1996)

SC 4 requires that an AP document may have an informative annex containing a fully attributed EXPRESS specification of the ARM.

An ARM implementation may be taken as validation evidence at DIS according to resolution 226.

RESOLUTION 305: (Toronto - October 1996)

SC 4 recognises this first successful step between WG 2 and WG 10 on a solution for

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

the problem of interoperability between STEP (ISO 10303) and P-LIB (ISO 13584) within ISO/TC 184/SC 4 and encourages WG 2 and WG 10 to complete this work. SC 4 encourages all STEP AP developers to study the feasibility of describing their library and dictionary data using the P-LIB standards.

RESOLUTION 306: (Toronto - October 1996)

SC 4 authorizes a preliminary stage 0 project on “Design and Configuration of Manufacturing Systems” and appoints Torsten Kjellberg as project leader.

RESOLUTION 307: (Toronto - October 1996)

SC 4 thanks its Canadian hosts and US meeting sponsors: The Standards Council of Canada, National Research Council of Canada, and US PRO. Particular thanks go to Melanie Varn, Erinbeth Massie, Kristi Adams, and Chuck Stark.

RESOLUTION 308: (Approved by mail ballot October 1996)

SC 4 notes the following technical SEDS reports as found in SC 4 N 458, for incorporating into the affected part of ISO 10303:

10303-203: Reports 24, 25, and 27.

SC 4 requests the SC 4 Secretariat ensure these reports are made publicly available on SOLIS (SC 4 ON-Line Information Service), and incorporated into technical corrigenda, amendments, or new editions when appropriate.

RESOLUTION 309: (Approved by mail ballot October 1996)

STEP abstract test suites (ATSs) shall not place additional requirements on the implementors of application protocol (AP) parts. AP parts may however make normative references to an ATS and other conformance testing-related parts for terminology. AP parts should not invoke requirements defined in ATS parts of STEP.

RESOLUTION 310: (Approved by mail ballot March 1997)

SC 4 approves replacing clauses 6.2 and 6.3 of the SC 4 Organization Handbook with the text found in ISO/TC 184/SC 4 N516. SC 4 further authorizes editorial changes to other clauses of the handbook to maintain consistency with this new text. This resolution supersedes SC 4 Resolutions #80 and #292.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 311: (San Diego, CA USA - June 1997)

SC 4 approves the following new project leaders and/or project editors for the associated SC 4 projects:

ISO Project #	Project Lead Document Editor
Java programming language	Lisa McCabe
EXPRESS mapping language	Ian
Bailey 10303-11E2	Phil Spiby
10303-24	Peter Dupont
10303-26	Lisa McCabe
10303-208	Chuck Amaral
Burkett 10303-216	Bill
10303-224	Frank Stolte
10303-227	William Gruttke
10303-307	Mark Palmer
10303-308	Prajakta Ketkar
10303-321	Chuck Amaral
10303-324	Sheila Lewis
10303-501	William Gruttke
10303-502	Bill Anderson
10303-503	Bill Anderson
10303-510	Bill Anderson
10303-515	Bill Anderson
10303-517	Martin Philipp
13584-24	Eric Sardet
13584-26	Gerd Ehinger
13584-31	Robert Schuler
13584-101	Lutz R. Hiene
13584-102	Eric Sardet
	Robert Schuler

RESOLUTION 312: (San Diego, CA USA - June 1997)

SC 4 supports working group representation at the ISO/IEC JTC 1 Joint Workshop on Registration of Metadata and requests the following delegates to attend:

WG 2: Guy
Pierra WG 10: Julian
Fowler

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Delegates are requested to prepare a written report on the workshop and submit the reports to the SC 4 Secretary for distribution to p-members.

RESOLUTION 313: (San Diego, CA USA - June 1997)

SC 4 approves the following be included in the SC 4 Organization Handbook under Clause 5.5 (reference ISO/TC 184 SC 4 N 492):

5.5.1 SC 4 Policy for all Liaisons

SC 4 standards need to interact efficiently with other relevant standards, as well as leverage standardization efforts outside SC 4 that is related to the work of SC 4. To do this effectively, SC 4 should focus upon liaisons which are critical for the success of the SC 4 work or for the acceptance of SC 4 standards.

Before establishing a liaison, an appropriate mode of cooperation should be agreed upon by both parties. That mode could include one or more of the following:

- creating joint working group,
- establishing a memorandum of understanding,
- appointing people to actively liaise,
- exchanging information via E-mail,
- exchanging information by participating in the SC 4 Liaison Plenary,
- exchanging documents,
- presenting topical or technical training sessions by either organization.

Establishing any liaisons must follow the ISO directives which distinguishes between another technical committee liaison, an A-liaison or a B-liaison. (IDP1, 1.13-1.15) It is the responsibility of the SC 4 Secretariat to maintain the list of liaisons consistent with ISO Central Secretariat.

RESOLUTION 314: (San Diego, CA USA - June 1997)

SC 4 wishes to thank Ms. Barbara Warthen for her liaison role with ISO/TC 211, and accepts her resignation from this position. SC 4 appoints Mr. Bernd Wenzel as the person to actively liaise on behalf of SC 4 to ISO/TC 211.

RESOLUTION 315: (San Diego, CA USA - June 1997)

SC 4 requests its Secretary to establish an active two-way liaison with ISO/TC 213, “Dimensional and geometrical product specification and verification”, and approves Mr. Alan Jones as the person to actively liaise to ISO/TC 213 on SC 4’s behalf.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 316: (San Diego, CA USA - June 1997)

SC 4 requests an active liaison with ISO/TC 29/WG 34, “Small tools: Computerized machining data exchange”, and requests the SC 4 Secretary confirm this liaison with ISO Central Secretariat and request ISO/TC 29/WG 34 identify its liaison to SC 4. SC 4 approves Guy Pierra and Saburo Tanatsugu as the persons who actively liaise with ISO/TC 29/WG 34 on behalf of SC 4/WG 2; and Roland Jochem as the person who actively liaises with ISO/TC 29/WG 34 on behalf of SC 4/WG 8.

RESOLUTION 317: (San Diego, CA USA - June 1997)

SC 4 accepts the Industry Alliance for Interoperability’s (IAI) request for A-liaison status to SC 4, and recognizes Jeffrey Wix as the person to actively liaise from IAI to SC 4.

SC 4 approves Wolfgang Haas as the active liaison from SC 4 to IAI. SC 4 encourages other P-members, where there is a national/regional IAI chapter, to nominate other SC 4-to-IAI liaisons and provide such nominations to the SC 4 Secretary by 1997-08-01.

RESOLUTION 318: (San Diego, CA USA - June 1997)

SC 4 cancels SC 4 Project #102 as New Work Item: (San Diego, CA USA - June 1997) ISO 10303-228, “Application protocol: Building services: Heating, ventilation, and air conditioning”, and it’s associated Abstract Test Suite.

RESOLUTION 319: (San Diego, CA USA - June 1997)

SC 4 approves “Application protocol: Building services: Heating, ventilation, and air conditioning” and “Abstract test suite: Building services: Heating, ventilation, and air conditioning as preliminary work items, stage 0, in recognition of the initiatives’ continued merit within SC 4.

RESOLUTION 320: (San Diego, CA USA - June 1997)

SC 4 authorizes a new Preliminary Work Item (stage 0) to develop a NODIF AP and appoints Timothy Wise as project leader.

RESOLUTION 321: (San Diego, CA USA - June 1997)

SC 4 authorizes preparatory (stage 0) work on product life-cycle support, to develop

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

a high-level planning model and if appropriate to formulate NWI proposals by October 1997. John Dunford and Rick Bsharah are appointed as joint project leaders, and the project is assigned to WG 3/T8 in addition to their existing approved projects.

RESOLUTION 322: (San Diego, CA USA - June 1997)

SC 4 supports the concept of all three MANDATE projects under the new work item, ISO 15531, and requests its Secretary to distribute to its P-members, a second CD ballot of 15531-31 along with the initial committee drafts of 15531-1, 15531-21, and 15531-41.

RESOLUTION 323: (San Diego, CA USA - June 1997)

SC 4 cancels SC 4 Project #62 as New Work Item: ISO 10303-220, "Application protocol: Process planning, manufacture, and assembly of layered electronic products" and its associated abstract test suite.

RESOLUTION 324: (San Diego, CA USA - June 1997)

SC 4 approves "Application protocol: Process planning, manufacture, and assembly of layered electronic products" and "Abstract test suite: Process planning, manufacture, and assembly of layered electronic products" as preliminary work items, stage 0, in recognition of the initiatives' continued merit within SC 4.

RESOLUTION 325: (San Diego, CA USA - June 1997)

SC 4 endorses continued ISO leadership of all the JWG 9 standards, and requests its secretary to forward this endorsement to ISO for the ISO Technical Management Board and IEC Committee of Action approval. During stages 0 and 1 there shall be strong attempts to coordinate and communicate with the appropriate IEC Technical Committees. The remaining procedures for initiating, reviewing, balloting, publishing, and revisions of electrical/electronic standards within ISO/TC 184/SC 4 will be unchanged.

RESOLUTION 326: (San Diego, CA USA - June 1997)

SC 4 cancels SC 4 Project # 114, ISO 10303-1 Amendment, "Overview and fundamental principles".

RESOLUTION 327: (San Diego, CA USA - June 1997)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 approves a Preliminary Work Item to define the amendment to ISO 10303-1 by May 1998 and appoints Howard Mason as Project Leader. SC 4 creates a STEP Overview Editing committee and invites P-member nominations to that committee.

RESOLUTION 328: (San Diego, CA USA - June 1997)

SC 4 approves the Preliminary Work Item on SGML and industrial data described in document WG 10 N114, as well as in WG 10 N118 attachment 1, and appoints Nigel Shaw and Daniel Rivers-Moore as joint Project Leaders. This project is assigned to WG 10.

RESOLUTION 329: (San Diego, CA USA - June 1997)

SC 4 approves the Preliminary Work Item on the SC 4 Data Architecture described in document WG 10 N113 and appoints Matthew West as Project Leader. This project is assigned to WG 10.

RESOLUTION 330: (San Diego, CA USA - June 1997)

SC 4 cancels project #7, 10303-33, Conformance testing methodology and framework: Structure and use of abstract test suites.

RESOLUTION 331: (San Diego, CA USA - June 1997)

SC 4 accepts the QC/WG 11 proposal to limit coverage requirements to Application Objects and to leave to the ATS developers the choice of selecting which combinations of attributes and assertions are of primary interest for the conformance testing.

SC 4 approves the following metrics in order for an Abstract Test Suite (ATS) to proceed to Technical Report Type II or Committee Draft ballot. An ATS must satisfy the following minimum coverage requirements:

1. All application object test purposes (TPs) must be covered (i.e. the ATS must contain test cases in which these TPs are verdicted).
2. Each test case must be complete. Each preprocessor input specification must have at least one accompanying postprocessor mirror. Each input specification must represent a complete and valid exchange scenario.

The AP/ATS team is responsible to decide necessary additional coverage

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

requirements in its ATS other than those specified in the minimum requirements.

SC 4 recognizes that 10303-302, 303, 304, 305, and 307 were into the Quality Committee by 1997-05-15 and ready for Technical Report Type II ballot prior to the above metrics being approved. SC 4 supports their release for Technical Report ballot, as previously agreed upon, at the completion of the Quality Committee's review.

SC 4 requests WG 11 and Quality Committee provide the corrective text to all appropriate SC 4 Standing Documents (by clause) to reflect these new metrics for assessing the minimum coverage. The Quality Committee Production Support Team shall perform spot checks on all ATSs to verify that the metrics are satisfied.

RESOLUTION 332: (San Diego, CA USA - June 1997)

SC 4 thanks Phillip Kennicott for his dedicated service and technical contributions to the work of ISO/TC 184/SC 4, and wishes him well in his new endeavors.

RESOLUTION 333: (San Diego, CA USA - June 1997)

SC 4 approves Greg Paul as Convener of WG 12, SC 4 Common Resources.

RESOLUTION 334: (San Diego, CA USA - June 1997)

SC 4 requests WG 12 to produce consensus-built documents of revisions to 10303-41, 43, and 44 to be completed and into Quality Committee for qualification no later than 1997-08-15. Quality Committee is to give the review and signoff of these parts its highest priority. P-members are actively encouraged to provide technical experts to both WG 12 and Quality Committee to meet these goals.

In the case that WG 12 does not reach consensus by the end of the [revisions development] workshop (1997-07-25), the SC 4 Chair will decide by 1997-07-31 which documents are sent to Qualification by 1997-08-15. The Change Management Team is requested to provide a clear business framework for guiding the technical debate prior to the workshop (1997-07-21).

RESOLUTION 335: (San Diego, CA USA - June 1997)

SC 4 approves the changes to the Abstract Test Suite Guidelines, SC 4 N 536, and requests the SC 4 Secretariat take appropriate action to publish and disseminate this

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 Standing Document to the P-members.

RESOLUTION 336: (San Diego, CA USA - June 1997)

SC 4 establishes an ad-hoc Requirements Team to update the SC 4 documentation on the industry requirements for SC 4 standards and the terminology for describing these requirements. This work will build on the work of the SC 4 Policy and Planning Committee. Mark Palmer is requested to serve as team leader. The Requirements Team will deliver their initial report to SC 4 at their 1997-10-24 meeting.

SC 4 encourages SC 4 P-members to provide information and resources to support this team.

RESOLUTION 337: (San Diego, CA USA - June 1997)

SC 4 creates an ad hoc group "Shared Enterprise Data" to:

- Review the developments in other standards organisations related to data modelling and standardisation of enterprise information
- Establish an SC 4 position with respect to these activities that can be presented to other standards bodies and to the proposed coordinating group.

P-members and A-liaisons are invited to nominate experts for the group, and SC 4 WGs may also nominate experts if not already included in the list of nominations.

SC 4 further requests its Chairman and Secretary to ensure that ISO/TC 184 communicates to ISO the interest of SC 4 in any joint coordination activity with other standards bodies.

RESOLUTION 338: (San Diego, CA USA - June 1997)

SC 4 thanks its United States US PRO sponsor. Particular thanks go to Melanie Ridenhour, Erinbeth Massie, Kristi Adams, Susan Gilley, and Chuck Stark.

RESOLUTION 339: (Florence, Italy - October 1997)

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

ISO Project #

**Project Lead
Document Editor**

PWI: "10303-220/320"

John DeLoof

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

10303-24	David Price	David Price
10303-207	Prajakta Ketkar	Prajakta Ketkar
10303-221	Theo Arntz	
10303-223/323		Somendra Singh
10303-229/329		Somendra Singh
10303-317	Burt Gischner	Burt Gischner
10303-518	Jochen Haenisch	Jochen Haenisch

RESOLUTION 340: ISO/TC 184/SC 5 Liaison to/from SC 4 (Florence, Italy - October 1997)

SC 4 wishes to thank Mr. Neal Laurance for his liaison role with ISO/TC 184/SC 5, and endorses SC 5's liaison representative, Mr. Meinolf Groepper, as a joint active liaison between SC 5 and SC 4.

RESOLUTION 341: (Florence, Italy - October 1997)

SC 4 cancels Resolution 319 due to lack of leadership and activity on these preliminary work items.

RESOLUTION 342: Expectations for Normative References to SC 4 Parts (Florence, Italy - October 1997)

SC 4 requires that by the time a part is being registered as a Draft International Standard (DIS) its associated normative references shall also be registered at least as Draft International Standards.

By the time a part is being registered as a Final Draft International Standard (FDIS) its normative references shall be at least registered as Draft International Standards (DIS).

RESOLUTION 343: WG 12 Deputy Convener (Florence, Italy - October 1997)

SC 4 approves Michael Endres as Deputy Convener of WG 12, SC 4 Common Resources.

RESOLUTION 344: Appreciation for Service as JWG 9 Convener (Florence, Italy - October 1997)

SC 4 wishes to thank Shaun Devlin for his service to SC 4 as Convener of JWG 9, ISO/IEC Electrical/Electronics Applications.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 345: Appreciation for Service as WG 10 Convener (Florence, Italy - October 1997)

SC 4 wishes to thank Julian Fowler for his service to SC 4 as Convener of WG 10, Technical Architecture.

RESOLUTION 346: Quality Committee Deputy Convener (Florence, Italy - October 1997)

SC 4 approves Julian Fowler as Deputy Convener of the Quality Committee.

RESOLUTION 347: Updates To Guidelines For The Development Of Abstract Test Suites (Florence, Italy - October 1997)

SC 4 approves the changes to 'Guidelines for the development of abstract test suites' as provided in SC 4 N637.

RESOLUTION 348: PWI - Requirements Definition for Future Editions of AP 203 (Florence, Italy - October 1997)

SC 4 establishes a Preliminary Work Item within WG 3, to perform requirement definition of future editions of AP 203 and appoints Laurence J. McKee as project leader.

RESOLUTION 349: Appreciation for Service as WG 3 Convener (Florence, Italy - October 1997)

SC 4 wishes to thank Barbara Warthen for her service to SC 4 as Convener of WG 3, Product Modeling.

RESOLUTION 350: Numbering Designation of the Parametrics Efforts (Florence, Italy - October 1997)

SC 4 directs the parametrics team to number any proposed standards that result from this work as parts of ISO 10303. The 14959 designation shall be retired/recycled.

RESOLUTION 351 (Florence, Italy - October 1997)

SC 4 thanks its Italian UNINFO host and sponsor for the SC 4 meeting in Florence, Italy. Particular thanks go to Carla Sirocchi, Elena Carnevale, Sylvia Carnevale,

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Jenny Martellini, and Luciano Lauro.

RESOLUTION 352 (Approved by mail ballot resolution 1998-03-18)

SC 4 approves Gerald Radack as Convener of WG 3, Product Modeling.

RESOLUTION 353 (Approved by mail ballot resolution 1998-03-18)

SC 4 approves Bernd Wenzel as Convener of WG 10, Technical Architecture.

RESOLUTION 354 (Bad Aibling, Germany – June 1998)

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

ISO Project # Editor	Project Lead	Document
10303-221	David Leal (primary) Hans Teijgeler.(alt.)	
10303-501, -502, -503, -510, -515	Len Slovensky	
10303-232	Greg Paul	
10303-314	Manfred Fischer	
TR: "Parametric representation and Exchange"	Richard Wittenoom	

RESOLUTION 355 (Bad Aibling, Germany – June 1998)

SC 4 instructs the ISO 10303-221 project leader and the ISO 15926 PWI team leader to develop the technical and business proposal for approval by SC 4 before submission for approval by ISO.

SC 4 instructs its secretary to notify the ISO/CEO, in accordance with the ISO Directives, that the establishment of a registration authority for class libraries will be necessary to support the use of ISO 10303-221 and ISO 15926. SC 4 asks the project leaders of ISO 10303-221 and the PWI Team for ISO 15926 to work with the ISO CEO to create a suitable registration authority.

SC 4 instructs the project teams to report on progress at its meeting in San Francisco

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

and present the final technical and business proposal for approval by SC 4 before submission for approval by ISO.

SC 4 asks that any other projects that may be interested in this activity work with the two project leaders to incorporate their requirements in the proposal.

RESOLUTION 356 (Bad Aibling, Germany – June 1998)

ISO/TC 184/SC 4 instructs its secretariat and PPC to publish the SC 4 organizational handbook in a Web-based format which does not duplicate the ISO directives, but provides hypertext links to the electronic version of the document.

RESOLUTION 357 (Bad Aibling, Germany – June 1998)

SC 4 wishes to thank Ms. Sharon Kemmerer for her contributions as SC 4 Secretary. SC 4 presents a plaque of gratitude in honor of Sharon Kemmerer which shall read:

June, 1998

ISO/TC 184/SC 4

Recognizes Ms. Sharon

Kemmerer

In Appreciation
for

Outstanding Performance and Contributions
as SC 4 Secretary
1995-1997

RESOLUTION 358 (Bad Aibling, Germany – June 1998)

SC 4 resolves to add the following language to the SC 4 Handbook, Section 2.3 The Proposal Stage. The language shall be added as a new paragraph after paragraph 3.

For any normative references, their status at the time the new work item is expected to be registered for circulation as a Draft International Standard shall be provided. For normative references whose status is not at least Draft International Standard (4.00), Final Draft International Standard (5.00) or International Standard (6.00) at the time of

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

the proposal, the expected date for registration as a Draft International Standard shall be provided. If the normative reference has not been circulated as a Committee Draft at the time of the proposal, the expected date for circulation of the Committee Draft shall be also be provided. The source of dates shall be the Project Master Schedule Data Base maintained by the Secretariat.

RESOLUTION 359 (Bad Aibling, Germany – June 1998)

SC 4 approves acceptance of the PWI proposal (SC 4 N715) entitled 'History-based shape modeling', and directs the Secretariat to take appropriate action to initiate the Work Item.

RESOLUTION 360 (Bad Aibling, Germany – June 1998)

SC 4 approves Bruno Schilli as Deputy Convener of ISO/TC 184/SC 4/WG 3.

RESOLUTION 361 (Bad Aibling, Germany – June 1998)

SC 4 resolves that ISO 10303 Release 2.0 comprises the following parts and instructs the project leaders to complete the work necessary to make them international standards by April, 2000.

- Part 1E2 Overview and Fundamental Principles
- Part 11* Description Methods: The EXPRESS Language Reference Manual
- Part 21* Implementation methods: Clear text encoding of the exchange structure
- Part 22 Implementation methods: Standard data access interface specification
- Part 23 Implementation methods: C++ language binding to the standard data access interface
- Part 24 Implementation methods: C language binding to the standard data access interface
- Part 26 Implementation methods: Interface definition language binding to the standard data access interface
- Part 31* Conformance testing methodology and framework: General concepts
- Part 32 Conformance testing methodology and framework: Requirements on testing laboratories and clients
- Part 34 Conformance testing methodology and framework: Abstract test methods
- Part 41E2 Integrated generic resource: Fundamentals of product description and support
- Part 42E2 Integrated generic resource: Geometrical and topological representation

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Part 43E2 Integrated generic resource: Representation structures
- Part 44E2 Integrated generic resource: Product structure configuration
- Part 45 Integrated generic resource: Materials
- Part 46 * Integrated generic resource: Visual presentation
- Part 47 * Integrated generic resource: Shape variation tolerances
- Part 49 Integrated generic resource: Process structure and properties
- Part 101* Integrated application resource: Draughting
- Part 104 Integrated application resource: Finite element analysis
- Part 105* Integrated application resource: Kinematics
- Part 106 Integrated application resource: Building construction core model
- Part 201* Application protocol: Explicit draughting;
- Part 202* Application protocol: Associative draughting;
- Part 203* Application protocol: Configuration controlled design
- Part 204 Application protocol: Mechanical design using boundary representation;
- Part 205 Application protocol: Mechanical design using surface representation;
- Part 207 Application protocol: Sheet metal die planning and design;
- Part 208 Application protocol: Life cycle management - Change process;
- Part 209 Application protocol: Composite and metallic structural analysis and related design;
- Part 210 Application protocol: Electronic assembly, interconnect, and packaging design;
- Part 212 Application protocol: Electrotechnical design and installation;
- Part 213 Application protocol: Numerical control process plans for machined parts
- Part 214 Application protocol: Core data for automotive mechanical design processes;
- Part 221 Application protocol: Functional data and their schematic representation for process plant;
- Part 222 Application protocol: Exchange of product data for composite structures
- Part 223 Application protocol: Exchange of design and manufacturing product information for casting parts;
- Part 224 Application protocol: Mechanical product definition for process plans using machining features;
- Part 225 Application protocol: Building elements using explicit shape representation;
- Part 227 Application protocol: Plant spatial configuration;
- Part 501 Application interpreted construct: Edge based wireframe
- Part 502 Application interpreted construct: Shell-based wireframe
- Part 503 Application interpreted construct: Geometrically bounded 2D wireframe

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Part 504 Application interpreted construct: Draughting annotation
- Part 505 Application interpreted construct: Drawing structure and administration
- Part 506 Application interpreted construct: Draughting elements
- Part 507 Application interpreted construct: Geometrically bounded surface
- Part 508 Application interpreted construct: Non-manifold surface;
- Part 509 Application interpreted construct: Manifold surface
- Part 510 Application interpreted construct: Geometrically bounded wireframe
- Part 511 Application interpreted construct: Topologically bounded surface
- Part 512 Application interpreted construct: Faceted boundary representation
- Part 513 Application interpreted construct: Elementary boundary representation
- Part 514 Application interpreted construct: Advanced boundary representation
- Part 515 Application interpreted construct: Constructive solid geometry
- Part 517 Application interpreted construct: Mechanical design
geometric presentation
- Part 518 Application interpreted construct: Mechanical design
shaded representation
- Part 519 Application interpreted construct: Geometric tolerances
- Part 520 Application interpreted construct: Associative draughting elements

Also included are the technical report 10303-12 Description Methods: The EXPRESS-I language reference manual and the technical reports 10303-302 to -305, -307 to -310, -312 to -314, -321 to -325, and -327 which are the abstract test suites for the corresponding application protocols. An asterisk (*) denotes parts that are already international standards. The E2 notation after a part number denotes a second edition.

Additional parts may be considered for Release 2, however, no additional part will be included if its inclusion will result in a schedule slippage.

Moreover, if there is slippage in the adopted schedule of any part in the above list, then SC 4 may vote to defer that part to a future release.

SC 4 requests project leaders to confirm that their projects can achieve the proposed dates and instructs them to provide firm milestones to the secretariat.

SC 4 requests that the Secretariat report on any slippage of these parts at each SC 4 meeting so that appropriate action can be taken.

RESOLUTION 362 (Bad Aibling, Germany – June 1998)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 approves extending the scope of the SEDS procedures to cover:

- SC 4 standing documents;
- SC 4 technical reports.

SC 4 requests its secretariat to incorporate this change into the SC 4 handbook along with the new SEDS procedures as described in SC 4 N 714.

RESOLUTION 363 (Bad Aibling, Germany – June 1998)

SC 4 requests an active two-way liaison with European Marine STEP Association, and requests the SC 4 Secretary confirm this liaison with ISO Central Secretariat. SC 4 approves Mr. Tim Turner as the person to actively liaise with EMSA on SC 4's behalf.

RESOLUTION 364 (Bad Aibling, Germany – June 1998)

SC 4 requests an active two-way liaison with ISO/IEC JTC 1/WG 4 (Document Description Languages), and requests the SC 4 Secretary confirm this liaison with the ISO Central Secretariat. SC 4 approves Nigel Shaw as the person who actively liaises with ISO/IEC JTC 1/WG 4 on behalf of SC 4; and Elliot Kimber as the person who actively liaises with SC 4 on behalf of ISO/IEC JTC 1/WG 4.

RESOLUTION 365 (Bad Aibling, Germany – June 1998)

SC 4 resolves to forward "Technical Corrigendum 2, Industrial automation systems and integration — Product data representation and Exchange — Part 203: Application protocol: Configuration controlled 3D designs of mechanical parts and assemblies" to ISO for publication, including ISO/TC 184/SC 4 N 731.

RESOLUTION 366 (Bad Aibling, Germany – June 1998)

SC 4 approves the Preliminary Work Item on Process Plant Operations and Maintenance described in the proposal document (SC 4 N 733). SC 4 assigns this Preliminary Work Item to WG 3, and appoints Dr Graham Bird as Project Manager, Dr. Ming Liang Lu as Technical Leader and N. K. Nakada as Technical Co-Leader.

RESOLUTION 367 (Bad Aibling, Germany – June 1998)

SC 4 wishes to thank Mr. Groepper for his active work within the SC 4 community, for

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

his service on the SC 4 Policy and Planning Committee, both as a member and as Rapporteur and for his work as joint active liaison between ISO/TC 184/SC 5 and SC 4.

RESOLUTION 368 (Bad Aibling, Germany – June 1998)

SC 4 notes the work of IEC Sector Board 3 and the encouragement given by ISO/TC 184 (Paris Resolution 1) to the ISO/TC 184 Chairman. SC 4 encourages its Chairman and Secretary to continue their supporting efforts to establish a comprehensive, industry-driven global strategy for industrial automation standards.

RESOLUTION 369 (Bad Aibling, Germany – June 1998)

SC 4 approves the establishment of a Preliminary Work Item on Dimensional Inspection Information Exchange (SC 4 N734).

RESOLUTION 370 (Bad Aibling, Germany – June 1998)

SC 4 notes the progress to extend the ISO/IEC UN/ECE Memorandum of Understanding into the domain of electronic business and the specific invitation for the participation of the SC 4 Chairman as a member of the MoU Management Group. SC 4 endorses the Paris 21 and 22 resolutions of ISO/TC 184 and encourages the SC 4 Chairman to act as representative and point of contact.

RESOLUTION 371 (Bad Aibling, Germany – June 1998)

SC 4 notes the Paris 3 resolution of ISO/TC 184 and requests the SC 4 Secretariat to prepare a report for the Beijing meeting describing the anticipated electronic operation of SC 4 by September 1999. P-members, O-members, and Liaisons that will require special considerations or exemptions from electronic operation should make those known to the Secretariat by September 1, 1998 so that they may be included in the report.

RESOLUTION 372 (Bad Aibling, Germany – June 1998)

SC 4 approves the MoU as documented in SC 4 N 732.

RESOLUTION 373 (Bad Aibling, Germany – June 1998)

SC 4 instructs its Change management team to check all Proposals for New Work Items against timescale and resources.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

The following questions for check have to be considered before a proposal goes out for ballot to P-members for vote:

- General scope of the standard/part
- What other standards/parts are affected?
- What other standards/parts have to change for this work to be successful?
- What other standards/parts can be improved by this work?
- What other PWI/NWI will need to be developed?
- What is the estimated schedule?
- What resources are committed
- What schedules need to be coordinated

- Is there an overlap with the existing projects?
- Should scopes be modified?
- What other projects should be consulted?

A report based on these question shall be added to the New Work Item Proposal to inform P-members.

RESOLUTION 374 (Bad Aibling, Germany – June 1998)

SC 4 approves the Preliminary Work Item on STEP Modularization as described in document WG 10 N185 (SC 4 N 736) and appoints David Price as Project Leader. This project is assigned to WG 10.

SC 4 requests the projects to address validation of the process across multiple application protocols and the organization required to support implementation of the concept.

SC 4 directs the project to work in close cooperation with the data architecture project.

RESOLUTION 375 (Bad Aibling, Germany – June 1998)

SC 4 approves the PWI in furniture product data and project data as described in SC 4 N 735.

SC 4 assigns the project to WG 3.

SC 4 appoints Miguel Borràs and Ricardo Gonçalves as projects leaders.

RESOLUTION 376 (Bad Aibling, Germany – June 1998)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

In order to provide a firm foundation for the next release of SC 4 standards tentatively scheduled for 2003, SC 4 requests:

- P-members to supply statements of user requirements for this timescale, taking into account the strategic plan for SC 4 and the requirements document WG 10 N 173, in writing to the Secretariat by 15 September 1998;
- Working groups and the Quality Committee, as well as the leaders of any PWI work, to supply statements of any input and key milestones for technology and development processes, in writing to the QC Change Management Team leader by 15 September 1998.

The PPC and Change Management Team are requested to make a progress report for consideration by the SC 4 meeting in Beijing.

RESOLUTION 377 (Bad Aibling, Germany – June 1998)

SC 4 resolves to ensure that the international standard parts of STEP are Year 2000 compliant and to include necessary revisions for compliance as a part of the Technical Corrigendum for EXPRESS errors to be completed prior to the Beijing meeting. SC 4 directs the Convenors of WG 3, WG 11, and WG 12 to further examine other Parts and to propose changes where necessary by Beijing.

RESOLUTION 378 (Bad Aibling, Germany – June 1998)

SC 4 wishes to thank Bernd and Ramona Wenzel and the EuroSTEP support staff including the following members for their excellent service and hospitality in facilitating the Spring 1998 SC 4 meeting in Bad Aibling:

- Karl Brandt
- Patrick Leicht
- Malanie Mohaupt
- Andreas Saliseh

RESOLUTION 379 (Beijing, China – October 1998)

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

ISO Project #	Project Lead	Document Editor
ISO 10303-43e2	Julian Fowler	Julian
Fowler ISO 10303-209	Adnan Yucel	
ISO 10303-224	Len Slovensky	Len
Slovensky ISO 10303-314		Manfred Fischer
Dana Tripp Secretary, ISO/TC 184/SC 4 trippdsc4@gmail.com		



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

	Manfred Fischer	
ISO 10303-324	Len Slovensky	Len
Slovensky ISO 10303-504		Bill
Anderson		
ISO 10303-505		Bill Anderson
ISO 10303-506		Bill Anderson
ISO 10303-507		Jochen Haenisch
ISO 10303-508		Jochen Haenisch
ISO 10303-509		Jochen Haenisch
ISO 10303-517		Martin Philipp
ISO 10303-519		Martin Philipp
ISO 10303-520		Martin
Philipp ISO 13584-1		Ekkehard
Zwicker	Robert Schuler	
ISO 13584-10		Ekkehard Zwicker
ISO 13584-24		Eric Sardet
ISO 13584-26	Gerry Radack	Gerry
Radack ISO 13584-31	Gerd Ehinger	Ben Kassel

RESOLUTION 380 (Beijing, China – October 1998)

SC 4 agrees that the ability to maintain a previous edition of a part when a new edition is published may be useful in managing the evolution of SC 4 Standards.

SC 4 further agrees that the decision to exercise this ability should be made on a part- by-part basis.

SC 4 requests its Secretariat to request a waiver from ISO Central Secretariat which would enable the ability to retain up to two editions of a part, to inquire as to what guidelines should apply, and to present an appropriate procedure for ratification at the next SC 4 meeting

RESOLUTION 381 (Beijing, China – October 1998)

SC 4 approves the appointment of Matthew West as Deputy Convener of ISO/TC 184/SC 4/WG 10.

RESOLUTION 382 (Beijing, China – October 1998)

SC 4 requests an active two-way liaison with NAFEMS, and requests the SC 4

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Secretary to confirm this liaison with the ISO Central Secretariat. SC 4 approves David Leal as the person to liaise with NAFEMS on SC 4's behalf.

RESOLUTION 383 (Beijing, China – October 1998)

SC 4 notes the approval of the NWI "Ship Operational Logs, Records, and Messages" as ISO 10303-234. SC 4 recognizes that the information requirements are derived in part from the scope of ISO/TC 8/SC 10, Ships and Marine Technology/Computer Applications. SC 4, therefore, approves the creation of a joint working group (JWG) between ISO/TC 8/SC 10 and ISO/TC 184/SC 4.

The scope of the JWG shall be the development of Application Protocol ISO 10303-234 including documentation that supports the usage of this AP.

In order to ensure a development in accordance with the STEP procedures for Application Protocols and harmonisation with the current ship APs the joint working group shall be placed as JWG 14, "Ship Operational Logs", under ISO/TC 184/SC 4. To draw maximum benefit from the expertise of ISO/TC 8/SC 10, SC 4 encourages the active participation of ISO/TC 8/SC 10 experts in the JWG. The JWG shall collaborate closely with ISO/TC 184/SC 4/WG 3/T23 Ships, ISO/TC 8/WG 6, Maritime Navigation and Radio Communication Equipment and Systems / Digital Interfaces for Navigational Equipment within a Ship, and IEC/TC 18, Electrical Installations of Ships and of mobile and fixed Offshore Units.

RESOLUTION 384 (Beijing, China – October 1998)

SC 4 resolves to disband its working group 6, noting that any residual work is transferred to WG 11.

RESOLUTION 385 (Beijing, China – October 1998)

Further to Resolution 376, SC 4 requests its Secretariat to distribute the attached outline capability definition for the April 2003 release of SC 4 standards (SC 4 N776) to P and O-members and liaisons, with an invitation to provide written comments by the end of 1998, and further to nominate experts for an ad hoc meeting to refine the definition, to be held during the January 1999 meeting of SC 4.

RESOLUTION 386 (Beijing, China – October 1998)

ISO/TC 184/SC 4 is pleased to see that ISO/TC 211 is ready to harmonize its geometry representation with the one published in ISO 10303-42 and ISO 10303-43, as indicated

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

by ISO/TC 211 Resolution 78.

ISO/TC 184/SC 4 instructs its secretariat to announce a meeting on collaborative use of geometry between ISO 10303-42/43 and ISO 15046-7/11, to be held 1999-02-01/05 in the USA.

ISO/TC 184/SC 4 encourages all its working groups and projects and requests especially the Shape Representation project in its WG 12 and the Building and Construction Team in its WG 3 to participate in this meeting.

RESOLUTION 387 (Beijing, China – October 1998)

SC 4 approves the WG 8 request for a re-initialization of its time schedule for MANDATE based on the proposal of the WG 8 September 1998 slippage report (ISO/TC 184/SC 4/WG 8 N190) and the ISO directives.

RESOLUTION 388 (Beijing, China – October 1998)

SC 4 wishes to thank the China State Bureau of Quality and Technical Supervision, the China Standardization and Information Classifying and Coding Institute, the China STEP Center, and Mr. Ping Wang and his staff for their great hospitality, graciousness, and high quality of service in hosting the SC 4/WGs Meeting in Beijing, China, 1998-10- 05/09.

RESOLUTION 389 (San Francisco, California - January 1999)

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

ISO Project #

10303-208
Tim Turner
John Kendall

**Project Lead
Document Editor**

Janice Chinn

RESOLUTION 390 (San Francisco, California - January 1999)

SC 4 directs its Secretariat to provide SC 4 ballot documents to P-members, O-members and A-liaisons in electronic format as follows:

- SC 4 ballot documents will be made available on SOLIS in their original format,

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

PDF and ASCII text.

- SC 4 ballots will be posted to SOLIS, and announced via the SC 4 and P-Member exploder two weeks prior to the official start date of the ballot.
- Paper copies of SC 4 ballot documents will be provided to P-members on a per-request basis only.

SC 4 directs the secretariat to deliver an annex to the handbook detailing this process by April 9 1999

RESOLUTION 391 (San Francisco, California - January 1999)

SC 4 recognizes that the version of the SC 4 Organization Handbook held on SOLIS is the master copy of the document, and amends the approval procedure for changes as follows (changes highlighted):

5.7.5.2 Approval of the Minutes and Handbook Changes of the Preceding Meeting

Approval of the minutes of the preceding meeting confirms that resolutions appearing in the minutes properly reflect all the editorial changes made during the meeting. If resolutions caused changes in the SC 4 Organization Handbook, the proper inclusion and publication of those changes on SOLIS shall be separately confirmed.

5.11 Updates to the ISO/TC 184/SC 4 Organization Handbook

Updates to the ISO/TC 184/SC 4 Organization Handbook for changes to the organization or the procedures require a regular resolution paper (see 5.7.3.1). The resolution paper must contain the exact wording of the text to be included or changed in the SC 4 Organization Handbook. Changes to the SC 4 Organization Handbook take effect immediately after their decision in the SC 4 Meeting. While the minutes of the meeting report the outcome of the decision, the amended form of the Handbook will be published on SOLIS for confirmation at the next SC 4 Meeting. Because frequent changes can be expected in the Appendices, changes will be made at the discretion of the Secretariat, published on SOLIS and distributed at the next SC 4 meeting.

RESOLUTION 392 (San Francisco, California - January 1999)

SC 4 approves the following amendment to the Organisation Handbook to permit the

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

discussion of position papers at SC 4 meetings (changes highlighted):

5.7.1 ling an SC 4 Meeting

SC 4 meeting dates and locations shall be published at least 18 months ahead in order to allow member bodies to plan national meetings and travel budgets. An SC 4 meeting is officially called by transmitting the invitation to the meeting and the agenda at least four months prior to the date of the meeting. A shorter period is permissible if no P- member objects. Update pages to the SC 4 Organization Handbook, ordinary resolutions, position papers and papers for decision shall be mailed at least six weeks prior to the date of the meeting (IDP1, 3).

5.7.2 SC 4 Meeting Agenda

The SC 4 Meeting Agenda follows a general sequence, in which a number of items are prescribed.

The content of the Agenda is the following:

- 1. Opening of the Meeting*
- 2. Roll Call of Delegates*
- 3. Appointment of Drafting Committee and of two vote tellers*
- 4. Adoption of the Agenda*
- 5. Minutes and Handbook Changes of the Preceding Meeting*
- 6. Administrative Reports*
 - 6.1 Report of the Secretariat*
 - 6.2 Report from Liaisons*
- 7. Committee and WG reports*
 - 7.1 Reports from WG Conveners and Project Leaders*
 - 7.2 Reports from Quality Committee*



N4118

2026 – 05 – 15

7.3 Report from the PPC

7.4 Reports from Ad hoc Committees

8. Resolution Papers

8.1 Ordinary Resolution Papers

8.2 Last-Day Resolution Papers

8.3 Position Papers for discussion

9. Approving Conveners

10. Future Meetings

11. Closing the Meeting

5.7.3 Adding Items to the Agenda

The Chairman of SC 4, the SC 4 Secretary, the P-members, and the Conveners of all Committees and WGs can add items to the agenda by submitting resolution papers and position papers for discussion.

- Resolution papers shall contain background information stating the reasons for the resolution, the aims envisaged and the text of the resolution. Resolutions shall be carefully worded in order to achieve maximum clarity. See Annex G.3 for the format of a resolution paper.*
- Position papers shall contain background information for a particular issue, to permit an informed discussion at an SC 4 meeting and to allow the creation of a resolution on any consensus that may result from the discussion.*

5.7.3.1 Issuing a Resolution Paper

All resolution papers define a specific action, which may include a change to the organization or the procedures in the SC 4 Organization Handbook (see 5.11). Ordinary resolution papers, together with the agenda) must be available to P-members at least six weeks prior to the SC 4 Meeting. They must, therefore, be in the hands of the SC 4 Secretariat at least eight weeks prior to the SC 4 Meeting.



N4118

2026 – 05 – 15

5.7.3.2 Issuing a Last-Day Resolution Paper

Last-day resolution papers may address urgent issues resulting from immediately preceding WG meetings for which a high level of acceptance may be expected. Last-day resolution papers cannot cause changes to the Organization Handbook, however, a temporary action may be initiated, which may lead to a regular resolution paper in a subsequent meeting. Copies of last-day resolution papers must be handed to the SC 4 Secretary or meeting desk by 10.30 in order to be available for pick-up by delegation leaders at 15.30 on the day preceding the start of the SC 4 meeting. A camera-ready hard copy and digital ASCII file shall be provided by the originating national body or convener.

A last-day resolution will not be considered at the meeting if an objection is raised by a P-member. The resolution will be circulated as a letter ballot.

5.7.3.3 Issuing a Position Paper

Position papers provide the background information for the discussion of an issue at the SC 4 meeting. They should clearly identify the issue and its implications, together with any other relevant information for the discussion. Position papers should be available to

P-members at least six weeks prior to the SC 4 Meeting. They must, therefore, be in the hands of the SC 4 Secretariat at least eight weeks prior to the SC 4 Meeting.

Position papers submitted after this deadline must be handed to the SC 4 Secretary or meeting desk by 10.30 in order to be available for pick-up by delegation leaders at 15.30 on the day preceding the start of the SC 4 meeting. A camera-ready hard copy and digital ASCII file shall be provided by the originating national body or convener. Such a position paper will not be considered at the meeting if an objection is raised by a P-member.

RESOLUTION 393 (San Francisco, California - January 1999)

SC 4 adopts SC 4N777 as its strategic plan and requests the Secretariat to publish the document on SOLIS, and forward it to the ISO/TC 184 Secretariat.

SC 4 invites the WG and QC Conveners to review the proposed tactics against their current procedures, and to identify any discrepancies.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 394 (San Francisco, California - January 1999)

Further to Resolution 374, SC 4 resolves to allow nominated STEP projects to adopt the modular approach as defined in the following deliverables of the WG 10 STEP Modularization PWI:

1. WG 10 N221 Guidelines for the content of application modules Revision 0.6
2. WG 10 N222 Guidelines for the content of application protocols using application modules Revision 0.6
3. WG 10 N219 Industrial Framework Model Version 0.95
4. WG 10 N223 Application Module Development Points within the Application Protocol Development Process Draft 0.1
5. WG 10 N246 The Proposed Approach to STEP Modularization (Slide Presentation)
6. WG 10 N247 WG 10 STEP Modularization PWI Proposal for Standardization of Modules and Extensions Draft 0.3 (Slide Presentation)
7. WG 10 N226 WG 10 STEP Modularization PWI Proposal for Standardization of Modules and Extensions Draft 0.1 (White Paper)
8. WG 10 N227 WG 10 STEP Modularization Requirements Specification Revision 0.1
9. WG 10 N228 STEP Modularization PWI Issue Log
10. WG 10 N229 The STEP Modules Catalogue Demo

SC 4 resolves that the results of the nominated projects may be published as PASs or TSs.

SC 4 nominates the Engineering Analysis Core Model and the PDES Inc modular extension projects to endeavor to develop their deliverables using the above process.

SC 4 directs the WG 10 STEP Modularization PWI to support these projects and continue the development of the required guidelines and processes until SC 4 approves the documents and the supporting SC 4 organizations are in place.

SC 4 requests that its members, liaisons and other organizations provide additional resources to support this effort in the following areas:

11. finalizing requirements;
12. resolving the remaining technical and organizational issues;
13. reviewing and commenting on guideline documents;
14. using the modular approach and providing feedback to WG 10;

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

15. Addressing the use of ITAs, PASs and TSs within SC 4.

SC 4 requests WG 10 to report on progress on the pilot projects to the next SC 4 meeting.

RESOLUTION 395 (San Francisco, California - January 1999)

a. SC 4 requests a Category A active two-way liaison with the Object Management Group, and requests that the SC 4 Secretary confirm this liaison with the ISO Central Secretariat.

b. SC 4 approves Peter Denno and Evan Wallace (both of NIST, U.S.) as the parties who will be responsible for maintaining this active liaison, with Evan Wallace designated the recipient for formal SC 4 mailings and other formal communications between SC 4 and the OMG.

RESOLUTION 396 (San Francisco, California - January 1999)

SC 4 requests its Secretary to modify section 5.7.3 of the SC 4 Handbook to include A-liaisons in the list of organizations and individuals permitted to submit resolutions to SC 4.

RESOLUTION 397 (San Francisco, California - January 1999)

SC 4 accepts the A-liaison of IEC/TC 93 on JWG 9. SC 4 recommends that JWG 9 prepare a plan by the Lillehammer meeting for increased co-operation with IEC/TC 93.

RESOLUTION 398 (San Francisco, California - January 1999)

SC 4 welcomes the nomination of a point of contact within the ISO CS, to work with the current IEC Technical Officer, noting that ISO/TC 184 is the only ISO Technical Committee with such an arrangement.

SC 4 notes the increasing importance of coordination with other ISO Technical Committees and Subcommittees, particularly in the context of the Memorandum of Understanding between ISO, IEC, UN/CEFACT and the participating International User Groups, and also seeks to fully exploit the new ISO deliverables and electronic publication methods.

Accordingly, SC 4 requests ISO/TC 184 and the ISO CS to take the necessary steps

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

to provide ISO/TC 184 with a Technical Officer from the ISO staff.

RESOLUTION 399 (San Francisco, California - January 1999)

SC 4 resolves to cancel the following projects:

10303-222 - Application Protocol: Exchange of product data for composites
10303-322 – Abstract Test Suite: Exchange of product data for composites

RESOLUTION 400 (San Francisco, California - January 1999)

SC 4 resolves to remove the following AP projects and their associated Abstract test suites from ISO 10303 release 2.0

10303-208/308 - Application Protocol: Lifecycle management - Change process
10303-222/322 - Application Protocol: Exchange of product data for composites

RESOLUTION 401 (San Francisco, California - January 1999)

SC 4 cancels SC 4 project 10303-229/329 as New Work Item: “Application protocol: Exchange of design and manufacturing product information for forged parts”; but in recognition of the initiative’s continued merit within SC 4, approves it as a preliminary work item, stage 0.

RESOLUTION 402 (San Francisco, California - January 1999)

SC 4 requests QC to revise the SC 4 methods documents to allow the use of parts of other ISO/TC 184/ SC 4 standards as Common Resources in the development of STEP Application Interpreted Models, and other SC 4 standards.

SC 4 requests WG 12 to develop procedures to enable and control the use of other ISO/TC 184/SC 4 parts as Common Resources in the development of STEP Application Interpreted Models, and other SC 4 standards.

RESOLUTION 403 (San Francisco, California - January 1999)

SC 4 directs the Quality Committee and WG 11 to develop procedures to ensure that EXPRESS models in SC 4 standards conform to the requirements of ISO 10303-11

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

and to the guidelines for EXPRESS usage specified in the relevant methods documents.

RESOLUTION 404 (San Francisco, California - January 1999)

SC 4 wishes to thank US PRO, Chuck Stark and his tireless support staff for their quality services and generous hospitality in hosting the SC 4/WGs meeting 1999-01-25/29.

RESOLUTION 405 (Lillehammer, Norway - June 1999)

SC 4 Project Leadership and Editorship Approvals

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

<u>ISO Project #</u>	<u>Project Lead</u>	<u>Document Editor</u>
10303-35	Lothar Klein	
10303-41 ed 2	Pascal Huau,	Neal
Laurence 10303-50	Philip Kraushar	Ray Goult
10303-108	Mike Pratt	
10303-214		Martin Philipp
10303-233		Ian McDonanld
10303-504	Markus Maier	Markus Maier
10303-505	Markus Maier	Markus Maier
10303-506	Markus Maier	Markus Maier
10303-515	Markus Maier	Markus Maier

RESOLUTION 406(Lillehammer, Norway - June 1999)

SC 4 A-Liaison with PIEBASE

SC 4 approves a Category A active two-way liaison with the PIEBASE, approves Mr. Alan Doniger as the person to actively liaise to PIEBASE on SC 4's behalf, and requests that the SC 4 Secretary confirm this liaison with the ISO Central Secretariat.

RESOLUTION 407 (Lillehammer, Norway - June 1999)

FDIS Ballot Cycle

SC 4 request its Secretariat to ask ISO-Central Secretariat for a waiver of the FDIS vote for SC 4 standards subject to the following conditions:

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

After completion of the DIS balloting, the SC 4 Chairman, WG Convener, and Project Leader shall meet to decide if an FDIS ballot is required (the SC 4 Secretariat also attends this meeting but does not participate in the decision process). Currently the exemption from the formal YES/NO vote in an FDIS is established as 100% approval of member bodies. If that is not the case, an FDIS ballot will not be required if there are no technical issues submitted against the DIS document. If there are technical issues submitted, the decision as to whether to proceed to the FDIS ballot stage shall be at the discretion of the SC 4 Chairman, WG Convener, and Project Leader. The decision from these three parties must be unanimous for a document to bypass the FDIS ballot.

RESOLUTION 408 (Lillehammer, Norway - June 1999)
New Quality System

SC 4 resolves to accept the amendments to the SC 4 Organization Handbook as defined in SC 4 QC N111, with immediate effect. SC 4 directs the Quality Committee to circulate the SC 4 Quality Manual (QC N112) for SC 4 Standing Document ballot.

RESOLUTION 409 (Lillehammer, Norway - June 1999)

Procedures for the Preparation of Technical Corrigenda and Changes to Standing Documents

SC 4 instructs its secretariat to modify the SC 4 organizational handbook to reflect new procedures for incorporating minor changes to standards and standing documents as documented in SC 4N862

RESOLUTION 410 (Lillehammer, Norway - June 1999)
Documentation of ISO Time Limits for Project Deliverables

SC 4 instructs its secretariat to modify the SC 4 organizational handbook to include the ISO time limits for project deliverables and requirements for handling project slippage as documented in SC 4N863:

RESOLUTION 411 (Lillehammer, Norway - June 1999)
Template for Last-day resolution papers

SC 4 instructs its secretariat to modify the SC 4 organizational handbook to include a reference to the resolution papers template as documented in SC 4N864.



N4118

2026 – 05 – 15

RESOLUTION 412(Lillehammer, Norway - June 1999)

Approval of PWI for Process Specification Language

Subject to discussions with other interested groups in ISO/TC 184, SC 4 approves acceptance of the PWI proposal (SC 4 N875) entitled 'Process Specification Language', and directs the Secretariat to take appropriate action to initiate the Work Item.

RESOLUTION 413 (Lillehammer, Norway - June 1999)

Appointment of Convenor for Quality Committee

SC 4 approves the appointment of Thomas Warren as Convenor of the ISO/TC 184/SC 4 Quality Committee.

RESOLUTION 414 (Lillehammer, Norway - June 1999)

Appreciation for Service as Quality Committee Convenor

SC 4 wishes to express its thanks to Allison Barnard Feeney for her leadership of the ISO/TC 184/SC 4 Quality Committee and her contributions to the implementation of the new quality process for the SC 4 organisation.

RESOLUTION 415 (Lillehammer, Norway - June 1999)

Appreciation for Service as SC 4 Secretary

SC 4 wishes to express its thanks to Lisa Phillips for her efforts as Secretary of the ISO/TC 184/SC 4 organisation from 1997-9.

RESOLUTION 416 Lillehammer, Norway - June 1999)

SC 4 Contribution to the Integration of Enterprise Information

SC 4 invites its WG 12 to lead an ad hoc group to deliver the SC 4 information requirements into the BSR activity and the proposed MoU/MG summit, and participate in the construction of a global synthesis of enterprise information. National bodies are invited to nominate representatives to the ad hoc group by 31 July 1999.

RESOLUTION 417 (Lillehammer, Norway - June 1999)

Creation of Product Structure Team

SC 4 invites interested parties in the collection and synthesis of the requirements for product structure to participate in the WG 10 workshop on STEP Modularization in

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

September 1999.

Participants are specifically invited from projects including, but not restricted to, PLCs, the proposed Revision to AP 203, the PDM Schema, Ship Common Model and individual national initiatives such as PDML.

RESOLUTION 418 (Lillehammer, Norway - June 1999)

Change to Content of STEP Release 2

SC 4 decides to remove Part 221 and its associated Abstract Test Suites as well as Part 106 from the list of parts planned for Release 2 of ISO 10303.

RESOLUTION 419 (Lillehammer, Norway - June 1999)

PDML

SC 4 directs WG 11 to take into account the technical aspects of PDML within its work for the XML representation of EXPRESS-driven data, specifically in the development of a specification for an early bound form.

Furthermore, SC 4 recognizes and accepts the proposal and resolution of SC 34 to collaborate in the NWI being considered by WG 11: XML representation for EXPRESS- driven data.

RESOLUTION 420 (Lillehammer, Norway - June 1999)

Rainer Glatz

On the occasion of his last meeting, SC 4 recognizes the contribution of Rainer Glatz to the work of the Subcommittee since its inception in July 1984.

RESOLUTION 421 (Lillehammer, Norway - June 1999)

Appreciation of the Host

SC 4 wishes to thank Knut Lindelien, NTS, and his tireless support staff: Bjørnhild Sæterøy, Hans Backer, Ulf Leirstein, Ann-Kristin Nilsen, Camilla Svendsen, for their quality services and generous hospitality in hosting the SC 4/WGs meeting 1999-06-05/11.

RESOLUTION 422 (New Orleans, LA, USA - November 1999)

SC 4 Project Leadership and Editorship Approvals

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ISO Project #	Project Lead	Document Editor
10303-11 Edition 2	Don Sanderson	
10303-325	Julian Fowler	
10303-AP221	Andries van Renssen.	

RESOLUTION 423 (New Orleans, LA, USA - November 1999)
Retention of Dual Editions of ISO 10303-41, -42, -43, and -44

SC 4 requests that the ISO Central Secretariat retain the 1994 editions of ISO10303 – 41, -42, -43, and –44. In order to support parts 104, 204, 205, 209, 213 and 518 and the corresponding ATs. SC 4 directs the Project Teams of the respective parts to modify their documents to reflect this decision.

RESOLUTION 424 (New Orleans, LA, USA - November 1999)
EXPRESS Amendment

SC 4 resolves that the proposed amendment (WG 11N084 as amended) to ISO 10303- 11(1994) EXPRESS to support modularization be balloted as a DIS and incorporated in a new document with Technical Corrigendum 1 and Technical Corrigendum 2 for publication.

RESOLUTION 425 (New Orleans, LA, USA - November 1999)
Recognition of Joint OMG/SC 4 Harmonization Activity

SC 4 recognizes and supports the continuation of the harmonization activity by the OMG Manufacturing Domain Task Force and members of the SC 4 community. SC 4 requests that members disseminate information about this white paper and the OMG/ harmonization activity in general.

RESOLUTION 426 (New Orleans, LA, USA - November 1999)
Minor Revision Process Update for SC 4 Handbook

SC 4 requests the Secretary to make the following changes to Clause 3.1.3 Revision of the SC 4 Handbook.

1. Replace Clause 3.1.3 with the following text:

A revision to an International Standard results in the publication of a new edition of the standard. Revisions may take one of two forms.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Revision
Minor
Revision

3.1.3.1 Revision

A revision is developed when the extent and scope of the changes being made make it impractical to publish the changes in the form of an amendment.

The procedure for authorizing, developing and approving a revision shall be as described in clauses 2.3 through 2.7 of this handbook.

3.1.3.2 Minor Revision

A minor revision is developed when there are no fundamental changes to the existing International Standard and when a) the use of a Technical Corrigenda is not justified as described in IDP1, 2.10.1 or b) the cost and time for a full revision as described in clause

3.1.3.1 of this handbook is not justified.

The procedure for authorizing, developing and approving a minor revision is not described in the ISO Directives. The ISO Technical Officer for SC 4 describes the process as: a) Develop a completely new document, b) Forward the document to ISO with a certification by the SC 4 Secretary that there are not fundamental changes and request publication as a minor revision, c) ISO will ballot as an FDIS and upon a successful ballot publish as a new edition of the International Standard.

Within SC 4, the proposer shall seek the advice of the Change Management Advisory Committee. They will provide to the committee: a) An explanation of the need for the minor revision as documented in SEDS reports, b) an explanation of the expected changes to the standard and why the resulting documents will not make a fundamental change to the existing international standard and c) the schedule and resources for producing the required document.

Upon acceptance of the information provided, the Change Management Committee will advise the Secretary to draft a resolution for SC 4 to authorize the commencement of work on the minor revision.

Approval by SC 4 shall require the same level of approval as the current level of approval for an FDIS ballot.

When the document is completed, complying with SC 4 procedures and check-off, the

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Secretary will forward the document to ISO CS for ballot circulation as a minor revision.

2. Replace the reference in clause 3.2 to the QC Change Management Committee and the reference in clause 3.1 to the Change Management Team of the Quality Committee in clause 3.1 with a reference to the Change Management Advisory Committee

RESOLUTION 427 (New Orleans, LA, USA - November 1999)
Electronic Balloting

SC 4 instructs the Secretariat to identify actions necessary to migrate to a totally electronic operation of SC 4 as soon as feasible and to prepare any necessary proposals for approval at Melbourne.

RESOLUTION 428 (New Orleans, LA, USA - November 1999)
Balloting of Technical Specifications

SC 4 approves the following amendments to the Organization Handbook (changes in italics):

Clause 2.1:

Amend the current text: 30 Committee Stage: Consensus is achieved on a Committee Draft (CD) *or preliminary Draft Technical Specification (PDTs)*;
Insert: *Technical Specifications pass directly from the Committee Stage to the Publication Stage*

Clause 2.5:

Amend the current text:

The Committee Stage begins with the circulation for formal ballot of the CD *or TS* document by the SC 4 Secretariat. The SC 4 Secretariat shall not begin the Committee Stage until an electronic form is available for review by SC 4. A four-month voting period, which may be extended to six months, is used for the first ballot.

After each ballot, the Project team and Editor prepare a revised document that resolves the ballot comments received. A technical and editorial review of the revised document is conducted within the Project. The SC 4 Quality Manual provides procedures and guidelines for preparation of the document and for review of its content. The Project Leader and the Working Group Convener review and approve the document using the checklists provided by the Quality Committee before the next

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

stage of balloting starts.

For a TS, the Chairman and Secretariat, in conjunction with the Project Leader, assess the nature of any comments and decide whether to either

(a) submit the revised document to ISO for publication as a TS, and prepare a report for SC 4 how all the comments have been addressed, or

(b) resubmit the document for a further CD ballot.

Clauses 6.2.4, 7.6, 8.2: Add TS to CD in the list of documents.

SC 4 further requests its secretariat to make the corresponding changes to the SC 4 web pages.

RESOLUTION 429 (New Orleans, LA, USA - November 1999)

Reconfirmation of Conveners

David Price as Deputy Convener WG 11

Christophe Viel WG 11

Jean Jacques Michel WG

8 Gerd Ehinger WG 2

RESOLUTION 430 (New Orleans, LA, USA - November 1999)

Liaison with PLCS, Inc.

SC 4 accepts the request of PLCS, Inc. to establish a Category liaison from PLCS, Inc. to SC 4 and appoints John Dunford as liaison officer.

RESOLUTION 431 (New Orleans, LA, USA - November 1999)

Additional Liaison to OMG

SC 4 authorizes Bernd Wenzel to act as the representative of SC 4 in his contacts with OMG regarding modelling language requirements.

RESOLUTION 432 (New Orleans, LA, USA - November 1999)

Liaison between ISO/TC 184/SC 4 and INCOSE

SC 4 approves a Category A liaison with INCOSE, appoints Mr. David Oliver as the liaison officer, and requests that the SC 4 secretary confirm this liaison with the ISO Central Secretariat.

RESOLUTION 433 (New Orleans, LA, USA - November 1999)

Dana Tripp

Secretary, ISO/TC 184/SC 4

trippdsc4@gmail.com



N4118

2026 – 05 – 15

ISO 10303-518 Preparation as a DIS

SC 4 authorizes the Project Leader for ISO 10303-518 to prepare the document as a DIS.

RESOLUTION 434 (New Orleans, LA, USA - November 1999)

Industrial Requirement to Introduce into ISO 10303 Structures for controlling and Changing the Flow of a Process

SC 4 notes the SC 1 requirements for the functionality's including, but not limited to looping, branching, conditionals and parallel execution.

SC 4 invites SC 1 WG 7 to work jointly with SC 4 to specify these requirements.

SC 4 requests SC 1 WG 7 to defer the DIS ballot on ISO 14649 in order to resolve issues related to integration of its information models with SC 4 Standards.

RESOLUTION 435 (New Orleans, LA, USA - November 1999)

WG 12 Deputy Convenor

SC 4 thanks Michael Endres Jr. for his efforts as Deputy Convenor WG 12 and approves Christian Donges as Deputy Convenor of ISO/TC 184/SC 4/WG 12 "Common Resources".

RESOLUTION 436 (New Orleans, LA, USA - November 1999)

Proposal to allow Projects to enter Stage 3 ballot in HTML format

SC 4 resolves to allow SC 4 projects to enter their Stage 3 ballot with drafts documented using HTML. SC 4 resolves that the project be required to provide a printable review version of the materials. SC 4 resolves that these printable review versions and the html versions of the materials be excepted from conformance to the presentation requirements of the Supplementary Directives for ISO 10303. The provisions of Resolution 395 with respect to the use of guidelines development by the WG 10 STEP Modularization PWI remain in effect. SC 4 instructs the Quality Committee and the WG 10 STEP Modularization PWI Project to provide advice on the use of html to present documents.

RESOLUTION 437 (New Orleans, LA, USA - November 1999)

Proposal to split ISO 10303-27 into two parts and publish as Technical Specification



N4118

2026 – 05 – 15

SC 4 resolves that the technical content of ISO 10303-27 shall be published as two separate parts and as Technical Specifications rather than as a single part as a Draft International Standard. Conformance Class 3 of ISO 10303-27 shall be published as ISO 10303-27 under the title of the current ISO 10303-27 and Conformance Class 1 and 2 shall be published as ISO 10303-29 under the title of "Lightweight SDAI (insert trademark) Java Programming Language Binding".

SC 4 directs the Secretariat to update the URL containing the titles of all ISO 10303 parts to reflect this change.

RESOLUTION 438 (New Orleans, LA, USA - November 1999)

Impact of New EXPRESS Document

SC 4 requests its WGs to investigate the impact and benefits of the new Express document (see resolution C) and to put forward NWIs for revising parts, as appropriate.

RESOLUTION 439 (New Orleans, LA, USA - November 1999)

Expression of Appreciation

SC 4 wishes to express their appreciation:

- To the US for hosting the New Orleans SC 4 meeting. A special thanks goes to US PRO, Chuck Stark, and his staff for their assistance and support
- To the Central Secretariat for providing support in the person of Ms Sophie Clivio, and
- To ISO/TC 184 Secretariat, Ms. Christine Hermetet-Filez, for participating in the SC 4 meeting.

RESOLUTION 440 (Melbourne, Australia - February 2000)

Project Leader/Editor Changes

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

ISO Project	Project Lead	Document Editor
10303-108		Mike Pratt
10303-28	Peter Bergstrom	Robin
Lafontaine 10303-215		Tim

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

Turner		
10303-216	Tim Turner	Tim Turner
10303-218		Len Slovensky
10303-318	Yong Dae Kim	Yong Dae Kim
10303-234	Stian Ruud	Jochen Haenisch
10303-334	Stian Ruud	Jochen
Haenisch 10303-14	Martin Hardwick	Peter Denno
10303-11: amd	Phil Spiby	Phil Spiby
10303 11: Ed 2	Phil Spiby	Don Sanderson
10303-21: amd ed 2:	David Loffredo	David
Loffredo 10303-24	David Price	David Price
10303-27	Lothar Klein	Lothar Klein
10303-29	Tom Rando	Tom Rando
10303-34	Chritophe Viel	Bob
Mathews 10303-35	Lothar Klein	Lothar Klein
10303-223		Gerald Radack
10303-323		Gerald Radack
10303-229 (PWI)		Gerald Radack
10303-329 (PWI)		Gerald Radack
10303-321	Andries van Renssen	David
Leal		

RESOLUTION 441 (Melbourne, Australia - February 2000)

Cancellation of AP230 and ATS 330

SC 4 resolves to cancel AP230 and its associated ATS 330- Building Structural Frame: Steelwork.

RESOLUTION 442 (Melbourne, Australia - February 2000)

Clarification of the SEDS process

SC 4 directs its Secretariat to make the following changes to the SC 4 Handbook.

Add the following to the first sentence of 3.3 (added text italicized):

... the use of the SC 4 Standards and Standing Documents, and to gather requirements and proposals for enhancements to future editions, the Standard Enhancement and

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Discrepancy System (SEDS) tracking system has ...

Add the following paragraph after the bullet list in 3.3.1:

Any issues submitted during the ISO ballot process that are classified as “deferred” during the ballot resolution process are transferred into the SEDS process by the responsible Project Leader and/or Convener. This procedure ensures that all relevant issues are considered when amendments or new editions are being proposed or developed. All issues transferred from ballots are assigned priority 0 or priority 1, as defined in 3.3.3.2 below.

In the first bullet point of 3.3.2, correct the spelling of “ensure”.

RESOLUTION 443 (Melbourne, Australia - February 2000)

Revisions to the operation of the Quality Committee

SC 4 directs its Secretariat to make the following changes to the SC 4 Handbook (SC 4 N935).

replace 6.3.1 by the following:

6.3.1 Tasks

The Quality Committee performs the following tasks and activities.

Conducting periodic reviews of the SC 4 Quality System with Working Group
Conveners;

Coordinating tasks that revise or extend the components of the SC 4 Quality System, in response to the needs of SC 4 projects, issues (including those raised through the SEDS process), and the requirements of the ISO Technical Management Board and Central Secretariat;

Liaising with the SC 4 Secretariat and the ISO Central Secretariat on matters related to quality;

Coordinating audits of SC 4 projects;

Establishing and implementing a formal process for collecting quality related issues that fall outside the scope of the SEDS process;

Managing the assignment of and ensuring closure for resolutions of any issue; and

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Providing revisions to the SC 4 Organization Handbook related to quality matters.

The Quality Committee's coordination tasks are primarily undertaken using electronic communications and at QC Plenary meetings held as part of the main SC 4 Working Group meetings. Additional task-related meetings and workshops are arranged as required.

replace 6.3.2 by the following:

6.3.2 The Coordinator

The Coordinator of the Quality Committee is appointed by SC 4. The Coordinator is responsible for the following tasks.

Receiving and logging requests for creation or revision of documents that comprise the SC 4 Quality System.

Requesting resources to work on QC tasks from SC 4 projects and P-members; the Coordinator works with the PPC, SC 4 Chair and SC 4 Secretariat to encourage projects, P-members and A-liaisons to provide resources to undertake the tasks.

Ensuring consistency amongst the documents that comprise the SC 4 Quality System.

Coordinating audits of SC 4 projects.

Reporting QC activities to SC 4.

Convening QC plenary meetings.

Scheduling and facilitating technical meetings and workshops as required.

A Deputy Coordinator is appointed by SC 4 to assist the Coordinator. The Coordinator and Deputy Coordinator agree to and document their individual responsibilities. If the Coordinator is unable to attend a particular meeting of SC 4, the Deputy Coordinator will fulfil the Coordinator's responsibilities at that meeting.

Assessment of the performance of the Coordinator / Deputy Coordinator is the responsibility of SC 4. Input on the performance may be provided by the SC 4 Chair and the SC 4 Secretariat and assisted by the PPC.

The Coordinator / Deputy Coordinator must have resources to travel to at least three

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

meetings of SC 4 and its Working Groups each year; access to electronic mail; good communication skills, knowledge of the English language, and technical writing abilities; knowledge of ISO standards processes and procedures; quality systems (ISO 9000), and familiarity with the domains of SC 4's standardization efforts.

replace 6.3.3 by the following:

6.3.3 The Membership

The P-Members and A-Liaisons of SC 4 nominate the members of the Quality Committee. The SC 4 Secretariat maintains a list of members of the Quality Committee (names and addresses, project affiliation, fax, phone and e-mail). Additional resources participate in the work of the Quality Committee on a task-by-task basis; these resources are provided by SC 4 Projects, P-members, and A-liaisons.

Delete 6.4.

SC 4 also directs the Quality Committee to ensure that the SC 4 Quality Manual is consistent with the revised description of the QC's mode of operation as described above.

RESOLUTION 444 (Melbourne, Australia - February 2000)

Appointment of QC Coordinator and Deputy Coordinator

SC 4 appoints Tom Warren as Coordinator of the Quality Committee, and Julian Fowler as Deputy Coordinator of the Quality Committee.

RESOLUTION 445 (Melbourne, Australia - February 2000)

UML

SC 4 encourages OMG to accept interoperability between UML and EXPRESS (ISO 10303-11) as a mandatory requirement in the UML 2.0 development effort.

SC 4 directs its WGs 10 and 11 to work closely with OMG, especially the UML 2.0 WG in OMG's Analysis and Design PTF, in support of this goal.

SC 4 asks its member bodies and liaison organisations to support these efforts adequately in OMG through their member companies and organisations.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 446 (Melbourne, Australia - February 2000)

Appreciation Change of Project Leader for ISO 10303-28

SC 4 thanks Nigel Shaw and Daniel Rivers-Moore for their effort as project leader under WG 11.

RESOLUTION 447 (Melbourne, Australia - February 2000)

Title Change for ISO 10303-28

SC 4 accepts the recommendation of WG 11 to change the title of ISO 10303-28 to:
Product data representation and exchange: Implementation methods: XML
representation of EXPRESS schemas and data.

RESOLUTION 448 (Melbourne, Australia - February 2000)

Rationalization of SC 4 Liaisons

SC 4 notes that a number of its liaisons are inactive, due to evolving requirements and changes in other organizations, and resolves to terminate its direct liaisons with the following organisations:

CAM-I
UN/ECE
ISO/IEC JTC 1 SC 24
ISO/IEC JTC 1/SC 14
ISO/IEC JTC 1/SC 21/WG 3
ISO/IEC JTC 1/SC 21/WG 3/CSMF
ISO/IEC JTC 1/SC 6
ISO/IEC JTC 1/SC 7
ISO/IEC JTC 1/SC 7/WG
11 ISO/IEC JTC 1/WG 4

SC 4 notes the importance of its relation with UN/CEFACT which will be continued as part of participation in the MoU management group for electronic business.

RESOLUTION 449 (Melbourne, Australia - February 2000)

Balloting procedures for Technical Specifications

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 requests its secretariat to amend the Organisation Handbook as follows:
Amend Clause 2.3 to add reference to Technical Specifications "..... (CD) or Preliminary Draft Technical Specification (PDTS), (2).....

Amend Clause 2.4 by replacing the reference to "CD" by "ballot"

Amend Clause 2.5 by replacing all references to "CD ballot" by "ballot".

SC 4 also requests its secretariat to make any corresponding changes to the standardisation process summary on the SC 4 web site.

RESOLUTION 450 (Melbourne, Australia - February 2000)

Recognition of Sophie Clivio services to SC 4

SC 4 thanks the ISO/CEO for the nomination of Mrs. Sophie Clivio as Technical Officer for ISO/TC 184 and ISO/TC 184/SC 4.

SC 4 recognizes the excellence of the support and advice of Mrs. Sophie Clivio in the interpretation and explanation of the ISO Directives and the many TMB decisions complementing the Directives.

SC 4 recommends the ISO/CS to publish a new edition of the ISO Directives Parts 1 and 3 incorporating the various ISO/TMB decisions, in order to facilitate the access to the rules governing the Committee work.

RESOLUTION 451 (Melbourne, Australia - February 2000)

Establish Liaison with ISO/TC 207/SC 5

SC 4 accepts the request from ISO/TC 207/SC 5 to establish a liaison and appoints Dr. Anna Moreno as the liaison officer from SC 4 to ISO/TC 207/SC 5.

RESOLUTION 452 (Melbourne, Australia - February 2000)

Establish Liaison with VAMAS

SC 4 requests the Versailles Project on Advanced Materials and Standards (VAMAS) to establish a Category A liaison with SC 4, directed particularly at Technical Working Area (TWA) 10. SC 4 appoints Dr. Anna Moreno as the liaison officer from SC 4 to VAMAS.

RESOLUTION 453 (Melbourne, Australia - February 2000)

ISO/TC 184/SC 4 QC Resolution Paper

SC 4 resolves to accept the amendments to the SC 4 Organization Handbook

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

as defined in SC 4 QC N123, with immediate effect.

RESOLUTION 454 (Melbourne, Australia - February 2000)

Cooperation Between SC 4 and Object Management Group (OMG)

The SC 4 Secretariat has received a communication from the Object Management Group (OMG) concerning mutual interest and potential cooperative activities. SC 4 fully supports and looks forward to working with the OMG organization to achieve mutual benefits in the development, balloting, and accreditation of appropriate specifications which qualify as ISO deliverables.

RESOLUTION 455 (Melbourne, Australia - February 2000)

Appreciation Resolutions

SC 4 expresses its appreciation to Ms. Nicole Watkinson, Director of Manufacturing Industry, Victoria, in the Department of Industry and Regional Development, speaking on behalf of the Victorian Minister for Manufacturing and Industry, for her welcoming remarks and expressions of support for the work of SC 4.

Acclamation

SC 4 expresses its appreciation to its meeting host, Standards Authority of Australia (SAA), for the excellent meeting facilities, good food, and superb desserts as well as the wonderful banquet on Tuesday evening. SC 4 also expresses its appreciation to Mr.

David Bruce-Steer, Belinda Sparks, Sylvia Minton, Liam Wallwork and Helen Gomes for their hospitality and outstanding support both before and during the meeting.

Acclamation

SC 4 expresses its appreciation to the Convenors, Project Leaders and Rapporteurs for their efforts in support of the work of SC 4 and for the preparation of the reports that reflect these activities.

Acclamation

SC 4 expresses its appreciation to Rob Anderson of BSI and Howard Mason of BAE SYSTEMS for their efforts in support of the clean-up, verification, and validation of the SC 4 database. This is a major contribution that is key to the successful operation of SC 4.

Acclamation

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 expresses its appreciation to the Drafting Committee for the completion of its resolutions.

Acclamation

SC 4 expresses its appreciation to its Chairman, Mr. Dick Wandmacher, for his fine efforts in bringing about another successful SC 4 Plenary Meeting. His continuous leadership, patience, good humor and attention to getting the work done are critical to the success of SC 4.

Acclamation

SC 4 expresses its appreciation to Gabriel Barta for his service and contributions as Technical Officer for ISO/TC 184/SC 4.

RESOLUTION 456 (Bordeaux, France - June 2000)

SC 4 Project Leadership and Editorship Approvals

Project:	Project Leader	Project Editor
ISO 10303-24		David Loffredo
ISO 10303-214		Ullrich Pfeifer
ISO 10303-215		Len Slovensky
ISO 10303-216		Len Slovensky
ISO 10303-227ed1		Robin Rodenhauer
ISO 10303-227ed2	Mark Palmer	Burt Gischner
ISO 10303-332	Jim Clark	Jim Clark

RESOLUTION 457 (Bordeaux, France - June 2000)

ISO 8601-conformance

SC 4 resolves that all SC 4-related documents, standards, and announcements shall conform to ISO 8601 (Data elements and interchange formats – Information interchange

– Representation of dates and times). In particular, all calendar dates shall be in the form yyyy-mm-dd, utilizing the four-digit date, as prescribed in clause 5.2.1.1 of ISO 8601. Thus, the date February 29, 2000 would be represented as 2000-02-29.

SC 4 directs the Quality Committee to add ISO 8601 conformance to the appropriate standing documents (including the "Supplementary Directives for the Drafting and Presentation of ISO 10303") and to the checklists for approval of SC 4 documents. SC 4 directs the Quality Committee to revise the Supplementary Directives, wherever

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

applicable, to reflect this date policy.

This requirement shall be effective immediately, and shall be binding on all SC 4-related standards, documents, and announcements, regardless of whether it is documented in an approved standing document or checklist.

It is the responsibility of each person within SC 4 who produces a document to comply with this resolution. Project leaders, conveners, the SC 4 secretariat, and SC 4 management are directed to withhold signoff and publication of any documents that do not conform to the requirements stated in this resolution. The Quality Committee shall incorporate checks of compliance into its audit program.

RESOLUTION 458 (Bordeaux, France - June 2000)

Revised - Changes to AP Guidelines supporting mapping table improvements

SC 4 directs the Quality Committee to modify *Guidelines for the development and approval of STEP application protocols* to reflect the changes in the specification of mapping tables as documented in SC 4 N1048.

RESOLUTION 459 (Bordeaux, France - June 2000)

External Specifications

SC 4 requests the Secretariat to establish a process for adopting the appropriate specifications from outside ISO and IEC. These specifications shall be related to the ISO/TC 184/SC 4 scope and should be potential candidates for ISO accreditation.

The Secretariat is requested to submit a proposal for approval by SC 4 for the process and the criteria to be applied to such specifications.

RESOLUTION 460 (Bordeaux, France - June 2000)

Creation of a Joint Working Group with ISO/TC 184/SC 5

According to its previous resolution #412 at the Lillehammer meeting, and taking into account SC 5 KYOTO meeting resolution #346, SC 4 endorses the proposal itemized below concerning the creation of a Joint working group with SC 5.

- The SC 4/WG 8 shall be transformed into a new Joint working group (JWG 8 - Manufacturing Process and Management Information) between SC 4 and SC 5 with an enlarged scope to take into account the standardization work on PSL, as well as its existing work program.
- The JWG 8 shall be placed under SC 4 for administrative purposes.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- The convenor of SC 4/WG 8, Jean-Jacques Michel, shall be the convenor of the JWG 8.
- The JWG 8 shall ideally meet four times a year in conjunction with meetings of SC 4 and SC 5 or their WGs.
- The scope for the JWG 8 shall be as follows:
To enable the interchange of information needed to specify a manufacturing process in terms of a standardized representation of data, to exclude products and parts data, that express the information exchanged, shared or archived within, or between, industrial plants.
- Process Specification Language (PSL) standardization work, currently existing as a PWI (project 18629) of the SC 4 program of work, shall be one of the projects of the JWG 8 and is envisioned to lead to a new standard or multi-part standard. The scope of the PSL standardization work is the semantic based interchange of a manufacturing process description between different applications.
- PSL work in this JWG 8 shall be conducted by a project team. Michael Gruninger (SC 5/WG 1 and SC 4/WG 8 expert, University of Toronto and NIST) shall be the project leader, David Shorter (SC 5/WG 1 expert, IT-FOCUS) shall be deputy project leader, and Anne-Françoise Cutting-Decelle (SC 4/WG 8 expert, University of Savoie) shall be the project editor.

RESOLUTION 461 (Bordeaux, France - June 2000)

Convenor Reappointment

SC 4 re-appoints Greg Paul as convenor of WG 12 for a further 3-year term.

RESOLUTION 462 (Bordeaux, France - June 2000)

Proposal to create a STEP Modules Harmonization Team in WG 10

A new team is proposed under WG 10 (with the expectation that it will move to the appropriate WG as STEP Modularization is adopted as a mainstream practice) called the "STEP Modules Harmonization Team" with the following responsibilities:

- identify commonality across existing and proposed STEP APs, based

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- on industry requirements
- facilitate the exploitation of commonality through the development and use of modules
- support the development and standardization of modularized or extended application protocols, application modules and extensions and common resources by teams working in WG 3, JWG 9, and WG 12

The STEP Modules Harmonization team will perform the following tasks:

- facilitate harmonization workshops (typically 2-4 per year);
- build and maintain knowledge of the full scope of AP development within SC 4 (i.e., "keep the big picture")
- create and maintain tools such as a catalogue of modules (covering those in development as well as published modules), usage by APs and other modules, UoF repository, module repository, etc.
- maintain an issues log for development and use of modules, and work with other teams to resolve issues
- report to SC 4 on the level of harmonization achieved within and between specific module suites
- as requested by module/AP development teams, to act as a neutral third party, or arbitrator, in making technical decisions when disputes arise between and within the development teams

SC 4 requests nominations to this team from P-members and A-liaisons, to fulfil the following roles:

- Team leader
- Industry/application/domain expert
- Information requirements analysis expert
- Interpretation/interoperability expert
- Harmonization infrastructure developer

SC 4 resolves to establish this team prior to the next meeting in Charleston, and directs them to commence work upon completion of the necessary guidelines by the PWI Project (see resolution 374). SC 4 directs the PWI to complete the guidelines and submit them for standing document ballot no later than 2000-10-30.

RESOLUTION 463 (Bordeaux, France - June 2000)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4/SC 5 Liaison officer

According to the need of liaison reinforcement between ISO/TC 184/SC 4 and ISO/TC 184/SC 5, SC 4 approves the nomination of A. Bezos of ISO/TC 184/SC 4 as liaison officer with ISO/TC 184/SC 5.

RESOLUTION 464 (Bordeaux, France - June 2000)

A-Liaison FSIG

ISO/TC 184/SC 4 approves the A-liaison with funStep Interest Group (FSIG), and appoints Mr. Ricardo Gonçalves (UNINOVA-Portugal) and Mr. Miguel Borrás (AIDIMA – Spain) as the liaison officers, and requests the SC 4 secretary to confirm this liaison with the ISO Central Secretariat.

RESOLUTION 465 (Bordeaux, France - June 2000)

Change of title of ISO 15926

SC 4 approves a change of name of ISO 15926 from "Integration of life-cycle data for oil and gas production facilities" to "Integration of life-cycle data for process plants including oil and gas production facilities". SC 4 directs its Secretariat to update its database and web-pages to show this change of name, and to ensure that this is coordinated with the ISO Central Secretariat's records.

RESOLUTION 466 (Bordeaux, France - June 2000)

ISO/TC 184/SC 4 Liaison Officer to ISO/TC 213

According to the need of liaison reinforcement between ISO/TC 184/SC 4 and ISO/TC 213, SC 4 approves the nomination of V. Srinivasan of ISO/TC 184 as two-way liaison officer with ISO/TC 213.

RESOLUTION 467 (Bordeaux, France - June 2000)

Publication of Part 28 as Technical Specification

SC 4 confirms the decision of the Part 28 Project to process ISO 10303-28 as a Technical Specification, instead of as a Committee Draft as specified in the NWI Proposal. This TS is expected to be revised as necessary to accommodate new developments.

RESOLUTION 468 (Bordeaux, France - June 2000)

Proposal to split the EXPRESS/UML/XMI mapping from ISO 10303-28 into a new part

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

and publish as a Technical Specification

SC 4 resolves that the development of *A mapping of EXPRESS into the physical meta- model of UML enabling OMG XMI* use shall be removed from Part 28 and published as a new part and as a Draft Technical Specification ISO 10303-25 *EXPRESS to OMG XMI binding*. SC 4 directs the new Part 25 and Part 28 technical development teams to continue coordination efforts as appropriate. SC 4 names Peter Bergström as the Part 25 Project Leader and names David Price as Project Editor.

SC 4 directs the Secretariat to update the URL containing the titles of all ISO 10303 parts to reflect this change.

RESOLUTION 469 (Bordeaux, France - June 2000)

Changes to ATS requirements

Part-1

SC 4 cancels its resolutions #168, #169, #257, and #278 and establishes the following policy for development and publication of test cases and abstract test suites.

SC 4 requires that all projects develop appropriate test cases to validate the draft specifications during their development, and to deliver such test cases with the CD and DIS, or DTS. For Parts that have already been registered as DIS, such test cases should be made available at the earliest opportunity.

Test cases shall be developed on the basis of usage scenarios and sample data reflecting the priorities of industry users. The industry priority for test cases should be reviewed by the appropriate Implementor's Forums, International Certification Boards, and/or industrial liaison organizations. The test cases shall be published as SC 4 documents that are freely available on the SC 4 Web site. The test cases may be included in an AP Usage Guide, in a recommended practices document, or in a separate document. Test cases for APs should be in a form suitable for use as a basis for an Abstract Test Suite.

SC 4 directs project leaders and working group convenors to assess the availability of suitable test cases as part of the pre CD and DIS ballot approval process, and also directs the Quality Committee to update the quality checklists accordingly.

SC 4 directs the convenors of WG 3 and JWG 9 to identify all abstract test suite parts that are under active development for publication as Technical Specifications, and thereby produce a list of ISO 10303-3xx parts that can be cancelled due to lack of

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

activity.

SC 4 directs the Convenors and the QC Coordinator to establish an ad hoc task group to develop simple guidelines for the documentation of these test cases, taking into account the existing ATS Guidelines. The ad hoc task group shall distribute the draft test case guidelines before the 2000-10 SC 4 meeting and conduct a review meeting at the 2000-10 meeting.

Part-2

SC 4 requests its AP projects to consider the industrial requirement for developing Abstract Test Suites, in order to facilitate conformance testing.

The industry priority for Abstract Test Suites should be reviewed by the appropriate Implementor's Forums, International Certification Boards, and/or industrial liaison organizations.

Abstract Test Suites may be based on test cases published by the AP projects as part of their development process.

Abstract Test Suites shall be published as Technical Specifications.

RESOLUTION 470 (Bordeaux, France - June 2000)

Enhanced liaison with ISO/TC 29/WG 34

SC 4 extends its existing liaison between SC 4/WG 2 and ISO/TC 29/WG 34 to cover all SC 4 activities. SC 4 nominates Norman Swindells as its liaison officer, and welcomes the nomination of a liaison from ISO/TC 29/WG 34.

RESOLUTION 471 (Bordeaux, France - June 2000)

Enhanced liaison with ISO/TC 10

SC 4 extends its existing liaison between SC 4 and ISO/TC 10/SC 1 to cover all ISO/TC 10 activities related to the scope of SC 4. SC 4 nominates Alain Bezos as its liaison officer, and welcomes the nomination of a liaison from ISO/TC 10.

RESOLUTION 472 (Bordeaux, France - June 2000)

Change of content for release 2

SC 4 decides to remove Parts 26, 213 and 313, and 223 and 323 from the list of parts planned for Release 2 of ISO 10303.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 473 (Bordeaux, France - June 2000)

Replacement for SC 4 Chair

SC 4 endorses the nomination by the United States of Howard Mason for the position of Chair of ISO/TC 184 / SC 4.

RESOLUTION 474 (Bordeaux, France - June 2000)

Appreciation's

Acclamation

SC 4 expresses its appreciation to its former Chairman, Mr. Dick Wandmacher, for his leadership, patience, un-tiring dedication and brilliant leadership.

Acclamation

SC 4 expresses its deep appreciation to Christine HERMETET-FILEZ, AFNOR, ISO/TC 184 Secretary, for her outstanding support both before and during the meeting; and for her positive and responsive attitude. We especially appreciate her welcoming remarks, expressions of support for SC 4, her patience, hospitality.

We address a special recognition to Emmanuelle Taulet and Isabelle Chrétien from BEC and to the hostesses, Beatrice, Lida and Marjorie, for their hospitality and outstanding support both before and during the ISO/TC 184/SC 4 meeting.

Acclamation

SC 4 expresses its appreciation to its meeting host, AFNOR, for the excellent meeting facilities, good food, as well as the wonderful Outstanding Social Event on Tuesday evening. SC 4 also expresses its appreciation for their hospitality and outstanding support both before and during the meeting.

We especially thank AFNOR representatives: Alain Durand, Deputy manager general and Pascal Poupet, Head of ICT Department, for taking the time to participate and share some pleasant moment with the SC 4 activities.

Acclamation

SC 4 expresses its appreciation to Sophie Clivio for taking time from her busy schedule to participate in our meeting and for her valuable guidance and support for the SC 4 work.

Acclamation

SC 4 expresses its appreciation to Jacques Pons for taking time from his busy

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

schedule to participate in our meetings and for offering kind words during the welcome party.

Acclamation

SC 4 expresses its appreciation to Rob Anderson BSI and Howard Mason BAE Systems for the tremendous progress achieved they have achieved in the database cleanup. Their efforts in support of the clean-up, verification, and validation of the SC 4 database is a significant and major contribution that is key to the successful operation of SC 4 and maintaining valid data for all members of SC 4.

Acclamation

SC 4 wishes to express its appreciation to Greg Paul for his dedicated effort, attention to details, superb management of the organization and work of WG 12. He is also commended for his support of other related SC 4 activities.

Acclamation

SC 4 wishes to expressed its appreciation to Gaylen Rinaudot for her years of service as the SOLIS administrator and for SC 4.

Acclamation

SC 4 express its appreciation for Tom Warren for his service as the Acting Chair of ISO/TC 184/SC 4.

RESOLUTION 475 (Charleston, SC - October 2000)

Permission to Include Photographs in SC 4 Documents

SC 4 resolves that photographs may be included in SC 4 normative documents when such photographs will enhance the understandability and use of those documents, in accordance with ISO directives.

SC 4 directs the Quality Committee to exclude any restriction in the Supplementary Directives on the inclusion of photographs.

RESOLUTION 476 (Charleston, SC - October 2000)

Replacement for Chair of Implementor's Forum

SC 4 endorses the nomination of Martin Hardwick as Chair of the Industrial Data Implementor's Forum to serve for a period of three years.

RESOLUTION 477 (Charleston, SC - October 2000)

Integrated Manufacturing Task Force

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 creates a task force on Integrated Manufacturing for mechanical components, with the following objectives:

- Define the scope of Integrated Manufacturing
- Create an Integrated AAM covering the whole scope of Integrated Manufacturing for mechanical components, taking into account the overlapping requirements of other sectors
- Evaluate the current SC 4 and other relevant normative documents in relation to the Integrated AAM, and to identify the overlaps and the missing links
- Create requirements for enhancing current SC 4 and other relevant normative documents- Survey the evidence of industrial needs for this effort

The results of the technical evaluation will be submitted to SC 4 at the next meeting, and the survey on industrial needs at the following meeting.

SC 4 appoints Pr. Takeshi Kishinami as leader of the task force

SC 4 directs its secretariat to circulate to P-members and A-liaisons a call for experts.

RESOLUTION 478 (Charleston, SC - October 2000)

New ISO Number for Express Edition 3

SC 4 requests ISO Central Secretariat to provide a new standard number for SC 4 project #32952.

SC 4 resolves that SC 4 project #32952 be removed as a part of ISO 10303 and published as a draft technical specification part under the new standard number.

SC 4 directs WG 11 and Express team to continue efforts to deliver this in a timely fashion and to take into account any requirements for dynamic capabilities arising from SC 5.

SC 4 requests WG 11 to identify a new name for the standard, emphasizing its derivation from EXPRESS.

RESOLUTION 479 (Charleston, SC - October 2000)

UML-EXPRESS Cooperation

SC 4 resolves that WG 11 create a team of people lead by Bernd Wenzel and including, but not limited to, the following individuals:

Dave Price

Phil Spiby

Dana Tripp

Secretary, ISO/TC 184/SC 4

trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 directs this team to work with the UML 2.0 submission teams to achieve the harmonization of EXPRESS and UML and act as a liaison with ISO/IEC JTC 1/SC 7.

SC 4 requests SC 4 secretariat to send a call for experts for the team.

RESOLUTION 480 (Charleston, SC - October 2000)

Cancellation of Inactive ATS Projects

SC 4 cancels the following inactive Abstract Test Suite Parts of ISO 10303, and instructs its Secretariat to remove them from the work programme, in the absence of any firm development commitment by 20 February 2001:

301, 309, 310, 313, 315, 316, 317, 321, 322, 323, 326, 327, 334, 335

RESOLUTION 481 (Charleston, SC - October 2000)

Appreciations

Acclamation

SC 4 expresses its appreciation to its meeting host, US PRO, for the excellent meeting facilities, good food, as well as the wonderful Social Event on Tuesday evening. SC 4 also expresses its appreciation for their hospitality and outstanding support both before and during the meeting.

SC 4 commends Chuck Stark for the outstanding arrangements that contributed to the successful completion of SC 4 business.

A special recognition to Stacy Watts, Jill Vaning, and Linda Beck for their hospitality, patience and outstanding support and all around good humor during crises both before, during, and after the ISO/TC 184/SC 4 meeting.

Acclamation

SC 4 expresses its appreciation to Sophie Clivio for taking time from her busy schedule to participate in our meeting and for her valuable guidance and support for SC 4 work.

Acclamation

SC 4 expresses its appreciation to Larry McKee for many years of outstanding leadership, and his many contributions to the development of SC 4 standards. Larry will be remembered for his patience and his valuable assistance to all. Last but not

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

least Larry will be remembered for his special “personal” humor and he will be missed.

Acclamation

SC 4 expresses its appreciation to those who participated for their outstanding contributions and professionalism this week to the work of ISO/TC 184/SC 4.

RESOLUTION 482 (Funchal, Portugal - February 2001)

SC 4 Project Leadership and Editorship Approvals

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Project Leader	Editor
10303-1001	Gerry Graves	
10303-1002	Gerry Graves	Darla Nettles
10303-1003	Gerry Graves	Darla Nettles
10303-1004	Gerry Graves	Darla Nettles
10303-1005	Gerry Graves	Darla Nettles
10303-1006	Gerry Graves	Darla Nettles
10303-1007	Gerry Graves	
10303-1008	Gerry Graves	
10303-1009	Gerry Graves	
10303-1025		Peter Denno
10303-0035		Gintaras Palabeckis

RESOLUTION 483 (Funchal, Portugal - February 2001)

Common Resources:

SC 4 resolves that ISO 10303 Generic and Application Integrated Resources and Application Integrated Constructs are Common Resources.

SC 4 resolves that ISO/TS 10303-1001 through 1009 are Common Resources.

SC 4 resolves that ISO 13584-20 (Logical resource: Logical model of expressions) is a Common Resource.

SC 4 resolves that ISO 15531-42 (Time model) is a Common Resource.

SC 4 directs WG 12 to prepare an SC 4 Standing Document for ballot, based upon

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

WG 12 N613 and this resolution, to identify current Common Resource documents, to define Common Resource Salient Characteristics, and to define Common Resource Unique Naming Requirements. The SC 4 Standing Document shall authorize WG 12 to modify the SC 4 Standing Document to update the list of Common Resources as they are approved.

SC 4 resolves that the Project Leader, during the Change Management review of a document, shall identify if a document is a candidate to be a Common Resource (or should be converted to a Common Resource). This decision shall be reviewed with WG 12 for compliance with Common Resources requirements. SC 4 directs Change Management to modify their procedures accordingly.

RESOLUTION 484 (Funchal, Portugal - February 2001)

Classification of requirements for ASN.1 identifiers

SC 4 approves the procedure for using ASN.1 identifiers described in document QC N177. SC 4 directs its project teams to apply this procedure to their work immediately. SC 4 also directs the Quality Committee to include this procedure in an updated version of the Supplementary Directives as soon as possible.

RESOLUTION 485 (Funchal, Portugal - February 2001)

Procedures for changes to the SC 4 Quality Manual

SC 4 approves the changes to clause 5.11 of the SC 4 Organization Handbook (SC 4 N1087) described in document QC N178 and requests its Secretariat to make the corresponding change to clause 5.7.3.2.

RESOLUTION 486 (Funchal, Portugal - February 2001)

Clarification of approval procedures

SC 4 approves the changes to the SC 4 Quality Manual specified in document QC N186.

RESOLUTION 487 (Funchal, Portugal - February 2001)

Externally Developed Standards

SC 4 resolves to adopt the procedures described in ISO/TC 184/SC 4 N_Funchal 16 "*Procedures For Transposing Externally Developed Specifications Into ISO Deliverables*" as a standing document to enable SC 4 to process Externally Developed Specifications, that meet certain criteria, for transposition for acceptance and

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

accreditation as an International Standard (IS), Technical Specification (TS), or Publicly Available Specification (PAS) and instructs the secretariat to circulate the document for approval as a standing document.

RESOLUTION 488 (Funchal, Portugal - February 2001)

Publicly Available Specification for Systems Engineering Data Representation

SC 4 resolves that the current contents of AP 233 be published as a Publicly Available Specification “Reference Model for Systems Engineering” with a new ISO number. SC 4 requests ISO Geneva to issue a new number for this Publicly Available Specification

SC 4 directs the Part 233 project team to prepare a publicly available specification for ballot, based upon WG 3 N908, N909, N910, N983 and this resolution.

SC 4 expects this PAS to be withdrawn within 3 years of publication and be replaced by ISO 10303-233 as initiated in Resolution H.

RESOLUTION 489 (Funchal, Portugal - February 2001)

Revision of PAS for Part 233

SC 4 directs the Part 233 project team to start the revision of the PAS defined in Resolution “Publicly available specification for systems engineering data representation”, to be published as Application Protocol 233.

RESOLUTION 490 (Funchal, Portugal - February 2001)

Use of HTML for publishing SC 4 standards

SC 4 welcomes the statement from ISO CS that SC 4 standards may now be published in HTML without any need for other digital formats such as PDF. SC 4 notes that use of HTML has many advantages, include the following:

- Ease of navigation for the end user using a hyperlinked document (or set of documents).
- Ease of preparation of documents, using automated tools to generate drafts as at present.
- Simpler review of drafts in HTML format, leading to a better quality product, which would not need to be converted into a different form for publication.
- Improved cross-references between documents, giving shorter documents



N4118

2026 – 05 – 15

- and avoiding transcription errors from replicating the same material.
- Improved support for modules.

SC 4 also notes that use of HTML does introduce risks associated with publishing standards in a user-editable form. SC 4 considers that these risks are outweighed by the advantages identified above.

In order to exploit the advantages identified above, SC 4 recognizes the need for guidelines for the use of HTML. SC 4 therefore directs its Quality Committee to develop an initial set of guidelines for the use of HTML. These guidelines should be available as a draft for comment by the San Francisco meeting in June 2001, and should at least address the following topics:

- version(s) of HTML to be used;
- HTML tags to be used for the different elements of SC 4 standards;
- validation of HTML documents;
- naming conventions for files and directories;
- hyperlinks within documents and between documents.

SC 4 also requests the Quality Committee to develop a roadmap for other issues regarding publication of SC 4 standards using HTML.

These guidelines should also take into account the issues and requirements identified during the joint WG 3 and QC meeting on Publishing Standards using XML/HTML and any relevant guidelines from the W3C and elsewhere.

SC 4 instructs the Quality Committee to liaise closely with Mrs. Joanna Goodwin and her colleagues at ISO CS to ensure that the guidelines developed for SC 4 are consistent with ISO's policies and production requirements.

SC 4 requests its P-members and projects to provide resources to the Quality Committee to work on this task, and encourages experts to attend or provide input to the QC meeting at Oklahoma State University in March 2001.

SC 4 directs any projects wishing to publish standards in HTML to work with the Quality Committee to provide requirements and to ensure that their work is consistent with the emerging guidelines for the use of HTML.



N4118

2026 – 05 – 15

The requirement for printable files defined in resolution 436 needs to be considered by the Quality Committee. SC 4 encourages the development of appropriate software tools to provide this capability.

RESOLUTION 491 (Funchal, Portugal - February 2001))

Appointment of David Loffredo as Deputy convenor of WG 11

SC 4 thanks David Price for his efforts as WG 11 Deputy Convenor and the STEP Modularization Project and appoints David Loffredo to serve as Deputy Convenor of its Working Group 11

RESOLUTION 492 (Funchal, Portugal - February 2001)

Cancellation of ISO 10303-231, Application protocol: Process engineering data:
Process design and process specifications of major equipment and cancellation of
ISO 10303-331

SC 4 cancels ISO 10303-231, Application protocol: Process engineering data: Process design and process specifications of major equipment, and its associated Abstract Test Suite, ISO 10303-331.

RESOLUTION 493 (Funchal, Portugal - February 2001)

Resignation WG 2

SC 4 thanks Gerd Ehinger for his leadership of the work in WG 2 and wishes him all the best for his future.

RESOLUTION 494 (Funchal, Portugal - February 2001)

Resignation of convenor of WG 10

SC 4 thanks Bernd Wenzel for his efforts within the whole SC 4 community and for his work to promote harmonization of activities in and around the work of SC 4, including his liaisons with ISO/TC 211 and OMG, and wishes him all the best for his future.

RESOLUTION 495 (Funchal, Portugal - February 2001)

Appointment WG 2

SC 4 appreciates and accepts the offer of Dr. Wolfgang Wilkes to serve as Convener of WG 2. SC 4 approves the nomination of Dr. Wolfgang Wilkes to serve for a period of three years.



N4118

2026 – 05 – 15

RESOLUTION 496 (Funchal, Portugal - February 2001)

Appointment JWG 9

SC 4 appreciates and accepts the offer of Lothar Klein to serve as Convener of JWG 9. SC 4 approves the nomination of Lothar Klein as Convener of the JWG 9 to serve for a period of three years.

RESOLUTION 497 (Funchal, Portugal - February 2001)

Part 28 processing

SC 4 endorses the following approach to publication of Part 28 as a Technical Specification:

- - the late binding and both early bindings will be kept in this version of the Tech. Spec.
- - provision for bi-directional inter-operability between the early bindings using XSLT will be added
- - objective is to revise the document for publication by June meeting

Following publication of the initial Part 28, work on an XML Schema binding will begin as requested in the ballot comments. The XML Schema solution will not be constrained to be upward compatible with the DTD solutions.

SC 4 requests Japan, France and USA to re-consider their votes based on the above actions to achieve consensus. In addition, member countries are requested to provide resources to complete the ballot resolution and editing process.

RESOLUTION 498 (Funchal, Portugal - February 2001)

Reappointment of WG 3 convenor

SC 4 approves the nomination of Gerry Radack as Convenor of the WG 3 to serve for a period of three years.

RESOLUTION 499 (Funchal, Portugal - February 2001)

Appreciations

Acclamation

SC 4 thanks Bruno Schilli for his efforts within the whole SC 4 community and for his



N4118

2026 – 05 – 15

work as WG 3 Deputy Convenor and PPC member and wishes him all the best for his future.

Acclamation

SC 4 thanks Reinhard Nerke for his efforts for his work as JWG 9 Deputy Convenor and wishes him all the best for his future.

Acclamation

SC 4 thanks David Price for his efforts for his work as WG 11 Deputy Convenor, his many project and technical contributions, and wishes him all the best for his future. His innovative and creative leadership is greatly appreciated.

Acclamation

SC 4 thanks Jim Nell of ISO/TC 184/SC 5, for his conscientious and dedicated assistance for updating STEP on a Page (SOAP) for SC 4.

Acclamation

SC 4 expresses its appreciation to Sophie Clivio for taking time from her busy schedule to participate in our meeting and for her valuable guidance and support for the SC 4 work, particularly the exploitation of HTML for publication of the standards.

Acclamation

SC 4 expresses its appreciation to Laura Thommen Vidale, the English technical editor for ISO/TC 184/SC 4, for the support she provides to SC 4's project teams and the Quality Committee in facilitating the delivery of SC 4 standards.

SC 4 expresses its deep appreciation to the Instituto Português da Qualidade (IPQ), Instituto de Desenvolvimento de Novas Tecnologias (UNINOVA), the Departamento de Engenharia Electrotécnica Faculdade de Ciências e Tecnologia Universidade Nova de Lisboa, and the staff of the Carlton Madeira for their efforts in organizing this meeting, arranging the facilities, excellent food, and numerous social events. SC 4 also expresses its appreciation for their hospitality and outstanding support both before, during, and after the meeting. SC 4 appreciates the attendance and support of Adolfo Steiger Garção, Presidente do UNINOVA and Carlos Ganopa, Presidente do IPQ.

We address a special recognition to Valdemar Lopes (IPQ), Ermelinda Barros (IPQ), Rubina Mendonça (IPQ), Lízia Martins (IPQ), André Mora (UNINOVA), Pedro Maló (UNINOVA), and Rui Tavares (UNINOVA) for their hospitality and outstanding support both before and during the ISO/TC 184/SC 4 meeting.

Acclamation

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 expresses its appreciation to the meeting sponsors: the Governo Regional da Madeira; the Secretaria Regional do Turismo e Cultura da Madeira; Câmara Municipal do Funchal; Caldeira, Costa & Companhia Lda.; CiberDimensão; Cabo TV Madeirense; and Banco Comercial Português without whom we would not have had as many opportunities to see and appreciate the people and land of the Island of Madeira.

SC 4 expresses its sincere appreciation to Ricardo Goncalves for his dedication, untiring efforts, enthusiasm, and creativity in achieving one of the most successful and memorable meetings in the history of the committee.

Acclamation

SC 4 expresses its sincere and deep appreciation to Alberto João Jardim, Presidente da Madeira, the Secretary for Tourism and Culture, and the Secretary for Social Affairs, and the Regional Director for Tourism and Culture for their attendance at the opening plenary of the conference and their hospitality at the Wednesday night social.

RESOLUTION 500 (San Francisco, USA - June 2001)

SC 4 Project Leadership and Editorship Approvals

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Project Leader	Editor
10303-0011ed3		Jochen Haenisch
10303-0024	Dave Loffredo	Dave Loffredo
10303-1025		David Price
10303-0215		Pete Lazo
10303-0233	Jim U'Ren	
10303-0234	Peik Jenssen	

RESOLUTION 501 (San Francisco, USA - June 2001)

Cancellation of ISO 10303-217

SC 4 cancels ISO 10303-217, Application Protocol: Ship Piping.

RESOLUTION 502 (San Francisco, USA - June 2001)

Development of Test Case Guidelines

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

In accordance with Resolution 469, SC 4 requests all Project Leaders to (1) submit a summary of the industrial demand for test cases in their implementable standards and technical specifications projects and (2) indicate the status, technical approach, and any issues on the development of test cases for their implementable standards and technical specifications projects. These should be sent to the Secretariat by 2001-09-10 in order to support the further development of the Test Case Guidelines (QC n-206).

RESOLUTION 503 (San Francisco, USA - June 2001)

Relationship with IAI

SC 4 welcomes the cooperation with the Industry Alliance for Interoperability on processing its existing specifications as ISO deliverables, and the efforts to ensure consistency between IAI and SC 4 deliverables.

SC 4 encourages its member bodies to develop cooperation with national chapters and experts of the IAI in order to facilitate the international activities.

RESOLUTION 504 (San Francisco, USA - June 2001)

Appointment of Deputy Convenor for WG 12

SC 4 thanks Christian Donges for his efforts as Deputy Convenor of WG 12 and approves Jochen Haenisch as Deputy Convenor of ISO/TC 184/SC 4/WG 12 “Common Resources”.

RESOLUTION 505 (San Francisco, USA - June 2001)

Cancellation of Preliminary Work Item on SGML and Industrial Data

SC 4 cancels its project 10303-pwi 0033 “Preliminary work item: SGML and industrial data”.

RESOLUTION 506 (San Francisco, USA - June 2001)

Creation of an ad hoc group on “Industrial data on the web”

SC 4 creates an ad hoc group to investigate how SC 4’s standards can be used for communication of industrial data using the internet and the World Wide Web. The primary focus of the ad hoc group shall be the use of standards developed by SC 4 as specifications of the content of transactions, messages, product data sets, and other aspects of internet and intranet based communications.



N4118

2026 – 05 – 15

This group shall deliver to SC 4 recommendations that identify the actions necessary to exploit the applications of SC 4's standards to the internet and to align SC 4's standards with those used for internet communications. These recommendations may include (but are not limited to) the following:

- establishment of formal liaisons with industrial initiatives, consortia, professional organisations, or standards bodies;
- revisions to existing SC 4 standards;
- initiation of new SC 4 projects (preliminary work items or new work items);
- marketing initiatives that promote the applicability of SC 4's standards to the internet.

SC 4 directs the ad hoc group to liaise with the team established by WG 3 and the QC to create guidelines for the use of XML and HTML for development and publication of SC 4 standards. The ad hoc group should also take into consideration the results of the PWI on SGML and Industrial Data (WG 10 n324 Annex B).

SC 4 directs the ad hoc group to present its initial results to the SC 4 meeting in Fukuoka (October 2001) and its final results to the SC 4 meeting in Myrtle Beach SC (March 2002).

SC 4 appoints Julian Fowler as the leader of this ad hoc group. SC 4 requests the conveners of WG 2, WG 3, JWG 8, JWG 9, WG 10, WG 11, and WG 12, and the coordinator of the Quality Committee to each nominate at least one expert to the ad hoc group. SC 4 requests its P-members and A-liaisons to nominate additional experts to the ad hoc group by the end of July.

The group will be disbanded in March 2002.

RESOLUTION 507 (San Francisco, USA - June 2001)

Ballot Software

SC 4 invites its member bodies to install and pilot the PDES, Inc. Ballot Software for the PDM Module Ballot.

The results of the pilot use of this software will be considered in developing future electronic balloting strategy.

RESOLUTION 508 (San Francisco, USA - June 2001)

Appreciations

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Acclamation

SC 4 expresses its appreciation to US Pro and the staff of the Cathedral Hill Hotel for their efforts in organizing this meeting, arranging the facilities, excellent food, and social events. SC 4 also expresses its appreciation for their hospitality and outstanding support before, during, and after the meeting.

We address a special recognition to Chuck Stark (US Pro), Stacy Watts (ATI), Sarah Hartline (ATI), and Martha Graham (NIST) for their hospitality and outstanding support both before and during the ISO/TC 184/SC 4 meeting.

Acclamation

SC 4 expresses its appreciation to Mr. Jim Mandracchia, Chief Strategist, EXOSTAR, for his insightful presentation during the Monday luncheon meeting.

Acclamation

SC 4 expresses its appreciation to Mr. Ricardo Gonçalves for the video summary of the SC 4 Funchal meeting.

RESOLUTION 509 (Fukuoka, Japan - October 2001)

SC 4 Project Leadership and Editorship Approvals

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Project Leader	Editor
20542 Systems Engineering PAS	Jim U'ren	

RESOLUTION 510 (Fukuoka, Japan - October 2001)

Project Planning Requirements

SC 4 amends clause 2.3 of the Organization Handbook to define the NWI requirements for an accompanying work plan as follows:

"The NWI shall contain a work plan identifying target dates for the following events, noting the suggested elapsed times from NWI approval:

ISO Stage	Milestone	Normal time from NWI
20.20	First working draft available on SOLIS	+12 months
30.20	CD/TR/TS ballot initiated by SC 4 Secretariat	+18 months



N4118

2026 – 05 – 15

40.00	DIS document accepted for ballot by ISO	+28 months
50.00	FDIS ballot registered for formal approval (if FDIS required)	+36 months
60.00	Final text accepted for publication by ISO	+40 months

The originator of the NWI shall nominate a Project Leader."

RESOLUTION 511 (Fukuoka, Japan - October 2001)

Harvesting of IAI IFC2x

SC 4 creates an ad hoc group to assess the proposal from the IAI to transpose the IFC2x platform into a Publicly Available Specification (PAS).

SC 4 invites WG 3, WG 10, WG 11, WG 12, PPC and the Quality Committee (QC) to nominate representatives by 2001-10-30. The ad hoc group is requested to assess the specification and explanatory report (N1211), negotiate with IAI on any issues, and make a recommendation to SC 4 by 2002-01-15 on how to process the proposal.

RESOLUTION 512 (Fukuoka, Japan - October 2001)

Harvesting of ATA Spec2000 Chapter 9, Bar Coding

SC 4 creates an ad hoc group to assess the proposal from the ATA to transpose the ATA Spec 2000 Chapter 9, Bar Coding, into an International Standard (IS).

SC 4 appoints King Yee as the Convener of the group and invites WG 3, WG 12, PPC and the Quality Committee (QC) to nominate representatives by 2001-10-30. The ad hoc group is requested to assess the specification and explanatory report (N1212), negotiate with ATA on any issues, and make a recommendation to SC 4 by 2002-01-15 on how to process the proposal.

RESOLUTION 513 (Fukuoka, Japan - October 2001)

Reappointment of Matthew West as WG 10 Deputy Convener

SC 4 appoints Matthew West as Deputy Convenor of WG 10 to serve for a period of three years.

RESOLUTION 514(Fukuoka, Japan - October 2001))

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

AP227ed2 Ballot Period

SC 4 authorizes AP227 to use a three-month ballot cycle for the initial CD ballot of second and subsequent editions of the part.

RESOLUTION 515 (Fukuoka, Japan - October 2001)

AP Module Referencing

SC 4 directs WG 10, in cooperation with WG 3, JWG 9, WG 11, WG 12, and the Implementors Forum, to assess the feasibility of allowing Application Modules to accurately and unambiguously reference content from IS Application Protocols rather than re-working the AP content, to identify and resolve any issues arising from this process, and to identify the changes required to the following documents:

- ◆ Guidelines for the contents of Application Modules
- ◆ Application Module development points within AP development process
- ◆ Guidelines for the contents of APs that use Application Modules

The work should address issues related to configuration management and the context, constraints, and rules associated with reference elements.

SC 4 invites its member bodies to nominate appropriate resources to WG 10 to accomplish the task. WG 10 is invited to report on this issue at the February 2002 meeting.

RESOLUTION 516 (Fukuoka, Japan - October 2001)

Publication of SC 4 Standards in HTML

SC 4 cancels Resolution 436 and establishes the following policy with regard to publication of standards in HTML.

SC 4 resolves to allow SC 4 projects to enter any stage ballot with drafts documented using HTML. It is the responsibility of the Project Leader to ensure that any documents entering Stage 4 or higher are in a form acceptable to the ISO Central Secretariat. The provisions of Resolution 395 with respect to the use of guideline development by the WG 10 STEP Modularization PWI remain in effect.

SC 4 instructs the Quality Committee to develop guidelines for the use of HTML for publication of documents, with reference to guidance provided by ISO CS and the



N4118

2026 – 05 – 15

ITSIG "Guide for the use of IT in the development and delivery of standards".

RESOLUTION 517 (Fukuoka, Japan - October 2001)
Harmonization of ISO 10303-219 and ISO 22093 (DMIS)

SC 4 respectfully urges its members to coordinate their national experts on ISO 10303- 219, ISO 10303-238, and other relevant STEP APs and ISO 22093 in order to avoid inconsistencies between these ISO/TC 184 standards.

RESOLUTION 518 (Fukuoka, Japan - October 2001)
Conformance testing of XML implementations

Following the completion of ISO 10303-28 edition 1, SC 4 asks WG 11 to establish the requirements for conformance testing of XML implementations of STEP conforming to ISO 10303-28 editions 1 and 2.

WG 11 is requested to prepare an NWI proposal and national bodies are invited to nominate appropriate resources.

RESOLUTION 519 (Fukuoka, Japan - October 2001)
ISA SP95 – Enterprise Control Integration

SC 4 respectfully urges its members to review carefully IEC 62264-1, 2 and 3 against the work of SC 4 and send their comments to Secretariat of ISO/TC 184.

RESOLUTION 520 (Fukuoka, Japan - October 2001)
Resolution from Integrated Manufacturing Task Force

SC 4 asks its secretariat to circulate the report of the Task Force on Integrated Manufacturing (N1214-N1216) to P-members, O-members, and Liaisons for comments and proposals for further requirements for manufacturing tasks. Responses should be received by the Secretariat not later than 2002-01-15.

The Final Report of Task Force on Integrated Manufacturing should contain specific recommendations for any new work required in SC 4 or other Committees.

RESOLUTION 521 (Fukuoka, Japan - October 2001)
Review of Impact of Express Edition 2

SC 4 notes the review of the new Edition of Part 11 (WG 11N179) undertaken by WG 11 members at the Fukuoka meeting, and requests WG 11 to circulate WG 11N179

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

to the full membership of the Working Group and to the Implementors for further review.

SC 4 requests its Secretariat to circulate WG 11N179 to P-members for information in considering their responses to the DIS ballot on ISO 10303-11 Edition 2.

SC 4 directs WG 11 to prepare a proposal for the implementation of the necessary changes to the 10303-2x series parts and 10303-14, in consultation with affected Working Groups, for consideration at the February 2002 SC 4 meeting. This should include estimates of timescales and resource requirements.

RESOLUTION 522 (Fukuoka, Japan - October 2001)

Appreciations

Acclamation

SC 4 expresses its deep appreciation to the Electronic Commerce Promotion Council of Japan (ECOM) and the staff of the Grand Hyatt Fukuoka for their efforts in organizing this meeting, arranging the facilities, excellent food, and social events. SC 4 also expresses its appreciation for their hospitality and outstanding support before, during, and after the meeting.

SC 4 addresses a special recognition to the staff and associates of ECOM, Mr. Hotsuma Miyakawa, Director of ECOM, Ms. Tomoko Fukugawa, Chief, International Policy Planning and Research Division, Kyushu Bureau of the Ministry of Economy, Trade and Industry (METI); Mr. Takashi Inouye, Mr. Yoshitaka Terakado, Ms. Chie Kouchi, Mr. Masahiko Chida, and Ms. Hiroko Tawara for their hospitality and outstanding support both before and during the ISO/TC 184/SC 4 meeting.

Acclamation

SC 4 expresses its deep appreciation to Mr. Julian Fowler for his untiring efforts as Deputy Coordinator of the Quality Committee. It is with great regret that SC 4 accepts his resignation and his announcement of reduced contributions to the SC 4 world.

Julian has been a major pillar to the organization since its earliest days, and his contributions in developing consensus and convergence has always been valued. His willingness to take on unpopular, but essential tasks, has been exemplary, and the challenge of finding a replacement will not be trivial.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 wishes him well in his new role!

Acclamation

SC 4 expresses its deep appreciation to Mr. Christophe Viel for his efforts as WG 11 Convenor, his many project and technical contributions, and wishes him all the best for his future. His innovative and creative leadership was greatly appreciated

Acclamation

SC 4 expresses its sincere thanks to Chris Kreiler for his untiring efforts and excellent performance as acting Secretary before and during the meeting, in the absence of the regular secretariat.

SC 4 also thanks Ms Allison Barnard-Feeney for her support of the Secretariat from NIST both before and during the meeting.

SC 4 sends its good wishes to Teresa Brittingham for a rapid recovery.

RESOLUTION 523 (Myrtle Beach, SC - March 2002)

SC 4 Project Leadership and Editorship Approvals

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303-28e2	Joe Trausch	Thomas
Liebich ISO 10303-51	David Leal	
ISO 10303-236		Ms. Maria José Nuñez
ISO 13584-25		(CO) Guy Pierra
20542 Systems Engineering PAS		Phil Spiby

RESOLUTION 524 (Myrtle Beach, SC - March 2002)

CD Ballot Period for Second and Subsequent Editions

SC 4 directs the PPC to modify the SC 4 Handbook and the appropriate guidelines to permit a three-month ballot cycle for the CD and TS ballots of SC 4 standards. A longer ballot period may be requested by any P-member, the Convenor or the project leader.



N4118

2026 – 05 – 15

RESOLUTION 525 (Myrtle Beach, SC - March 2002)

Appointment of David Loffredo as Convenor of WG 11

SC 4 appoints David Loffredo to serve as Convenor of its Working Group 11 for a period of three years.

RESOLUTION 526 (Myrtle Beach, SC - March 2002)

Ballot Comment Resolution for Technical Specifications

SC 4 agrees that the following procedure shall be applied to the resolution of comments submitted against successful Technical Specification ballots.

- 1) Resolution of comments shall be agreed with the originators, where possible.
- 2) The revised document incorporating the agreed comment resolutions shall be circulated within the development team and to the commentators for internal review to ensure agreement before signoff by the Convenor.
- 3) If the Convenor, in conjunction with the Project Leader, considers that the revised document contains significant change, other than corrections to the use of English, the document shall be passed to the Secretariat for a confirmatory four-week ballot.

This procedure is effective immediately, and SC 4 requests the Change Management Committee to prepare an advance resolution for the June 2002 meeting to make the necessary changes to the SC 4 Handbook.

RESOLUTION 527 (Myrtle Beach, SC - March 2002)

Integrated Manufacturing Task Force

SC 4 accepts the Report of the Integrated Manufacturing Task Force (IMTF) including all the correlating documents presented, and requests its Secretariat to circulate the final results to P-members.

SC 4 thanks the IMTF and its Chairman, Professor Takeshi Kishinami, for all their efforts in presenting such a detailed overview, and disbands the Integrated Manufacturing Task Force.

SC 4 invites the submission of appropriate PWI and NWI to address the requirements identified in the report.

RESOLUTION 528 (Myrtle Beach, SC - March 2002)

ISO/TC 10/SC 1 Request for Clarification in ISO 3098

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ISO/TC 184/SC 4 requests that the ISO/TC 10/SC 1 Liaison representative present the ISO 3098 clarifications identified in ISO/TC 184/SC 4/WG 12 N1016 for consideration at the next ISO/TC 10/SC 1 meeting.

RESOLUTION 529 (Myrtle Beach, SC- March 2002)

Appreciations

Acclamation

SC 4 expresses its deep appreciation to US Pro and the staff of the Wyndham Myrtle Beach Resort for their efforts in organizing this meeting, arranging the facilities, excellent food, computer facilities and social events. SC 4 also expresses its appreciation for their hospitality and outstanding support before, during, and after the meeting.

SC 4 addresses a special recognition to Chuck Stark, Sarah Hartline Martha Graham of NIST for their hospitality, outstanding support, and patience both before and during the ISO/TC 184/SC 4 meeting.

Acclamation

SC 4 expresses its deep appreciation to Mr. Peter Müller, Chairman ISO/TC 184/SC 1, for his participation.

Appreciation to our SC 4 Liaison Presenters

SC 4 expresses a special thank you to our the SC 4 Liaison Plenary presenters Peter Müller, Francois Grobler, Norman Andersen, Phil Spiby, Bill Anderson, Matthew West, and Jochen Hænisch for their insightful and comprehensive presentations.

Acclamation

SC 4 expresses its appreciation to those who participated for their outstanding contributions and professionalisms this week to the work of ISO/TC 184/SC 4.

Acclamation

SC 4 wishes to congratulate Tim Turner (WG 3/T23 - Shipbuilding) and Christy Haas on their wedding last November. SC 4 is delighted to have provided the opportunity for the happy couple to meet and grow their love during the period that Christy was working with US PRO and providing support to SC 4 and associated meetings. SC 4 wishes Tim and Christy a long and blossoming marriage as they make their home in Charleston, SC. (picture in Annex A of Myrtle Beach SC 4 Minutes)

RESOLUTION 530 (Stockholm, Sweden - June 2002)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 Project Leadership and Editorship Approvals

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303-235		Norman Swindells
ISO 10303 – PDM Modules		Pascal Huau

RESOLUTION 531 (Stockholm, Sweden - June 2002)

ISO/TC 184/SC 4 Resolution Paper for SC 4 Committee Meeting ISO/TC 184/SC 4: A-liaison with ASAM

SC 4 approves a Category A liaison with the Association for Standardisation of Automation and Measuring Systems (ASAM), accepts and welcomes Dr. Helmut HELPENSTEIN as the liaison officer between SC 4 and ASAM for two years, and requests that the SC 4 secretary confirms this liaison with the ISO Central Secretariat.

RESOLUTION 532 (Stockholm, Sweden - June 2002)

SC 4 agrees that the title and scope of AP239 shall be changed to read as follows:

Title: Application Protocol: Product life cycle support

Scope: This part of ISO 10303 will provide extension to the capabilities of AP203 and AP214 (and hence the PDM Schema/Modules) to address the requirements for Configuration Change Management over the full product life. This part of ISO 10303 will address the information requirements needed to define and deliver life cycle support to complex assets. This includes specification of

- the identification and composition of a product design from a support viewpoint (as an extension of the PDM Modules);
- the definition of documents and their applicability to products and support activities (as an extension of the PDM Modules);
- the identification and composition of realized products;
- configuration change management activities, over the complete life cycle;



N4118

2026 – 05 – 15

- the properties, states and behaviour of products for life cycle support;
- activities required to sustain product function;
- the resources needed to perform such activities;
- the planning and scheduling such activities;
- the capture of feedback on the performance of such activities, including the resources used;
- the capture of feedback on the usage and condition of a realized product;
- the definition of the support environment in terms of people, organizations, skills, experience and facilities;
- the definition of classes of product, activities, states, people, organizations and resources.

RESOLUTION 533 (Stockholm, Sweden - June 2002)

Open positions

SC 4 reminds its member bodies, that the following leadership positions are currently open in the organization:

- QC coordinator
- QC deputy coordinator
- WG 3 deputy convenor
- JWG 9 deputy convenor
- WG 11 deputy convenor
- "Industrial data on the Web" task force leader
- "Module harmonization" task force leader
- WG 3/T19 "Mechanical products" team leader
- Test Case Guidelines Project Leader

SC 4 respectfully urges its member bodies to submit nominations to its Secretariat accordingly.

RESOLUTION 534 (Stockholm, Sweden - June 2002)

Disbanding of WG 10

SC 4 resolves to disband WG 10, immediately after publication of TS 18876-1 and TS 18876-2.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 resolves to share the responsibility of the Module Harmonization task force as a joint activity between WG 3/JWG 9/WG 12.

SC 4 resolves to transfer responsibility for UML interfaces with the OMG to WG 11.

SC 4 notes that further projects on architecture, such as updates to the SC 4 Framework project, will be handled by Ad Hoc groups.

SC 4 expresses its recognition of the technical achievements of WG 10, and thanks all the former members of WG 10.

SC 4 expresses a special recognition to Matthew West for his dedication in the position of acting convenor of the group.

RESOLUTION 535 (Stockholm, Sweden - June 2002)

Support of ISO/TC 184 Business Strategy and Development task force

SC 4 strongly supports the objectives of the task force created by ISO/TC 184/AG. SC 4 member bodies and liaisons are encouraged to provide their views relative to the scope of the ISO/TC 184 Business Strategy and Development task force, especially in terms of:

- relevance of the current SC 4 work items to their industrial needs,
- fields of industrial data not yet covered,
- priorities and available expertise.

SC 4 strongly objects to any subdivision of SC 4 without an adequate business case subject to approval by the member bodies.

Input should be forwarded to SC 4 Chair and Secretariat, no later than end of September 2002.

RESOLUTION 536 (Stockholm, Sweden - June 2002)

STEP AP219 NWI

SC 4 requests WG 3 to submit a NWI proposal on AP219 – Inspection planning, no later than March 2003.

SC 4 requests WG 3 to establish the necessary liaisons and co-operations in the field

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

of AP219 scope

RESOLUTION 537 (Stockholm, Sweden - June 2002)

SC 4 new Website

SC 4 expresses its appreciation of the work done by its Secretariat on the new web site, SC 4ONLINE, and specifically thanks Mr. Robert Kidwell, Mr. Joe Brazy, Mr. Ted Kula, Mr. Steve Mealey, Ms. Teresa Brittingham, Mr. Mark Bryant, Mr. Bob Cox, Mr. Rob Addair, and Ms. Perye Letsinger for their dedication to that task.

SC 4 encourages its Secretariat to continue its effort in this development.

RESOLUTION 538 (Stockholm, Sweden - June 2002)

Transposing the IAI IFC2x Specification as an ISO PAS

SC 4 invites the IAI to submit the IFC2x Platform Specification, along with the final version of the explanatory report, to the Secretariat for letter ballot for publication as an SC 4 PAS, under the SC 4 Harvesting procedure.

SC 4 requests that a joint activity be established between its working groups 3/11/12 and the IAI to harmonize the IFC2x Platform Specification and existing SC 4 standards, prior to the 3 year mandatory review of the PAS.

SC 4 proposes that its activity be focused within ISO/TC 184/SC 4 WG 3/T22, and invites the Convenors of WG 3/11/12 to nominate appropriate representatives to support this work.

SC 4 thanks the ad hoc group for their efforts, and disbands the group.

RESOLUTION 539 (Stockholm, Sweden - June 2002)

Communication about document availability to members of committees and working groups

SC 4 directs the Secretariat and WG convenors to comply with the TMB request for following elements of document notifications sent by email, as specified in TMB communiqué no. 16 May 2002.

SC 4 directs the Secretariat to publish the guidelines on SC 4ONLINE, to make any necessary changes to the SC 4 Handbook, to define a text template for e-mail messages, and to provide automated weekly digests of new additions to the SC

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

4ONLINE website.

RESOLUTION 540 (Stockholm, Sweden - June 2002)

Abstracts for ISO/TC 184 – Specification

SC 4 directs Project Leaders to submit to the Secretariat an abstract for each document, prior to publication by ISO, to comply with the following requirements defined by ISO and published on the SC 4ONLINE website.

SC 4 directs the QC to make the necessary changes to the SC 4 Supplementary Directives, and any other affected documents.

RESOLUTION 541 (Stockholm, Sweden - June 2002)

Improving accessibility to SC 4 Standards

It is resolved that the ISO funding mechanism for the delivery of standards produced by SC 4 should be revised to be consistent with the ISO funding mechanisms for the delivery of other standards that are freely available to the general public.

This resolution shall be forwarded to the ISO Central Secretariat for review and consideration by the Commercial Policies Steering Group.

RESOLUTION 542 (Stockholm, Sweden - June 2002)

Submission of AIC 522 for DIS Ballot

SC 4 accepts the recommendation of the Change Management Committee to separately publish AIC 522, “Machining Features”, based on the approved NWI for AP 224 and to progress AIC 522 direct to DIS ballot, and instructs its Secretariat to submit the necessary documents to ISO Central Secretariat.

RESOLUTION 543 (Stockholm, Sweden - June 2002)

Appreciations

Acclamation

SC 4 expresses its deep appreciation to Eurostep, the Swedish Institute of Standards and the staff of Skogshem & Wijk for their efforts in organizing this meeting, arranging the facilities, excellent food, and social events, and excellent accompanying events. SC 4 also expresses its appreciation for their hospitality and outstanding support before, during, and after the meeting.

SC 4 addresses a special recognition to Helena Lindström, Magnus Nerman, Håkan

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Kården, Gunilla Sivard, Lars Flink, and Jalle Magnusson for their hospitality, outstanding support, and patience both before and during the ISO/TC 184/SC 4 meeting.

Acclamation

Appreciation to our SC 4 Liaisons. We appreciate your representation to and for ISO/TC 184/SC 4.

A special thank you to the SC 4 Liaison Plenary presenters for their insightful and comprehensive presentations:

ASAM	Hans Bothe
PDES, Inc - next phase	Bob
Kiggans VAMAS -	Anna
Moreno IAI - Nordic chapter	Väino
Tarandi PLCS -	John
Dunford	
ProSTEP -	Max Ungerer
OMG	David Price

Acclamation

SC 4 expresses its appreciation to those who participated for their outstanding contributions and professionalisms this week to the work of ISO/TC 184/SC 4.

Acclamation

SC 4 expresses its sincere thanks to Tom Warren for his outstanding contribution to the development of quality processes within ISO/TC 184/SC 4 and expresses its hope that he will be successful in obtaining support for resuming his activities.

RESOLUTION 544 (Seoul, Korea - November 2002)

SC 4 Project Leadership and Editorship Approvals

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303 – 28ed2	Heidi Preston	

RESOLUTION 545 (Seoul, Korea - November 2002)

Cancellation of ISO 10303-29

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 cancels ISO 10303-29, Implementation methods: Java programming language binding to the EXPRESS modeling language from its work program.

RESOLUTION 546 (Seoul, Korea - November 2002)

Cancellation of AP226

SC 4 resolves to cancel 10303-226, Ship mechanical systems

RESOLUTION 547 (Seoul, Korea - November 2002)

Cancellation of Inactive ATS Projects

SC 4 resolves to cancel 10303 parts 302, 318, 324ed2, and 333.

RESOLUTION 548 (Seoul, Korea - November 2002)

Cooperation with OAG

SC 4 welcomes the request for a formal membership of the Open Applications Group. Inc. (OAG) to ISO IEC ITU UN/ECE MoU. Based on the library described in OAGIS 8.0 SC 4 sees a complementary activity to its own STEP development. Therefore SC 4 invites the OAG to give a presentation at the next meeting on OAGs work. After the presentation SC 4 shall consider inviting OAG to nominate an official Liaison Officer, and initiation of the SC 4 Harvesting Procedure for OAGIS 8.0.

SC 4 invites National Member Bodies to recommend other similar groups that should be invited to present their work programs to SC 4 and to be considered for liaison.

RESOLUTION 549 (Seoul, Korea - November 2002)

Part Marking Standard

In anticipation of a request from ISO/TC 20 to take over the task of expanding the part marking standard to cover other industry sectors, SC 4 directs its Secretariat to respond positively to such a request. SC 4 requests the ad hoc group established by Resolution

#512 to assess the specification and to prepare an appropriate NWI proposal and circulate it to SC 4 and the ISO/IEC/ITU/CEFACT MoU/MG.

RESOLUTION 550 (Seoul, Korea - November 2002)

Establishing an SC 4 Education and Outreach Rapporteur

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Recognizing the need to make the public, research community, and industry knowledgeable of STEP, SC 4 International Standards, and SC 4 activities, SC 4 establishes an Education and Outreach Rapporteur as a focal point for SC 4 outreach, education, and promotional activities. This should be closely linked to International STEP Centers (ISC) related marketing and promotional activities.

SC 4 appoints Mr. Ricardo Gonçalves as the first SC 4 Education and Outreach Rapporteur.

RESOLUTION 551 (Seoul, Korea - November 2002)

Minor Revision of ISO 10303-214:2001

SC 4 approves the creation of a new edition of ISO 10303-214 as a minor revision to - correct errors in EXPRESS, Mapping Tables, Application Object definitions, and orthography.

SC 4 appoints Max Ungerer as Project Leader and Ulrich Pfeifer-Silberbach as Part Editor.

SC 4 instructs its Secretariat to submit the completed document to ISO CS for FDIS ballot as a Minor Revision when available.

RESOLUTION 552 (Seoul, Korea - November 2002)

International promotion of Parts Library fasteners data dictionary

SC 4 requests national member bodies to promote the achievements to date by CNIS and the proposed project for development of “Mechanical systems and components for general use – Reference dictionary for fasteners” as part 511 of ISO 13584. In particular, SC 4 recommends national member bodies to establish links with the corresponding national technical committees involved with fastener technologies and standards to promote the work and seek to attract potential ballot reviewers for the expected ISO 13584-511

RESOLUTION 553 (Seoul, Korea - November 2002)

Deletion of part 21, 33, and 45 of MANDATE

SC 4 cancels ISO 15531-21, ISO 15531-33 and ISO 15531-45, and removes ISO 15531-21 from the MANDATE P1 project, ISO 15531-33 from the P2 project and ISO 15531-45 from the P3 project. Nevertheless the potential provision to BSR of specific

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

manufacturing BSU's for data exchange is kept in the JWG 8 program of work and will become a new work item if needed after clarification of evolution of the BSR project.

RESOLUTION 554 (Seoul, Korea - November 2002)

SC 4 position on ISO/IEC 62264

ISO/TC 184/SC 4 notices some overlaps and inconsistencies between ISO/IEC 62264 and some of its existing developed standards or work in progress. Although some normative references have been added in part 1 in the framework of ballot comment resolution, SC 4 is concerned that insufficient consideration of the SC 4 approach, work and constructs have been made in ISO/IEC 62264-2 which is currently under ballot through a fast-track procedure.

This especially concerns the objects product, material, equipment, resource, personnel, product segment, personnel segment, capability segment in ISO/IEC 62264-2 versus product, resource, capability, event, time as they are defined in STEP, MANDATE and other SC 4 standards. In addition, some of the objects as they are defined in ISO/IEC 62264-2 are not independent one from the other and contains their relationship to the other. Time is not managed independently and appears in each object. That is also incompatible with the SC 4 approach.

Since ISO/TC 184 standards and ISO/IEC 62264 address strongly related and overlapping area of work, this situation may lead to conflict, difficulties and to the unnecessary duplication of work. SC 4 asks its member bodies to take appropriate actions to ensure that these comments will be taken into account in their answer to the ballot, from the ISO side as well as from the IEC side.

SC 4 also asks ISO/TC 184 to clarify as far as possible the situation with IEC/TC 65 and to improve the collaboration already set up by the SC 5/JWG 15 to include SC 4 participation to that working group.

Moreover SC 4 asks ISO/TC 184 to take appropriate actions to ensure that ISO and IEC committees with scopes that may overlap or relate to its own subcommittees are aware of the standards already published and the work in progress in ISO/TC 184.

RESOLUTION 555 (Seoul, Korea - November 2002)

SC 4 position on ISO/IEC 62264 – AP

SC 4 invites JWG 15 to consider the possibility of developing a STEP Application Protocol with the scope of the information defined in ISO/IEC 62264 Part 2.

RESOLUTION 556 (Seoul, Korea - November 2002)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Comparative assessment of ISO 14649 and ISO 10303-238

SC 4 notes resolution 145 of SC 1 and requests SC 1 to ensure adequate representation of SC 4 development activities within the proposed studies, noting the existing participation of SC 4 experts in SC 1 meetings.

SC 4 requests WG 3/T24 experts to report on the status of this activity at the SC 4 meeting in San Diego in March 2003, and give initial recommendation on the necessary convergence between AP238 and ISO 14649.

SC 4 invites ISO/TC 184 to note these requirements.

RESOLUTION 557 (Seoul, Korea - November 2002)

Publication of Modular APs

SC 4 notes the significant progress made in the definition of the structure and content of modular APs, which shall be published as an AP with a complementary AP module.

SC 4 requires all projects developing modular APs and Application Modules to use the XSL development environment in the STEP Repository. Deviations from this requirement shall be approved by the appropriate Convenors and the Quality Committee.

SC 4 requests its secretariat to register new AP modules as 10303 parts in the 400 series. The AP module number for an AP 2xy shall be 4xy.

SC 4 permits modular APs to be submitted directly to DIS ballot following successful completion of the TS ballots for their component modules.

SC 4 requests its secretariat to negotiate with ISO for the publication of standard products comprising APs and all their supporting modules.

SC 4 requests its member bodies and the affected projects to nominate resources to an ad hoc team to develop revised AP Guidelines and Supplementary Directives

RESOLUTION 558 (Seoul, Korea - November 2002)

Appreciations

Acclamation

SC 4 expresses its deep appreciation to KATS (Korean Agency for Technology and

Dana Tripp

Secretary, ISO/TC 184/SC 4

trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

 Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

Standards) and KSTEP (Korea STEP Center) and the staff of Seoul COEX Center for their efforts in organizing this meeting, arranging the facilities, excellent food, social events, and excellent accompanying events. SC 4 also expresses its appreciation for their hospitality and outstanding support before, during, and after the meeting.

SC 4 expresses its deep appreciation and thanks for the warm welcome by **Mr. Dong-Chul Kim**, Administrator of KATS; and to the ISO/TC 184/SC 4 Meeting Organizing Committee: **Mr. Jong-Hee Hong**, Director General, Department of Manufacturing Technology and Standards, Korean Agency for Technology and Standards Ministry of Commerce, Industry and Energy; **Mr. Se-Jin Kim**, Director of Automation and Components Division, Chief Organizer of ISO/TC 184/ SC 4 Seoul meeting, Korean Agency for Technology and Standards (KATS); **Dr. Soonhung Han**, Chairman of KSTEP.

SC 4 addresses a special recognition to Ju Hyun Yi (CES), Eunsook Kim (KATS), Youchon Oh (KATS), Byeong Huee Lee (KATS), Su Jeong Baek (KSTEP), Du Hwan Mun (KSTEP) and Sang Uk Cheon (KSTEP) for their hospitality, outstanding support, and patience both before and during the ISO/TC 184/SC 4 meeting.

SC 4 also expresses its sincere appreciation and thanks for the following ISO/TC 184/SC 4 meeting sponsors: **Jaebon Choi** and **Byungwhui Lee**, Automation and Components Division in KATS; **Dr. Yongdae Kim**, Korea Ocean Research & Development Institute (KORDI); **Professor Inhan Kim**, Laboratory for Information Technology in Architecture (ITalab); **Mr. Seongsig Kim**, Senior Researcher, Korea Institute of Construction Technology (KICT); **Professor Hyunchan Lee**, Hongik University Department of I.I.E. CAD/PAM Laboratory, and **Mr. Kukhwan Lee**, Integrated Forum on Electronic Commerce.

Acclamation

SC 4 expresses its sincere thanks to the French GOSSET organisation for its contributions to the work of SC 4 since 1989. SC 4 expresses its particular thanks for the leading roles played by Alain Bezous and Pascal Huau within the SC 4 organisation, and for their outstanding contributions to the results of SC 4. We wish both of them every success in their future careers, and hope for their continued participation in the work of SC 4.

Acclamation

Appreciation to our SC 4 Liaisons. We appreciate your representation to and for ISO/TC 184/SC 4.

A special thank you to the SC 4 Liaison Plenary presenters for their insightful and

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

comprehensive presentations:

ASAM -	Gerald
Sammer IAI - Nordic chapter	
	Jeff Wix
PDES, Inc -	Gerry Graves
PLCS -	John Dunford
SC 1 -	Peter Mueller
ISO/TC 29/WG 34 -	Norman Swindells

Acclamation

SC 4 extends its sincere appreciation for the efforts of Ms. Teresa Brittingham during her tenure as Deputy Secretary. SC 4 wishes her well in her future endeavors. SC 4 welcomes the new Deputy Secretary, Mr. Ron Kolakowski, and looks forward to working with him to ensure a successful and productive organization.

Acclamation

SC 4 expresses its sincere thanks to Joe Brazy (ManTech) and Chris Kreiler (CACI) for their untiring efforts and excellent performance as acting Deputy Secretary before and during the meeting.

Acclamation

SC 4 expresses its appreciation to NIST for their support in the development of the STEP repository.

RESOLUTION 559 (San Diego, CA, United States - March 2003)

SC 4 Project Leadership and Editorship Approvals

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303 – 1348	Jim U'Ren	Ian Bailey
ISO 10303 – 1227	Bill Anderson	Darla Nettles
ISO 10303 – 1228	Bill Anderson	Darla Nettles
ISO 15531 – 0032	Robert Young	
ISO 10303 – 0210		Darla Nettles

RESOLUTION 560 (San Diego, CA, United States - March 2003)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Review of Active Experts

SC 4 requests its secretary to request its national bodies to review their list of active persons for SC 4 and send input for update of database to SC 4 secretary previous to SC 4 meetings.

RESOLUTION 561 (San Diego, CA, United States - March 2003)

Meetings Frequency

SC 4 reviews the necessity of 3 meetings per year and asks its member bodies whether they support the current situation. If the member bodies agree to reduce the number of meetings to two meetings a year, then this shall be installed as soon as possible. The existing schedule and invitations from some countries need to be considered because they have already done some preparatory work

RESOLUTION 562 (San Diego, CA, United States - March 2003)

Appointment of Keith Hunten as Convenor of WG 12

SC 4 approves Keith Hunten to serve as Convenor of its Working Group 12 for a period of three years.

SC 4 expresses a special recognition and thanks to Greg Paul for his exceptional dedication and service for the past six years in the position of Convenor of its Working Group 12.

RESOLUTION 563 (San Diego, CA, United States - March 2003)

Appointment of Len Slovensky as Deputy Convenor of WG 3

SC 4 approves Len Slovensky to serve as Deputy Convenor of its Working Group 3 for a period of three years.

RESOLUTION 564 (San Diego, CA, United States - March 2003)

Cancellation of ISO/TC 184/SC 4 Resolution 462

SC 4 cancels its Resolution 462 that established a STEP Modules Harmonization Team and urges its Working Groups 3, 8, 9 and 12 along with the active module development projects to cooperate in module harmonization.

SC 4 requests the conveners of WG 3, 8, 9, and 12 to report on module harmonization activities to plenary meetings of SC 4 and its Working Groups. SC 4 also urges its Working Groups 3, 8, 9 and 12 to complete and update the

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

guidelines documents for module creation.

In particular, SC 4 requests that guidance on module harmonization be added to the Module Development Guidelines, and that consideration be given to the use of ISO 18876 in this context.

RESOLUTION 565 (San Diego, CA, United States - March 2003)

Extension of Task Force for “Industrial Data on the Web”

SC 4 directs the ad hoc group for Industrial Data on the Web (IDW) to report regularly to SC 4 plenaries concerning the impact of established and emerging Web technologies on the work of SC 4. These technologies include web services, such as WSDL, UDDI and LDAP.

IDW will be a mechanism by which web-related technologies developed by individual SC 4 WGs are shared.

IDW will deliver SC 4 resolutions, tutorials, N docs and technology demonstrations.

SC 4 urges the ad hoc group to recommend liaisons with W3C committees, and to coordinate their activities with the Education and Outreach Committee.

SC 4 appoints Jim U'Ren as the leader of this ad hoc group, and David Leal as the deputy.

IDW will deliver a report to the last SC 4 meeting in 2004 with a recommendation as to the continuation of the ad hoc group.

RESOLUTION 566 (San Diego, CA, United States - March 2003)

Establishment of Liaison with SASIG

SC 4 proposes the establishment of a Category A liaison with the Strategic Automotive product data Standards Industry Group (SASIG) and nominates Oscar Rocha as its liaison officer for the next two years.

It is requested that the SC 4 secretary confirm this liaison with the ISO Central Secretariat.

SASIG is similarly invited to select a liaison officer from their organization to ISO/TC 184/SC 4

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 567 (San Diego, CA, United States - March 2003)

Minor Revision of ISO 10303-210:2001

SC 4 approves the creation of a new edition of ISO 10303-210 as a minor revision to correct errors in EXPRESS, Mapping Tables, Application Object definitions, and orthography.

SC 4 requests its Secretariat to submit the completed document to ISO CS for FDIS ballot as a Minor Revision when available

RESOLUTION 568 (San Diego, CA, United States - March 2003)

Liaison with ISO/TC 67

SC 4 proposes a Category A liaison with ISO/TC 67, nominates Nils Sandmark as the liaison officer for a period of two years, and requests that the SC 4 secretary confirms this liaison with the ISO Central Secretariat.

SC 4 invites ISO/TC 67 to nominate a liaison to SC 4.

SC 4 also requests its Secretariat to forward the NWIP and Change Management forms for the proposed work on 15926 Parts 4-7 to ISO/TC 67, in parallel with review by SC 4.

RESOLUTION 569 (San Diego, CA, United States - March 2003)

Requirements for Education and Outreach

SC 4 seeks to effectively spread and enhance the perception of SC 4 standards in industrial, academic and research communities.

SC 4 invites its experts, member bodies and liaisons to provide input to the Education and Outreach Rapporteur about the major problems of perception that have been identified and the needs and opportunities for outreach, education, and promotional activities.

Responses are requested by 2003-05-15 to allow a summary report to be made available at the Stuttgart meeting.

RESOLUTION 570 (San Diego, CA, United States - March 2003)

Public Availability of SC 4 Standards and Specifications

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 requests its Secretariat to request approval from the ISO Council to make the following documents publicly available, free of charge, on the World Wide Web:

- ISO 10303 (STEP)
- ISO 13584 (PLIB)
- ISO 15926 (Process Plant)
- ISO 18876 (IIDEAS)

The rationale to be included with this request is contained in document SC 4N1456.

RESOLUTION 571 (San Diego, CA, United States - March 2003)

Proposal to VAMAS for the Testing of Material Standards

SC 4 endorses the proposal N1457 to VAMAS for the comparison and validation of the different standards for the representation and the exchange of data for Life Cycle Assessment (LCA), as part of the liaison between ISO/TC 184/SC 4 and VAMAS.

SC 4 recognizes the key role that the national research centers, represented in VAMAS, could play in the exploitation and validation of the International and industry Standards identified in the proposal.

SC 4 requests its secretariat to forward the proposal to VAMAS, and to invite material domain experts to contribute to the further development of these SC 4 standards.

RESOLUTION 572 (San Diego, CA, United States - March 2003)

Reappointment of JWG 8 Convenor

SC 4 re-appoints Jean-Jacques MICHEL as convenor of JWG 8 for an additional three- year term.

RESOLUTION 573 (San Diego, CA, United States - March 2003)

Appreciations

Acclamation

SC 4 expresses its deep appreciation to US PRO, its Members and the staff of the Hyatt Islandia Hotel for their efforts in organizing this meeting, arranging the facilities, excellent food, social events, and excellent accompanying events. SC 4 also expresses its appreciation for their hospitality and outstanding support before, during, and after the meeting.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Acclamation

SC 4 appreciates the participation and support of the new ISO/TC 184 Chair Mr. Jean-Marc CHATELARD and the Secretary Ms Christine HERMETET-FILEZ.

Acclamation

SC 4 expresses its sincere thanks to Chris Kreiler (CACI) for his untiring efforts and excellent performance in supporting the work of the Secretariat.

Acclamation

SC 4 expresses its sincere thanks to Mark Palmer for his long-standing, dedicated efforts and excellent performance as the leader of T20, WG 3.

Acclamation

Appreciation to our SC 4 Liaisons. We appreciate your representation to and for ISO/TC 184/SC 4.

A special thank you to the SC 4 Liaison Plenary presenters for their insightful and comprehensive presentations:

UBL -	Jon Bosak
PDES, Inc. -	Bob Kiggans & Simon Frechette
PLCS -	John Dunford
EPISTLE -	Matthew West
OAGI	Tony Blazej

Acclamation

SC 4 expresses its particular thanks to Mr. Chuck Stark, Ms Sarah Hartline, Ms Charlotte Lewis and Mrs. JoAnn Norek for their superior administrative support, outstanding hospitality, and patience both before and during the ISO/TC 184/SC 4 meeting.

RESOLUTION 574 (Stuttgart, Germany, - June 2003)

SC 4 Project Leadership and Editorship Approvals

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303-0055	Mike Pratt	Mike Pratt
ISO 10303-0523	Lothar Klein	Ray Goult

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

ISO 10303-0238
ISO 10303-0054
ISO 18629-4x

Alan Crawford
John Dunford

Dave Loffredo
David Leal
Line Pouchard

RESOLUTION 575 (Stuttgart, Germany, - June 2003)

Appointment of WG 2 Deputy Convener

SC 4 appoints Guy Pierra as Deputy Convener of WG 2 for an additional three-year term.

RESOLUTION 576 (Stuttgart, Germany, - June 2003)

Appointment of Quality Coordinator

SC 4 appoints Mats Elgh from Sweden as the new Quality Coordinator, for a three year term.

SC 4 expresses its appreciation to Rob Anderson for his commitment and efforts in leading quality activities in SC 4.

RESOLUTION 577 (Stuttgart, Germany, - June 2003)

SC 4 Liaison Officer with ISO/TC 10

SC 4 approves Tatsuo Terai to serve as the Liaison Officer to ISO/TC 10 for a period of two years and requests its Secretariat to make the appointment accordingly.

SC 4 expresses its appreciation to Alain Bezos for his liaison role with ISO/TC 10 and accepts his resignation from that position.

RESOLUTION 578 (Stuttgart, Germany, - June 2003)

SC 4 LIAISON WITH ISO/TC 211

SC 4 wishes to thank Bernd Wenzel for his liaison role with ISO/TC 211 and the ISO/IEC Joint Spatial Group (JSG) since June 1997 and accepts his resignation from this position.

SC 4 approves Jerry Smith as the liaison from SC 4 to ISO/TC 211 - Geographic Information Systems/Geomatics and the ISO/IEC Joint Steering Group on Spatial Standardization and Related Interoperability (JSG) and requests its Secretariat to make the appointment accordingly.

RESOLUTION 579 (Stuttgart, Germany, - June 2003)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ESTABLISH LIAISON WITH ISO/IEC JTC 1/SC 31

In accordance with ISO/TC 184 resolution # 320, SC 4 requests its Secretariat to take the necessary actions to implement a liaison between ISO/TC 184/SC 4 and ISO/IEC JTC 1/SC 31 and its WG 2, and nominate Dr. King Yee (UID ad hoc chair) and Mr. Frank Goodell (USA) as the physical liaisons.

RESOLUTION 580 (Stuttgart, Germany, - June 2003)

ESTABLISHMENT OF LIAISON WITH OASIS

SC 4 requests its Secretariat to establish a liaison to the OASIS consortium, nominates David Leal as the liaison officer, and requests the Secretariat to make the appointment.

SC 4 invites OASIS to make a reciprocal nomination, and to invite any of its ISO/TCs with activities in the scope of SC 4 to establish liaison arrangements.

RESOLUTION 581 (Stuttgart, Germany, - June 2003)

CANCELLATION OF XI'AN MEETING

SC 4 endorses the decision of the Chair and Secretariat to cancel the Xi'an meeting scheduled for October 2003, due to natural events beyond the control of the organisers, and instructs its secretariat to offer October 2005 as an alternative date.

SC 4 offers its best wishes for good health to the Chinese delegation, and it's apologies for any inconvenience caused.

RESOLUTION 582 (Stuttgart, Germany, - June 2003)

HARVESTING OF ASAM ODS 5.0

SC 4 creates an ad hoc group led by Len Slovensky to assess the proposal from ASAM to transpose the ODS 5.0 specification into a Publicly Available Specification (PAS).

SC 4 invites WG 3, WG 11, WG 12, PPC and the Quality Committee (QC) to nominate representatives by 2003-08-31. The ad hoc group is requested to assess the specification and explanatory report (N1508), negotiate with ASAM on any issues, and make a recommendation to SC 4 by 2003-09-30 on how to process the proposal.

RESOLUTION 583 (Stuttgart, Germany, - June 2003)

REGISTRATION OF SC 4 MIME-TYPES

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 requests its Secretariat to register STEP Part 21 MIME-types with the Internet IANA Registration Authority as .stp, and .step.for files according to ISO 10303 Application protocols and .p21 for general use including those according to ISO 10303, ISO 13584 and ISO 15926.

SC 4 requests its Secretariat to register STEP Express MIME-types with the Internet IANA Registration Authority as .exp, and .expr.

Additionally, WG 11 is requested to amend Parts 11 and 21 to recommend the use of these extensions in all STEP applications.

RESOLUTION 584 (Stuttgart, Germany, - June 2003)
ISO/IEC GUIDE FOR COMMON USE OF PARTS LIBRARY DEFINITIONS

ISO/TC 184/SC 4 strongly supports the German initiative to create an ISO/IEC Guide for the common use of parts library definitions to encourage all IEC and ISO committees dealing with the standardization of products to use the description and definitions of IEC 61360 and ISO 13584 for the development of product libraries.

RESOLUTION 585 (Stuttgart, Germany, - June 2003)
LIAISON BETWEEN ISO/TC 184 SC 4 WG 2 AND ISO/TC 37

To avoid duplication or contradictory work, SC 4 wishes to encourage ISO/TC 37 to inform ISO/TC 184/SC 4/WG 2 about the actual status on "Basic principles of multilingual classification for electronic commerce" and related activities. SC 4 strongly encourages ISO/TC 37 to consider adoption of the SC 4 standards as a basis for their ongoing activities in related fields. Based on that response, ISO/TC 184/SC 4 may wish to establish a liaison with the committee within ISO/TC 37 working on that work item.

Required actions following approval of this resolution:

To submit this resolution directly to ISO/TC 37 secretariat with a request to provide information about their results and plans and to contact ISO/TC 184/SC 4/WG 2 with the goal to establish a liaison to identify overlapping or contradicting work (Secretariat).

RESOLUTION 586 (Stuttgart, Germany, - June 2003)
EXPRESSION OF APPRECIATION TO PROFESSOR KIMURA

SC 4 would like to thank Professor Kimura for his significant contributions and support

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

to the SC 4 community since its inception in 1984. His untiring efforts and excellent performance as Head of the Japan National Committee in supporting the work of the SC 4 sets the high water-mark for the SC 4 community.

SC 4 looks forward to continued cooperation with the new leader of the Japan National Committee Mr. Ohtaka.

RESOLUTION 587 (Stuttgart, Germany, - June 2003)

Appreciations

Acclamation:

SC 4 expresses its deep appreciation to the team of Büro Management Mohrmann, its Members and the staff of the Kongresshotel Europe for their efforts in organizing this meeting, arranging the facilities, excellent food, social events, and excellent accompanying events. SC 4 also expresses its appreciation for their hospitality and outstanding support before, during, and after the meeting.

Acclamation:

Appreciation to our SC 4 Liaisons. We appreciate your representation to and for ISO/TC 184/SC 4.

A special thank you to the SC 4 Liaison Plenary presenters for their insightful and comprehensive presentations:

PDES, Inc	Tony Ranger
PLCS	Margaret Christison
OMG	Rob Bodington
ASAM	Gerald Sammer
PROSTEP	Max Ungerer
ISO/TC 29 / WG 34	Norman Swindells
FSIG	Maria José
Nunyez	
IAI	Thomas Liebich
VAMAS & ISO/TC 207	Anna
Moreno PIEBASE	Paul van
Exel	
MoU/MG	Howard Mason

Acclamation:

SC 4 expresses its particular thanks to Mrs Bärbel Mohrmann and her staff, Sabine

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Ludwig, Ulrike Müller, Amaya Steinhilber and Christian Schwinge for their superior administrative support, outstanding hospitality, and patience both before and during the ISO/TC 184/SC 4 meeting.

RESOLUTION 588 (Poitiers, France, - October 2003)

SC 4 Project Leadership and Editorship Approvals

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303-0058	Rolf Lervik	
ISO 10303-0240	Chiaki Sakamoto	Len Slovensky
ISO 15531-0041	Anne-Françoise Cutting-Decelle	Jean-J. Michel
ISO 15531-0043	Anne-Françoise Cutting-Decelle	Jean-J. Michel

RESOLUTION 589 (Poitiers, France, - October 2003)

Cancellation of STEP Part 314

SC 4 agrees to cancel STEP Part 314 due to lack of progress.

RESOLUTION 590 (Poitiers, France, - October 2003)

Timescales for ISO standard development stages

SC 4 notes the objectives of the new timeframes and accepts the overall timescales proposed, although it must be recognised that reduced timescales are often incompatible with adequate quality, consensus and market relevance. However, SC 4 notes that the intermediate milestones proposed in the new timeframe do not take sufficient account of the need for technical discussion and consensus building, while allowing extended periods for publication and administration. The use of publication tools to accelerate the administrative tasks means that SC 4 does not need to exploit the full time allowed for administration, but requires the maximum time possible for technical work.

SC 4 therefore requests ISO/CS to confirm that the intermediate milestones of the new timeframes can be adjusted to allow appropriate work planning, within the overall time limits given.

SC 4 also seeks confirmation that ISO/CS can achieve its timescales for processing, noting that there are examples of FDIS processing exceeding three months.



N4118

2026 – 05 – 15

RESOLUTION 591 (Poitiers, France, - October 2003)

ISO/PAS 12006-3 Building construction — Organization of information about construction works — Part 3: Framework for object-oriented information exchange — revised version 2

SC 4 asks its secretariat to submit a liaison statement to ISO/TC 59 "Building construction" explaining the area of work of SC 4, especially to point out the activities of its WG 2 and WG 3 work on modelling within the area of building construction.

SC 4 also requests ISO/TC 59, especially ISO/TC 59/SC 13/WG 6, to identify areas of need for harmonization with SC 4 standards such as ISO 10303-225 and ISO 15926 in its work programme, in order to foster cooperation and collaboration in the future.

SC 4 appoints Tatsuo Teraï as a physical liaison to ISO/TC 59/SC 13/WG 6 in order to support the required cooperation and requests its Secretariat to establish the liaison.

Action: National Bodies of SC 4 should examine this initiative to promote ISO 12006 as a DIS and assess it against the existing procurement regulations in their countries.

RESOLUTION 592 (Poitiers, France, - October 2003)

ISO/DIS 16792 Technical product documentation - digital product definition data practices

SC 4 asks its secretariat to inform ISO/TC 10 that the planned publication of ISO 16792 will lead to significant changes in already existing and published standards by SC 4. SC 4 encourages ISO/TC 10 and ISO/TC 213 to reach a clear consensus on unambiguous and consistent representations of 2D and 3D annotation that can be referenced by the SC 4 standards for digital representation of product information when they are revised, since it will not be viable to support conflicting forms of annotation.

Further, SC 4 asks ISO/TC 10 to provide a clear strategic statement on their future development plans; especially because SC 4 has referenced several ISO/TC 10 standards on annotation and drafting.

Action: The teams of AP 202, 203, 210, 212, 214 and are requested to check the potential impact of the content of ISO16792 against their projects, with a view to directly referencing the standard in future



N4118

2026 – 05 – 15

RESOLUTION 593 (Poitiers, France, - October 2003)
Industrial Data Framework

SC 4 invites member bodies to review the proposal to update the Industrial Data Framework as an ISO Technical Report and to indicate to the Secretariat by the end of December 2003 whether there will be sufficient industry interest to justify the creation of a NWI, and whether they would be willing to provide the necessary resources.

RESOLUTION 594 (Poitiers, France, - October 2003)
ISO/IEC Guide for the common use of parts library definitions

SC 4 notes the ad hoc meeting to develop the ISO/IEC Guide on "Product properties and classification" in Berlin on Wednesday, 2003-12-10 and nominates Wolfgang Wilkes, Guy Pierra, Karlheinz Rosan, Nils Sandmark, Tim King and Norman Swindells to attend the meeting.

SC 4 invites its Working Groups to nominate additional delegates to this meeting
RESOLUTION 595 (Poitiers, France, - October 2003)
Liaison to OAGI

SC 4 appoints Günter Staub to serve as the liaison officer to OAGI and requests its Secretariat to make the appointment accordingly

RESOLUTION 596 (Poitiers, France, - October 2003)
Public Availability of SC 4 Standards and Specifications

SC 4 notes with disappointment the ISO Council's rejection of the request to make some SC 4 Standards freely available on the Web.

SC 4 appreciates the freedom to permit distribution of these standards to other ISO and IEC groups for further standards development.

SC 4 endorses the response of its Chair (Document SC 4 N1606) which notes that the ISO Council has not addressed several of the issues raised in the previous SC 4 Resolution.

SC 4 further urges its member national bodies to approach their ISO Council and TMB members and impress upon them the urgency and importance of this issue.



N4118

2026 – 05 – 15

Attached documents: SC 4 N1606, SC 4 N1456

RESOLUTION 597 (Poitiers, France, - October 2003)

Task Force for Dictionaries, part libraries and reference data libraries

SC 4 establishes a task force to produce a white paper on the scope, relationship between ISO 10303, ISO 13584 and ISO 15926 and their industrial usage to meet business requirements for the definition of dictionaries, part libraries and reference data libraries. The white paper will be issued by the 15th of February 2004 and the team will report to SC 4 at the next ISO/TC 184 SC 4 meeting in Ft Lauderdale, spring 2004.

The results shall be clearly understandable by SC 4 experts and stakeholders, in order to promote consistent presentation of SC 4 products to the market place.

ISO/TC 184/SC 4 appoints the following task force team members:

- Gerry Radack (Task force team leader)
- Guy Pierra
- Wolfgang Wilkes
- Karlheinz Rosan
- Hiroshi Murayama
- Margaret Christison
- Ricardo Gonçalves
- Nils Sandsmark
- Magne Valen-Sendstad
- Andries van Renssen
- Paul van Exel Norman Swindells
- Yamine AIT-AMEUR
- Junfeng Zhan

RESOLUTION 598 (Poitiers, France, - October 2003)

Liaison with ECCMA

SC 4 accepts the request from the Electronic Commerce Code Management Association (ECCMA) to form a two-way Category A liaison, and appoints Gerry Radack as liaison officer from SC 4 to ECCMA.

RESOLUTION 599 (Poitiers, France, - October 2003)

Appreciations

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Acclamation:

SC 4 expresses its deep appreciation to the AFNOR and ENSMA/LiSi team, its Members, Sponsors and the staff of the Clarion Hotel FUTUROSOCPE for their efforts in organizing this meeting, arranging the facilities, excellent food, social events, and excellent accompanying events. SC 4 also expresses its appreciation for their hospitality and outstanding support before, during, and after the meeting

Acclamation:

SC 4 expresses its particular thanks to Prof Guy Pierra and his staff, Yamine AIT AMEUR, Aïda AIT AMEUR, Frédéric CARREAU, Claudine RAULT, Ladjel BELLATRECHE, Youcef AKLOUS, Sofiane CHALLAL, Hondjack DEHAINSALE, and Zum DUNG for their superior administrative support, outstanding hospitality, and patience both before and during the ISO/TC 184/SC 4 meeting

Acclamation:

SC 4 would like to thank Mr. John Dunford for his significant contributions and support to the SC 4 community. His untiring efforts as Working Group 3, Team 8 (Product Life- Cycle) Team Leader, his many project and technical contributions, and wishes him all the best for his future. The significant contributions due to his innovative and creative leadership will be greatly missed.

RESOLUTION 600 (Ft. Lauderdale, FL, USA - March 2004)

SC 4 Project Leadership and Editorship Approvals

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303-0235	Anna Moreno	
ISO 10303-1143	Ricardo Goncalves	
ISO 10303-1144	Ricardo Goncalves	
ISO 10303-1145	Ricardo Goncalves	
ISO 10303-1146	Ricardo	
Goncalves		

RESOLUTION 601 (Ft. Lauderdale, FL, USA - March 2004)

Reappointment of WG 3 Convener

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 appreciates and accepts the offer of Dr. Radack to serve as Convener of WG 3.
SC 4 approves the nomination of Dr. Radack to serve for an additional three year term

RESOLUTION 602 (Ft. Lauderdale, FL, USA - March 2004)

Appointment of Liaison Officer to IEC SC 65B/WG 10

SC 4 nominates Hiroshi Murayama as primary liaison officer, with Dr. Wolfgang Wilkes as alternate, to IEC SC 65B/WG 10 for a period of two years, and welcomes Dr. Jean-Philippe Herzog as liaison officer from SC 65B.

RESOLUTION 603 (Ft. Lauderdale, FL, USA - March 2004)

Retention of ISO 10303-203 ed1 in the ISO Catalogue of Available International Standards

SC 4 requests the ISO Central Secretariat to retain the listing of ISO 10303-203 edition 1 in the ISO catalogue of available International Standards, after the on publication of the ISO 10303-203 edition 2 as a Technical Specification.

RESOLUTION 604 (Ft. Lauderdale, FL, USA - March 2004)

Liaison with NATO AC/135

SC 4 proposes a Category A liaison with NATO AC/135, nominates Steven Arnett, US Defense Logistics Information Service, as the liaison officer, and requests that the SC 4 secretary confirms this liaison with the ISO Central Secretariat.

SC 4 invites AC/135 to make a reciprocal nomination, and to invite any of its constituent bodies with activities in the scope of SC 4 to establish liaison arrangements.

RESOLUTION 605 (Ft. Lauderdale, FL, USA - March 2004)

Investigation of Use of Languages in Coordination with EXPRESS

SC 4 supports the proposed investigation by the Industrial Data on the Web Group (IDW) of the relationship between SC 4 standards and OASIS, W3C and OMG standards

Use of languages such as UML and OWL in coordination with EXPRESS is to be included as part of this investigation.



N4118

2026 – 05 – 15

SC 4 requests the IDW team to present a report on this activity at the Bath meeting.

RESOLUTION 606 (Ft. Lauderdale, FL, USA - March 2004)

Extension of Task Force for Dictionaries, Part Libraries, and Reference Data Libraries

SC 4 notes the work done to date by the Task Force for Dictionaries, Part Libraries, and Reference Data Libraries and acknowledges the need for additional time to achieve consensus. SC 4 directs the Task Force to circulate a draft report to the SC 4 community by 2004-06-18 and to report to SC 4 at the next ISO/TC 184/SC 4 meeting in Bath, UK.

RESOLUTION 607 (Ft. Lauderdale, FL, USA - March 2004)

Acknowledgements and Appreciations

Acclamation:

SC 4 expresses its deep appreciation to US PRO, its Members and the staff of the Bonaventure Resort and Spa Hotel for their efforts in organizing this meeting, arranging the facilities, excellent food, and excellent social events. SC 4 also expresses its appreciation for their hospitality and outstanding support before, during, and after the meeting

Acclamation:

Appreciation to our SC 4 Liaisons. We appreciate your representation to and for ISO/TC 184/SC 4.

A special thank you to the SC 4 Liaison Plenary presenters for their insightful and comprehensive presentations:

ECCMA eOTD	Peter Benson & Steve
Arnett SASIG	Oscar Rocha
EPISTLE	Dalip Sud
UCC	John McInerney
EMSA	Tony Fry

Acclamation:

SC 4 expresses its particular thanks to Mr. Chuck Stark, Ms Sarah Hartline and Ms Tina Conlon for their superior administrative support, outstanding hospitality, and

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

patience both before and during the ISO/TC 184/SC 4 meeting

RESOLUTION 608 (Bath, United Kingdom - July 2004)

SC 4 Project Leadership and Editorship Approvals

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303-0239		Nigel Shaw

RESOLUTION 609 (Bath, United Kingdom - July 2004)

Reappointment of WG 12 Deputy Convener

SC 4 appreciates and accepts the offer of Jochen Haenisch to serve as Deputy Convener of WG 12. SC 4 approves the nomination of Mr. Haenisch to serve for an additional three year term.

RESOLUTION 610 (Bath, United Kingdom - July 2004)

Appointment of SC 4 PPC Convenor

SC 4 appoints Mr. Torbjorn Holm as ISO/TC 184/SC 4 PPC Convenor, in succession to Mr. Alain Bezos, with immediate effect.

RESOLUTION 611 (Bath, United Kingdom - July 2004)

Nomination of SC 4 delegates to ISO/IEC JWG 1

SC 4 nominates Hiroshi Murayama, Dr. Wolfgang Wilkes, Professor Guy Pierra, Dr Gerry Radack, Dr Tim King, Dr Norman Swindells, Dr Nils Sandmark and Mr. Ping Wang as delegates to the first meeting of ISO/IEC JWG 1 on 8 September 2004 in Berlin, and requests its Secretariat to notify DIN of the participation.

SC 4 requests its P-members to ensure that their delegations to the meeting are fully briefed on the SC 4 priorities.

RESOLUTION 612 (Bath, United Kingdom - July 2004)

Enhancement of Expertise for ISO/IEC JWG 1 work

ISO/TC 184/SC 4 respectfully request ISO/TMB to encourage the venue of future ISO/IEC JWG 1 (Product properties and families) meetings to be planned in conjunction with ISO/TC 184/SC 4/WG 2 and IEC SC 3 D meetings.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

The major expected benefit from such a request is to enhance technical participation in ISO/IEC JWG 1 work.

RESOLUTION 613 (Bath, United Kingdom - July 2004)

Reappointment of WG 2 Convener

SC 4 appreciates and accepts the offer of Dr. Wilkes to serve as Convener of WG 2. SC 4 approves the nomination of Dr. Wilkes to serve for an additional three year term.

RESOLUTION 614 (Bath, United Kingdom - July 2004)

Establishment of Category “A” Liaison with UNSPSC

ISO/TC 184/SC 4 welcomes the approach from UNSPSC (United Nations Standard Products and Services Codes), and agrees to establish a Category A liaison with UNSPSC, and nominates John McInerney as liaison officer. Further, SC 4 directs its Secretary to confirm this liaison with the ISO CS.

RESOLUTION 615 (Bath, United Kingdom - July 2004)

Establishment of Category “A” Liaison with USPI-NL

SC 4 accepts the request from the Uitgebreid Samenwerkingsverband Proces Industrie Nederland (USPI-NL) to form a two-way Category A liaison, requests the Secretariat to confirm their eligibility with ISO CS, and appoints Paul van Exel as liaison officer from SC 4 to USPI-NL.

RESOLUTION 616 (Bath, United Kingdom - July 2004)

Establishment of Category “A” Liaison with POSC Caesar Association

SC 4 accepts the request from the POSC Caesar Association to form a two-way Category A liaison, requests the Secretariat to confirm their eligibility with ISO CS, and appoints Nils Sandmark as liaison officer from SC 4 to POSC Caesar Association.

RESOLUTION 617 (Bath, United Kingdom - July 2004)

Meeting Fees

As part of the ongoing programme of action to reduce meeting costs, all hosts of future SC 4 meetings are required to provide a budget plan for the meeting to the secretariat

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

of SC 4 as part of their request for the approval of hosting the particular meeting. The budget plan shall include estimates for meeting fees, local travel costs and hotel costs in accordance with the SC 4 reporting form, so that the Secretariat and PPC can ensure that the overall costs to delegates are reasonable and allowable under ISO rules.

RESOLUTION 618 (Bath, United Kingdom - July 2004)

Liaison to ISO/TC 37

SC 4 asks its secretariat to initiate a liaison to ISO/TC 37 with a statement describing the ongoing work within SC 4/WG 2.

RESOLUTION 619 (Bath, United Kingdom - July 2004)

Requirements for Dictionaries on Measuring Instruments

SC 4 requests WG 2 to narrow the scope of ISO 13584-501 to deal only with environment measuring instruments and laboratory measuring instruments.

SC 4 kindly requests that its Chair and secretariat establish the appropriate contacts in order to create a joint activity to develop a consistent dictionary for industrial process control measuring instruments

RESOLUTION 620 (Bath, United Kingdom - July 2004)

Acknowledgements and Appreciations

APPRECIATION:

SC 4 expresses its deep appreciation to EMSA, its Members and the support staff for their outstanding efforts in organizing this meeting, arranging the facilities, excellent food, and excellent social events. SC 4 also expresses its appreciation to the University of Bath for their hospitality and support during the SC 4 meetings.

APPRECIATION:

Appreciation to our SC 4 Liaisons. We appreciate your representation to and for ISO/TC 184/SC 4.

A special thank you to the SC 4 Liaison Plenary presenters for their insightful and comprehensive presentations:

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

IAI	Jeffrey Wix
ISO/TC 29/WG 34	Jonas Nordstrom
ASAM	Gerald Sammer
ISO/TC 172/SC 1/WG 4	Prudence
Wormwell ISO/TC 211	Jerry Smith
FSIG	Ricardo
Gonçalves VAMAS & ISO/TC 207	Anna Moreno
PDES, Inc	Bob Kiggans
NATO AC/135/ECCMA	Steve Arnett & Peter
Benson SASIG	Sven Hjelm

APPRECIATION:

SC 4 expresses its particular thanks to Mr. Tony Fry, Mr. John Kendall, Mrs. Louise Fry, Ms Marie Nesbitt and Mr. Mike Barter for their superior administrative support, outstanding hospitality, and patience both before and during the ISO/TC 184/SC 4 meeting. Additionally, we would like to thank Bernard Smalley and Collin Williams for maintaining the computer systems for the conference

RESOLUTION 621 (Seattle, WA, United States, October 2004)

SC 4 Project Leadership and Editorship Approvals

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303-0058	Nils Sandsmark	

RESOLUTION 622 (Seattle, WA, United States, October 2004)

Progress on Modular APs

SC 4 requests ISO to consider the progress of individual modules when deciding whether it is appropriate to cancel an AP Project for lack of progress.

In particular, ballot submission of a substantive set of the new implementation and foundation modules that are included within the scope of an AP (as defined in its NWI), should also be considered as a change in the status for the overall AP document, and should therefore preclude its cancellation for lack of progress.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 requests its Secretariat to ask ISO to apply this approach to the development of AP233 as a test case, noting the substantive progress on an initial set of modules, and the threat of cancellation of the AP project in January 2005.

RESOLUTION 623 (Seattle, WA, United States, October 2004)

Retention of ISO TS 10303-28 ed1 in the ISO Catalogue of Available ISO Standards

SC 4 requests the ISO Central Secretariat to retain the listing of ISO TS 10303-28 edition 1 in the ISO catalogue of available International Standards, after the publication of the ISO TS 10303-28 edition 2 "Implementation methods: XML representations of EXPRESS schemas and data, using XML schemas".

Note: Action 124 to notify ISO CS of name change in the database.

RESOLUTION 624 (Seattle, WA, United States, October 2004)

Appointment of a New SC 4 Education and Outreach Rapporteur

Recognizing the continuing need to make industry, the public, the research community, and other stakeholders knowledgeable of SC 4 standards and SC 4 activities, SC 4 appoints Mr. Darren Meister as the new SC 4 Education and Outreach Rapporteur to act as the focal point for SC 4 outreach, education, and promotional activities. These activities should be closely linked to International Industry STEP Centers (ISC) related marketing and promotional activities.

RESOLUTION 625 (Seattle, WA, United States, October 2004)

PWI for Lifecycle support of AEC facilities

The goal of this PWI is to create a NWI for a model for lifecycle support of AEC facilities. The following tasks will be performed:

- Detailed in scope, out of scope analysis of the information requirements of AEC facilities including information requirement for facility management.
- Review of capability of existing SC 4 standards to cover lifecycle aspects such as facility management. Describe to which extent they cover the information requirements of AEC facilities
- Describe gaps and necessary adaptations
- Explore feasibility and best approach to combine existing standards. The different approaches of developing a modular AP or a conventional approach will be assessed.

SC 4 directs WG 3 to create the NWI within 6 months forward it to SC 4 for review and



N4118

2026 – 05 – 15

approval, and appoints Inhan Kim and Wolfgang Haas as co project leaders

Attached documents: Description of the PWI, ISO/TC 184/SC 4/WG 3 N1505, change management project review questionnaire, ISO/TC 184/SC 4/WG 3 N1506.

RESOLUTION 626 (Seattle, WA, United States, October 2004)

Harvesting of SASIG - PDQ 2.0

SC 4 creates an ad hoc group led by Oscar Rocha to assess the proposal from SASIG to transpose the PDQ 2.0 guidelines into a Publicly Available Specification (PAS). SC 4 invites WG 3, WG 11, WG 12, PPC and the Quality Committee (QC) to nominate representatives by 2004-11-30. The ad hoc group is requested to assess the specification and explanatory report (SC 4 N1792), negotiate with SASIG on any issues, and make a recommendation to SC 4 by 2005-01-31 on how to process the proposal.

RESOLUTION 627 (Seattle, WA, United States, October 2004)

Appointment of Liaison to ISO/IEC JWG 1 on Product properties and families

SC 4 appoints Dr. Wolfgang Wilkes and Dr. Gerry Radack as the SC 4 Rapporteurs to ISO/IEC /JWG 1.

SC 4 extends the mandate of the delegates to JWG 1 as nominated in Resolution 611.

RESOLUTION 628 (Seattle, WA, United States, October 2004)

Liaison statement to IEC SC 65B to ensure the rapid development of dictionaries compliant to ISO 13584 / IEC61360 in the area of measuring instruments

SC 4 decides to split the scope of the registered dictionary defined by ISO 13584 Part 501 (ISO id 39753) into two parts, covering (a) laboratory and environment measuring instruments and (b) industrial process measuring instruments.

SC 4 directs WG 2 to stop the work on item (b) and directs its Secretariat to offer the work on this item to IEC SC 65B for potential publication either as a joint ISO/IEC standard or as an IEC standard.

SC 4 directs WG 2 to continue the work on item (a), and authorizes a new delivery date for the DIS ballot of 2004-11-30.

SC 4 invites IEC experts to contribute to the work on item (a) in WG 2, both during the further development of the standard and during the operation of the registration

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

authority. SC 4 encourages WG 2 experts who are also members of IEC to participate in the work of IEC SC 65B on item (b).

SC 4 instructs WG 2 to take into account forthcoming results of ISO/IEC JWG 1 concerning the maintenance of dictionaries during the preparation of the next versions of the ISO 13584-501 standard.

RESOLUTION 629 (Seattle, WA, United States, October 2004)

Liaison to OASIS PLCS Technical Committee

SC 4 accepts the nomination of Dr. Timothy King as the OASIS PLCS Technical Committee Liaison, and nominates Dr. Timothy King as the SC 4 liaison to the OASIS PLCS Technical Committee. SC 4 would like to express its appreciation to Ms. Margaret Christison who has previously acted in these roles

RESOLUTION 630 (Seattle, WA, United States, October 2004)

Appreciation to Chuck Stark

SC 4 expresses its utmost appreciation to Chuck Stark for his many years of dedication organizing all SC 4 meetings held in North America for the last 14 years.

Chuck has provided an extremely high level of support and hospitality to make these meetings successful.

SC 4 wishes Chuck every success in his future endeavours.

RESOLUTION 631 (Seattle, WA, United States, October 2004)

Appreciation to Tony Fry

SC 4 expresses its sincere thanks to Tony Fry for his long-standing, dedicated efforts and excellent performance in the leadership of WG 3/T23, and wishes him all the best for his future. In particular, it would like to commend him for his efforts leading to the adoption of four Shipbuilding Application Protocols as ISO Standards and for his leadership in organizing the ISO/TC 184/SC 4 meeting in Bath, U.K.

RESOLUTION 632 (Seattle, WA, United States, October 2004)

Acknowledgements and Appreciations

ACCLAMATION:

SC 4 expresses its appreciation to Ricardo Goncalves for his dedication, untiring

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

efforts, enthusiasm, and creativity as the leader for the SC 4 Education and Outreach Initiative. Ricardo will be remembered for his willingness to take on unpopular, but essential tasks and his valuable assistance to all. The challenge of finding his replacement will not be trivial.

SC 4 wishes him well in his future endeavors, and looks forward to his continued contribution to the work of SC 4.

APPRECIATION:

SC 4 wishes to give its unanimous thanks to US PRO, its Members and the staff of the Seattle Renaissance Hotel for their efforts in organizing this meeting, arranging the facilities, excellent food, and excellent social events.

APPRECIATION:

SC 4 commends Chuck Stark for the outstanding arrangements that contributed to the successful completion of SC 4 business.

SC 4 recognizes the unique talents and demanding requirements necessary to support the SC 4 meeting workload. A special recognition to Ms Sarah Hartline and Ms Karri McGough for their superior administrative support, outstanding hospitality, and patience before and during the ISO/TC 184/SC 4 meeting.

RESOLUTION 633 (Lillehammer, Norway - March, 2005)

Reappointment of WG 11 Convenor

SC 4 appreciates and accepts the offer of David Loffredo to serve as Convenor of WG 11. SC 4 approves the nomination of David Loffredo to serve for an additional three year term.

RESOLUTION 634 (Lillehammer, Norway - March, 2005)

Liaison with ISO/TC 37/SC 1

SC 4 requests an active two-way liaison with ISO/TC 37/SC 1 and requests the SC 4 Secretary to confirm this liaison with the ISO Central Secretariat. SC 4, appoints Gerald Radack and Eric Sardet, as the liaison officers from SC 4 to ISO/TC 37/SC 1, and requests them to provide feedback to WG 2, WG 3 and relevant teams on the specific activities of ISO/TC 37/SC 1 that are of relevance to SC 4 work.

SC 4 requests its secretariat to provide appropriate accreditation to alternate liaison officers when nominated by the appointed officers.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 635 (Lillehammer, Norway - March, 2005)

Liaison with ISO/IEC JTC 1/SC 32

SC 4 appoints Gerald Radack as the liaison officer from SC 4 to ISO/IEC JTC 1/SC 32, and related activities and invites ISO/IEC JTC 1/SC 32, to notify SC 4 of its reciprocal liaison officer

RESOLUTION 636 (Lillehammer, Norway - March, 2005)

PWI for Process Monitoring Data

SC 4 approves the PWI set out in SC 4 N1865, with a view to creating a common resource for SC 4.

RESOLUTION 637 (Lillehammer, Norway - March, 2005)

Use of OWL in SC 4 standards

SC 4 requests WG 11 to conduct a review of the deployment of OWL technology in SC 4, with contributions from the other WGs, and make recommendations for its consistent use.

RESOLUTION 638 (Lillehammer, Norway - March, 2005)

Making Some SC 4 Standards Freely Available on the Web

Based on recent investigations within US and UK standards organizations, SC 4 notes that very little revenue is being generated from the sale of SC 4 standards despite very significant potential for industrial benefit from their use.

SC 4 believes that the free of charge availability of selected SC 4 standards will further promote the adoption of those standards and increase their visibility in a highly competitive marketplace particularly for small and medium enterprises.

SC 4 would like to follow the path of ISO/IEC JTC 1/SC 32 in which certain of their standards, including ISO 11179, are freely available on the Web, and revenue is generated through tutorials and implementer sessions.

SC 4 requests the PPC to examine new business models for generating alternative sources of revenue based on SC 4 standards.

SC 4 requests the PPC to prepare a justification to the ISO Council for approving that ISO 10303 Application Protocol 203 and its normative references within ISO 10303

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

should be made freely available on the Web.

RESOLUTION 639 (Lillehammer, Norway - March, 2005)

Procedures for an SC 4 Maintenance Agency

SC 4 creates an ad hoc group with the task of developing a unified approach to specify the procedures for a Maintenance Agency administered by the SC 4 secretariat to support reference data databases for SC 4 standards available on the web.

This group will create a proposal for the Maintenance Agency to be submitted to the ISO Technical Management Board (TMB) and will develop a business case to present to the ISO Council as to why these reference data libraries can be accessed free of charge from an appropriately serviced site.

Chris Kreiler is appointed as the Chair of this ad hoc group. Other members should be nominated by PPC, WG 2, WG 3, and other interested SC 4 committees nominated by end of March 2005. SC 4 request the group report back to the SC 4 meeting in Valencia.

RESOLUTION 640 (Lillehammer, Norway - March, 2005)

Product Data Quality (PDQ) Strategy

SC 4 has identified the need to create a Task Force to develop a Strategy for the PDQ Requirements of SC 4 concerns. The title of the Task Force will be: Product **Data Quality (PDQ) Strategy for Industrial Data**. SC 4 asks its Secretariat to seek nominations from member bodies, relevant liaisons such as SASIG and WG 2, WG 3, JWG 8 in order to create this Task Force at the next meeting (Valencia 2005).

RESOLUTION 641 (Lillehammer, Norway - March, 2005)

Use of translated standards

SC 4 requests the NSBs to review their policies for translating standards which might be a basis for software applications, in order to avoid creating incompatibilities between implementations.

The following wording shall be incorporated in any introduction of the SC 4 standards. "Attention:

This standard provides a specification which can be used for the development of software. Any implementation of software based on a translation of this standard into

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

national languages other than the official languages of ISO may cause incompatibility in machine-machine communication. In order to avoid such incompatibilities, the agreed international standard is the governing version.”

SC 4 requests its Quality Committee to insert this language into the Supplementary Directives for incorporation in all future publications

RESOLUTION 642 (Lillehammer, Norway - March, 2005)

Acknowledgements and Appreciations

ACCLAMATION:

Appreciation to our SC 4 Liaisons. We appreciate your representation to and for ISO/TC 184/SC 4.

A special thank you to the SC 4 Liaison Plenary presenters for their insightful and comprehensive presentations:

USPI-NL	Leo van Ruijven
EPISTLE	Per Christian Malmin,
Statoil Integrated Operations in Oil & Gas	Thore
Langeland ECCMA	Peter Benson

ACCLAMATION:

SC 4 expresses its particular thanks to Mr. Knut Lindelien and his staff, Rune Bergwitz, Berit Holthe, Audun Lona and Bjørnhild Sæterøy for their superior administrative support, outstanding hospitality, and patience both before and during the ISO/TC 184/SC 4 meeting.

RESOLUTION 643 (Valencia, Spain - June 2005)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303-0111	Michael Pratt	Michael Pratt
ISO 13584-0026 TC 1	Hiroshi Murayama	Noriko Minamino

RESOLUTION 644 (Valencia, Spain - June 2005)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Appointment of a New SC 4 Education and Outreach Rapporteur

SC 4 appoints Anna Moreno as Rapporteur of CEO. Recognizing the continuing need to make industry, the public, the research community, and other stakeholders knowledgeable of SC 4 standards and SC 4 activities, SC 4 appoints Mrs. Anna Moreno as the new SC 4 Education and Outreach Rapporteur to act as the focal point for SC 4 outreach, education, and promotional activities. These activities should be closely linked to International Industry STEP Centers (ISC) related marketing and promotional activities.

RESOLUTION 645 (Valencia, Spain - June 2005)

Nomination of Wolfgang Wilkes as Liaison officer to ISO/TC 172 " Optics and Photonics

SC 4 nominates Wolfgang Wilkes as liaison officer to ISO/TC 172

RESOLUTION 646 (Valencia, Spain - June 2005)

Nomination of Experts to ISO/TC 184/SC 5/WG 7 "Diagnostic and maintenance applications integration"

SC 4 sees a potential need of cooperation with its projects ISO 10303-239 "Product Life Cycle Support" (PLCS), ISO 15926 and the ASAM ODS PAS 22720. To avoid any overlap or gaps in the development of these projects SC 4 proposes that Tim King, Leif Gyllström, Magne Valen-Sendstad and Nigel Shaw be nominated as experts to SC 5/WG 7 by their respective National Bodies.

SC 4 appoints them as liaison officers to ISO/TC 184/SC 5 and invites them to participate in the joint SC 1, SC 4 and SC 5 review on "Diagnostics and maintenance applications integration".

RESOLUTION 647 (Valencia, Spain - June 2005)

Liaison officer to ISO/TC 37 "Terminology and other language resources

SC 4 notes the request of ISO/TC 37 "Terminology and other language resources" for nominating an additional point of contact for the SC 4 work on Classification. SC 4 appoints Wolfgang Wilkes, Eric Sardet, Magne Valen-Sendstad, Peter Benson and Gerry Radack as liaison officers to ISO/TC 37.

RESOLUTION 648 (Valencia, Spain - June 2005)

Format and Layout of SC 4 Standards

SC 4 adopts the following policy with regard to format and layout of SC 4 standards:

- Parts delivered to ISO in HTML shall follow the format and layout as given in

Dana Tripp

Secretary, ISO/TC 184/SC 4

trippdsc4@gmail.com



N4118

2026 – 05 – 15

the HTML Guidelines.

- Parts delivered to ISO in PDF shall follow the appropriate format and layout as given in the current clause 5, Format and layout of SC 4 standards, of the Supplementary Directives.
- Parts delivered to ISO in Microsoft® Word shall be prepared with one of the ISO templates for drafting standards.

This policy shall go into effect immediately. SC 4 directs the Quality Committee to make the appropriate changes to the Supplementary Directives to implement the above policy.

RESOLUTION 649 (Valencia, Spain - June 2005)

Progress of ISO/IEC/JWG 1

SC 4 expresses its concerns at the apparent lack of progress in the work of JWG 1 on providing Guidance on broader exploitation of ISO 13584 and IEC 61360. SC 4 directs its nominated liaison officers and experts as shown in Resolution 611 & 627 to endeavor to expedite the completion of the JWG 1 tasks and report progress to SC 4. The Chairman of SC 4 is asked to write to the convener of JWG 1 expressing the urgency of the SC 4 requirement for completion of the JWG 1 work.

RESOLUTION 650 (Valencia, Spain - June 2005)

Assignment of a URN to an object within an ISO standard

SC 4 requests ISOCS to complete the 'tcdefined' section of SC 4/N1916, including guidance on the use of this section for the identification of an object within an ISO standard, and circulate the resulting document to the relevant ISO/TCs for review.

SC 4 further requests ISO Central Secretariat to agree to host the equivalent URL, which document SC 4/N1916 says 'should be possible', and to publish documents where the following criteria are satisfied:

- it has been agreed that the standard, or the object defined within the standard, shall be freely available on the Web;
- the Technical Committee provides the document that has the URL or contains the URL as an object within it.

RESOLUTION 651 (Valencia, Spain - June 2005)

Installation of a Registration Authority according to ISO 13584 Part 501

In the absence of any specific recommendation from JWG 1, SC 4 requests its secretariat to take the necessary steps with ISO to request the establishment of a

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Registration Authority as defined in 13584 Part 501. SC 4 nominates the Japan Electric Measuring Instruments Manufacturers Association to act as the Registration Authority. After 3 years, the way of maintaining the dictionary shall be reviewed taking into account advice from JWG 1 and new rules and regulations coming from ISO or IEC.

RESOLUTION 652 (Valencia, Spain - June 2005)

NWI on Product properties and reference hierarchies, proposed by ISO/TC 10/SC 10/WG 11

SC 4 notes the ISO/TC 10 proposal to create a multi-part standard for defining dictionaries of properties with class reference hierarchies. This work appears to have a large overlap with the published ISO 13584, ISO 15926 and IEC 61360 standard series.

SC 4 also notes that the proposed scope of part 101 of this ISO/TC 10 multi-part standard appears to have a large overlap with its ongoing and published work on component libraries and reference hierarchies for the oil and gas industry.

In order to achieve synergy and to avoid duplicate and conflicting standards SC 4 invites ISO/TC 10 and ISO/IEC JWG 1 to participate in an open discussion with SC 4 on the technical content of the proposed standard in order to agree the necessary scope of work to be undertaken in the respective committees.

RESOLUTION 653 (Valencia, Spain - June 2005)

Acknowledgements and Appreciations

Acclamation:

Appreciation to our SC 4 Liaisons. We appreciate your representation to and for ISO/TC 184/SC 4.

A special thank you to the SC 4 Liaison Plenary presenters for their insightful and comprehensive presentations:

PDES, Inc
ProSTEP iViP
ISO/TC 211
USPI-NL
ECCMA / NATO AC 135

Bob Kiggans
Max Ungerer
Jerry Smith
Dalip Sud
Peter Benson
Steve Arnett

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

eCI@ss
ISO 13399
FSIG-SMART-fm
R & D and Standards
Lehouck

George
Sturrock
Raymond Betz
Norman Swindells
Ricardo Gonçalves
Anne

Acclamation:

The SMART-fm project is a funStep initiative that has been providing support for the development of AP236, and evaluated by the European Commission representatives during this SC 4 meeting.

FunStep would like to thank the SC 4 community for its outstanding support, and specifically recognize the continuous support and special efforts provided by Mr. Howard Mason, Mr. Robert Kiggans, Mr. Jerry Smith, Mr. Peter Benson and Mr. Keith Hunten during the final review process in conjunction with the European Commission.

Acclamation:

SC 4 expresses its particular thanks to Mrs Amparo Roca de Togores and her staff, Ivan Duro, Tiago Ferreira and Michele Vavallo for their superior administrative support, outstanding hospitality, both before and during the ISO/TC 184/SC 4 meeting. Our Special thanks to Mr. José Chaparro and Angel Dominguez, AIDIMA's system administrators, and from El Corte Ingles, Maria Jesus Grijalba and Erika Hvidkjaer for their excellent support and patience with the SC 4 Delegation.

RESOLUTION 654 (HangZhou, China -October 2005)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303-0059	Hiroyuki Hiraoka	Akihiko Otaka

RESOLUTION 655 (HangZhou, China -October 2005)

Termination of the Liaison between SC 4 and PLCS, Inc

Recognizing the successful completion of the development of AP239, and the closure

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

of the PLCS, Inc consortium, SC 4 terminates its liaison with PLCS, Inc.

The liaison with OASIS established under resolutions 580 and 629 remains in place, in order to address any SC 4 standardisation requirements arising from activities under the OASIS PLCS Technical Committee.

RESOLUTION 656 (HangZhou, China -October 2005)

Liaisons

SC 4 requests an active two-way liaison with ISO/TC 37/SC 2, 3 and 4, and requests the SC 4 Secretary to confirm these liaisons with the ISO Central Secretariat. SC 4 appoints Gerald Radack, Peter Benson, Magne Valen-Sendstad and Guy Pierra as liaison officers from SC 4 to ISO/TC 37/SC 2, 3 and 4, and requests them to report back on any relevant implications for SC 4 standards and projects, after each ISO/TC 37 and subcommittee meeting.

RESOLUTION 657 (HangZhou, China -October 2005)

Nomination of Experts to ISO/TC 184/SC 5/WG 7 "Diagnostic and maintenance applications integration"

In addition to the appointments and nominations stated in SC 4 Resolution # 646, SC 4 proposes that the ISO/TC 184/SC 4/JWG 8 Convener, Jean-Jacques MICHEL, be nominated as expert to SC 5/WG 7 by its National Body.

SC 4 appoints him as liaison officer to ISO/TC 184/SC 5 and invites him to participate in the joint SC 1, SC 4 and SC 5 review on "Diagnostics and maintenance applications integration".

RESOLUTION 658 (HangZhou, China -October 2005)

JWG 1

SC 4 notes the potential impact on many ISO Technical Committees of the proposed content of the JWG 1" Guide for the Specification of Product Properties and Classes".

SC 4 recommends that National Bodies ensure that any ballot on these documents is reviewed by their representatives to SC 4 and other affected Technical Committees.



N4118

2026 – 05 – 15

RESOLUTION 659 (HangZhou, China -October 2005)

Task Force on Dictionaries, Part Libraries and Reference Data Libraries

SC 4 appreciates the content of the report of the Task Force on Dictionaries, Part Libraries and Reference Data Libraries (SC 4 N2016), since it supports the acceptance and knowledge about the SC 4 standards and their relations in industrial use. SC 4 thanks the Task Force for its efforts. SC 4 disbands the Task Force.

RESOLUTION 660 (HangZhou, China -October 2005)

Allocation of the Project ISO 22745 "Open Technical Dictionary"

SC 4 assigns the Open Technical Dictionary Project to WG 12.

RESOLUTION 661 (HangZhou, China -October 2005)

Liaison to ISO/TC 60 "Gears"

SC 4 establishes a liaison to ISO/TC 60 Gears to enable a close cooperation on data exchange for gears. To avoid inconsistencies and incompatibilities with existing SC 4 standards, SC 4 nominates Alan Crawford and Len Slovensky as liaison officers.

RESOLUTION 662 (HangZhou, China -October 2005)

Nomination of representative to TMB ad hoc meeting on "Publication of Standards as databases" - Geneva 10/11 November.

SC 4 nominates its chairman Howard Mason to attend the ISO/TMB ad hoc meeting on 2005-11-10/11 in Geneva. The objective of this nomination is to reflect the interest of SC 4 projects and users of SC 4 publications to make these publicly available and free of charge, in order to facilitate their use.

The position of SC 4 should be compiled and circulated to SC 4 for review prior to the meeting.

RESOLUTION 663 (HangZhou, China -October 2005)

Acknowledgements and Appreciations

Acclamation:

Appreciation to our SC 4 Liaisons. We appreciate your representation to and for ISO/TC 184/SC 4.

A special thank you to the SC 4 Liaison Plenary presenters for their insightful and comprehensive presentations:

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

IAI
ISO/TC 207
ECCMA / NATO AC 135
Radack ISO/TC 37

Thomas Liebich
Anna Moreno
Peter Benson & Gerry
Christian Galinski

Appreciation:

SC 4 wishes to give its unanimous thanks to CNIS, its Members and the staff of the HangZhou Huagang HNA Resort Hotel for their efforts in organizing this meeting, arranging the facilities, excellent food, and excellent social events.

Appreciation:

SC 4 expresses its thanks to Mr. Bisong Liu and his staff, Mr. Guangli Qin, Mr. Junfeng Zhan, Mr. Wenfeng Sun, Mr. Hanjing Hu, Mr. Wenwu Li, Mr. Yan Hong, Ms. Xiaojing Zhang and Ms. Ping Xu for their superior administrative support, outstanding hospitality, both before and during the ISO/TC 184/SC 4 meeting. Our Special thanks to Mr.

Zhiqiang Wang, CNIS system administrator, and Ms. Jianghua Liao and Ms. Shanshan Sun for their excellent support with printing and copying requirements for the SC 4 Delegation. Finally, SC 4 expresses its particular thanks to Mr. Ping Wang for providing the overall excellent coordination for the entire SC 4 Meeting in HangZhou.

RESOLUTION 664 (Vico Equense, Italy - March 2006)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303-0224 ed3		Len Slovensky

RESOLUTION 665 (Vico Equense, Italy - March 2006)

Reappointment of WG 3 Deputy Convenor

SC 4 appreciates and accepts the offer of Len Slovensky to serve as Deputy Convenor of WG 3. SC 4 approves the nomination of Len Slovensky to serve for an additional three year term.

RESOLUTION 666 (Vico Equense, Italy - March 2006)

Reappointment of JWG 8 Convenor

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 appreciates and accepts the offer of Jean-Jacques Michel to serve as Convenor of

JWG 8. SC 4 approves the nomination of Jean-Jacques Michel to serve for an additional three year term

RESOLUTION 667 (Vico Equense, Italy - March 2006)

Reappointment of WG 12 Convenor

SC 4 appreciates and accepts the offer of Keith Hunten to serve as Convenor of WG 12. SC 4 approves the nomination of Keith Hunten to serve for an additional three year term

RESOLUTION 668 (Vico Equense, Italy - March 2006)

Liaisons

SC 4 requests an active two-way liaison with ISO/TC 229, Nanotechnology, and requests the SC 4 Secretary to confirm these liaisons with the ISO Central Secretariat. SC 4 appoints Ms Nonna Bond and Dr. Lars Ericson as the liaison officers from SC 4 to ISO/TC 229. The SC 4 Secretary is also requested to solicit additional nominations from the ISO/TC 184/SC 4 P-members.

RESOLUTION 669 (Vico Equense, Italy - March 2006)

Launch of AP 203 ed 2 ballot

SC 4 accepts the proposal of the AP 203 ed2 project team to launch a DIS ballot on the revised Technical Specification ISO/TS 10303-203, and requests its Secretariat to register the necessary New Work Item with ISO.

Referenced documents: See WG 2N1591 – issues log for AP203 and WG 12N3277 issues log for GD&T modules for additional information.

Action item: Secretariat to request the AP203 ed2 team to confirm that they will use the current part 10303-55 and the revised 10303-111, currently under DIS ballot, to support construction history, prior to submitting the DIS ballot for AP203 ed2

RESOLUTION 670 (Vico Equense, Italy - March 2006)

Reinstatement of Work Items for Parts ISO 10303-52, 53, and 110

SC 4 requests its Secretariat to register the necessary New Work Items for 10303-52, 10303-53, and 10303-110 with ISO, in order to allow them to proceed to DIS ballot.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 671 (Vico Equense, Italy - March 2006)

Assignment of ISO 8000 to WG 12

SC 4 assigns the current project ISO 8000 – (Catalogue Management Systems - Requirements) to its Working Group 12 on a temporary basis until the establishment of a new Working Group 13 for Industrial Data Quality.

Action to Secretariat: Secretariat is asked to inform the National Member Bodies of the new project and request them to nominate experts to the project.

RESOLUTION 672 (Vico Equense, Italy - March 2006)

Extension of New Work Item Ballot for AP241 – AEC Facilities

Based on the comments from IAI and SC 4 members, SC 4, as an exception, extends the ballot closing date for the New Work Item proposal on Generic Model for Life Cycle Support of AEC Facilities until July 31, 2006.

SC 4 encourages IAI and WG 3 Teams 8, 22, and 23 to refine the NWI proposal so that it is mutually acceptable.

RESOLUTION 673 (Vico Equense, Italy - March 2006)

Theme of SC 4 Meetings

SC 4 invites host organizations to work with the International Industry STEP Centers and the SC 4 Chairman and Secretariat to organize industry sector themes for SC 4 meetings as appropriate, which may be reflected in the content of OTF, Liaison Plenary, training, tutorials and Implementation activities.

RESOLUTION 674 (Vico Equense, Italy - March 2006)

Cooperation with ISO/IEC JTC 1/SC 31 on RFID

SC 4 expresses its interest in taking responsibility for Industrial Data application standards for radio frequency identifiers (RFIDs), in cooperation with ISO/IEC JTC 1/SC

31. SC 4 appoints the following as delegates to attend the meeting of the joint ad hoc group to consider how to take ISO 21849 forward to multiple industries to be held in Lawrenceville, NJ, USA, 2006-04-11/12.

King Yee

Gerald

Dana Tripp

Secretary, ISO/TC 184/SC 4

trippdsc4@gmail.com



N4118

2026 – 05 – 15

Radack Peter
Benson Nils
Sandsmark
David Leal
Norman Swindells
Howard Mason

SC 4 requests the delegation to provide a written report within two weeks of the meeting, and an oral report at the next SC 4 meeting.

Note: Resolution amended by RESOLUTION 688 (Toulouse, France - June 2006)

RESOLUTION 675 (Vico Equense, Italy - March 2006)

Technical Corrigenda for AP236 modules

SC 4 requests its Secretariat to submit the Technical Corrigenda for the following STEP modules to ISO for publication as second editions.

ISO/TS 10303—1103, 1104, 1108, 1109, 1110, 1111, 1112, 1129, 1341, 1342, 1343, 1344, 1346, 1347, 1351, 1352

RESOLUTION 676 (Vico Equense, Italy - March 2006)

Creation of new WG 13 on Industrial Data Quality

SC 4 resolves to establish a new Working Group 13, Industrial Data Quality.
The task of this working group will be to:

- Establish a coherent definition of the structure of Quality Standards to be produced by SC 4, based on the results of the Product Data Quality Strategy meeting held on 2006-03-07 (SC 4 N2078)
- Identify NWI and develop the necessary standards for accreditation of organizations and processes, and the quality criteria for information complying with each of the SC 4 standards, all to be published under the ISO 8000 series.
- The current ISO 8000 work item will become part of the ISO 8000 series.

SC 4 requests its Secretariat to invite National Member Bodies to nominate experts

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

and candidates for Convenorship for the group.

RESOLUTION 677 (Vico Equense, Italy - March 2006)
Cooperation with ISO/TC 171/SC 2 on PDF/E WD24517

SC 4 expresses its interest to participate in the proposed JWG with ISO/TC 171/SC 2 on PDF/E. SC 4 nominates as participants:

- Max Ungerer,
- Paul van Exel

SC 4 directs the delegation to provide written reports on progress with the project.

RESOLUTION 678 (Vico Equense, Italy - March 2006)
Initiating a Gear Design Data Exchange Project

SC 4 welcomes the proposition from Germany to initiate a gear design project based upon the work in specification VDMA 23900, and invites them to submit any necessary NWI to SC 4 for consideration.

RESOLUTION 679 (Vico Equense, Italy - March 2006)
Appointment of QC Coordinator

SC 4 appreciates and accepts the offer of Peter Benson to serve as Quality Coordinator. SC 4 approves the nomination of Peter Benson to serve for a three year term.

RESOLUTION 680 (Vico Equense, Italy - March 2006)
ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION:

Appreciation to our SC 4 Liaisons. We appreciate your representation to and for ISO/TC 184/SC 4.

A special thank you to the SC 4 Liaison Plenary presenters for their insightful and comprehensive presentations:

ISO/TC 184

Jean-Marc Chatelard

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

 Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

JWG 1	Wolfgang Wilkes
OASIS	Mats Nilsson
ECCMA	Peter Benson
USPI-NL	Leo van Ruijven
ISO/TC 67	Nils Sandsmark
ENEA	Anna Moreno

APPRECIATION:

SC 4 wishes to give its unanimous thanks to ENEA, its Members and the staff of the Oriente Hotel for their efforts in organizing this meeting, arranging the facilities, excellent food, and excellent social events.

APPRECIATION:

SC 4 expresses its particular thanks to Dr. Anna Moreno for providing the overall excellent coordination for the entire SC 4 Meeting in Vico Equense.

SC 4 also wishes to recognize the members of her staff for their superior administrative support, outstanding hospitality, both before and during the ISO/TC 184/SC 4 meeting. They are; Mario Montanino, Anna Amato, Celestina Coccia, Augusto Massari, Enzo Moreno and Imma e Francesco Arpino.

Finally, SC 4 expresses its particular thanks to Rosaria Savarese from the Town Hall for taking time to speak to the SC 4 members.

RESOLUTION 681 (Toulouse, France - June 2006)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 approves the following new project leaders or project editors for the associated SC

4 projects:

Project	Leader	Editor
ISO 10303-0214ed3		Joselito Henriques

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ISO 10303-58

Tore Hartvigsen

RESOLUTION 682 (Toulouse, France - June 2006)

Reappointment of SC 4 Chair

SC 4 endorses the nomination by the United States of Howard Mason for an additional three year term (January 2007-December 2009) as the Chair of ISO/TC 184 / SC 4 and instructs the Secretary to forward his nomination to ISO/TC 184 within 30 days.

RESOLUTION 683 (Toulouse, France - June 2006)

Convenership of WG 13 "Industrial Data Quality"

SC 4 appoints Dr. Tim King as the Convener of ISO/TC 184/SC 4/WG 13 for a three year term.

RESOLUTION 684 (Toulouse, France - June 2006)

Start of work of WG 13 "Industrial Data Quality"

SC 4 requests WG 13 to hold its first meeting no later than the next meeting of SC 4 (October 2006) in Hershey, PA, USA and asks the convener to give at this meeting a status report to SC 4 on its work programme based on the Resolution 676 .

RESOLUTION 685 (Toulouse, France - June 2006)

Revision of NP Ballot on AP 241

SC 4 withdraws the existing NWI on AP 241 (SC 4 N 2038). SC 4 requests the SC 4/WG 3/T22 and the IAI to prepare a revised NWI for a new NWI ballot. The revision of the document and the revised change management form shall be sent to the SC 4 secretariat for circulation no later than 2006-08-31. The NWI should include details of the work partition between IAI and SC 4.

RESOLUTION 686 (Toulouse, France - June 2006)

Reappointment of WG 2 Deputy

SC 4 appreciates and accepts the offer of Prof. Guy Pierra to serve as Deputy Convener of WG 2. SC 4 approves the nomination of Prof. Guy Pierra to serve for an additional three year term.

Parallel processing of AP DIS and module TS ballots

SC 4 requires that SC 4 modules normatively referenced by modular STEP Application Protocols (APs) shall complete their Committee Draft Technical

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Specification (CD-TS) ballot prior to the AP completing its Draft International Standard (DIS) ballot. Modules need not be published by ISO to be normatively referenced. This Resolution clarifies and extends Resolutions 557 and 342.

RESOLUTION 688 (Toulouse, France - June 2006)

Cooperation with ISO/IEC JTC 1/SC 31 on AIDC

SC 4 replaces the first paragraph of resolution 674 with the following:

"SC 4 expresses its interest in taking responsibility for Industrial Data application standards for all automatic identification data capture (AIDC) technologies (not limited to radio frequency identification [RFID]), in cooperation with ISO/IEC JTC 1/SC 31."

SC 4 agrees to extend the ad hoc group until SC 31 and ISO/TC 184 and its SC 4 define and agree the scope of the work required within their respective committees, and jointly prepare and agree any necessary New Work Item Proposals for consideration by national bodies.

SC 4 recognizes that formal liaison arrangements with SC 31 may need to be implemented at some future date.

SC 4 requests its delegation to provide a written report within two weeks of each physical meeting, and an oral report at each SC 4 meeting.

Note: Resolution amends RESOLUTION 674 (Vico Equense, Italy - March 2006)

RESOLUTION 689 (Toulouse, France - June 2006)

ISO/IEC JTC 1/SC 7 NWI on Systems Engineering

SC 4 notes the NWI on Systems Engineering from ISO/IEC JTC 1/SC 7. (N3513, N3521, N3529) SC 4 requests its WG 3/T8 to prepare a liaison statement to SC 7 drawing their attention to the scope of the current work on Systems Engineering with a view to avoiding duplicated or inconsistent effort. If a liaison is required, SC 4 authorizes its Secretariat to initiate the liaison.

RESOLUTION 690 (Toulouse, France - June 2006)

Change of title for ISO 22745

SC 4 resolves to change the title of ISO 22745 to: "Open technical dictionaries and their application to catalogues".

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 691 (Toulouse, France - June 2006)
ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION:

Appreciation to our SC 4 Liaisons. We appreciate your representation to and for ISO/TC 184/SC 4.

A special thank you to the SC 4 Liaison Plenary presenters for their insightful and comprehensive presentations:

OMG	David Price
ISO/TC 29/WG 34	Norman
Swindells ECCMA/NATO AC/135	
	Steve Arnett
IAI	Thomas Liebich
ISO/TC 29 WG 34	Jonas Nordstrom
ISO/IEC JTC 1/SC 32	Gerry
Radack ISO/TC 37	Gerry Radack
ASD (LOTAR)	Jean-Yves Delaunay

We also offer thanks to those external projects who presented their material at the meeting.

eCI@ss – PFI- ORCHID	Raymond Betz
PROMISE	Dimitris Kiritsis

ACCLAMATION:

SC 4 expresses it's thanks to Alain Brail for organizing a most productive industry day on the future of STEP for manufacturing, and to all the speakers who contributed to the event, which has produced a wide range of information and guidance on future standards development activities.

ACCLAMATION:

SC 4 expresses its thanks to Airbus for its sponsorship and the memorable social event, particularly for the efforts of Antoine Scotto d'Appollonia, Francesco Sperandio, Jean- Bernard Hentz and Jean-Yves Delaunay and Alain Brail.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ACCLAMATION:

SC 4 expresses its thanks to Paul Sabatier University, Pierre Lagarrigue and Michele Grimal for preparing and hosting the machining demonstration,

ACCLAMATION:

SC 4 expresses its particular thanks to Christine Hermetet-Filez and her staff, Delphine Michel and Marie-France (CTA Events), Isabelle Issanchou and Jerome (Mercure Atria), Benoit and Cyrile (LCD Vision) and Martine Dégardin (AFNOR) for their superior administrative support, outstanding hospitality, both before and during the ISO/TC 184/SC 4 meeting.

RESOLUTION 692 (Hershey, PA, USA - October 2006)

Appointment of Quality Committee deputy coordinator

SC 4 appreciates and accepts the offer of Dr. Junfeng Zhan to serve as Quality Committee deputy coordinator. SC 4 approves the nomination of Dr. Junfeng Zhan to serve for a three year term.

RESOLUTION 693 (Hershey, PA, USA - October 2006)

Co-Leaders for the ad hoc Group Industrial Data on the Web (IDW)

SC 4 appreciates and accepts the offer of David Leal and David Price to serve as co-leaders of the ad hoc group for Industrial Data on the Web (IDW). SC 4 approves the nomination of David Leal and David Price to serve as co-leaders of the ad hoc group for Industrial Data on the Web, and requests them to produce an updated workplan by the next meeting.

SC 4 expresses its utmost appreciation to Jim U'Ren for his enthusiastic work in leading the SC 4 ad hoc Group for Industrial Data on the Web and wishes him well in his new endeavours.

RESOLUTION 694 (Hershey, PA, USA - October 2006)

Liaison with The American Society for Quality (ASQ)

SC 4 proposes a Category A liaison with the American Society for Quality (ASQ). SC 4 requests that the SC 4 secretary confirm this liaison with the ISO Central Secretariat.

SC 4 invites the American Society for Quality (ASQ) to make a reciprocal nomination,

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

and to invite any of its constituent bodies with activities within the scope of SC 4 to establish liaison arrangements.

The SC 4 Secretary is also requested to solicit liaison officer nominations from the ISO/TC 184/SC 4 P-members.

RESOLUTION 695 (Hershey, PA, USA - October 2006)

Liaison officer to Marine e-Business Standards Association (EMSA)

SC 4 nominates Nils Sandmark as liaison officer to the Marine e-Business Standards Association (EMSA). SC 4 expresses its appreciation for the work of John Kendall as the previous liaison officer.

RESOLUTION 696 (Hershey, PA, USA - October 2006)

Liaison with The International Association for Information and Data Quality (IAIDQ)

SC 4 proposes a Category A liaison with the International Association for Information and Data Quality (IAIDQ). SC 4 requests that the SC 4 secretary confirm this liaison with the ISO Central Secretariat.

SC 4 invites the International Association for Information and Data Quality (IAIDQ) to make a reciprocal nomination, and to invite any of its constituent bodies with activities within the scope of SC 4 to establish liaison arrangements.

RESOLUTION 697 (Hershey, PA, USA - October 2006)

Nomination of Hiroshi Murayama as Liaison officer to IEC/TC 3 “Information structures, documentation and graphical symbols”

SC 4 nominates Hiroshi Murayama as liaison officer to IEC/TC 3 “Information structures, documentation and graphical symbols”.

RESOLUTION 698 (Hershey, PA, USA - October 2006)

Nomination of David Price as Liaison officer to Object Management Group (OMG)

SC 4 nominates David Price as an additional liaison officer to the Object Management Group (OMG).

RESOLUTION 699 (Hershey, PA, USA - October 2006)

Appreciation for Sophie Clivio

SC 4 expresses its appreciation and gratitude to Ms Sophie Clivio for all her support of

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

the work of SC 4 over the last decade, and wishes her well in her new role in ISO.

RESOLUTION 700 (Hershey, PA, USA - October 2006)

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION:

Appreciation to our SC 4 Liaisons. We appreciate your representation to and for ISO/TC 184/SC 4.

A special thank you to the SC 4 Liaison Plenary presenters for their insightful and comprehensive presentations:

PDES, Inc	Bob Kiggans
PDES, Inc	Jack Harris
NIST	Steve Ray
Naval Surface Warfare Center	Jim Mays
Renault	Oscar Rocha
OLF	Thore Langeland
Sandvik Coromant	Jonas Nordström
DLIS	Steve Arnett

ACCLAMATION:

SC 4 expresses its particular thanks to Peter Benson and his staff, Yvette Thomas, Dan King, Amber Davis and Melissa Schieb for their superior administrative support, outstanding hospitality, both before and during the ISO/TC 184/SC 4 meeting. Well done.

ACCLAMATION:

SC 4 expresses its thanks to ECCMA and PiLog for its sponsorship and the memorable social event, particularly for the efforts of Mr. Ed Seguire, Vice President of Research and Development and Quality Assurance for the Guittard Chocolate Company. His presentation, overview and instruction of the correct etiquette while tasting chocolate will long be remembered.

ACCLAMATION:

SC 4 wishes to give its unanimous thanks to Mr. Ted Maloy, The Hotel Hershey Conference Planning Manager and staff for their efforts in organizing this meeting,

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

arranging the facilities and excellent food. Their considerable efforts created a very memorable event.

RESOLUTION 701 (Funchal, Madeira Island, Portugal - March 2007)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303-0233	Harold Frisch & Phil Spiby (Deputy)	Allison Barnard-Feeney
ISO 10303-43ed3	Simon Frechette	Lothar Klein
ISO 10303-46ed2	Simon Frechette	Lothar Klein
ISO 10303-101ed2 Klein	Simon Frechette	Lothar

RESOLUTION 702 (Funchal, Madeira Island, Portugal - March 2007)

Reappointment of WG 3 Convener

SC 4 appreciates and accepts the offer of Dr. Radack to serve as Convener of WG 3. SC 4 approves the nomination of Dr. Radack to serve for an additional three year term.

RESOLUTION 703 (Funchal, Madeira Island, Portugal - March 2007)

Establishment of an Internal Liaison with ISO/TC 59/SC 13

SC 4 accepts the request from ISO/TC 59/SC 13 to establish an internal two-way liaison and appoints Mr. Tatsuo Terai as the liaison officer from SC 4 to ISO/TC 59/SC 13.

RESOLUTION 704 (Funchal, Madeira Island, Portugal - March 2007)

Establishment of a Category A Liaison with European Computer Manufacturers Association (Ecma) International Technical Committee 43

SC 4 accepts the request from Ecma (European Computer Manufacturers Association) International Technical Committee 43 to establish a Category A liaison and appoints King Yee as the liaison officer from ISO/TC 184/SC 4 to Ecma TC 43.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 welcomes Tom-Michael Thamm as liaison officer from Ecma TC 43 to ISO/TC 184/SC 4.

RESOLUTION 705 (Funchal, Madeira Island, Portugal - March 2007)
Development of IS version of ISO 13584-35

SC 4 requests its Secretariat to initiate a DIS ballot on ISO 13584-35 as soon as the DIS document is available.

Nominations:

Project leader: Hiroshi Murayama

Project editor: Nikolaus Ondracek and Noriko Minamino

RESOLUTION 706 (Funchal, Madeira Island, Portugal - March 2007)
Liaison to IEC SC 65 E

SC 4 requests its secretariat to establish a liaison to IEC SC 65 E and appoints Hiroshi Murayama as liaison officer.

RESOLUTION 707 (Funchal, Madeira Island, Portugal - March 2007)
Establishment of an Internal Liaison with ISO/IEC JTC 1/SC 32/WG 2 "Common Logic"
SC 4 establishes a liaison from SC 4 to ISO/IEC JTC 1/SC 32/WG 2 and nominates Matthew West as the liaison officer. SC 4 requests JWG 8 to nominate an additional liaison officer to cover PSL matters.

RESOLUTION 708 (Funchal, Madeira Island, Portugal - March 2007)
Minor Revision for STEP IR 43, 46, & 101

SC 4 directs WG 12 to prepare FDIS documents for ISO 10303 parts 43, 46, & 101 and submit them to ISO for processing as minor revisions.

RESOLUTION 709 (Funchal, Madeira Island, Portugal - March 2007)
Initiation of FDIS ballot for ISO 13584-26 amendment

SC 4 asks its secretariat to initiate the FDIS ballot on the amendment to ISO 13584-26 as soon as the document is available.

RESOLUTION 710 (Funchal, Madeira Island, Portugal - March 2007)
Termination of Liaison to ISO/IEC JTC 1/SC 34

Dana Tripp

Secretary, ISO/TC 184/SC 4

trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 terminates its liaison to ISO/IEC JTC 1/ SC 34.

RESOLUTION 711 (Funchal, Madeira Island, Portugal - March 2007)

Ad Hoc Group on Visualization

SC 4 creates an ad hoc group to define requirements for 3D visualization specifications. Oscar Rocha is appointed as chair. Members of this group shall include King Yee, Martin Hardwick and David Leal. National bodies shall be invited to nominate additional representatives. The Group shall report back to the Chairman and PPC not later than 2007-06-12.

RESOLUTION 712 (Funchal, Madeira Island, Portugal - March 2007)

Exemption from ISO Standards Target dates

SC 4 requests its chair and Secretariat to negotiate a waiver from the TMB to restore the use of planned target dates rather than limit dates in the ISO work programme for SC 4 standards. SC 4 requests its P-members to support the proposal with their TMB representatives.

RESOLUTION 713 (Funchal, Madeira Island, Portugal - March 2007)

Clarification of copyright statement for executable electronic inserts

SC 4 requests its chair and Secretariat to agree with ISO CS a form of words which can be used where necessary in SC 4 deliverables to ensure that electronic inserts containing processable content, such as XML, EXPRESS, PSL, STEP part 21 files, OWL, RDF, Common Logic, UML models and reference data in digital form may be used without restriction from the standard ISO copyright statement.

SC 4 requests its QC to ensure that the necessary changes are made to the Supplementary Directives.

RESOLUTION 714 (Funchal, Madeira Island, Portugal - March 2007)

Establishment of an Internal Liaison with ISO/IEC JTC 1/SC 7 "Software and Systems Engineering"

SC 4 establishes a liaison from SC 4 to ISO/IEC JTC 1/SC 7 and nominates Phil Spiby as the liaison officer.

RESOLUTION 715 (Funchal, Madeira Island, Portugal - March 2007)

Establishment of an Internal Liaison with ISO/IEC JTC 1/SC 31 "AIDC"

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 reconfirms a liaison from SC 4 to ISO/IEC JTC 1/SC 31 and nominates Howard Mason and Peter Benson as additional liaison officers. JWG 8 is invited to nominate an appropriate expert for manufacturing matters.

RESOLUTION 716 (Funchal, Madeira Island, Portugal - March 2007)

ACKNOWLEDGEMENTS and APPRECIATIONS

APPRECIATION:

Norman Swindells has played a pivotal role in the many successes of ISO/TC 184/SC 4.

SC 4 expresses its sincere appreciation and gratitude to Norman Swindells for all his support of the work of SC 4 over the last two decades, and wishes him the very best in his retirement.

APPRECIATION:

Appreciation to our SC 4 Liaisons. We appreciate your representation to and for ISO/TC 184/SC 4.

ECCMA/AC 135
ISO/TC 29/WG 34
Swindells PDES Inc
Kox
FSIG

Peter Benson & Steve Arnett
Norman
Dennis
Maria Jose Nunez

We also offer thanks to those external projects who presented their material at the Open Technical Forum.

SOA and PLCS PLM Web Services

Phil Spiby

SEINE

Jean Brange/Nicolas Figay

A Framework for STEP Based
Harmonization of conceptual
Models

Marco Delgado

Practical use of a Reference Data

Leo van

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Ruijven Library

ACCLAMATION:

SC 4 expresses its special thanks to the Government of Madeira, its President, Alberto Joao Jardim, and Secretario Regional do Turismo e Cultura Joao Carlos Abreu, Secretaria Regional dos Assuntos Sociais, Conceicao Estudante, for its outstanding support and sponsorship.

Also special thanks to the Director Regional de Turismo Maria Graca Luis for its superior assistance and organization of memorable social events.

SC 4 also recognizes IMS for support of the SC 4 Conference and the close involvement of the Honorable Mr. Dan Nagy.

XECOMPLEX-XEROX Portugal, Mr. Joao Rodrigues, for outstanding technical support and conference equipment support.

Cabo TV Madeirense, Mr. Nuno Aguiar, for outstanding technical support and conference equipment support.

Universidade da Madeira, Mr. Nuno Jardim Nunes and Jorge Cardoso, for outstanding technical support and conference equipment support.

ACCLAMATION:

SC 4 expresses its particular thanks to Mr. Fernando Santana, Director of Faculdade de Ciencias e Tecnologia da Universidade Nova de Lisboa, Mr. Adolfo Steiger-Garcia, Mr. Ricardo Jardim Gonçalves and Mrs. Cristina Gorba, and their on-location support staff;

Bruno Almeida

Maria Brazão

Maria Marques

Ruben Costa

Moisés Dutra

Fátima Brazão

Agostinho Hugo Vieira

João Pedro Mendonça

João Sarraipa

Marco Delgado

Sérgio Onofre

Pedro Maló

Nelson Alves

Miguel Beça

Carlos

Gustavo Fernandes

Licínia Santos

Dana Tripp

Secretary, ISO/TC 184/SC 4

trippdsc4@gmail.com



N4118

2026 – 05 – 15

for superior administrative support and outstanding hospitality, both before and during and after the ISO/TC 184/SC 4 meeting.

RESOLUTION 717 (Ibusuki, Japan - July 2007)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303-241	Inhan Kim	Tatsuo Terai

RESOLUTION 718 (Ibusuki, Japan - July 2007)

Appointment of WG 2 Convener

SC 4 appreciates and accepts the offer of Nikolaus Ondracek to serve as Convener of WG 2. SC 4 approves the nomination of Mr. Ondracek to serve for a three year term.

SC 4 thanks Dr. Wolfgang Wilkes for his significant contributions in his role as convener, lead technical expert and also his successes as a consensus builder throughout his appointments.

RESOLUTION 719 (Ibusuki, Japan - July 2007)

Reappointment of WG 12 Deputy Convener

SC 4 appreciates and accepts the offer of Jochen Haenisch to serve as Deputy Convener of WG 12. SC 4 approves the nomination of Mr. Haenisch to serve for an additional three year term.

RESOLUTION 720 (Ibusuki, Japan - July 2007)

Appointment of Replacement Liaison officer to ISO/TC 172 "Optics and Photonics"

SC 4 nominates Nikolaus Ondracek as liaison officer to ISO/TC 172.

SC 4 expresses its special thanks to Dr. Wilkes, for his outstanding support and sponsorship.

RESOLUTION 721 (Ibusuki, Japan - July 2007)

Appointment of Alternate Liaison Officer to IEC SC 65E

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 nominates Nikolaus Ondracek as alternate liaison officer to IEC SC 65E System engineering and management.

SC 4 again expresses its special thanks to Dr. Wilkes for his outstanding support and sponsorship.

RESOLUTION 722 (Ibusuki, Japan - July 2007)

Demonstration of planning, machining and measurement

Recognizing the success of the demonstrations at the Toulouse and Ibusuki meetings, SC 4 requests its Secretariat to invite additional countries to join France, Japan, Sweden, UK and USA in the planning, machining and measurement demonstration to be given in Dallas, with a deadline for response of 2007-09-15.

RESOLUTION 723 (Ibusuki, Japan - July 2007)

Ad Hoc Group on Visualization

SC 4 recognizes the progress made by the ad hoc group on visualization and encourages completion of the requirements by the end of July. SC 4 requests its Secretariat to circulate the completed requirements statement for a two month ballot as soon as it is available, and to initiate in parallel a call for experts to participate in a new ad hoc group to undertake an independent evaluation of candidate 3D visualization formats against the agreed requirements. Oscar Rocha is appointed to chair the new group.

RESOLUTION 724 (Ibusuki, Japan - July 2007)

ACKNOWLEDGEMENTS and APPRECIATIONS

Acclamation:

Appreciation to our SC 4 Liaisons. We appreciate your representation to and for ISO/TC 184/SC 4.

buildingSMART
INCOSE

Thomas Liebich
Phil Spiby

We also offer thanks to those external projects who presented their material at the Open Technical Forum.

Light-SC 4 and Industry Data Readiness Paul van Exel

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Agile STEP Implementation David

Price Semantic Interoperability for ezSTEP

Soonhung Han

Simple Implementation of Part
21 in the Japanese Construction Industry Minoru Akiyama

Building and Interchanging Product Akira
Hosokawa Catalogue Information via
Spreadsheets and Web-Service

A Report on CEO Activity Anna Moreno

Acclamation

SC 4 expresses thanks to all the organizers, presenters and session chairs at the three Open Seminars held during the course of the week. The seminars provided technical insight through a combined total of 27 presentations across the three separate themes of "Data Quality", "Dictionary/Ontology Register & Maintenance" and "Planning, Machining & Measurement using STEP & STEP-NC". SC 4 appreciates the significant contribution that such events make to identifying a solid foundation on which SC 4 can develop standards embodying international consensus and industrial best practice.

Acclamation

SC 4 expresses its deep appreciation to the Japan Information Processing Development Corporation (JIPDEC) and the staff of the Hakusuikan Hotel for their efforts in organizing this meeting, arranging the facilities, excellent food, and social events. SC 4 also expresses its appreciation to Mr. Tadataka Simotakehara (Executive Vice President), Mr. Hiroo Nagata (General Manager of Hakusuikan), and Mr. Kichinori Takata (Manager in charge of the ISO conference), for their hospitality and outstanding support. Their quality of service has set a new standard of excellence for SC 4 meetings.

SC 4 addresses a special recognition to Mr. Hiroshi Murayama, Organizer, Mr. Masaru Suzuki, Chief of Staff and staff members Mr. Shinichi Nakagawa, Ms Chie Kouchi and,

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Mr. Akira Hosokawa for their superior administrative support, and outstanding hospitality both before and during the ISO/TC 184/SC 4 meeting.

RESOLUTION 725 (Irving (Dallas), TX USA - November 2007)

Appointment of Liaison officer between ISO/TC 118 “Compressors, pneumatic tools and pneumatic machines” and ISO/TC 184/SC 4 “Industrial data”

SC 4 requests the Secretariat to establish a liaison with ISO/TC 118. SC 4 invites ISO/TC 118 to nominate Dr. Peter Langer as liaison officer to ISO/TC 184/SC 4.

RESOLUTION 726 (Irving (Dallas), TX USA - November 2007)

Liaison to Industry Alliance on Interoperability (IAI)

SC 4 thanks Mr. Inhan Kim for his offer to act as the liaison officer to the International Alliance for Interoperability (IAI). SC 4 appoints Mr. Inhan Kim as the liaison officer to the IAI.

RESOLUTION 727 (Irving (Dallas), TX USA - November 2007)

Liaison to ISO/TC 8 “Ships and marine technology”

SC 4 thanks Mr. Soonhung Han for his offer to act as the liaison officer to ISO/TC 8 “Ships and marine technology”. SC 4 appoints Mr. Soonhung Han as the liaison officer to ISO/TC 8 and its SC 10.

RESOLUTION 728 (Irving (Dallas), TX USA - November 2007)

Creation of a “Big Picture” to show the relation of Standards to be used within and between Enterprises

SC 4 requests the PPC to continue and extend its existing work to generate a “Big Picture” illustrating the relationship and interaction of standards created by ISO/TC 184 in an operational environment, and seeks participation from other ISO/TC 184 Subcommittees in this work. The PPC may also invite additional SC 4 experts to participate in this work.

The target groups for the results will be:

- SC 4 experts - to reach a common understanding
- Standardizers of other committees - to show how ISO/TC 184 standards can be used within their standards activities
- Industry - to understand which standards might be used to solve their problems

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

in areas such as collaboration, interoperability, data exchange

SC 4 requests its Chair and Secretariat to encourage the other SCs of ISO/TC 184 to participate in this work.

SC 4 requests the group to take full account of the results of previous initiatives in this area.

A presentation of results is expected for the meeting of ISO/TC 184 in Oct. 2008.

Action item: A SharePoint service will be installed by the SC 4 Secretariat to facilitate cooperation on this topic.

RESOLUTION 729 (Irving (Dallas), TX USA - November 2007)

Common development of data standards for manufacturing applications

SC 4 expresses its extreme concern at the violation of the common ITU-R/ITU-T/ISO/IEC patent policy by the issuance of patents on the implementation of ISO 14649, and the resulting proposal for further similar patents being registered to protect implementation of SC 4 standards. All committees within ISO, IEC and ITU may be placed into a similar situation by granting of patents that restrict implementation of ISO standards. Therefore SC 4 requests ISO/CS to undertake the necessary actions to avoid this problem in the future, and invites its P-members to pursue the matter through their national bodies.

RESOLUTION 730 (Irving (Dallas), TX USA - November 2007)

Common development of data standards for manufacturing applications

SC 4 notes that its manufacturing team wishes to continue its successful collaboration with SC 1 on the further development of ISO 10303-238 and ISO 14649. SC 4 also notes that SC 1 is working to resolve the issue of the 14649 related patent by the end of 2007. SC 4 therefore requests its Secretariat to invite SC 1 to collaborate on the harmonization, subject to successful resolution of the current patent issue. In recognition of the substantial damage that the patent issue has caused, and to mitigate further damage, and in the event that the patent issue is not resolved, SC 4 requests WG 3/T24 to remove any elements of AP238 which are affected by the patent.

RESOLUTION 731 (Irving (Dallas), TX USA - November 2007)

PWI for AP 210 ed3

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 initiates a Preliminary Work Item for AP 210 ed3 for the purpose of minor extensions and maintenance

RESOLUTION 732 (Irving (Dallas), TX USA - November 2007)

Extending the scope of AP 203 ed2 to include technical drawings

SC 4 approves the extension of the scope of AP203 ed2 to include the two extra modules required for technical drawings. The Preliminary Work item for AP201 ed2 is cancelled.

RESOLUTION 733 (Irving (Dallas), TX USA - November 2007)

Change of Title, Clarification of Scope, and Designation of Series for ISO 8000, Catalogue Management Systems – Requirements

Resolution: SC 4 resolves to change the title of the ISO work item 42919 from Catalogue management systems – Requirements to Data quality – Master data. SC 4 clarifies that the scope of the work item is master data, notes the revised scope of ISO 8000-110 as defined in WG 13 N44, and requests WG 13 to develop a title for ISO 8000-110 that correctly reflects the restrictions in its scope. Upon approval of the new

NWI for ISO 8000, Data Quality, SC 4 assigns the work item 42919 to the ISO 8000-100 series of parts.

RESOLUTION 734 (Irving (Dallas), TX USA - November 2007)

PWI for “Business documents for product transaction and management”

SC 4 initiates a Preliminary Work Item for “Business documents for product transaction and management” as described in SC 4 N2253.

RESOLUTION 735 (Irving (Dallas), TX USA - November 2007)

ACKNOWLEDGEMENTS and APPRECIATIONS

Acclamation

Thank you to our SC 4 short report presenters at the opening plenary. We appreciate your representation to and for ISO/TC 184/SC 4.

ECCMA

Peter Benson

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

NATO AC/135
PDES, inc

Steve Arnett
Jack Harris

We also offer thanks to those external projects who presented their material at the Open Technical Forum.

SLSS	Anthony E. Trovato
Getting Quality Master Data from Suppliers	Peter Benson &
Gerry Radack	
"ALOES", a Logistic Support Analysis System	Lothar Klein
Army's PLCS implementation of FALCON	Raj Iyer
Linking Logistics to Design Data using PLCS	Nigel Shaw
Building support solutions using PLCS	Jochen
Haenisch Early implementations of PLCS	Tim King

Acclamation

Appreciation to our SC 4 Liaisons and those projects who presented at the Liaison Plenary.

Hiroshi Murayama –	"IEC Liaisons – ISO/TC 3, SC 3D, SC
65E" Gerry Radack –	"ISO/TC 37, ISO/IEC JTC 1/SC 32"
Dalip Sud –	"Future extension of ISO15926-4 dictionary content in cooperation with discipline ISO/TC's and
SC's" Reinhard Nerke –	"Current activities of eClass"
Francois Grobler –	"IS 16739 under ISO/TC 184/SC 4"
Anna Moreno –	"DEPUIS results on the eco-management
perspective	
in	relation with the activity of ISO/TC
207" Tom-Michael Thamm –	"Ecma /TC 43"

Acclamation

SC 4 expresses its deepest appreciation to SOLE – The International Society of Logistics (SOLE) for its hosting of the 53rd ISO/TC 184/SC 4 Plenary Meeting at the Omni Mandalay Hotel at Las Colinas, Irving, Texas. Ms. Sarah R. James, Executive Director and Past President; Mr. Jeffrey L. Schafer, President; and Mr. Anthony E.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Trovato, CPL, Past President and those logisticians jointly interacting with the SC 4 community are recognized for not only their initiative to bring a standards user perspective to the open technical session and those joint activities, but also in SOLE's engagement of members of the SC 4 community – particularly Peter Benson, Gerry Radack and Nigel Shaw – as instructors in SOLE's Logistics Standards professional development workshops.

Special acknowledgement is extended to the Omni Mandalay Hotel at Las Colinas and its staff: particular thanks go to Mr. Cameron McDonald, Director of Catering and Conference Planning; and Ms. Deborah Nsi, Set Up Supervisor for their seamless support. AV One, the meeting's audio-visual provider, is recognized for their outstanding support. A final thanks goes to the Irving Convention and Visitors' Bureau and their staff – especially Mrs. Mary Hammett – for providing registration support and gracious hospitality.

RESOLUTION 736 (Louisville, KY, USA - March 2008)

Reappointment of WG 11 Convener

SC 4 appreciates and accepts the offer of David Loffredo to serve as Convener of WG 11. SC 4 approves the nomination of David Loffredo to serve for an additional three year term.

RESOLUTION 737 (Louisville, KY USA - March 2008)

Schedule for AP 203

SC 4 notes that ISO/NP 10303-203 has passed the limit date for its DIS ballot. SC 4 agrees to the following dates:

2008-03-21: target date for start of CD-TS ballot of AP 203 modules

2008-03-31: target date for submission of AP 203 ed 2 to ISO for DIS ballot

2008-05-23: requested date for national bodies to submit copies of their ballot

comments on the AP 203 modules to the AP 203 project

leader 2008-06-06: closing date for CD-TS ballot of AP 203 modules

2008-06-08 to 13: AP 203 module ballot comment resolution at SC 4 Sun City meeting

2009-02-28: target date for stage 50 of AP 203 ed2

SC 4 agrees that minor EXPRESS errors in AP 203 modules can be addressed through ballot comments.



N4118

2026 – 05 – 15

SC 4 requests that technical comments on the AP 203 modules should be submitted through the AP 203 module CD-TS ballot and not through the AP 203 DIS ballot.

SC 4 requests its secretariat to update the ISO database with these target dates in accordance with TMB Resolution 56/2007.

RESOLUTION 738 (Louisville, KY USA - March 2008)

Liaison to MIMOSA

SC 4 decides to create a Category A liaison to the MIMOSA organization. SC 4 requests its secretariat to initiate the necessary actions. Subject to confirmation of the liaison by ISO, SC 4 welcomes the offer of Nils Sandsmark to act as the liaison officer from SC 4 to MIMOSA. SC 4 appoints Nils Sandsmark as liaison officer.

The secretariat is requested to invite MIMOSA to nominate a liaison officer from MIMOSA to SC 4

RESOLUTION 739 (Louisville, KY USA - March 2008)

Liaison to OASIS

SC 4 welcomes the offer of Howard Mason to act as liaison officer to OASIS. SC 4 appoints Howard Mason as liaison officer to OASIS.

RESOLUTION 740 (Louisville, KY USA - March 2008)

Creation of a Task Force to describe requirements for “Cross-AP-Implementations”

SC 4 decides to establish an ad hoc Task Force to investigate how several modular and non-modular APs can be used in a single implementation. Members of the Task Force will be: Lothar Klein (Chair), Gerry Radack (WG 3), Keith Hunten (WG 12), Burt Gischner (PPC), and AP developers. Gerry Radack is requested to ask for members by a circular letter within WG 3.

A report shall be made available by 2008-09-15. A verbal report is expected during the meeting in November 2008.

RESOLUTION 741 (Louisville, KY USA - March 2008)

Advancement of ISO 22745 Initial Set of Parts

SC 4 approves the submission of ISO/TS 22745 parts 1, 2, 11, 13 and 20 for ballot as International Standards. SC 4 directs the project team to provide the Secretariat

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

with drafts for DIS balloting as soon as possible, and directs the Secretariat to submit these parts to ISO for DIS ballot.

RESOLUTION 742 (Louisville, KY USA - March 2008)

URI defined by an SC 4 Standard

SC 4 requests its Secretariat to submit SC 4 N2265 "URI defined by an SC 4 standard" with changes agreed at the Louisville meeting for ballot as a standing document.

RESOLUTION 743 (Louisville, KY USA - March 2008)

Appointment of a new Liaison officer between ISO/TC 118 "Compressors, pneumatic tools and pneumatic machines" and ISO/TC 184/SC 4 "Industrial data"

Further to Resolution 725, SC 4 welcomes Gerd Koziel as liaison officer from ISO/TC 118 to ISO/TC 184/SC 4, and extends the liaison to include ISO/TC 118/SC 1.

RESOLUTION 744 (Louisville, KY USA - March 2008)

Liaison to ISO/TC 171 "Document management applications"

SC 4 thanks Dr King Yee for his offer to act as the liaison officer to ISO/TC 171 "Document management applications" and its SC 2 "Application issues". SC 4 appoints Dr Yee as the liaison officer to ISO/TC 171 and SC 2.

RESOLUTION 745 (Louisville, KY USA - March 2008)

Liaison to ISO/TC 29/WG 34 "Cutting tool data representation and exchange"

SC 4 welcomes the offer of Bengt Olsson to act as liaison officer to ISO/TC 29/WG 34. SC 4 appoints Bengt Olsson as liaison officer to ISO/TC 29/WG 34.

SC 4 records its sincere thanks to Norman Swindells for his efforts and achievements in acting as liaison officer.

RESOLUTION 746 (Louisville, KY USA - March 2008)

ACKNOWLEDGEMENTS and APPRECIATIONS

Acclamation

Thank you to our SC 4 short report presenters at the opening plenary. We appreciate

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

your representation to and for ISO/TC 184/SC 4.

OMG
PDES, Inc

David Price
Jack Harris

We also offer thanks to those external projects who presented their material at the Open Technical Forum.

“Results of the S-TEN project to map STEP data models onto RDF/OWL”	Lothar Klein	“Beyond data models”	David Leal
“Binary, OMG EXPRESS, Ruby, OWL/RDF, DEXLib templates”	David Price	“Toward Data Quality Guarantee Data Exchange and Share”	Akihiko Ohtaka
“History-based parametrics interface between CAD & CAE”	Ilhwan Song		

Acclamation

Appreciation to our SC 4 Liaisons and those projects who presented at the Liaison Plenary.

Steve Arnett –	NATO AC/135 – “Building Bridges”
Peter Benson –	“ISO 8000-110 Certification Program”
Nils Sandsmark –	“Implementation of ISO 15926”
David Price –	“OMG Activities”

Acclamation

SC 4 expresses its appreciation to ECCMA – The Electronic Commerce Code Management Association for its hosting of the 54th ISO/TC 184/SC 4 Plenary Meeting at the Marriott Downtown, Louisville, Kentucky. Mr. Peter Benson, Executive Director, Mr. Dan King, Chief Operating Officer; and Ms Melissa Scheib, Marketing and Events Manager for an excellent venue and outstanding service to the SC 4 community .

Special acknowledgement is extended to the Marriott Downtown, Louisville and its staff for their seamless support.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 747 (Sun City, South Africa - June 2008)

Appointment of WG 13 Deputy Conveners

SC 4 approves the appointment of Ms. Emily Grantner and Mr. Leif Gyllström as Deputy Conveners for WG 13 for a period of three years.

RESOLUTION 748 (Sun City, South Africa - June 2008)

INCOSE Liaison

SC 4 thanks Dr. Phil Spiby for his offer to act as the liaison officer from SC 4 to INCOSE. SC 4 appoints Dr. Phil Spiby as the liaison officer to the INCOSE.

SC 4 expresses its appreciation to David Oliver for his liaison role with INCOSE and accepts his resignation from that position.

RESOLUTION 749 (Sun City, South Africa - June 2008)

ISO Project 22274

SC 4 expresses the following concerns regarding NWI 22274 from ISO/TC 37/SC 3:

- This proposal appears to violate the agreements made at the ISO/TC 37 Beijing 2006 and Provo 2007 meetings.
- SC 4's request that ISO/TC 37/SC 3 delay circulating a NWI proposal for 22274 until a harmonised position between ISO/TC 184/SC 4 and ISO/TC 37 could be agreed and until a draft could be produced that could be circulated as a combined NWI/CD was not followed.
- This proposal seems to have expanded to cover product classification schemes, which are within scope of SC 4 and not of ISO/TC 37.

SC 4 directs the Chair, Secretariat, and liaison officers to ISO/TC 37/SC 3 to jointly develop a liaison statement expressing SC 4's concerns about this project, for submission to ISO/TC 37/SC 3. SC 4 requests the liaison officers to coordinate an SC 4 review of the NWI and prepare a set of ballot comments, noting the ballot closing date of 2008-07-31. SC 4 encourages its member countries to submit corresponding ballot comments through their national bodies.

RESOLUTION 750 (Sun City, South Africa - June 2008)

Termination of Liaison to: IEC SB3 and ISO/TC 10/SC 6

SC 4 requests its secretariat to terminate the liaison to IEC SB3 and ISO/TC 10/SC 6.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 751 (Sun City, South Africa - June 2008)

Proposed new ISO/TC 184/SC 4 Scope

SC 4 requests its secretariat by 2008-06-30 to provide ISO/TC 184 with the following proposed new scope for SC 4:

Standardization of the definition, integration and quality management of information which is shared, exchanged and archived in the context of

- *the definition, description and classification of products and facilities throughout their lifecycle,*
- *industrial management and operations,*
- *other information requirements supporting automation systems.*

RESOLUTION 752 (Sun City, South Africa - June 2008)

Alignment of OMG activity on data quality with the work of WG 13

SC 4 invites the Object Management Group (OMG) to take account of the scope and deliverables of the approved ISO 8000 work programme on data quality (ISO/TC 184/SC 4 N2271) in moving forward with activities around data quality. SC 4 appoints Peter Benson as its delegate to attend the MIT Information Quality Industry Symposium (2008-07) and subsequent OMG Proposed Data Quality Standard Preparatory Meeting. SC 4 requests Mr Benson to promote the full scope of the ISO 8000 work programme at these meetings, to identify individuals with whom to establish working relationships and to report his findings back to SC 4.

RESOLUTION 753 (Sun City, South Africa - June 2008)

ISO 15926-7

SC 4 notes the successful second CD-TS ballot on ISO 15926-7 and resolves that the current content may be published as the following ISO 15926 parts:

- Part 7: Templates and methodology
- Part 8: OWL representation
- Part 9: Interfaces through facades
- Part 10: Test methods

SC 4 requests its Secretariat to inform ISO of this subdivision.

RESOLUTION 754 (Sun City, South Africa - June 2008)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ACKNOWLEDGEMENTS and APPRECIATIONS

By Acclamation:

We offer thanks to those external projects who presented their material at the Open Technical Forum.

“Combining Gellish and RDF” – Leo van
Ruijven “RDF Pilot for Materials” – David
Leal

By Acclamation:

Appreciation to our SC 4 Liaisons and those projects who presented at the Liaison Plenary.

Thomas Liebich	– IAI
Anna Moreno	– ISO/TC 207 and VAMAS
Oscar Rocha	– SASIG
Gerry Radack	– ISO/TC 37
Gerry Radack	– ISO/IEC JTC 1/SC 32
Tom Thurman	– PDES, Inc

By Acclamation:

SC 4 notes with regret the resignation of Len Slovensky from the positions of WG 3 Deputy Convenor and AP229 project leader and editor, and wishes him well in his future endeavours.

By Acclamation:

SC 4 expresses its appreciation to SABS, PiLog, SASOL, Transnet and Traces of Africa for their hosting and support of the 55th ISO/TC 184/SC 4 Plenary Meeting at the Sun City Hotel, Sun City, South Africa. We especially commend Dr Salomon de Jager – PiLog; Dr Hanneke du Preez, Mrs Charlotte du Plessis, and Ms Barrie-Anne Basson – Traces of Africa for an excellent venue and outstanding service to the SC 4 community.

A special acknowledgement is extended to Ms Erna Stadlander – Sun City Conference Coordinator and her Support Team.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 755 (Busan, South Korea - November 2008)
Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303-241		Tatsuo Terai; Seong-Sig Kim
Module 1388-1395		Tatsuo Terai; Kyungik An
ISO 15926-7	Peter Paul Pruijn	Onno Paap, Johan Wilhelm Kluewer
ISO15926-8	Peter Paul Pruijn	Onno Paap, Martin Georg Skjæveland
ISO15926-9	Peter Paul Pruijn	Onno Paap, Julian Bourne
ISO15926-10	Peter Paul Pruijn	Onno Paap

RESOLUTION 756 (Busan, South Korea - November 2008)
ISO/TC 154 Liaison

SC 4 thanks Mr. Peter Benson and Dr. Ricardo Goncalves for their offers to act as liaisons from SC 4 to ISO/TC 154. SC 4 appoints Mr. Peter Benson, Dr Wolfgang Wilkes and Dr. Ricardo Goncalves as its liaison officers to ISO/TC 154.

RESOLUTION 757 (Busan, South Korea - November 2008)
Registration Number for Work Item on Gears

ISO/TC 184/SC 4 decides to register the project “Usage Guide for ISO 10303-214 for gear units” as a separate ISO project, to be published as a TS. The project team is requested to work with WG3, WG12, and the Quality Committee on the development of a common format for usage guidance. SC4 requests its Secretariat to circulate the amended document for a confirmatory ballot.

This resolution cancelled by Resolution 781.



N4118

2026 – 05 – 15

RESOLUTION 758 (Busan, South Korea - November 2008)

ISO/TS 10303-1000

SC 4 resolves to manage the ISO 10303 application modules in the 400 and 1000 series using the ISO Procedures for the Management of standards as databases.

SC 4 requests its Secretariat to submit the operating procedures described in SC 4 N2401 for a one-month committee internal Yes/No ballot. Any negative votes must be accompanied by comments.

SC 4 requests its Secretariat to:

- Forward to ISO for publication an initial version of ISO/TS 10303-1000 consisting of all ISO 10303 application modules that have already been published or are ready for publication, with references to resource parts and AICs.
- Invite ISO to withdraw ISO 10303 parts 401 to 4xx and 1001 through 1xxx, marking each as having been replaced by part 1000, once Part 1000 is published.
- Propose that ISO modify its catalogue search function so that a search for any ISO 10303 part in the range 401 to 499 or 1001 to 1999 will return a message referring the user to part 1000.
- Propose to ISO that the publication strategy for ISO 10303 should offer a range of products, including:
 - part 1000 as defined;
 - individual APs and either the 1000 part or just the referenced application modules;
 - CD with all ISO 10303 parts.

SC 4 decides that no new ISO 10303 application modules or new editions of existing ISO 10303 application modules will be submitted for publication as separate parts, once Part 1000 is published.

SC 4 adopts the plan of action in SC 4 N2402.

RESOLUTION 759 (Busan, South Korea - November 2008)

Cancellation of ISO 10303-229

Having received no offer of resources to complete remaining work on ISO 10303-229, SC 4 requests its Secretary to inform ISO CS of the cancellation of the project.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 760 (Busan, South Korea - November 2008)

Progress of Visualization ad hoc Group

SC 4 notes that the 3D Visualization ad hoc group has made significant progress at this meeting in moving forward the harvesting process for visualization formats. In acknowledgement of this effort SC 4 directs the 3D Visualization ad hoc group to complete the assessment of the candidates and provide the summary report no later than 2009-03-31. SC 4 directs the Chair and secretariat to initiate the PAS ballots for any candidates which pass the SC 4 Harvesting qualification process immediately after the delivery of the committee report.

RESOLUTION 761 (Busan, South Korea - November 2008)

Appointment of TC 184/SC 4 delegation to the ISO 22274 project

For the improvement of communication and active participation in the project ISO 22274, "Systems to Manage Terminology, Knowledge and Content – Internationalization and Concepts – Related Aspects of Classification Systems" SC 4 appoints the following experts as delegates to the ISO 22274 project: Gerry Radack, Wolfgang Wilkes, Junfeng Zhan, and Guy Pierra. [ISO/TC 37 has circulated ISO 22274 as an NWIP within ISO TC37/SC3 for ballot.]

RESOLUTION 762 (Busan, South Korea - November 2008)

Extension of DIS Date for Parts 10303-52, -53, and -110

SC 4 requests its Secretariat to inform ISO that the target dates for the submission of 10303 parts -52, -53 and -110 for DIS ballot are extended to 2009-01-30, and that these parts should not be automatically cancelled as a recovery plan is in place.

RESOLUTION 763 (Busan, South Korea - November 2008)

Minor Revision of Part 10303-47

SC 4 directs WG12 to prepare FDIS documents for ISO 10303-47, 503, 504, 506, 517, 518, 519, 521, and 523 and submit them to ISO for processing as minor revisions.

This amendment has been superseded by Resolution 779.

RESOLUTION 764 (Busan, South Korea - November 2008)

PWI for Gellish-RDF

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 creates an Ad Hoc Group for Gellish-RDF with project leader Paul van Exel and editor Leo van Ruijven. The group is requested to clarify the position of the proposed standard with respect to SC 4 and W3C standards, and seek alignment where appropriate. SC 4 directs the group to develop any necessary NWI proposals for review by the Change Management Team at the next SC 4 meeting, in May 2009.

RESOLUTION 765 (Busan, South Korea - November 2008)

ISO 8000 Scope

SC 4 directs WG 13 to deliver a brief (1-3 page) draft scope statement and phased delivery plan for ISO 8000 as a whole, and a definition of other standardization requirements both inside and outside SC 4 for consideration by SC 4 at its next meeting (2009-05).

SC 4 requests its Secretariat to circulate the original content of this resolution, resolution #676 and #733, and SC4 N 2271, WG 13 N 0107, and SC4 N 2292 to member bodies with a request for comments by year end in order to provide further guidance to WG 13.

RESOLUTION 766 (Busan, South Korea - November 2008)

Maintenance of reference data using “Standards as databases” procedure

SC 4 requests its Secretariat to submit the operating procedures described in SC 4 N2407 for a two-month committee internal ballot. Any negative votes must be accompanied by comments.

SC 4 requests the ad hoc group to complete the detailed operational processes.

RESOLUTION 767 (Busan, South Korea - November 2008)

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

We offer thanks to those external projects who presented their material at the Open Technical Forum.

Jager

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

“Case Study of ISO 8000 Implementation”

Dr. Salomon de

“Data Consolidation Using PLCs”

Dr. Jochen Haenisch

“Product Information Explorer”

Dr. Timothy King “Compressor

Product Knowledge Model Business Case”

Leo van Ruijven

ACCLAMATION

Appreciation to our SC 4 Liaisons and those projects who presented at the Liaison Plenary.

SASIG

Harmonization of Part 59

ISO TC 211

DEPUIS

ISO/IEC JTC1/SC 32

NAFEMS

OMG

PDES, Inc

buildingSMART

ASD

Takamasa Tanaka

Takamasa Tanaka/Akihiko Ohtaka

Jerry Smith

Dr. David Leal

Dr. Gerry Radack

Dr. David Leal

David Price

David Price

Dr. Thomas Liebich

Howard Mason

Thank you to our SC 4 short report presenters at the Opening Plenary.

ISO TC 37

Dr. Gerry Radack

We appreciate your representation to and for ISO TC 184/SC 4.

ACCLAMATION

SC 4 expresses its appreciation to KATS – Korean Agency for Technology and Standards, Korea STEP Center, ECIF, Korea Database Promotion Center, PartDB, Busan Convention & Visitors Bureau, Korea Tourism Organization, and The Plan for their hosting and support of the 56th ISO TC 184/SC 4 Plenary meetings at the Paradise Hotel, Busan, Korea. We especially commend Prof. Inhan Kim – Kyung Hee University

and Mr. Kyungik An – Korea STEP Center for an excellent venue and outstanding service to the SC 4 community. A special acknowledgement is extended to Mr. Sejin Kim and Mr. Yongdeuk Kim, KATS.

Dana Tripp

Secretary, ISO/TC 184/SC 4

trippdsc4@gmail.com



N4118

2026 – 05 – 15

Special acknowledgement is extended to the Paradise Hotel, Busan and its staff for their seamless and attentive support.

RESOLUTION 768 (Parksville, British Columbia, Canada - May 2009)
ISO/PC 246 Liaison

Establishment of an Internal Liaison with ISO/PC 246 Anti-counterfeiting tools

SC 4 accepts the request from ISO/PC 246 Anti-counterfeiting tools to establish an internal liaison and appoints Robert Kidwell as the liaison officer from SC 4 to ISO/PC 246.

SC 4 welcomes Rainer Glatz as liaison officer from ISO/PC 246 to SC 4.

RESOLUTION 769 (Parksville, British Columbia, Canada - May 2009)
Re-Appointment of Convenor of Working Group 12

SC 4 approves Keith Hunten to serve as Convenor of its Working Group 12 for an additional period of three years.

SC 4 expresses its special recognition and thanks for his exceptional dedication and service for the past three years in the position of Convenor.

RESOLUTION 770 (Parksville, British Columbia, Canada - May 2009)
Re-Appointment of Convenor of Working Group 13

SC 4 approves Tim King to serve as Convenor of its Working Group 13 for an additional period of three years.

SC 4 expresses its special recognition and thanks for his exceptional dedication and service for the past three years in the position of Convenor.

RESOLUTION 771 (Parksville, British Columbia, Canada - May 2009)
Recognition of Dr. King Yee

SC 4 notes with regret the retirement of Dr. King Yee from the significant positions he has held over many years within SC 4, offers its thanks for his outstanding support and wishes him health and happiness in his well-earned retirement.



N4118

2026 – 05 – 15

RESOLUTION 772 (Parksville, British Columbia, Canada - May 2009)

Convener of JWG 8

SC 4 thanks Jean-Jacques Michel for his outstanding effort as convener of JWG 8 since 1996. SC 4 appoints Anne-Françoise Cutting-Decelle as convener of JWG 8 for a three-year term.

RESOLUTION 773 (Parksville, British Columbia, Canada - May 2009)

Duration of CD ballot

SC 4 resolves that any CD ballots that are not combined with NP ballots shall be 2 months, unless an extension to 3 months is requested prior to the ballot by any P- member, or at the discretion of the SC 4 Chairman.

RESOLUTION 774 (Parksville, British Columbia, Canada - May 2009)

Report of 3D Visualization ad hoc Group

SC 4 accepts the report SC4 N 2465 of the 3D visualization ad hoc group, SC 4 thanks the group for its effort, and disbands the group.

RESOLUTION 775 (Parksville, British Columbia, Canada - May 2009)

Initiation of Harvesting for 3D Visualization Specifications

SC 4 requests its Secretariat to request the bodies responsible for each of the candidate 3D visualization formats identified in SC4 N2465 that have successfully passed the assessment process and are not already subject to standardization elsewhere, to provide the complete specification, the harvesting proposal form and any necessary supporting material to the Secretariat.

SC 4 requests its Secretariat to initiate the ballots on receipt of the necessary material, and to include a copy of SC4 N2465 in the ballot package for information.

RESOLUTION 776 (Parksville, British Columbia, Canada - May 2009)

HTML Version of ISO/TS 10303-1005

SC 4 approves a minor revision of ISO/TS 10303-1005, comprising reformatting in HTML conforming to current guidelines, with the current EXPRESS. SC 4 requests the Secretariat to notify ISO CS that the revised document should be published within the first edition of ISO/TS 10303-1000.



N4118

2026 – 05 – 15

RESOLUTION 777 (Parksville, British Columbia, Canada - May 2009)

Extension of initiation of CD ballot of ISO 10303-241

SC 4 requests its Secretariat to inform ISO CS that SC 4 has decided to extend the initiation of the CD ballot for ISO 10303-241 until end of 2009. SC4 will review this decision during its next meeting in November.

RESOLUTION 778 (Parksville, British Columbia, Canada - May 2009)

Extension of DIS for ISO 10303-61

SC 4 requests its Secretariat to inform ISO/CS that the target date for the submission of 10303-61 for DIS ballot is extended to 2009-07-23, and that this part should not be automatically cancelled as a recovery plan is in place.

RESOLUTION 779 (Parksville, British Columbia, Canada - May 2009)

Minor Revision of Part 10303-47 (Amendment to Resolution 763)

SC 4 directs WG12 to prepare FDIS documents for ISO 10303-47, 503, 504, 506, 517, 518, 519, 520, 521, and 523 and submit them to ISO for processing as minor revisions. **Resolution 763 is superseded.**

RESOLUTION 780 (Parksville, British Columbia, Canada - May 2009)

Advancement of ISO 22745/29002 Parts

SC 4 approves the submission of ISO/TS 22745 parts 10, 14, 30, 35 and 40 and ISO/TS 29002 parts 4, 5, 6, 10, 20 and 31 for ballot as International Standards. SC 4 directs the project teams to provide the Secretariat with drafts for DIS balloting as soon as possible, and directs the Secretariat to submit these parts to ISO/CS for DIS ballot.

RESOLUTION 781 (Parksville, British Columbia, Canada - May 2009)

Registration Number for Work Item on Gears

ISO/TC 184/SC 4 decides to publish the "Usage Guide for ISO 10303-214 for gear units" as ISO 10303-5001. The format of the document shall be edited to align with an AP. New conformance classes shall not be introduced.

SC4 requests its Secretariat to circulate the amended document for a confirmatory ballot.

All members of ISO TC184/SC4 are encouraged to promote the development of usage guides, such as for the 5000-series of ISO 10303.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

This resolution cancels resolution 757 from Busan, South Korea - November 2008.

RESOLUTION 782 (Parksville, British Columbia, Canada - May 2009)

Change Title of Standard ISO/DIS 13584 Part 32

SC 4 approves the new title “Automation systems and integration — Parts library — Part 32: Implementation resources: OntoML: Product Ontology Markup Language”, advises SC4 secretariat to perform this change and requests WG2 to prepare the document for IS publication.

RESOLUTION 783 (Parksville, British Columbia, Canada - May 2009)

Change of ISO/TC 154 Liaison

SC 4 accepts the request from WG2 to replace Dr. Wolfgang Wilkes by Mr. Hiroshi Murayama as liaison officer to ISO/TC 154.

SC 4 appreciates the dedication of Dr. Wilkes as liaison officer to ISO/TC 154.

RESOLUTION 784 (Parksville, British Columbia, Canada - May 2009)

Creation of the Advisory Group “SC 4 Implementation Forum”

SC4 decides to create the Advisory Group “SC 4 Implementation Forum”, which will also incorporate the tasks and members of the existing Committee for Education and Outreach, which is now disbanded. The group has the following tasks:

- Gather feed back from implementers
- Gather success stories
- Promote business benefits
- Promote awareness of capabilities
- Facilitate implementations
 - Test model/Testing procedure
- Seek resources for sustainment
- Provide education and outreach

SC 4 requests its secretariat to invite participation from its member bodies and liaisons to undertake this work. Member bodies are invited to ensure the participation of STEP Centres and other supporting organizations

Paul van Exel is nominated as leader of this group

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Anna Moreno is nominated as Co-deputy leader of this group
Burton Gischner is nominated as Co-deputy leader of this group

The Advisory Group is invited to present its work program to the November 2009 SC 4 Plenary meeting in November.

RESOLUTION 785 (Parksville, British Columbia, Canada - May 2009)

Creation of a PWI on “Future SC 4 Architecture”

SC 4 approves a PWI on “Future SC 4 Architecture” to:

- Explore the range of options for exploiting new technologies to support interoperability between SC4 standards, and the work of other committees, operating across a range of platforms
- Recommend an overall architecture for future use by SC 4 standards
- Define migration paths for existing SC 4 capabilities
- Identify specific licensing, commercial or other administrative barriers
- Illustrate and validate any conclusions against a representative example.

SC 4 invites all of its member bodies and WGs to contribute to the work of the PWI.
SC 4 appoints David Leal and David Price as joint leaders for the PWI and invites them to report back no later than the November 2009 meeting.

RESOLUTION 786 (Parksville, British Columbia, Canada - May 2009)

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION:

We offer thanks to Dr. Gerry Radack, Leo van Ruijven, Rob Bodington, Howard Mason, and David Price who presented their material at the Open Technical Forum.

ACCLAMATION:

Appreciation to our SC 4 Liaisons and those projects who presented at the Liaison Plenary.

- PDES, Inc – George Sebes
- NATO AC/135 – Steve Arnett
- ISO/TC 154 – Ricardo Goncalves
- FSIG – Ricardo Goncalves
- VAMAS – Anna Moreno

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- OASIS PLCS – Rob Bodington
- BuildingSMART – Thomas Liebich
- European project ORCHID WS – Paul van Exel
- Work of the NAFEMS Simulation Data Management WG – Randy Cigel
- IEC liaisons – Hiroshi Murayama
- OMG – David Price
- ECCMA – Peter Benson

We appreciate your representation to and for ISO TC 184/SC 4.

ACCLAMATION:

SC 4 expresses its appreciation to ECCMA – Electronic Commerce Code Management Association for hosting and support of the 57th ISO TC 184/SC 4 Plenary meetings at the Tigh-Na-Mara Conference Centre in Parksville, British Columbia, Canada. We especially commend Mr. Peter Benson, Executive Director, and Ms Melissa Scheib, Events Manager, and Ms Brittany Marmo, Administrative Assistant of ECCMA for an excellent venue and outstanding service to the SC 4 community.

Special acknowledgement is extended to the Tigh-Na-Mara Hotel and Conference Centre, Parksville and its staff for their excellent cooperation, service and food presentation, which was instrumental in ensuring the effectiveness of the meeting.

RESOLUTION 787 (Rotterdam, The Netherlands - November 2009)

Reappointment of SC 4 Chair

SC4 endorses the nomination by the United States of Howard Mason for an additional three year term as the Chair of ISO TC 184 /SC 4 and instructs the Secretary to forward his nomination to ISO TC 184 within 30 days.

RESOLUTION 788 (Rotterdam, The Netherlands - November 2009)

Re-Appointment of Deputy Convenor of Working Group 2

SC 4 approves Guy Pierra to serve as Deputy Convenor of its Working Group 2 for an additional period of three years.

SC 4 expresses its special recognition and thanks for his exceptional dedication and service for the past three years in the position of Deputy Convenor.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 789 (Rotterdam, The Netherlands - November 2009)

Appointment of SC 4 Liaison To ISO TC 211

SC 4 thanks Mr. Jerry Smith for his work as liaison from SC 4 to ISO/TC 211 Geographic information/Geomatics. SC 4 thanks Mr. Jean Brange's offer to act as liaison from SC 4 to ISO/TC 211 and appoints him to serve in that capacity.

RESOLUTION 790 (Rotterdam, The Netherlands - November 2009)

Establishment of an informal liaison with CEN/ISSS Workshop ELSSI-EMD

SC4 accepts the request from CEN/ISSS Workshop ELSSI-EMD to establish an informal liaison and appoints Anna Moreno as the liaison officer from SC4 to CEN/ISSS Workshop ELSSI-EMD.

SC4 welcomes David Leal as liaison officer from CEN/ISSS Workshop ELSSI-EMD to SC4.

RESOLUTION 791 (Rotterdam, The Netherlands - November 2009)

SC4 Organization Handbook Update: Clause 6.4, Education and Outreach Rapporteur

Change SC4 Organization Handbook Clause 6.4 to read:

6.4 SC4 Implementation Forum advisory group

6.4.1 Tasks

The SC 4 Implementation Forum advisory group performs the following tasks and activities:

- Gathering feed back from implementers;
- Gathering success stories;
- Promoting business benefits;
- Promoting awareness of capabilities;
- Facilitating implementations;
 - Testing models and Testing procedures;
- Seeking resources for sustainment;
- Providing education and outreach.



N4118

2026 – 05 – 15

It incorporates the activities of the former Committee for Education and Outreach, which has been disbanded.

6.4.2 Membership

The P-members, O-Members and A-liaisons of SC 4 nominate the members of the SC 4 Implementation Forum advisory group. The SC 4 secretariat maintains a list of members of the SC 4 Implementation Forum (names and addresses, organization affiliation, fax, phone and e-mail).

RESOLUTION 792 (Rotterdam, The Netherlands - November 2009)

SC 4 Organization Handbook Update: New SEDS Process. Multiple Clauses

SC 4 requests its Secretariat to update the SC 4 Organization Handbook (SC4 N2411) to reflect the changes specified in document SC4 N2520, Changes to SC 4 Handbook for New SEDS Process (2009-11-12) (attached).

Attached documents: SC4 N 2520 “Changes to SC 4 Handbook for New SEDS Process” by Dr. Gerry Radack.

RESOLUTION 793 (Rotterdam, The Netherlands - November 2009)

ISO 8000-2

SC 4 decides to create ISO 8000-2 as a vocabulary part for the entire ISO 8000 series, through the merger of vocabulary from parts 1 “Overview” and 102 “Master data: Exchange of characteristic data: Vocabulary”.

SC 4 requests WG 13 to provide the Secretariat with a draft for FDIS balloting, and directs the Secretariat to submit this part to ISO/CS for FDIS ballot as soon as the document is available.

SC 4 decides that the normative version of all vocabulary used by the ISO 8000 series shall be in part 2, and replicated where the terms are used in individual parts.

RESOLUTION 794 (Rotterdam, The Netherlands - November 2009)

Advancement of ISO 8000 Parts

SC 4 approves the submission of ISO/TS 8000 parts 100 “Overview”, 120

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

“Provenance”, 130 “Accuracy”, and 140 “Completeness” for ballot as International Standards.

SC 4 directs the project teams to provide the Secretariat with drafts for DIS balloting as soon as possible, and directs the Secretariat to submit these parts to ISO/CS for DIS ballot.

RESOLUTION 795 (Rotterdam, The Netherlands - November 2009)
ISO 29002 Vocabulary Part

SC 4 decides to create 29002-2 as a vocabulary part for the entire 29002 series through merger of vocabulary from the TS versions of ISO 29002 parts 4, 5, 6, 10, 20, and 31.

SC 4 directs the project team to provide the Secretariat with a draft for DIS balloting as soon as possible, and directs the Secretariat to submit this part to ISO/CS for DIS ballot.

SC 4 decides that the normative version of all vocabulary used by the ISO 29002 series shall be in part 2, and replicated where the terms are used in individual parts.

RESOLUTION 796 (Rotterdam, The Netherlands - November 2009)
Extension of target date of ISO10303-26

SC 4 requests its Secretariat to inform ISO/CS that the target date for the submission of ISO TS 10303-26 “Implementation methods: Binary representation of EXPRESS-driven data” for publication is extended to 2010-06-30, and that this part should not be automatically cancelled as a recovery plan is in place.

SC 4 will review progress on this part during its next meeting in May 2010.

RESOLUTION 797 (Rotterdam, The Netherlands - November 2009)
ISO/NP 20140 “Environmental and energy efficiency evaluation method for manufacturing systems” within ISO/TC 184/SC 5

SC 4 notes the ISO/NP 20140 “Environmental and energy efficiency evaluation method for manufacturing systems” within ISO/TC 184/SC 5. Based on its work items, ISO/TC 184/SC 4-SC 5/JWG 8 is interested to participate in that project.

Therefore SC 4 decides to offer its participation to SC 5 via JWG 8 after the NP is accepted.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 798 (Rotterdam, The Netherlands - November 2009)

Extension of ISO 10303-1000 to cover integrated resources

SC 4 approves the inclusion of all ISO 10303 integrated resources, including application interpreted constructs, and of ISO 13584-20 “*Logical model of expressions*” into the STEP Modules Library.

SC 4 instructs the SC 4 secretariat to update the SC4 standing document “*Procedures for the maintenance of ISO/TS 10303-1000*” correspondingly and to add that changes to ISO 10303 integrated resources and ISO 13584-20 shall pass a WG 12 integration review before a STEP Module Library change request process is initiated.

RESOLUTION 799 (Rotterdam, The Netherlands - November 2009)

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

We offer thanks to those external projects who presented their material at the Open Technical Forum.

“Data requirements from the Korean nuclear and shipbuilding industries” – Soonhung Han

ACCLAMATION

Appreciation to our SC 4 Liaisons and those projects who presented at the Liaison Plenary.

“Mou/MG”	Howard Mason
“ISO TC 37”	Gerry Radack and Christian Galinski
“ISO/IEC JTC 1/SC 32”	Gerry Radack
“ISO TC 213”	Vijay Srinivasan
“CEN WS/ELSSI-EMD”	David Leal
“FIATECH activities on ISO 15926”	Robin
Benjamins “OMG Activities Report”	David Price

We appreciate your representation to and for ISO TC 184/SC 4.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ACCLAMATION

SC 4 expresses its appreciation to the Implementation Forum organizers, Paul van Exel and Anna Moreno, for their efforts in organizing the excellent event, speakers and venue.

ACCLAMATION

SC 4 expresses its appreciation to Paul van Exel, Stichting USPI-NL, and his team for their hosting and support of the 58th ISO TC 184/SC 4 Plenary meetings at the Erasmus University Conference Center in Rotterdam, Netherlands. We especially commend Mr Paul van Exel for an excellent venue and outstanding service to the SC 4 community. A special acknowledgement is extended to Ms. Suzanne Eijkemans, logistics coordination, Elsevier Congressen and to Antoinette Schipper, Elsevier Congressen, hostess for the week.

Special acknowledgement is extended to the Erasmus University Conference Center and its staff for their seamless and attentive support.

RESOLUTION 800: (Approved by electronic ballot May 2010)

Managing the ISO 15926 Reference Data Library as a database

SC 4 resolves to manage the ISO 15926 Reference Data Library (RDL) using the ISO Annex ST procedures for the management of standards as databases, as customized in the SC 4 standing document "Procedure for the maintenance of ISO/TC 184/SC 4 reference data".

SC 4 requests the WG 3/T 25 ISO 15926 part 3 and 4 projects to nominate one member each to the maintenance team. SC 4 requests its Secretariat to invite P-members, Category A liaisons, internal liaisons, and the WG 3 convener to nominate additional experts, as needed under the agreed procedure.

SC 4 requests its Secretariat to invite all P-members, Category A liaisons and internal liaisons to appoint one member each to the validation team.

SC 4 requests its Secretariat to:

- Forward to ISO for publication an initial version of the RDL consisting of all ISO 15926 reference data that have already been published formally or have been successfully balloted as ready for publication (ISO/TS 15926-3, -4 and am1).

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Request that ISO publish the RDL on standards.iso.org under the same access conditions as the existing electronic inserts on the SC 4 site.
- Ask ISO to add references in the ISO Catalogue entries for ISO 15926 parts 3 and 4 referring to the RDL.

SC 4 approves a minor revision to create new editions of ISO 15926 parts 3 and 4 that point to the reference data online, rather than including the reference data as electronic attachments.

RESOLUTION 801 (Bethlehem, Pennsylvania USA – June 2010)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 approves the following changes to project leaders and project editors for the associated SC 4 projects:

Project	Leader	Editor
ISO 10303-0026	Keith Hunten	Marco Delgado
ISO 10303-0203e2	Simon Frechette	
ISO 13584-0035		N. Ondracek removed
ISO 15926-0007		Johan Wilhelm Klüwer Onno Paap Hans Teijgeler
ISO 15926-0008		Onno Paap Hans Teijgeler
ISO 15926-0009		Rahul Patil Onno Paap Hans Teijgeler
ISO 15926-0010		Rahul Patil Onno Paap Hans Teijgeler

RESOLUTION 802 (Bethlehem, Pennsylvania USA – June 2010)

Reappointment of SC 4 Officers

SC 4 expresses its special recognition and thanks for the exceptional dedication and service for the past three years of these Officers listed below and re-appoints these

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

Officers to serve for an additional three year term:

Policy and Planning Committee (PPC) Chair	Torbjorn Holm
Quality Committee Chair	Peter Benson
Quality Committee Deputy Chair	Dr. Junfeng Zhan
WG03 Convener	Dr. Gerald Radack
WG02 Convener	Nikolaus Ondracek
WG12 Deputy Convener	Jochen Haenisch

RESOLUTION 803 (Bethlehem, Pennsylvania USA – June 2010)

International Association for Information and Data Quality (IAIDQ) Liaison

SC 4 welcomes Christian Walenta as liaison from the International Association for Information and Data Quality (IAIDQ) for SC 4. SC 4 nominates Peter Benson as its liaison to IAIDQ.

RESOLUTION 804 (Bethlehem, Pennsylvania USA – June 2010)

Withdrawal of ISO 8000-102

SC 4 directs the Secretariat to withdraw ISO 8000-102 “Master data: Exchange of Characteristic Data: Vocabulary”, upon publication of ISO 8000-2 “Data Quality - Vocabulary”.

RESOLUTION 805 (Bethlehem, Pennsylvania USA – June 2010)

Advancement of ISO 8000-1 & the Taxonomy of Industrial Data

SC 4 instructs its Secretariat to submit the revised version of ISO 8000-1 “Data Quality - Overview, principles and general requirements”, excluding the taxonomy, to ISO/CS for TS publication by 2010-09-15.

SC 4 approves the submission of ISO/TS 8000-1 for ballot as an International Standard. SC 4 directs the project team to provide the Secretariat with a draft for DIS balloting as soon as possible, and directs the Secretariat to submit this part to ISO/CS for DIS ballot.

SC 4 directs WG 13 to develop a taxonomy of industrial data to include contents that correspond to the scope of SC 4 and resolves that this taxonomy forms the contents of ISO 8000-3, with the title "Data quality – Taxonomy of industrial data" and to

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

include a reference to the new Part 3 in the published TS for Part 1.

SC 4 requests its Secretariat to inform all other Sub-Committees of TC 184 that WG 13 is developing taxonomy of industrial data, inviting those Sub-Committees to nominate experts who are willing to contribute to the development and review of the taxonomy.

RESOLUTION 806 (Bethlehem, Pennsylvania USA – June 2010)

Managing the ISO 8000 vocabulary as a database

SC 4 resolves to manage the ISO 8000 vocabulary using the ISO Annex ST procedures for the management of standards as databases, as customized in the SC 4 standing document “Procedure for the maintenance of ISO/TC 184/SC 4 vocabulary”, once ISO 8000-2 is approved for publication.

SC 4 requests the ISO 8000-2 “Data quality – Vocabulary” project to nominate a member to the maintenance team. SC 4 requests its Secretariat to invite P-members, Category A liaisons, internal liaisons, and the WG 13 convener to nominate additional experts, as needed under the agreed procedure.

SC 4 requests its Secretariat to invite all P-members, Category A liaisons and internal liaisons to appoint one member each to the validation team.

SC 4 requests its Secretariat to establish a Configuration Management Repository for the ISO 8000 vocabulary.

RESOLUTION 807 (Bethlehem, Pennsylvania USA – June 2010)

Ad Hoc Group on Mechatronics

SC 4 appoints Jean Brangé, Mike Zink, Tom Thurman, Peter Benson, and Gerry Radack as delegates to the TC 184 ad hoc group on mechatronics to ensure that effective use is made of existing SC 4 standards, and to assist in the identification of any specific new requirements.

SC 4 instructs its Secretariat to seek additional nominees from other P-members.

RESOLUTION 808 (Bethlehem, Pennsylvania USA – June 2010)

SC4 Nomination to Energy Efficiency Task Force

ISO/TC 184/SC 4 requests its Secretariat to nominate the convener of its JWG 8,

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Anne- Francoise Cutting-Decelle as the point of contact for the proposed ISO Strategic Advisory Group on Energy (SAG-E) Workshop and also to SC5 /WG10.

RESOLUTION 809 (Bethlehem, Pennsylvania USA – June 2010)

Appointment of delegates from SC 4 to Oil and Gas Asset Management PWI

SC 4 notes the launch of the TC 184 PWI on “Oil and gas asset management and operations and maintenance operations”, and requests its Secretariat to circulate an immediate call to member bodies for participation in the PWI.

RESOLUTION 810 (Bethlehem, Pennsylvania USA – June 2010)

Addition of modular APs to STEP Module and Resource Library (SMRL)

SC 4 approves the management of all ISO 10303 modular APs under the Annex ST procedure as part of the STEP Module and Resource Library. Snapshot builds of the APs shall be held in a separate directory. In order to ensure consistency, SC 4 approves the transfer of 400-series AP modules from the main Library directory to the appropriate snapshot directories.

SC 4 instructs the SC 4 secretariat to update the SC 4 standing document “*Procedures for the Maintenance of the STEP Module and Resource Library*”.

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

We offer thanks to those external projects who presented their material at the Open Technical Forum.

- “*NWI for PDQ-S extension and quality guarantee process by the use of PDQ-S*” – Akihiko Ohtaka
- “*A Validation Technology for 3D CAD Data*” – Atsuto Soma
- “*Master Data Ontology Manager*” - Peter Benson
- “*Data quality in UK Defence*” – Tim King

ACCLAMATION

Appreciation to our SC 4 Liaisons and those projects who presented at the Liaison Plenary

- “*Liaison Issues*” – Howard Mason
- “*ECCMA Update*” – Peter Benson

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- “CEN ELSSI-EMD” Workshop – David Leal
- “buildingSMART” - Thomas Liebich
- “VAMAS” - Anna Moreno

We appreciate your representation to and for ISO TC 184/SC 4.

ACCLAMATION

SC 4 recognizes and thanks Ron Kolakowski for his years of hard work and the excellence of his support to the community while serving as an administrator in the SC4 Secretariat. We wish him well in his new career opportunities.

ACCLAMATION

SC 4 expresses its appreciation to Mr. Peter Benson ECCMA, and his team for their hosting and support of the 59th ISO TC 184/SC 4 Plenary meetings at the Hotel Bethlehem in Bethlehem, Pennsylvania. We especially commend Mr. Peter Benson, Executive Director, and Ms. Melissa Scheib, Events Manager, of ECCMA for an excellent venue and outstanding service to the SC 4 community. A special acknowledgement is extended to Ms. Melissa Scheib, and to Ms. Krysten Wolf, outstanding hostesses for the week.

Special acknowledgement is extended to the Hotel Bethlehem and its staff for their seamless and attentive support, excellent cooperation, service and food presentation, all of which was instrumental in ensuring the effectiveness of the meeting.

RESOLUTION 812: (Approved by electronic ballot August 2010)

Managing the ISO 22745 vocabulary as a database

SC 4 resolves to manage the ISO 22745 vocabulary using the ISO Annex ST procedures for the management of standards as databases, as customized in the SC 4 standing document “Procedure for the maintenance of ISO/TC 184/SC 4 vocabulary”.

SC 4 requests the ISO 22745-2 project to nominate a member to the maintenance team. SC 4 requests its Secretariat to invite P-members, Category A liaisons, internal liaisons, and the WG 13 convener to nominate additional experts, as needed under the agreed procedure.

SC 4 requests its Secretariat to invite all P-members, Category A liaisons and internal

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

liaisons to appoint one member each to the validation team.

SC 4 requests its Secretariat to establish a Configuration Management Repository for the ISO 22745 vocabulary.

RESOLUTION 813: (Vico Equense, Italy – October 2010)

Managing the ISO 29002 vocabulary as a database

SC 4 resolves to manage the ISO 29002 vocabulary using the ISO Annex ST procedures for the management of standards as databases, as customized in the SC 4 standing document “Procedure for the maintenance of ISO/TC 184/SC 4 vocabulary”.

SC 4 requests the ISO 29002 project to nominate a member to the maintenance team. SC 4 requests its Secretariat to invite P-members, Category A liaisons, internal liaisons, and the WG 13 convener to nominate additional experts, as needed under the agreed procedure.

SC 4 requests its Secretariat to invite all P-members, Category A liaisons and internal liaisons to appoint one member each to the validation team.

SC 4 requests its Secretariat to establish a Configuration Management Repository for the ISO 29002 vocabulary.

RESOLUTION 814: (Vico Equense, Italy – October 2010)

Recycling of ISO 15926 Part 9 & 10

SC 4 requests its Secretariat to request ISO CS to return 15926 Part 9 & 10 to Stage 10.99.

RESOLUTION 815: (Vico Equense, Italy – October 2010)

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

We offer thanks to those external projects who presented their material at the Open Technical Forum.

- “Analysing the Supply Chain Using the Semantic Web” by David Leal
- “LCIA PLCS Translator Pilot” by Dr Timothy M. King

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- “*Enabling the Usage of Standards Within the Supply Chain*” by Xenia Fiorentini

ACCLAMATION

Appreciation to our SC 4 Liaisons and those projects who presented at the Liaison Plenary

“*Liaison issues*” by Howard Mason
“*Necessity of Open Standards*” by Prof Alfred Katzenbach
“*ISO/PAS 16739/IFC*” by Thomas Liebich
“*CEN ORCHID WS*” by Paul van Exel
“*Dutch Ship Building Project - Integral Cooperation*” by Leo van Ruijven
“*TC 29/WG 34 - Implementation of a Tool Library Services*” by Bengt Olsson
“*ISO TC 184/SC 5/WG 10*” by Jean Brange

We appreciate your representation to and for ISO TC 184/SC 4.

ACCLAMATION

SC 4 expresses its appreciation to the Implementation Forum organizers, Paul van Exel and Anna Moreno, for their efforts in organizing the excellent event, speakers and venue.

ACCLAMATION

SC 4 expresses its appreciation to Anna Moreno, her family, and her team for their hosting and support of the 60th ISO TC 184/SC 4 Plenary meetings at the Oriente Hotel in Vico Equense, Italy. We especially commend them for an excellent venue, outstanding hospitality, and valuable service to the SC 4 community. A special acknowledgement is extended to Celestina Coccia, Giangiacomo Ponzo, and Alessio Ubertini hosts for the week.

RESOLUTION 816: (Portland, Oregon USA – May 2011)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project	Leader
---------	--------

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Editor 15926-11

Leo van Ruijven

15926-6

Magne Valen-Sendstad

David Leal

10303-242

Darla Nettles

RESOLUTION 817 (Portland, Oregon USA – May 2011)

Reappointment of WG 11 Conveners

SC 4 appreciates and accepts the offer of David Loffredo to serve as Convener of WG 11. SC 4 approves the nomination of David Loffredo to serve for an additional three year term.

RESOLUTION 818 (Portland, Oregon USA – May 2011)

Rationalization of SC 4 Liaisons

SC 4 notes that a number of its liaisons are inactive, due to evolving requirements and changes in other organizations, and resolves to terminate its direct liaisons with the following organizations:

Marine e-Business Standards Association (EMSA)

European Process Industries Step Technical Liaison Executive (EPISTLE)

RESOLUTION 819 (Portland, Oregon USA – May 2011)

Liaison to Strategic Automotive product data Standards Industry Group (SASIG)

SC4 thanks Oscar Rocha for his outstanding effort as the SC 4 liaison to SASIG since 2003.

SC 4 welcomes the offer of Dr. Max Ungerer to act as liaison officer to the Strategic Automotive product data Standards Industry Group (SASIG). SC 4 appoints Dr. Max Ungerer as liaison officer to SASIG.

RESOLUTION 820 (Portland, Oregon USA – May 2011)

Liaison to Energistics

SC4 thanks Alan Doniger for his efforts as the Energistics (formerly POSC) liaison

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

representative to SC 4.

SC 4 accepts and welcomes Jana Schey as liaison officer from Energistics to SC 4.

RESOLUTION 821 (Portland, Oregon USA – May 2011)

Transposing ISO 8000-150 to the general data quality series

SC 4 decides to change the title of ISO 8000-150 to “Data Quality - Part 50: Quality management framework”, change the part number to 50, and no longer restrict the applicability of the standard to only master data. SC 4 directs the project team to implement this change to the document following completion of the CD ballot.

RESOLUTION 822 (Portland, Oregon USA – May 2011)

New ISO Directives

SC 4 notes the new ISO Supplement and requests its Secretariat to update the Standing Documents to refer to Annex SK at the next revision. Meanwhile they will continue to reference the Annex ST available at the time they were approved.

RESOLUTION 823 (Portland, Oregon USA – May 2011)

Approval of New Work Item Proposals

SC 4 requests its Secretariat to ensure that any Change Request resulting in the creation of a new Application Protocol or AP module, or a new edition of an existing Application Protocol or AP module containing significant change is submitted for an NP ballot, even if the component modules and resources in the Change Request are approved under the normal Annex ST procedure.

RESOLUTION 824 (Portland, Oregon USA – May 2011)

Call for nominations for WG 3 Convener

SC 4 requests its Secretariat to circulate a call to P-members for nominations for the post of WG 3 Convener.

RESOLUTION 825 (Portland, Oregon USA – May 2011)

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

We offer thanks to those external projects who presented their material at the Open

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Technical Forum.

“Open Simulation Data Management” – Jochen Haenisch – Jotne/EPM

“Use of AP 210 ed2 as an engineering data integration standard for design verification, based on industrial experience at Rockwell Collins” – Lothar Klein (reading Tom Thurman’s slides)

ACCLAMATION

Appreciation to our SC 4 Liaisons and those projects who presented at the Liaison Plenary

Energistics – Jana Schey

UPDM – Len Levine

ProSTEP – JT – Steven Vettermann

BuildingSMART – Øivind Rooth

Natural Real Property Identifier (NRPI) – Gerry

Radack MoU/MG and other liaisons – Howard Mason

We appreciate your representation to and for ISO TC 184/SC 4.

ACCLAMATION

SC 4 expresses its appreciation to Linda Kresl for her organization of the Industry Day, and to the speakers for attending and presenting on matters of interest to the SC 4 community.

ACCLAMATION

SC 4 expresses its appreciation to Melissa Scheib, and Kristen Wolf for their hosting and support of the 61st ISO TC 184/SC 4 Plenary meetings at the Crowne Plaza Hotel in Portland, Oregon.

Special acknowledgement is extended to the Portland Crowne Plaza Hotel and its staff for their seamless and attentive support.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 826: (Suzhou – Oct 2011)

Part titles for ISO 8000

SC 4 modifies resolution 821 by approving the publishing of ISO 8000-150 as a Technical Specification rather than transforming that part into ISO 8000-50 before publication.

SC4 approves the following titles for parts under the ISO 8000 Information and Data Quality project, each one being within the scope of the approved work item:

- Data quality – Part 8: Information and data quality: Concepts and measuring
- Data quality – Part 9: Information and data quality: Relationship to other standards
- Data quality – Part 10: Exchange of data: Syntax, semantic encoding, and conformance to data specification
- Data quality – Part 20: Exchange of data: Provenance
- Data quality – Part 30: Exchange of data: Accuracy
- Data quality – Part 40: Exchange of data: Completeness
- Data quality – Part 50: Quality management framework

SC4 also approves the following title for ISO 8000-311:

- Data quality – Part 311: Usage guide for ISO 10303-59 (PDQ-S)

SC4 instructs the SC4 Secretariat to ensure that ISO receives notification of the above information for these parts and also the information for ISO 8000-3 (“Data quality – Part 3: Taxonomy”), ensuring that the catalogue on the ISO.ORG website is complete with respect to the current WG 13 work programme.

RESOLUTION 827: (Suzhou – Oct 2011)

Approval of Project Leaders and Project Editors for SC 4 Projects

Objective: Approval of Project Leaders and Project Editors for SC 4 Projects

Resolution: SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

Project

Leader

Editor

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

RESOLUTION 828: (Suzhou – Oct 2011)

Appointment of Deputy Convener for SC 4/WG 13

SC 4 thanks Emily Grantner (Williams) and Leif Gyllström for their faithful service as joint Deputy Conveners for SC 4/WG 13 (Industrial Data Quality) since 2008 and regrets that neither is available to continue in these roles. SC 4 appoints Jean Brangé as Deputy Convener for WG 13, with appreciation for his offer.

RESOLUTION 829: (Suzhou – Oct 2011)

Appointment of additional liaison to ISO/TC 211

SC 4 appoints Peter Benson as an additional liaison to ISO/TC 211, with appreciation for his offer.

RESOLUTION 830: (Stockholm – June 2012)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 approves the following new project leaders or project editors for the associated SC 4 projects:

RESOLUTION 831: (Stockholm – June 2012)

Appointment of Deputy Convener for SC 4/WG 2

SC 4 thanks Guy Pierra for his faithful service as Deputy Convener for SC 4/WG 2 (Standard for the Neutral Representation of Standard Parts) since 2001 and regrets that he is not available to continue in these roles. SC 4 appoints Eric Sardet as Deputy Convener for WG 2, with appreciation for his offer.

RESOLUTION 832: (Stockholm – June 2012)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Appointment of JWG 8 Convenor

SC 4 re-appoints Anne-Françoise Cutting-Decelle, France as convener of JWG 8 for a three-year term.

SC 4 expresses its recognition and thanks for her exceptional dedication and service for the past three years in the position of Convenor.

RESOLUTION 833: (Stockholm – June 2012)

Appointment of WG 12 Convenor

Introduction: Appointment of WG 12 Convenor (Common Resources)

SC 4 approves Keith Hunten, United States, to serve as Convenor of its Working Group 12 for an additional period of three years.

SC 4 expresses its special recognition and thanks for his exceptional dedication and service for the past three years in the position of Convenor.

RESOLUTION 834: (Stockholm – June 2012)

Appointment of WG 13 Convenor

SC 4 approves Tim King, United Kingdom, to serve as Convenor of its Working Group 13 for an additional period of three years.

SC 4 expresses its special recognition and thanks for his exceptional dedication and service for the past three years in the position of Convenor.

RESOLUTION 835: (Stockholm – June 2012)

Approval of Project Leaders for SC 4 Projects



N4118

2026 – 05 – 15

SC 4 approves re-appointment of the following leaders for the **Advisory Group “SC 4 Implementation Forum”**:

Leader: Paul van Exel, USPI (The Netherlands)

Co-Deputy Leader: Dr. Anna Moreno, ENEA (Italy),

Co-Deputy Leader: Burton Gischner – General Dynamics Electric Boat (United States)

RESOLUTION 836: (Stockholm – June 2012)
Liaison to ISO/TC 261 “Additive Manufacturing”

SC 4 notes that the proposed work programme of ISO/TC 261 includes elements related to industrial data which already fall within the scope of ISO/TC 184/SC 4 such as product and manufacturing information.

SC 4 requests its secretariat to establish a liaison to TC 261.

RESOLUTION 837: (Stockholm – June 2012)
Scope of ISO/TC 261 “Additive Manufacturing”

SC 4 notes that the proposed work programme of ISO/TC 261 includes elements related to manufacturing information which already fall within the scope of ISO/TC 184 such as product modeling, NC programming and Robotic programming. SC 4 requests the TC 184 to take all necessary steps to avoid any duplication or inconsistency of deliverables.

RESOLUTION 838: (Stockholm – June 2012)
Result of ISO/DIS 14306 ballot – development of first edition

Noting the result of the ISO/DIS 14306 ballot, and the results of the ballot resolution

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

workshop SC4N2780, SC 4 requests WG 12 to develop a revised version of the specification incorporating the agreed changes from SC4N2780.
SC 4 requests its Secretariat to submit the revised version of the specification to ISO CS for FDIS ballot.

RESOLUTION 839: (Stockholm – June 2012)
Result of ISO/DIS 14306 ballot – development of second edition

Noting the result of the ISO/DIS 14306 ballot, and the results of the ballot resolution workshop SC4N2780, SC 4 requests WG 12 to develop a New Work Item Proposal for a new edition of ISO 14306 including but not limited to replacing XT Brep and JT Brep with STEP curve, surface and Brep entity definitions and extending the scope to include external references to support semantic PMI for assemblies and other requirements from AP 242.

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

We offer thanks to those external projects who presented their material at the Open Technical Forum.

- PLCS – Tor Arne Irgens
- AP 242 Business Object Model – Max Ungerer
- “*Business Object Models for Industrial Data Standards*” paper – Keith Huntten
- Business Objects Model & WEB Services IDEF0 – Jean Brange

ACCLAMATION

Appreciation to our SC 4 Liaisons and those projects who presented at the Liaison Plenary

- Liaison issues – Howard Mason
- IEC SC3D - Hiroshi Murayama
- buildingSMART - Thomas Liebich
- ISO/TS 14033 on Quantitative Environmental Information - Raul Carlson,

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

 Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

TC 207/SC5

- USPI-NL - Maintenance Knowledge Management (MKM) project in the Netherlands – Paul van Excel
- CFIHOS (Capital Facilities Information Hand Over Specification) – Paul van

Exel We appreciate your representation to and for ISO TC 184/SC 4.

SC 4 expresses its appreciation to the speakers and session leaders for attending and presenting on matters of interest to the SC 4 community.

- Howard Mason, TC 184/SC 4
- Per Forsgren, SIS
- Mikael Hedlind, KTH

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

 Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

- Dong Soo Ryu, KHNP
- Bengt Olsson, Sandvik
- Lothar Klein, LKSoft
- Sven Hjelm, Scania
- Lars Hanson, Scania
- Mats Nilsson, FMV
- Nils Sandsmark, DNV
-

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

 Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

- Martin Hardwick, STEP Tools Inc.
- Sune Horkeby, Siemens
- Paul van Exel, TC 184/SC 4
- Hikmet Hussain, SIS

We appreciate your representation to and for ISO TC 184/SC 4.

ACCLAMATION

SC 4 expresses its appreciation to Hikmet Hussain, Teres Osterberg, Susanna Tabell, Mikael Hedlind, Torbjorn Holm, Paul van Exel, Per Forsgren, and Sven Hjelm for organization and support of the Industry Day.

We appreciate your representation to and for ISO TC 184/SC 4.

ACCLAMATION

SC 4 expresses its appreciation to Hikmet Hussain, Teres Osterberg, Susanna Tabell, Mikael Hedlind, Torbjorn Holm, Torbjorn Cederholm for their hosting and support of the 63rd ISO TC 184/SC 4 Plenary meetings at the SIS Conference Center, Stockholm, Sweden.

We appreciate your representation to and for ISO TC 184/SC 4.

Special acknowledgement is extended to the SIS and the Conference Center and staff for their seamless and attentive support.

Resolution 840: (Florida – Nov, 2012)

Approval of Project Leaders and Project Editors for SC 4 Projects

Resolution: SC 4 approves the following new project leaders or project editors for the associated SC 4 projects

Resolution 841: (Florida – Nov, 2012)

Establishment of a Liaison and appointment of a Liaison officer to ISO/TC 261

Resolution: SC 4 requests its Secretariat to establish a Category a Liaison with

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ISO/TC 261 covering activities related to Geometry and information exchange which fall in the scope of SC 4. SC 4 appoints Dr. Martin Hardwick as the Liaison Officer from SC 4 to ISO/TC 261.

Resolution 842: (Florida – Nov, 2012)
Revision of the scope of SC 4/WG 12

Resolution: SC 4 revises the title and scope of WG 12 to:

Title: ***STEP Product Modeling and Resources***

Scope:

- Develop and maintain ISO 10303 (STEP) application protocols (APs)
- Develop and maintain the information models in the STEP Module and Resource Library (SMRL) to support the APs, together with other common resources for SC 4
- Develop and maintain guidance for the use of STEP APs
- Develop and maintain standards for 3D visualization of product models
- Establish liaison with other standards activities working on information models useful as common resources for SC 4

SC 4 transfers the following teams and their work from WG 3 to WG 12:

- T1 Shape Representation
- T4 Materials
- T8 Product Life Cycle
- T9 Engineering Analysis
- T10 Electrical/Electronics
- T19 Mechanical Products
- T22 Building and Construction
- T24 Manufacturing

Existing WG 12 teams such as Parametrics remain unchanged.

SC 4 requests WG 12 to utilize WG 21 to execute the tasks necessary to maintain the SMRL.

SC 4 requests WG 12 to utilize WG 22 to execute the tasks necessary to maintain reference data.

SC 4 invites member bodies to nominate additional experts for the new scope of WG

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

12.

Attached documents: ISO/TC 184/SC 4 Reorganizations

Resolution 843: (Florida – Nov, 2012)
Revision of the scope of SC 4/WG 3

Resolution: SC 4 revises the title and scope of WG 3 to:

Title: Oil, Gas, Process and Power

Scope:

- Develop and maintain ISO 15926
- Develop and maintain guidance for the implementation and use of ISO 15926
- Establish liaison with other standards activities working on relevant information models and reference data

SC 4 appoints Paul van Exel as Convener of WG 3 for a three-year term.

SC 4 transfers the work of WG 3/T25 “Oil, Gas, Process and Power” to WG 3. SC 4 requests WG 3 to utilize WG 22 to execute the tasks necessary to maintain the ISO 15926 reference data.

SC 4 invites member bodies to nominate additional experts for the new scope of WG 3.

Attached documents: ISO/TC 184/SC 4 Reorganizations

Resolution 844: (Florida – Nov, 2012)
Revision of the scope of SC 4/WG 2

Resolution: SC 4 revises the title and scope of WG 2 to:

Title: ***Product characteristics and libraries***

Scope:

- Develop and maintain International Standards for defining and exchanging product characteristics, based on open dictionaries
- Develop and maintain International Standards for defining and exchanging product libraries based on product characteristics
- Develop guidance for the implementation and use of the standards

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Establish liaisons with other standards groups that require services for the representation of product libraries and characteristics

The scope excludes reference data libraries.

SC 4 transfers work on ISO 22745 and 29002 to WG 2, and invites WG 2 to consider the establishment of teams focused on each standard.

SC 4 invites member bodies to nominate additional experts for the new scope of WG 2.

Attached documents: ISO/TC 184/SC 4 Reorganizations

Resolution 845: (Florida – Nov, 2012)
Revision of the scope of SC 4/WG 11

Resolution: SC 4 revises the title and scope of WG 11 to:

Title: Implementation and conformance methods

Scope:

- Develop and maintain International Standards describing EXPRESS and its extensions.
- Develop and maintain International Standards for conveying information described by EXPRESS schemas and methodologies for determining conformance to these standards
- Develop and maintain International Standards and guidance for conveying information defined in EXPRESS schemas using other representation forms
- Develop and maintain International Standards for the implementation of other SC 4 standards

SC 4 invites member bodies to nominate additional experts for the new scope of WG 11.

Attached documents: ISO/TC 184/SC 4 Reorganizations

Resolution 846: (Florida – Nov, 2012)

Creation of WG21 “SMRL Validation Team”

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Resolution: SC 4 creates WG 21 “SMRL Validation Team” to execute the tasks required for managing the STEP Module and Resource Library under ISO Directives Annex SK in accordance with SC4N2617 “Procedure for the maintenance of the STEP Module and Resource Library” or subsequent revisions. SC 4 requests WG 21 to work in close cooperation with WG 12 to support their requirements for maintenance of the SMRL.

SC 4 invites member bodies to ensure that nominated experts for the Validation Team are registered in the Global Directory as members of both SC 4/WG 21 and SC 4/WG 12, and that experts participating in the Maintenance Team are registered as observers.

SC 4 appoints Keith Hunten as Convenor of the group for a three-year term.

Attached documents: ISO/TC 184/SC 4 Reorganizations

Resolution 847: (Florida – Nov, 2012)
Creation of WG22 “Reference Data Validation Team”

Resolution: SC 4 creates WG 22 “Reference Data Validation Team” to execute the tasks required for managing SC 4 reference data from ISO 15926 and other standards under ISO Directives Annex SK in accordance with SC4N2591 “Procedure for the maintenance of ISO/TC 184/SC 4 reference data” or subsequent revisions. SC 4 requests WG 22 to work in close cooperation with WG 3 and WG 12 to meet the needs of the ISO 15926 and STEP development and implementation communities, and to establish liaisons with other groups responsible for reference data that can be used by SC 4.

SC 4 invites member bodies to ensure that nominated experts for the Validation Team are registered in the Global Directory as members of SC 4/WG 22, and also WG 3 and WG 12 as appropriate, and that experts participating in the Maintenance Team are registered as observers.

SC 4 appoints Nils Sandsmark as Convenor of the group for a three-year term.

Attached documents: ISO/TC 184/SC 4 Reorganizations

Resolution 848: (Florida – Nov, 2012)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Creation of WG23 “Vocabulary Validation Team”

Resolution: SC 4 creates WG 23 “Vocabulary Validation Team” to execute the tasks required for managing SC 4 vocabularies of terms and definitions from ISO 8000 and other standards under ISO Directives Annex SK in accordance with SC4N2612 “Procedure for the Maintenance of Vocabulary Standards” or subsequent revisions. SC 4 requests WG 23 to work in close cooperation with WG 13 and WG 2 to meet the needs of the ISO 8000 and ISO 22745 development communities.

SC 4 invites member bodies to ensure that nominated experts for the Validation Team are registered in the Global Directory as members of SC 4/WG 23 and also WG 2 (for ISO 22745) or WG 13 (for ISO 8000), and that experts participating in the Maintenance Team are registered as observers.

SC 4 appoints Dr Gerald Radack as Convenor of the group for a three-year term.

Attached documents: ISO/TC 184/SC 4 Reorganizations

Resolution 849: (Florida – Nov, 2012)
Transfer of ISO 16739 from ISO/TC 184/SC 4 work programme to ISO/TC 59/SC 13 work programme

Resolution: SC 4 notes TC 59 Resolution 327 and requests participation in the development of the document to be created by TC 59 to facilitate the collaboration between SC 4, TC 59/SC 13 and buildingSMART, in order to establish the necessary operational procedures. SC 4 asks its PPC, WG 11 and WG 12 to consider the appointment of some of its members as participants of the development activity.

SC 4 agrees to establish a Joint Working Group with ISO/TC 59/SC 13 for the maintenance of ISO 16739 and development of related standards, with appropriate liaison with buildingSMART.

SC 4 requests a report from the joint working group at the Paris meeting.



N4118

2026 – 05 – 15

Attached documents: See below extract from resolutions of TC 59 meeting

TC 59 Resolution 327 (Tokyo 12)

TC59 supports all the resolutions and recommendations made by different SCs of TC 59 during the plenary week, 14-19 October 2012 in Tokyo. In particular, TC 59 acknowledges:

SC13:

Resolution 146 (Tokyo 6) Transfer of responsibility for ISO 16739 from ISO/TC 184/SC 4 to ISO/TC 59/SC 13

SC 13 welcomes the transfer of responsibility for ISO 16739 from ISO/TC 184/SC 4 should the ISO/TC 184/SC 4 resolve this at their next Plenary meeting in Miami in November 2012. SC 13 is positive to the idea of establishing a JWG between ISO/TC 184/SC 4/WG 12 and SC 13.

Resolution 147 (Tokyo 7) Development of formal document on SC 13 and buildingSMART International collaboration

SC 13 asks the SC 13 secretariat to develop a formal document to facilitate the collaboration between SC 13 and buildingSMART International by describing the ways in which documents and information are exchanged between the two and provide some guidelines as to how meetings can be arranged. This document should also address the relation to ISO/TC 184.

TC 59 requests that the collaboration document follows the requirements of the ISO Directives and requests that once finished, the document should be applicable also at the TC level.

Resolution 850: (Florida – Nov, 2012)
Appointment of SC 4 Liaison Officer

Resolution:

SC 4 welcomes the offer of Jean Brangé to act as an liaison officer to ProSTEP iViP.. SC 4 appoints Jean Brangé as liaison officer to ProSTEP iViP's JT Implementor Forum, and its joint meetings together with the JT Workflow Forum.

Resolution 851: (Florida – Nov, 2012)
Reappointment of SC 4 Chair

Resolution: SC4 endorses the nomination by the Secretariat of Howard Mason for an

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

additional three year term as the Chair of ISO/TC 184 /SC 4 and instructs the Secretary to forward his nomination to ISO TC 184 within 30 days.

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

We offer thanks to Dr Tim King for leading the Forum on “*Explaining the Business Benefits of SC 4 Standards*” and to the SC 4 experts who contributed to a successful session.

ACCLAMATION

Appreciation to our SC 4 Liaisons and those projects who presented at the Liaison Plenary

- MOU/MG Report – Howard Mason
- X3D - Soonhung Han
- The Oil and Gas Interoperability (OGI) Project - David Leal
- Liaison Issues – Howard Mason

We appreciate your representation to and for ISO/TC 184/SC 4.

ACCLAMATION

SC 4 expresses its appreciation to Melissa Hildebrand and Krysten Wolf for their hosting and support of the 64th ISO/TC 184/SC 4 Plenary meetings at the Embassy Suites Hotel in Miami, Florida.

Special acknowledgement is extended to the Miami Embassy Suites Hotel and its staff for their seamless and attentive support.

Resolution 852: (St Denis – June, 2013) **Appointment of Coordinator Quality Committee**

Resolution: SC 4 expresses its special recognition and thanks to Peter Benson for his exceptional dedication and service for the past three years in the position of

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Coordinator, Quality Committee. SC 4 thanks Dr Radack's offer to act as Coordinator, Quality Committee and appoints him to serve in that capacity. SC 4 approves Dr Gerald Radack as Coordinator, Quality Committee for a period of three years.

Resolution 853: (St Denis – June, 2013)
Cancellation of ISO 8000-050

Resolution: SC 4 approves removal of ISO-8000-050 from the list of standards in the SC 4 work programme.

SC 4 requests its Secretariat to update the SC 4 database and inform ISO CS of this change to the work programme

Resolution 854: (St Denis – June, 2013)
Launch of new edition 3 of 10303-210

Resolution: SC 4 approves a NP to create a new edition 3 of 10303-210 with the same scope as 10303-210 ed2, using the accelerated timescale, with Thomas Thurman as the project leader. SC 4 decides that the project should proceed direct to DIS ballot.

Resolution 855: (St Denis – June, 2013)
Harvesting "SysML" and "UPDM"

Resolution: SC 4 welcomes the proposed submission from OMG of SysML1.3 for harvesting as NWI/CD and UPDM 2.1 for harvesting as DIS under the SC4 harvesting procedures.

Resolution 856: (St Denis – June, 2013)
Secretariat for JWG with ISO/TC 59/SC 13

Resolution: SC 4 agrees that Standards Norway should take the administrative lead for the Joint Working Group, on the understanding that the first task of the JWG will be to create the operational procedures for the support of ISO 16739, for approval by the next plenary meetings of SC 4 and SC 13.

NOTE: France proposes Bernard Ferries



N4118

2026 – 05 – 15

Resolution 857: (St Denis – June, 2013)
ISO 8000 Programme Dates

Resolution: SC 4 notes that the revised dates for delivery are within the SC 4 planned timescales and requests its Secretariat to inform ISO CS of the requirement to place ISO 8000-8, 9, 10, 20, 30, 40 on the ISO extended timescale

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION – St Denis

Appreciation to those who presented at the Change Management Plenary:

ISO 8000 part 62 – Ismael Caballero and Sunho Kim

SysML – Andreas Korff, OMG

UPDM – Andreas Korff, OMG

ACCLAMATION- St Denis

SC 4 expresses its appreciation to Isabelle Rimbert, Stephane Mouliere, Ghislaine Ines Magnan and Martine Degardin for their hosting and support of the 65th ISO TC 184/SC 4 Plenary meetings at AFNOR in St Denis, France. We especially commend AFNOR and Jean Brange for an excellent venue and outstanding service to the SC 4 community. A special acknowledgement is extended to Christophe Mouton and to Bruno Schultz, for their support during the week. Special acknowledgement is extended to AFNOR and its staff for their seamless and attentive support.

Resolution 858: (Gyeongju, Nov - 2013)
Reappointment of WG 2 Convener

Resolution: SC 4 appoints Nikolaus Ondracek as Convener for WG2 for a further three-year term, with appreciation for his offer to serve in that capacity

Resolution 859: (Gyeongju, Nov - 2013)
Reappointment of WG 12 Deputy Convener

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Resolution: SC 4 appoints Jochen Haenisch as Deputy Convener for WG 12 for a further three-year term, with appreciation for his offer to serve in that capacity

Resolution 860: (Gyeongju, Nov - 2013)

Approval of Co-Leader for SC 4 Implementation Forum

Resolution: SC 4 appoints Dr Martin Hardwick as Co- leader for the Implementation Forum.

Resolution 861: (Gyeongju, Nov - 2013)

ISO/TC 184/SC 4 – ISO/TC 59/SC 13 JWG 12

Resolution:

ISO/TC 184/SC 4 agrees to the request of ISO/TC 59/SC 13 to transfer the Standard ISO 16739 to the administration of SC 13.

SC 4 requests its secretary to arrange any necessary transfer of the standard to the SC 13 work program.

SC 4 requests SC 13 to ensure that SC 4 Secretariat is informed of all ballot actions in order to circulate the documents to the members of SC 4.

SC 4 encourages SC 13 to work with SC 4 on other data-related standards.

SC 4 is prepared to nominate representatives to the proposed JWG 12 maintenance and validation teams.

SC 4 welcomes the proposal of SC 13 to alternate co-located JWG 12 meetings between the two parent committees.

SC 4 encourages its members to nominate appropriate additional technical experts to participate in the work of JWG 12.

NOTE:

The necessary HTML-documentation Guidelines requested by JWG 12 are publicly available on the SC 4 eCommittees site.



N4118

2026 – 05 – 15

Resolution 862: (Gyeongju, Nov - 2013)
Unified Architecture Framework (UAF)

Resolution: SC 4 notes the proposal from the US to offer the Unified Architecture Framework (UAF) for standardization in ISO/TC 184/SC 4, possibly in collaboration with other Standard Development Bodies (SDOs) such as OMG, OASIS, JTC1, and INCOSE, , and invites the US member body to submit a Preliminary Work Item (PWI) for consideration.

NOTE: The PWI should clearly demonstrate the relationship between SysML, UPDM and UAF

Resolution 863: (Gyeongju, Nov - 2013)
ISO/PAS 18828

Resolution: SC 4 establishes an ad hoc group with the task to improve the ISO/PAS 18828 document for submission at the next meeting of SC 4 on 2014-05-04/09.

Resolution 864: (Gyeongju, Nov - 2013)
Appreciation for Service of Dr Burton
Gischner

Resolution: SC 4 expresses its special recognition and thanks to Dr Burton Gischner for his exceptional dedication and outstanding service for many years in the international standardization community. SC 4 especially thanks Dr Gischner for his long standing service as the U.S. Head of Delegation.

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

Appreciation to our SC 4 Liaisons and those projects who presented at the Opening Technical Plenary:

TC59 / SC13 / JWG12 Liaison Report - Thomas Liebich, buildingSMART International



N4118

2026 – 05 – 15

OASIS PLCS Liaison Report - Howard Mason, Co-chair, PLCS TC

ACCLAMATION

Appreciation to those who presented at the Open Technical Forum:

Jean Brange, BOOST - *“BoostAeroSpace – Competitiveness and Innovation for the European Aerospace & Defence Industry”*

Jochen Haenisch, JOTNE – *“CLOUDFLOW: Computational Cloud Services And Workflows For Agile Engineering”*

David Leal- *“Industrial Data on the Web - Dereferencing”*

Howard Mason, AIA – *“Industrial Data in the Cloud: A US Aerospace and Defense Perspective”*

ACCLAMATION

SC 4 expresses its appreciation to Prof Soonhung Han for his organization of the Industry Day, and to the speakers for attending and presenting on matters of interest to the SC 4 community.

ACCLAMATION

SC 4 expresses its appreciation to Prof Soonhung Han, the Korean Agency for Technology and Standards (KATS), Korea STEP Center (KSTEP), and Korea Advanced Institute of Science and Technology (KAIST) for their hosting and support of the 66th ISO TC 184/SC 4 Plenary meetings at Hotel Hyundai in Gyeongju, Korea. We especially commend them for an excellent venue and outstanding service to the SC 4 community. A special acknowledgement is extended to Seung-joo Nam, Kyudong Kim and Jihye Kang for their outstanding support during the week. Special acknowledgement is extended to the KATS and its staff for their seamless and attentive support.



N4118

2026 – 05 – 15

Resolution 865: (Philadelphia, May – 2014)
Approval of Project Leaders and Project Editors for SC 4 Projects

Resolution: SC 4 appoints Sheron Koshy as Project Editor for Projects below:

- ISO 8000-10
- ISO 8000- 20
- ISO 8000- 30
- ISO 8000- 40
- ISO 8000- 100
- ISO 8000- 120
- ISO 8000- 130
- ISO 8000- 140

Resolution 866: (Philadelphia, May - 2014)
Appointment of Quality Committee Convener

Resolution: SC 4 requests its Secretariat to circulate a call for nominations for the post of Quality Committee Convener for a three-year term.

Resolution 867: (Philadelphia, May - 2014)
Reappointment of WG 11 Convener

Resolution: SC 4 appoints David Loffredo as Convener for WG 11 for a further three-year term, with appreciation for his offer to serve in that capacity

Resolution 868: (Philadelphia, May - 2014)
Appointment of WG 3 Deputy Convener

Resolution: SC 4 appoints Nils Sandmark as Deputy Convener for WG 3 for a three-year term, with appreciation for his offer to serve in that capacity

Resolution 869: (Philadelphia, May - 2014)
Appointment of WG 23 Vocabulary Validation Team Leader

Resolution: SC 4 requests its Secretariat to circulate a call for nominations for the post of WG 23 Vocabulary Validation Team Leader

Resolution 870: (Philadelphia, May - 2014)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Establish Liaison for OSLC

Resolution: SC 4 appoints Nicolas Figay as liaison to OSLC, with appreciation for his offer to serve in that capacity.

Resolution 871: (Philadelphia, May - 2014)
Establish Liaison for ISO 9000

Resolution: SC 4 appoints Tor Arne Irgens as liaison to ISO/TC 176 for ISO 9000, with appreciation for his offer to serve in that capacity

Resolution 872: (Philadelphia, May - 2014)
Liaison from Energistics

Resolution: SC 4 welcomes Jay Hollingsworth as liaison from Energistics, with appreciation for his offer to serve in that capacity

Resolution 873: (Philadelphia, May - 2014)
Cancellation of PAS 26183

Resolution: SC 4 agrees to cancel Publicly Available Specification 26183 “Product Data Quality Guidelines for the Global Automotive Industry”, and requests its Secretariat to inform ISO CS, and to retain all relevant documentation for future use

Resolution 874: (Philadelphia, May - 2014)
Establish Duration of New Work Item Proposals

Resolution: SC 4 requests its Secretariat to apply the 2-month duration to NPs for 15926-12 and 10303-242e2

Resolution 875: (Philadelphia, May - 2014)
Appoint Participants for Semantic Interoperability and Harmonization project

Resolution: SC 4 invites David Leal, Anne-Francoise Cutting-Decelle and Keith Hunten to participate in the MOU/MG Semantic Interoperability project

Resolution 876: (Philadelphia, May - 2014)
IEC/TC3 Scope

Resolution: SC 4 requests its Secretariat to send a liaison statement to IEC/TC 3



N4118

2026 – 05 – 15

before 2014-06-06 highlighting the potential overlap of scope, and to request the TC 184 Secretariat to re-establish a liaison to TC 3.

Resolution 877: (Philadelphia, May - 2014)
Decision and Support to ISO 14306ed2 "JT" development

Resolution: SC 4 directs the project group on ISO 14306ed2 "JT" to continue its work under the following conditions:

- The platform for document storage and providing information in written electronic form is the SC 4 space for eCommittees on the ISO internet platform.
- An updated timeframe with milestones shall be prepared with the strong request to meet the defined milestones

SC 4 requests its Secretariat to cancel the current CD ballot and to launch a new CD ballot with ISO/CD 14306 ed2 and the latest AP243 document, including a specific request for NBs to consider any proposed variations in scope.

Resolution 878: (Philadelphia, May - 2014)
Requirements for consistent approach to Product properties Creation of ad hoc group with IEC TC 65 representation

Resolution: Noting the successful collaboration between SC 4 and IEC TC 3 on the deployment of PLIB, SC 4 requests its WG 2 to participate in the proposed TC 184/TC 65 ad hoc group by showing and describing the practical uses of the standards ISO 13584 and ISO 22745 which shows the benefit in a productive environment, taking into account ISO/IEC Guide 77 and previous SC 4 work on this topic.

SC 4 supports the need for a common face of the standards program to potential users within industry. Therefore SC 4 invites WG2 to nominate members to participate in this ad hoc group by the end of May 2014

Resolution 879: (Philadelphia, May - 2014)
Liaison to IEC SC 65E

Resolution: SC 4 has noted possible overlaps of work to IEC SC 65E by reviewing

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

the scope of IEC 62714 AutomationML. SC 4 requests its liaison officers Messrs Murayama and Ondracek to contact IEC SC 65E and investigate possible areas of overlaps, which could be the subject of future cooperation between SC 4 and IEC SC 65E.

Resolution 880: (Philadelphia, May - 2014)
Liaison to IEC TC 171/SC 2 on ISO/CD
17911

Resolution: SC 4 requests its secretariat to send the above liaison statement to the secretariat of ISO/TC 171/SC 2

Resolution 881: (Philadelphia, May - 2014)
Appreciation for Service of Dr. Gerry Radack

Resolution: SC 4 expresses its special recognition and thanks to Dr. Gerry Radack for his exceptional dedication and outstanding service for many years in the international standardization community. SC 4 especially thanks Dr. Radack for his service as the
U.S. Head of Delegation

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

Appreciation to our SC 4 Liaisons and those projects who presented at the Opening Technical Plenary:

- Jochen Haenisch, JOTNE – “Open Data Exchange Standards for Infrastructure Opens the Door for Enterprise Resource Planning Systems”
- Thomas Liebich, Chair, ISO/TC 59/SC 13/JWG 12

ACCLAMATION

SC 4 expresses its appreciation to the US national TAG for their organization of the

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Industry Day, and to the speakers and panel members for attending and presenting on matters of interest to the SC 4 community.

ACCLAMATION

SC 4 expresses its appreciation to ECCMA for their hosting and support of the 67th ISO TC 184/SC 4 Plenary meetings at ACE Conference Center, Philadelphia, PA. We especially commend them for an excellent venue and outstanding service to the SC 4 community. A special acknowledgement is extended to Ms. Melissa Hildebrand and Ms. Vicky Falcone for their outstanding support during the week. Special acknowledgement is extended to the ACE Conference Center and its staff for their seamless and attentive support.

Resolution 882: (Belfort, France - Nov 2014)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 approves the following Project Leaders and Project Editors for the associated SC 4 projects:

Project	Project Leader	Editor
AP 242 ed 2	Jean Brange	Darla Nettles

Resolution 883: (Belfort, France - Nov 2014)

Reappointment of WG 13 Deputy Convenor

SC 4 appoints Mr. Jean Brangé as Deputy Convenor for WG 13 for a further three-year term, with appreciation for his offer to serve in that capacity.

Resolution 884: (Belfort, France - Nov 2014)

Appointment of WG 22 Deputy Convener

SC 4 appoints Mr. Paul Van Exel as Deputy Convener for WG 22 for a three-year term, with appreciation for his offer to serve in that capacity.

Resolution 885: (Belfort, France - Nov 2014)

Establish Duration of New Work Item Proposals

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 requests its Secretariat to apply the 2-month duration to NPs for:

- 18828-3 Information Flows within Production Systems Engineering;
- 15926-6e2 Scope and Representation for Additional Reference Data – OWL Extension;
- 15926-10 Conformance Testing;
- 15926: Initial Reference Data for the Maritime Industry;
- 15926 Integrated Lifecycle Asset Planning.

Resolution 886: (Belfort, France - Nov 2014)
Cancellation of UPDM

SC 4 requests its secretariat to cancel the project and remove ISO/PAS17729 from the SC 4 Work Program.

Resolution 887: (Belfort, France - Nov 2014)
Appointment of JWG 8 Deputy Convener

SC 4 appoints Mr. Bob Young as Deputy Convener for JWG 8 for a three-year term, with appreciation for his offer to serve in that capacity.

Resolution 888: (Belfort, France - Nov 2014)
Publication of ISO 8000 “Information and data quality — Part 8: Concepts and measuring” as IS
SC 4 note the results of the DIS ballot on ISO 8000 “Information and data quality — Part 8: Concepts and measuring” and request its secretary to submit the amended part to ISO CS for publication as IS omitting the FDIS ballot.

Resolution 889: (Belfort, France - Nov 2014)
Future scope of ISO/TC 59/SC 13/JWG 12

SC 4 notes the recommendations made by ISO/TC 59/SC 13/JWG 12 at their third meeting and the related resolutions 159 and 160 approved by ISO/TC 59/SC 13 at their plenary meeting in Toronto.

SC 4 welcomes the progress made to date by JWG 12 and their intention to turn ISO 16739 into a useful and widely adopted set of building information management

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

standards.

SC 4 encourages all members of JWG 12 to strive to maintain and where necessary even improve the interoperability of ISO 16739 and ISO 10303. Especially in light of the envisaged development of model view definitions for ISO 16739 SC 4 requests JWG 12 to leverage the experiences of SC 4 in producing and maintaining their application protocols over the last two decades.

SC 4 asks JWG 12 to report on their efforts to agree on a harmonized methodology for the presentation and implementation of ISO 16739 and ISO 10303 at the next SC 4 plenary meeting in Vico Equense in April 2015.

Resolution 890: (Belfort, France - Nov 2014)
Retirement of Secretariat

SC 4 expresses its great appreciation to Jerry Smith for his excellent service as Secretary to SC 4 since 1999, and wishes him a long and happy retirement. SC 4 commends his efforts to secure resources to support the original SC 4 website, and for the operation of the committee. SC 4 also thanks DISA, Mr. Dennis Devera and Ms. Bich Ngoc Nguyen for their support.

Resolution 891: (Belfort, France - Nov 2014)
ISO 14306 e2

SC 4 requests its Secretariat to apply the extended timeframe to the ISO 14306e2 project, with a DIS limit date of 2015-12-15.

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

We offer thanks to those external projects who presented their material at the Open Technical Forum.

“Visualization for the web for 3D CAD massive Datasets”
Behr “Industrial Data and the Circular Economy”
Swindells

Dr. Johannes
Dr. Norman

“Configuration Services for Global Production Networks”

Prof BobYoung

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

“Equivalence Validation”
Ohtaka

Mr. Akihiko

ACCLAMATION

We offer thanks to our liaison members who presented their material at the Liaison Plenary

“ISO/TC 176/SC 2”
“ISO/TC 59/SC 13/JWG 12 Buildings
Jochen Haenisch
and civil engineering works”
“Industry Views on ISO 15926”
Thostrup
“Liaison Issues”

Dr. Timothy King
Mr.

Mr. Anders

Mr. Howard Mason

ACCLAMATION

SC 4 expresses its appreciation to AFNOR – Association Française de Normalisation and the University De Technologie De Belfort-Montbéliard for their hosting and support of the 68th ISO TC 184/SC 4 Plenary meetings at the University De Technologie De Belfort-Montbéliard, Belfort France. We especially commend Ms Ghislaine Magnan and Mr. Christophe Mouton for an excellent venue and outstanding service to the SC 4 community.

ACCLAMATION

SC 4 expresses its appreciation to AFNOR for their organization of the Industry Day, and to the speakers and panel members for attending and presenting on matters of interest to the SC 4 community.

ACCLAMATION

SC 4 expresses its appreciation to Groupe Elephant for their excellent support of the 68th ISO TC 184/SC 4 Plenary meetings. Mr. David Ramos and Mr. Bruno Christmann and their staff are to be commended for their exceptional support and service to organization of the meetings.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Resolution 892: (Vico Equense – Apr 2015)

Reappointment of JWG 8 Convener

SC 4 appoints Anne-Françoise Cutting-Decelle as Convener for JWG 8 “Manufacturing process and management information” for a further three-year term, with appreciation for her offer to serve in that capacity.

Resolution 893: (Vico Equense – Apr 2015)

Reappointment of WG 12 Convener

SC 4 appoints Keith Hunten as Convener for WG 12 “STEP product modelling and resources” for a further three-year term, with appreciation for his offer to serve in that capacity.

Resolution 894: (Vico Equense – Apr 2015)

Reappointment of WG 13 Convener

SC 4 appoints Timothy King as Convener for WG 13 “Industrial data quality” for a further three-year term, with appreciation for his offer to serve in that capacity.

Resolution 895: (Vico Equense – Apr 2015)

Reappointment of Leader SC 4 Implementation Forum

SC 4 appoints Paul van Exel as Leader for the SC 4 Implementation Forum for a further three-year term, with appreciation for his offer to serve in that capacity.



N4118

2026 – 05 – 15

Resolution 896: (Vico Equense – Apr 2015)

Reappointment of Co-Deputy SC 4 Implementation Forum

SC 4 appoints Anna Moreno as Co-Deputy for the SC 4 Implementation Forum for a further three-year term, with appreciation for her offer to serve in that capacity.

Resolution 897: (Vico Equense – Apr 2015)

Appointment of SC 4 Liaison to Energistics

SC 4 appoints Nils Sandmark for a three-year term as the SC 4 Liaison to Energistics, with appreciation for his offer to serve in that capacity.

Resolution 898: (Vico Equense – Apr 2015)

Rationalization of SC 4 Liaisons

SC 4 requests its Secretariat to terminate its liaisons with the following organizations:

- CEN/ISSS Workshop ELSSI-EMD
- United Nations Standard Products and Services Code (UNSPSC)
- ISO PC 246 Anti-counterfeiting tools
- ISO TC 8 - Ships and marine technology
- ISO TC 10/SC 8 - Construction documentation
- ISO TC 60 – Gears

Resolution 899: (Vico Equense – Apr 2015)

Create new edition of 10303-235

SC 4 approves a NP to create a new edition of 10303-235 with the same scope as 10303-235 ed1 “Engineering properties for product design and verification”, using the accelerated timescale, with Norman Swindells as the project leader. SC 4 resolves that the project should proceed direct to DIS ballot.

Resolution 900: (Vico Equense – Apr 2015)

Create new edition of 15531-44

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 approves a NP to create a new edition 2 of 15531-44 with the same scope as 15531-44 “Information modelling for shop floor data acquisition”, using the accelerated timescale, with Anne-Francoise Cutting-Decelle as the project leader. SC 4 decides that the project should proceed direct to DIS ballot.

Resolution 901: (Vico Equense – Apr 2015)

Allocation of the ISO 18828 Project to JWG 8

ISO/DPAS 18828 “Reference process for seamless production planning” has been accepted for publication under the harvesting process, and it has been decided that further parts are necessary. SC 4 assigns the Reference process for seamless production planning project ISO 18828 to JWG 8.

Resolution 902: (Vico Equense – Apr 2015)

Launch of DIS ballot for ISO 18828-2

SC 4 requests its Secretariat to launch the DIS ballot for ISO 18828-2 – “Reference process for seamless production planning - Standardized procedure for production systems engineering.”

Resolution 903: (Vico Equense – Apr 2015)

Create new PWI for new Edition 3 of 10303 AP239

SC 4 approves a PWI to create a new edition 3 of 10303 AP239 “Product Life Cycle Support” as defined in SC 4 N2997 and invites member bodies to nominate experts to the project.

Resolution 904: (Vico Equense – Apr 2015)

Establish duration of New Work Item Proposals

SC 4 requests its Secretariat to apply the 2-month duration to the proposed NPs for

- Formal Semantic Models for the Configuration of Global Production Networks (SP-NET) – ISO 20534

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Equivalence Validation

Resolution 905: (Vico Equense – Apr 2015)

Proposed Integrated Model for Digital Manufacturing

A proposal is being developed for a new edition of AP238 which will include a series of bug fixes, incorporate new GD&T capabilities, improve integration with ISO 13399 for cutting tools, and ISO 14649 for NC machines, and support for additive manufacture. Other capabilities such as robotic machining may also need to be addressed. The responsibility for these standards is currently spread across TC 29/TC 39/TC 261/TC 184. There is therefore an opportunity to address this project through the coordination meeting proposed by ISO/TC 184 in its recent plenary meeting, as an example of the necessity for coordination.

SC 4 requests its Secretariat to inform ISO CS about the proposed Integrated Model for Digital Manufacturing project as an example of the work that should be addressed by the meeting suggested by ISO/TC 184 in its resolution 573.

Resolution 906: (Vico Equense – Apr 2015)

Appointment of WG 23 Convener

SC 4 appoints Aminata Mbengue as Convener for WG 23 "Vocabulary validation team" for a three-year term, with appreciation for her offer to serve in that capacity.

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

Appreciation to our SC 4 Liaisons and those projects who presented at the Liaison Plenary

- TC 184 Liaison Report
- TC 184 Big Picture
- MoU/MG

Howard Mason
Christophe Mouton
Howard Mason

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- ISO/IEC JTC 1/WG 10 Working Group on IoT
- Soonhung Han

ACCLAMATION

SC 4 expresses its appreciation to Anna Moreno and her team for hosting the 69th ISO/TC 184/SC 4 Plenary in Vico Equense, Italy, 2015-04-19/24

Special acknowledgement is extended to the Hotel Oriente, Vico Equense and its staff for their seamless and attentive support.

Resolution 907: (Baltimore – Oct 2015)

SC 4 approves the following new project leaders and/or project editors for the associated SC 4 projects:

ISO Project #	Project Lead	Document Editor
15926-10	Manoj Dharwadkar	Hiroshi Okada

Resolution 908: (Baltimore – Oct 2015)

Reappointment of Change Management Convener

SC 4 appoints Mr. Howard Mason as Convener for the Change Management Committee for a further three-year term, with an appreciation for his offer to serve in that capacity.

Resolution 909: (Baltimore – Oct 2015)

Reappointment of WG 03 Convener

SC 4 appoints Paul van Exel as Convener for WG 03 “Oil, Gas, Process and Power” for a further three year term, with appreciation for his offer to serve in that capacity.

Resolution 910: (Baltimore – Oct 2015)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Reappointment of WG 21 Convener

SC 4 appoints Keith Hunten as Convener for WG 21 “SMRL Validation Team” for a further three-year term, with appreciation for his offer to serve in that capacity.

Resolution 911: (Baltimore – Oct 2015)

Reappointment of WG 21 Deputy Convener

SC 4 appoints Jochen Haenisch as Deputy Convener for WG 21 “SMRL Validation Team” for a further three-year term, with appreciation for his offer to serve in that capacity.

Resolution 912: (Baltimore – Oct 2015)

Reappointment of WG 22 Convener

SC 4 appoints Nils Sandsmark as Convener for WG 22 “Reference Data Validation Team” for a further three-year term, with appreciation for his offer to serve in that capacity.

Resolution 913: (Baltimore – Oct 2015)

Reappointment of SC 4 Chair

SC 4 nominates Howard Mason for an additional two year term (January 2016-December 2017) as the Chair of ISO/TC 184 / SC 4 and instructs the Secretary to forward his nomination to ISO/TC 184 within 30 days.

Resolution 914: (Baltimore – Oct 2015)

Extension of Standard Development Track for ISO 10303-57 from 36 months to 48

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

months

SC 4 requests its Secretariat to change the development track of ISO 10303-57 from 36 months to 48 months.

Resolution 915: (Baltimore – Oct 2015)

Extension of Standard Development Track for ISO 18828-2 to 36 Months

SC 4 requests its Secretariat to change the development track of ISO 18828-2 from 24 months to 36 months.

Resolution 916: (Baltimore – Oct 2015)

Termination of SC 4 Liaison with ASAM

SC 4 requests its Secretariat to terminate its liaisons with the Association for Standardization of Automation and Measuring Systems (ASAM).

Resolution 917: (Baltimore – Oct 2015)

Establish Duration of New Work Item Proposals

SC 4 requests its Secretariat to apply the 2-month duration to the proposed NPs for

- ISO 10303 AP239 ed 3 PLCS
- ISO 18828-4 KPI in production planning processes
- ISO 8000-115 Master data: Quality identifiers

Resolution 918: (Baltimore – Oct 2015)

WG 2 Parts Libraries activities

SC 4 requests its secretariat to ask its P- and O-Members to nominate subject matter experts in the area of Part Libraries. The invitation should also be sent to ISO users of the PLIB standards.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Resolution 919: (Baltimore – Oct 2015)

Creation of a new Working Group on "Digital manufacturing"

SC 4 decides to establish Working Group 15 "Digital manufacturing". The goal of this working group is to coordinate the SC 4 internal efforts in this area with other standards organizations and to observe and interact with ongoing and upcoming developments.

Resolution 920: (Baltimore – Oct 2015)

Joint Task Force to explore the establishment of a database for product properties and classes

SC 4 appoints the following experts to assist in the work of the Joint Task Force:

- Wolfgang Wilkes ISO 13584 expert and user in other standards
- Norman Swindells User of ISO 13584 in other standards
- Nils Sandsmark ISO 15926 expert and user in oil and gas facility environments
- Nigel Shaw ISO 10303-239 and reference data expert and user
- Nicolas Figay ISO 10303-242 expert
- Peter Benson ISO 22745 expert and user
- Nikolaus Ondracek ISO 13584 expert and user
- David Leal ISO 15926 and RDL expert
- Paul van Exel ISO 15926 expert

SC 4 requests its PPC to arrange a coordination meeting for the nominees to ensure a consistent representation from SC 4 in this task force.

Resolution 921: (Baltimore – Oct 2015)

Create PWI for new Edition 3 of ISO 14306 JT

SC 4 notes the revised schedule for the delivery of a document for DIS ballot for edition

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

2 by the end of February 2016 and approves a PWI to create a new edition 3 of ISO 14306. SC 4 invites member bodies and other stakeholders such as the Global Automotive Action Group to nominate experts to the project and to solicit requirements to be addressed in the next edition by end March 2016.

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

SC 4 expresses its appreciation to our SC 4 Liaisons and those projects who presented at the Liaison Plenary

- Energistics – Jana Schey
- IAIDQ Conference – Tim King
- ECCMA – Peter Benson
- JTC 1/SC 24 X3D – Christophe Mouton
- ISO 9001:2015 – Tor Arne Irgens
- TC 184 update – Howard Mason

ACCLAMATION

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the 70th ISO/TC 184/SC 4 Industry Day in Baltimore, Maryland, 2015-10-21

- David Odendahl, Boeing Airplane
- Corey Dickman, Penn State & America Makes
- Mikael Hedlind, Scania
- Max Ungerer, ProSTEP
- Andy Byrd, Okuma Dan Finke, Penn State
- Larry Maggiano, Mitutoyo
- Bengt Olsson, Sandvik
- Tom Raun, Iscar Metals
- Sid Venkatesh, Boeing Airplane

ACCLAMATION

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 expresses its appreciation to Peter Benson, and Vicky Kondravy for hosting the 70th ISO/TC 184/SC 4 Plenary in Baltimore, Maryland, 2015-10-18/23

Special acknowledgement is extended to the Pier 5 Hotel, Baltimore, Maryland and its staff for their seamless and attentive support.

Resolution 922: (Sapporo – May 2016)

Appointment of SC 4 Liaison to IEC SC 3D

SC 4 notes that IEC/TC 3 has terminated its liaison and appoints Hiroshi Murayama for a three-year term as the SC 4 Liaison to IEC SC 3D, with appreciation for his offer to serve in that capacity.

Resolution 923: (Sapporo – May 2016)

Move ISO 8000-62 and ISO 8000-63 to Enlarged Timeframe

SC 4 requests its Secretariat to apply the enlarged timeframe to the ISO 8000-62 and ISO 8000-63 projects, with a DIS limit date of 2017-02-21.

Resolution 924: (Sapporo – May 2016)

Create PWI for Modelling and Simulation information in a collaborative Systems Engineering Context (MoSSEC)

SC 4 approves a PWI to create standards for modeling and simulation information in a collaborative systems engineering context as defined in SC4 N3086 and invites member bodies to nominate experts to the project.



N4118

2026 – 05 – 15

Resolution 925: (Sapporo – May 2016)

Create PWI for Manufacturing Change Management

SC 4 approves a PWI to create a standard for manufacturing change management as defined in SC4 N3087 and invites member bodies to nominate experts to the project. SC 4 requests its Secretariat to invite SC 5 to seek input from its P-members.

Resolution 926: (Sapporo – May 2016)

Establish Ballot Duration of the New Work Item Proposal for ISO 10303-238ed2

SC 4 requests its Secretariat to apply the 8 week ballot duration to the proposed NP for ISO 10303-238ed2.

Resolution 927: (Sapporo – May 2016)

Creation of Ad Hoc Group within SC 4 to support ISO/IEC DPPC TF

SC 4 creates an Ad Hoc group to shadow and support the ISO/IEC DPPC Task Force and appoints the following experts to the group:

- Hiroshi Murayama, Leader and ISO 13584 and IEC 61360 expert
- Wolfgang Wilkes, ISO 13584 expert and user in other standards
- Norman Swindells, User of ISO 13584 in other standards
- Nils Sandmark, ISO 15926 expert and user in oil and gas facility environments
- Nigel Shaw, ISO 10303-239 and reference data expert and user
- Nicolas Figay, ISO 10303-242 expert
- Peter Benson, ISO 22745 expert and user
- Nikolaus Ondracek, ISO 13584 expert and user
- David Leal, ISO 15926 and RDL expert
- Paul van Exel, ISO 15926 expert
- Hong Yan, ISO 13584-511 expert

SC 4 requests its Secretariat to invite member bodies to nominate other experts to participate in the group.



N4118

2026 – 05 – 15

Resolution 928: (Sapporo – May 2016)

Cooperation with the International Association of Oil and Gas Producers (IOGP)

SC 4 recognizes the request of the International Association of Oil and Gas Producers (IOGP) to become a liaison and asks its secretariat to initiate a liaison to this association as soon as possible.

SC 4 nominates Nils Sandsmark and Paul van Exel as the liaison officers from SC 4 to the IOGP.

Resolution 929: (Sapporo – May 2016)

Create a New Edition 4 of 10303-210

SC 4 approves a NP to create a new edition 4 of 10303-210 with the same scope as 10303-210 edition 3 “Electronic assembly, interconnection, and packaging design”, using the enlarged timescale, with Tom Thurman as the project leader.

Resolution 930: (Sapporo – May 2016)

Retirement of Secretariat

SC 4 expresses its great appreciation to Ken Virgil for his excellent service and hard work as Secretary to SC 4 since 2014 and wishes him the very best in his future endeavours.

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

SC 4 expresses its appreciation to our SC 4 Liaisons and the presentations that were provided:

- BuildingSMART – Thomas Liebich
- ISO/TC 59/SC 13 – Thomas Liebich
- ISO/IEC JTC 1/SC 24 – Christophe Mouton

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ACCLAMATION

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the 71st ISO/TC 184/SC 4 Opening Technical Forum in Sapporo, Japan 2016-05-23

- Martin Hardwick - Roadmap for deploying semantic GD&T in manufacturing
- Hiroshi Murayama – IEC SC3D and the CDD
- Jean Brange – Survey of STEP toolkits

ACCLAMATION

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the 71st ISO/TC 184/SC 4 AP242 Tutorial in Sapporo, Japan 2016-05-24

- Max Ungerer, ProSTEP AG
- Nicolas Figay, Airbus Group
- Jean Brange, Boost-Conseil
- Kevin Le Tutour, Boost-Conseil

ACCLAMATION

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the 71st ISO/TC 184/SC 4 Industry Day in Sapporo, Japan 2016-05-25

- Steven Vettermann, ProSTEP iViP
- Hideyuki Yoshimoto, JAMA
- Sangjin Jeong, ETRI
- Torbjörn Holm, Eurostep
- Junichi Takeuchi, FASOTEC Co., Ltd.
- Kevin Zhang, Shenzhen Pengrui Information Technology Co., Ltd.
- Nicolas Figay, Airbus Group
- Masahusa Matsui, Mitsubishi Aircraft Corporation
- Atsuto Soma, JEITA

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Larry Maggiano, Mitsutoyo America
- Takamasa Tanaka, Digital Process Ltd.

ACCLAMATION

SC 4 expresses its appreciation to Akihiko Ohtaka, Yoshihito Kikuchi, Hiroyuki Hiraoka, and the Manufacturing Science and Technology Center for hosting the 71st ISO/TC 184/SC 4 Plenary in Sapporo, Japan 2016-05-22/27

Special acknowledgement is extended to the Hokkai-Gakuen University, the Sapporo Park Hotel, and their staff for their seamless and attentive support.

Resolution 931: (Seattle – Oct 2016)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 approves the following Project Leaders and Project Editors for the associated Projects:

Project	Working Group	Project Leader/Editor
ISO 15926-10	WG 3	Paul van Exel (interim Project Leader)
ISO 15926-10	WG 3	Leo van Ruijven (co-editor)

Resolution 932: (Seattle – Oct 2016)

Appointment of Quality Committee Convenor

SC 4 appoints Hikmet Hussain as Convenor of the Quality Committee for a three year term, with appreciation for his offer to serve in that capacity.

Resolution 933: (Seattle – Oct 2016)

Reappointment of SC 4 Implementation Forum Co-Deputy Convenor

SC 4 appoints Martin Hardwick as Co-Deputy Convenor for the SC 4 Implementation Forum for a further three year term, with appreciation for his offer to serve in that

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

capacity.

Resolution 934: (Seattle – Oct 2016)

Reappointment of WG 12 Deputy Convenor

SC 4 appoints Jochen Haenisch as Deputy Convener for the WG 12 “STEP Product Modelling and Resources” for a further three year term, with appreciation for this offer to serve in that capacity.

Resolution 935: (Seattle – Oct 2016)

Appointment of SC 4 Chair-elect

SC 4 requests the Secretariat to nominate Kenny Swope from Boeing as the SC 4 Chair-elect, commencing January 2017, and to submit the required Chair nomination package to ANSI.

Resolution 936: (Seattle – Oct 2016)

Establish Ballot Duration of the New Work Item Proposal for ISO 18828-5

SC 4 requests its Secretariat to apply the 8 week ballot duration to the NP for ISO 18828-5 on Manufacturing Change Management.

Resolution 937: (Seattle – Oct 2016)

Establish Ballot Duration of the New Work Item Proposal for ISO 10303 MoSSEC

SC 4 requests its Secretariat to apply the 8 week ballot duration to the NP for ISO 10303 MoSSEC.

Resolution 938: (Seattle – Oct 2016)

Appointment of SC 4 Liaison to ISO/TC 307

SC 4 appoints Jean Brange for a three-year term as the SC 4 Liaison to ISO/TC 307 Blockchain and electronic distributed ledger technologies, with appreciation for his offer to serve in that capacity. The principal objective is to explore the potential applicability of the technology to distributed industrial data.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Resolution 939: (Seattle – Oct 2016)

IEC TC 65 Work to Adopt RAMI 4.0

SC 4 expresses concern over the IEC TC 65 work to adopt RAMI 4.0 as a PAS with undue haste contrary to the recommendation of the SAG-SM and the ISO TMB resolution 104/2016. SC 4 strongly supports the TMB proposal for a joint activity as already decided by TC 184 and invites the TMB to resolve this matter with the IEC SMB.

ACCLAMATION

SC 4 expresses its appreciation to our SC 4 Liaisons and the presentations that were provided:

- ISO/IEC JTC1/SC31 – Dan Kimball
- ISO/IEC JTC 1/SC 24 – Christophe Mouton

ACCLAMATION

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the 72nd ISO/TC 184/SC 4 Open Technical Forum in Seattle, WA USA 2016-10-03:

- Norman Swindells - Consolidation of property data for the life cycle of a composite product
- Sune Horkeby - Product Behaviour; Standards as a Framework for Life Cycle Management

ACCLAMATION

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the 72nd ISO/TC 184/SC 4 Industry Day in Seattle, WA, 2016-10-05:

- Dan Carnahan, CT 184 Consulting
- Alan Johnston, MIMOSA
- Ray Topping, Fiatech
- Martin Hardwick, STEP Tools
- Cliff Whitehead, ODVA/Rockwell Automation

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ACCLAMATION

SC 4 expresses its appreciation to Peter Benson, Brandi Fisher, and the rest of ECCMA's staff for hosting the 72nd ISO/TC 184/SC 4 Plenary in Seattle, Washington 2016-10-02/07. Special acknowledgement is extended to the Hilton Garden Inn Seattle North/Everett and its staff for their seamless and attentive support throughout the week.

ACCLAMATION

SC 4 expresses its sincere thanks to Kenny Swope and our Boeing hosts for the excellent VIP tour of the Everett production facility.

Resolution 940: (Oslo – May 2017)

Reappointment of WG 11 Convenor

SC 4 appoints David Loffredo as Convener for the WG 11 "Implementation Methods and Conformance Methods" for a further three year term, with appreciation for this offer to serve in that capacity.

Resolution 941: (Oslo – May 2017)

Reappointment of WG 03 Deputy Convenor

SC 4 appoints Nils Sandmark as Deputy Convener for the WG 03 "Oil, Gas, Process and Power" for a further three year term, with appreciation for this offer to serve in that capacity.

Resolution 942: (Oslo – May 2017)

Appointment of Liaison Officer to TC 171 and TC 171/SC 2

SC 4 appoints Kenny Swope as liaison officer to TC 171 "Document management applications" and its SC 2 "Document file formats, EDMS systems and authenticity of information" for a three year term, with appreciation for this offer to serve in that capacity.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Resolution 943: (Oslo – May 2017)

Establishment of Liaison with Web3D Consortium

SC 4 recognizes the request of the Web 3D Consortium to become a liaison and asks its Secretariat to initiate a liaison to this organization.

SC 4 nominates Prof. Soonhung HAN as the liaison officer from SC 4 to the Web3D Consortium.

Resolution 944: (Oslo – May 2017)

Move ISO 10303-239 to Enlarged Timeframe

SC 4 requests its Secretariat to apply the enlarged 48 month timeframe to the ISO 10303-239 Edition 3 project, with a DIS limit date of 2019-01-22.

Resolution 945: (Oslo – May 2017)

Move ISO 15926-10 to Enlarged Timeframe

SC 4 requests its Secretariat to apply the enlarged 48 month timeframe to the ISO 15926-10 project, with a DIS limit date of 2018-03-19.

Resolution 946: (Oslo – May 2017)

Establish Ballot Duration of the New Work Item Proposal for ISO 18828-1

SC 4 requests its Secretariat to apply the 8 week ballot duration to the NP for ISO 18828-1 on Standardized procedures for production systems engineering – Overview.

Resolution 947: (Oslo – May 2017)

Establish Ballot Duration of the New Work Item Proposal for ISO 8000-65

SC 4 requests its Secretariat to apply the 8 week ballot duration to the NP for ISO 8000-65 on Data quality - Process measurement questionnaire.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Resolution 948: (Oslo – May 2017)

Approval of Minor Revision to ISO/TS 15926-4

SC 4 requests WG 3 to prepare a new edition of ISO/TS 15926-4 with corrected URL links and incorporating Amendment 1.

SC 4 requests its Secretariat to submit the resulting document to ISO CS for FDIS ballot under the minor revision procedure.

Resolution 949: (Oslo – May 2017)

Closure of PWI Project ISO 14306 Edition 3

SC 4 closes the PWI project on ISO 14306 Edition 3.

SC 4 requests its Secretariat to circulate the resulting draft NP to WG12 for a four week final review. When completed, SC4 requests its Secretariat to circulate to National Bodies for ballot with an 8 week period.

Resolution 950: (Oslo – May 2017)

Proposed Joint Activity with JTC 1/SC 24

SC 4 notes the proposal for a NP and asks its Secretariat to request that the proposed NP be addressed through a JWG between JTC 1/SC 24 and TC 184/SC 4.

Resolution 951: (Oslo – May 2017)

Creation of ISO 15926-9 PWI for Facade Implementation

SC 4 approves a PWI to create the NWI for a new part ISO 15926-9 within WG 3, to identify methods for the exchange and integration of information between companies cooperating in the plant engineering supply chain through implementation of “facades”,

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

for delivery at the Jeju meeting.

SC 4 appoints Peter Paul Pruijn as Project Leader and invites member bodies to nominate experts to the project.

Resolution 952: (Oslo – May 2017)

Creation of preliminary work item for schematics

SC 4 approves the creation of a PWI to define an approach to schematics that will be applied across ISO TC 184/SC 4 for delivery at the spring 2018 meeting and invites member bodies to nominate experts to the project.

Resolution 953: (Oslo – May 2017)

Creation of PWI for Isogeometric Analysis and concise representation of analysis results and test data

SC 4 approves the creation of a PWI for Isogeometric Analysis and concise representation of analysis results and test data with delivery at the spring 2018 meeting and invites member bodies to nominate experts to the project.

Resolution 954: (Oslo – May 2017)

Review of DPPC Task Force Report

SC 4 notes and welcomes the report of the ISO/IEC DPPC Task Force.

SC 4 recognizes many potential use cases for consuming the content of the proposed future DPPC across multiple industry sectors, which will require various digital forms for extracting the content, including but not limited, to tables of values, OWL and CAD models.

SC 4 considers that it is vitally important to provide tools to facilitate the conversion of all product characteristics in current standards into digital form for incorporation into the proposed DPPC.

SC 4 notes the clear requirement to link the product properties to its existing product definition models, including shape representation, functionality and engineering properties.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 offers to support the further development of the requirements for the future DPPC as well as the proposed JWG to maintain the DPPC.

Resolution 955: (Oslo – May 2017)

Appointment of WG 13 Deputy Convenor

SC 4 appoints Anne Dourgnon as Convener for the WG 13 “Industrial Data Quality” for a three year term, with appreciation for this offer to serve in that capacity and thanks to Jean BRANGE’ for his service.

ACCLAMATION

SC 4 expresses its appreciation to our SC 4 Liaisons and the presentations that were provided:

- ISO/IEC JTC1/SC 24 – Christophe Mouton
- ISO/TC 171/SC 2 – Kenny Swope
- ISO/TC 307 – Jean Brange

ACCLAMATION

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the 73rd ISO/TC 184/SC 4 Open Technical Forum in Oslo, Norway 2017-05-08:

- ISO/TC 184/SC 5 – Charlotta Johnsson (Chair-elect)
- Adrian Murton – MoSSEC
- Howard Mason – BAE Systems
- Jean-Yves Delaunay – Airbus Group
- Mark Williams – The Boeing Company
- Nicolas Figay – Airbus Group



N4118

2026 – 05 – 15

ACCLAMATION

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the 73rd ISO/TC 184/SC 4 Industry Day in Oslo, Norway 2017-05-10:

- Jacob Mehus (CEO, Standards Norway)
- Jochen Haenisch (Jotne)
- Remi Lanza (NTNU/Jotne)
- Ole Fretheim (Norwegian Defence Research Establishment, FFI)
- Tor Dokken, Dr. Heidi Dahl (SINTEF)
- Mats Mareng Weum (OSL/T2, Aas-Jakobsen)
- Piotr Dabrowski, St. Olav Hospital
- Vic G. Samuel, IOGP- ISSC (Chevron)
- Anders Thostrup, USPI (Shell)
- Alan Johnston (MIMOSA)
- Robert Skaar (Statoil)
- Anders Gjerver (Aibel)
- Johan Wilhelm Klüwer (DNV GL)
- Jann Kåre Slettebakk (Aker Solutions)
- Dr. Timothy M. King (Babcock International)
- Dr. Salomon de Jager (Pilog)
- Dr. Boomann Choung (Department of Government 3.0, National Information Agency)

ACCLAMATION

SC 4 expresses its appreciation to Standards Norway for hosting the 73rd ISO/TC 184/SC 4 Plenary in Oslo, Norway 2017-05-07/12. Special thanks to Knut Aune – Standards Norway, Tor Arne Irgens – MtiK, Nils Sandsmark – POSC Caesar, and Jochen Haenisch – Jotne for co-organizing the event. Special acknowledgement is extended to the Thon Hotel Oslofjord and its staff for their seamless and attentive support throughout the week.

Resolution 956: (Jeu – November 2017)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 approves the following Project Leaders and Project Editors for the associated Projects:

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Project	Working Group	Project Leader/Editor
8000-3	WG13	Aminata Mbengue (PL)

Resolution 957: (Jeju – November 2017)

Reappointment of JWG 08 Deputy Convenor

SC 4 appoints Bob Young as Deputy Convenor for the JWG 08 “Joint ISO/TC 184/SC 4 - ISO/TC 184/SC 5 WG; Manufacturing process and management information” for a further three year term, with appreciation for his offer to serve in that capacity.

Resolution 958: (Jeju – November 2017)

Appointment of Liaison Officer to ECCMA

SC 4 appoints Peter Eales as liaison officer to ECCMA for a three year term, with appreciation for his offer to serve in that capacity.

Resolution 959: (Jeju – November 2017)

Appointment of Liaison Officer to ISO/TC 154

SC 4 appoints Torbjorn Holm as Liaison Officer to ISO/TC 154 “Processes, data elements and documents in commerce, industry and administration” for a three year term with specific interest in the Value Stream Management project, with appreciation for his offer to serve in that capacity.

Note: Peter Benson and Hiroshi Murayama are currently nominated as Liaison Officers and continue in that capacity.

Resolution 960: (Jeju – November 2017)

Establish category D Liaison with the A&D PLM Action Group

SC 4 welcomes the application of the A&D PLM Action Group as a category D Liaison to WG 12 and WG 15. SC 4 requests its Secretariat to submit its application to ISO CS for processing.

SC 4 appoints Kenny Swope as Liaison Officer for a three year term, with appreciation for his offer to serve in that capacity.



N4118

2026 – 05 – 15

Resolution 961: (Jeju – November 2017)

Establish Ballot Duration of the New Work Item for ISO 8000-116

SC 4 requests its Secretariat to apply the 8 week ballot duration to the NP for ISO 8000-116 on Authoritative Legal Entity Identifier (ALEI).

Resolution 962: (Jeju – November 2017)

Establish Ballot Duration of the New Work Item for 'Digital Twin Manufacturing Framework'

SC 4 requests its Secretariat to apply the 8 week ballot duration to the NP for 'Digital Twin Manufacturing Framework'.

Resolution 963: (Jeju – November 2017)

Establish Ballot Duration of the New Work Item for STEP geometry services

SC 4 requests its Secretariat to apply the 8 week ballot duration to the NP for STEP geometry services.

Resolution 964: (Jeju – November 2017)

Approve a NP to create a new edition 3 of ISO 10303-209

SC 4 approves a NP to create a new edition 3 of ISO 10303-209 with the same scope as ISO 10303-209 edition 2 "Multidisciplinary analysis and design", using SDT48, with Jochen Haenisch as the project leader.

Resolution 965: (Jeju – November 2017)

Establish WG 16

SC 4 requests its Secretariat to create WG 16 with the following title and scope

Title: ***Formats for Visualization and other derived forms of product data***

Scope:



N4118

2026 – 05 – 15

- Develop and maintain format and interface standards for 3D visualization of product models, including visualization of different classes of derived information such as geometry, product structure and others.
- Develop and maintain standards for consumption of data derived from and associated with product models.
- Develop and maintain standards for interfaces from SC 4 product definition standards.
- Establish liaison with other standards activities working on information models for 3D visualization.

SC 4 requests its Secretariat to assign any NP within scope to WG 16. SC 4 appoints Soonhung Han as Convenor and Christophe Mouton as Deputy Convenor each for a three year term, with appreciation for this offer to serve in that capacity.

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

SC 4 expresses its appreciation to our SC 4 Liaisons and the presentations that were provided:

- ISO/IEC JTC1/SC 24 – Christophe Mouton
- ISO/TC 171/SC 2 – Kenny Swope

ACCLAMATION

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the 74th ISO/TC 184/SC 4 Industry Day in Jeju, Korea 2017-11-08:

- Smart Manufacturing Track
 - Kenneth Swope (The Boeing Company, USA)
 - Max Ungerer (ProSTEP, Germany)
 - Donghag Choi (KATS, Korea)
 - Martin Hardwick (STEPtools, USA)
 - Meinolf Gröpper (VDMA, Germany)
 - Allison Barnard Feeney (NIST, USA)
 - Yongwoon Kim (ETRI, Korea)
- Nuclear Industry Track

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Hiroshi Seki (Hitachi, Japan)
- Yi Yong-Han (KHNP, Korea)
- Christophe Mouton (EDF, France)
- Weiqun Zhang (PlantResource Technology Co. Ltd., China)
- Jinsang Hwang (PartDB, Korea)
- Data Quality
 - Howard Mason (BAE Systems, UK)
 - Boomann Choung (National Information Society Agency, Korea)
 - JongYoul Joung (National Information Society Agency, Korea)
 - Tim King (Babcock International, UK)
 - Seokahn Han (National Information Agency Society Agency, Korea)
 - Peter Benson (ECCMA, USA)
 - Yong Kim (Public Procurement Service, Korea)
 - Sang Lee (KISTI, Korea)
 - Akihiko Ohtaka (Chairman, Japan National Committee for ISO TC 184/SC 4, Japan)
 - Sune Horkeby (Siemens, Sweden)
 - Sangsun Ahn (Money-Robo, Korea)
 - Ismael Caballero (UCLM, Spain)

ACCLAMATION

SC 4 expresses its appreciation to Korean Agency for Technology and Standards (KATS) for hosting the 74th ISO/TC 184/SC 4 Plenary in Jeju, Korea 2017-11-05/10. SC 4 also expresses thanks to the Korea STEP Center (KSTEP), KAIST, NIA and all other meeting organizers for their planning that ensured a successful meeting. Special acknowledgement is extended to the Jeju Booyoung Hotel and Resort and its staff for their seamless and attentive support throughout the week.

ACCLAMATION

SC4 expresses its appreciation to the group of individuals who prepared a tribute to Howard Mason for presentation during the social event banquet. This group consisted of: Sunho Kim, Allison Barnard Feeney, Nigel Shaw, Keith Hunten, Torbjorn Holm, Tom Thurman, Peter Benson, Matthew West, Akihiko Ohtaka, John Dunford and Tim King. SC4 also thanks the Korean hosts for preparing a gift and cake to mark the occasion for Howard and his wife.

ACCLAMATION

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 expresses its sincere appreciation to Howard Mason for 17 years of dedicated service as Chair of SC 4 with his extensive technical knowledge of Industrial Data standards, ISO Procedures and for his outstanding leadership of SC 4.

Resolution 966: (Beijing – May 2018)

Reappointment of WG 3 Convenor and IF Convenor

SC 4 appoints Paul van Exel as Convenor for the WG 3 “Oil, Gas, Process and Power” and “Implementation Forum (IF)” for a further three year term, with appreciation for his offer to serve in that capacity.

Resolution 967: (Beijing – May 2018)

Reappointment of WG 12 Convenor

SC 4 appoints Keith Hunten as Convenor for the WG 12, “STEP product modeling and resources” for a further three year term, with appreciation for his offer to serve in that capacity.

Resolution 968: (Beijing – May 2018)

Reappointment of WG 13 Convenor

SC 4 appoints Tim King as Convenor for the WG 13 “Industrial Data Quality” for a further three year term, with appreciation for his offer to serve in that capacity.

Resolution 969: (Beijing – May 2018)

Appointment of JWG 8 Convenor

SC 4 appoints Anne-Francoise Cutting-Decelle as Convenor for the JWG 08 “Joint ISO/TC 184/SC 4 - ISO/TC 184/SC 5 WG; Manufacturing process and management information” for a three year term, with appreciation for her offer to serve in that capacity.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Resolution 970: (Beijing – May 2018)

Appointment of Quality Committee Deputy Convenor

SC 4 appoints Changsoo Lee as Deputy Convenor for the Quality Committee for a three year term, with appreciation for his offer to serve in that capacity.

Resolution 971: (Beijing – May 2018)

Establish Ballot Duration of the New Work Item Proposal for Harvesting ANSI QIF

SC 4 requests its Chair to convene a group representing WG 12, WG 15, and WG 16 to review the explanatory report on QIF harvesting submitted to Change Management and recommend whether to proceed to an NP ballot. This review is expected to take four weeks and will be reported out at the next available PPC meeting.

SC 4 requests its Secretariat to apply the 8 week ballot duration to any resulting NP for harvesting the ANSI QIF standard.

Resolution 972: (Beijing – May 2018)

Establish Ballot Duration of the New Work Item Proposal for ISO 15926-14

SC 4 requests its Secretariat to apply the 8 week ballot duration to the NP for ISO 15926-14.

Resolution 973: (Beijing – May 2018)

Request 9 month extension for ISO 10303-242 ed2

SC 4 requests the ISO TMB to grant a 9 month extension to ISO 10303-242 ed2 in accordance with the attached justification.

Resolution 974: (Beijing – May 2018)

Move ISO 8000-64 to 48 Month Timeline

SC 4 requests its Secretariat to apply the 48 month development track to the ISO 8000-

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

64 project.

Resolution 975: (Beijing – May 2018)

Submit ISO 8000-110 ed2 for DIS Ballot

SC 4 notes that the recent ISO 8000-110 ed2 development has resulted in minor editorial changes with no change of scope and therefore requests ISO CS to register the project as SDT 18 and its Secretariat to submit the document straight for DIS ballot.

Target date for submission: 2019-02-28

Target date for publication: 2019-08-31

Project Leader: Peter Eales

Convenor: Tim King, WG13

Resolution 976: (Beijing – May 2018)

Development of Edition 2 of ISO 29002

SC4 approves development of ISO/DIS 29002. The document will consolidate all existing parts into a single document but will not expand the current scope of ISO 29002. The DIS will be submitted for ballot by a target date of 2019-02-28 and the IS for publication by a target date of 2019-08-31. SC 4 requests its Secretariat to initiate a call for experts and appoints Peter Eales (GBR) as the project leader.

Resolution 977: (Beijing – May 2018)

Add Experts to SC 4 Quality Committee

SC 4 requests each of its WG Convenors to nominate at least one of its experts to the SC 4 Quality Committee and to invite national bodies to review their membership on the Quality Committee.

Resolution 978: (Beijing – May 2018)

Proposed SC 4 Scope Change

SC 4 requests TC 184 to note the attached proposed draft new scope statement which

Dana Tripp

Secretary, ISO/TC 184/SC 4

trippdsc4@gmail.com



N4118

2026 – 05 – 15

will be subject to CIB after this meeting.

Resolution 979: (Beijing – May 2018)

Committee Innovation Proposal for ISO 15926-4

SC 4 agrees to support the Committee Innovation Proposal for ISO 15926-4. SC 4 requests its Secretariat to submit the proposal to ISO CS for consideration. SC 4 appoints Paul van Exel as project leader.

Resolution 980: (Beijing – May 2018)

Amend Title of ISO 15926-4 ed2

SC 4 agrees to amend the title of ISO 15926-4 ed2 from “Initial reference data” to “Core reference data”.

Resolution 981: (Beijing – May 2018)

Establish Category A Liaison with Small Business Standards

SC 4 welcomes the application of the Small Business Standards organization (www.sbs-sme.eu) as a category A Liaison. SC 4 requests its Secretariat to submit their application to ISO CS for processing.

SC 4 appoints Jochen Haenisch as Liaison Officer for a three year term, with appreciation for his offer to serve in that capacity.

Resolution 982: (Beijing – May 2018)

Appointment of Liaison Officer to ISO/TC 59/SC 13

SC 4 appoints Torbjorn Holm and Jean Brange as Liaison Officers to ISO/TC 59/SC 13 “Organization and digitization of information about buildings and civil engineering works, including building information modelling (BIM)” for a three year term.

Resolution 983: (Beijing – May 2018)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Approve Application for International Code Designator (To support application from KOIOS Master Data Limited for an International Code Designator)

SC 4 has reviewed the application from KOIOS Master Data Limited to be assigned an International Code Designator and agrees to act as the “sponsoring authority.”

Resolution 984: (Beijing – May 2018)

Additional Call for Experts and First JWG 24 Meeting

SC 4 requests its TPM to coordinate with other ISO TPMs to identify all TCs/SCs that may have interest in submission of their standards for inclusion in the IEC CDD, including TC 29, TC 59/SC13, TC131, and TC 172 and to invite the nomination of experts to JWG 24 by 2018-07-13. SC 4 also recommends the Convener of JWG 24 to hold its first JWG meeting at an appropriate date and location, either through face to face and/or teleconference, considering the nationalities and availability of the nominated experts from the call for experts mentioned above, and report the result to the SC 4 Chicago meeting to be held November 4-9, 2018.

Resolution 985: (Beijing – May 2018)

Request for digital twin manufacturing examples

SC4 invites its member countries, liaisons and other interested parties (i.e. OPC and MTConnect) to submit digital twin manufacturing examples, standards, technologies and use cases to the Convenor of WG15 (hardwick@steptools.com) by 2018-08-27 for consideration by WG15 at its special meeting in September.

ACCLAMATION (Beijing – May 2018)

SC 4 expresses its appreciation to our SC 4 Liaisons and the presentations that were provided:

- ISO/TC 171/SC 2 – Written Report Submitted
- ISO/TC 37/SC3 – Written Report Submitted
- ISO/TC 59/SC13 – Written Report Submitted

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- ISO/TC 251 – Written Report Submitted
- ISO/IEC JTC 1/SC24 – Christophe Mouton

ACCLAMATION (Beijing – May 2018)

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the 75th ISO/TC 184/SC 4 Open Technical Forum in Beijing, China 2018-05-21:

- Overview: Soonhung Han
- WG 3: Paul van Exel / David Leal
- WG 12: Jean Brange
- WG 13: Tim King / Peter Eales
- WG 15: Martin Hardwick

ACCLAMATION (Beijing – May 2018)

SC 4 expresses its appreciation to Beijing Research Institute of Automation for Machinery Industry (RIAMB), China Association for Mechatronics Technology and Application (CAMETA), and the Standardization Administration of China (SAC) for hosting the 75th ISO/TC 184/SC 4 Plenary in Beijing, China during 2018-05-21/25. Special acknowledgement is extended to the Jiuhua Resort & Convention Center and its staff for their seamless and attentive support throughout the week.

Resolution 986: (Chicago – Nov 2018)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 approves the following Project Leaders and Project Editors for the associated Projects:

Project	Working Group	Project Leader/Editor
ISO 15926-10	WG 3	Leader: Paul Van Exel
ISO 15926-10	WG 3	Editors: Leo van Ruijven, Hiroshi Okada
ISO 15926-6ed2	WG 3	Leader: Paul Van Exel
ISO 10303-1	WG 12	Leader: Keith Hunten

Resolution 987: (Chicago – Nov 2018)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Reappointment of WG 15 Convenor and Deputy Convenor

SC 4 appoints Martin Hardwick as Convenor and Bengt Olsson as Deputy Convenor for the WG 15 “Digital Manufacturing” for a further three year term, with appreciation for their offers to serve in those capacities.

Resolution 988: (Chicago – Nov 2018)

Reappointment of WG 22 Convenor

SC 4 appoints Nils Sandmark as Convenor for the WG 22, “Reference data validation team” for a further three year term, with appreciation for his offer to serve in that capacity.

Resolution 989: (Chicago – Nov 2018)

Appointment of WG 23 Convenor

SC 4 expresses its appreciation and gratitude to Aminata Mbengue for her service as Convenor of WG 23 since its inception. SC 4 wishes her all success in future endeavours. SC 4 appoints Tim King as Convenor for the WG 23 “Vocabulary validation team” for a three year term, with appreciation for his offer to serve in that capacity.

Resolution 990: (Chicago – Nov 2018)

Move ISO 10303-238ed2 to 48 Month Timeline

SC 4 requests its Secretariat to move the ISO 10303-238ed2 project from 36 months to the 48 month development track.

Resolution 991: (Chicago – Nov 2018)

Remove ISO 8000-3 from the ISO work programme

SC 4 cancels the project 60784 for ISO/NP TS 8000-3.

Resolution 992: (Chicago – Nov 2018)

Correct title of ISO 8000-116

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 approves change of the title “Data quality – Part 116: Application of ISO 8000-115 to the formatting of Authoritative Legal Entity Identifiers (ALEI) for individuals and organizations” to “Data quality — Part 116: Master data: Exchange of quality identifiers: Application of ISO 8000-115 to authoritative legal entity identifiers” and requests its Secretariat to contact ISO CS to update the ISO work programme.

Resolution 993: (Chicago – Nov 2018)

Establish Ballot Duration of the New Work Item Proposal for 8000-81

SC 4 requests its Secretariat to apply the 8 week ballot duration to the NP for 8000-81, Data quality assessment based on data profiling.

Resolution 994: (Chicago – Nov 2018)

Establish Ballot Duration of the New Work Item Proposal for 10303-59ed3

SC 4 requests its Secretariat to apply the 8 week ballot duration to the NP for 10303-59ed3, Integrated generic resource: Quality of product shape data.

Resolution 995: (Chicago – Nov 2018)

Establish Ballot Duration of a future New Work Item Proposal for 14306ed3

SC 4 requests its Secretariat to apply the 8 week ballot duration to the future NP for 14306ed3, JT file format specification for 3D visualization following a successful review by JWG 16 for scope per action item 195 from the Secretariat.

Resolution 996: (Chicago – Nov 2018)

Establish Ballot Duration of the New Work Item Proposal for 10303 parts (-15, -16, -17)

SC 4 requests its Secretariat to apply the 8 week ballot duration to the following NPs for 10303:

- 10303-15 (SysML XMI to XSD binding)
- 10303-16 (SysML XMI to EXPRESS binding)
- 10303-17 (EXPRESS to SysML XMI binding)



N4118

2026 – 05 – 15

Resolution 997: (Chicago – Nov 2018)

Establish PWI for “Industrial Requirements for Product Data Visualization”

SC 4 launches a PWI to update its requirements for visualization and to create a Technical Report on “Industrial Requirements for Product Data Visualization”. SC 4 requests its Secretariat to invite member bodies to nominate experts to JWG 16 to undertake this preliminary work.

Resolution 998: (Chicago – Nov 2018)

Submission of Quality Information Framework for fast track ballot

SC 4 requests its Secretariat to submit QIF 3.0, together with its harvesting explanatory report and the mapping between STEP and QIF MBD for DIS ballot under the fast track procedure.

Resolution 998: (Chicago – Nov 2018)

Request for digital twin manufacturing examples

SC 4 invites its member countries, liaisons, standards development organizations, and other interested parties (e.g. OPC and MTConnect) to submit digital twin manufacturing examples, standards, technologies and use cases to the Convenor of WG 15 (hardwick@steptools.com).

Resolution 999: (Chicago – Nov 2018)

Development of Edition 2 of ISO 10303-242

SC 4 agrees that the following points specified by WG 12 for the way forward with AP 242ed2 DIS and the extended architecture specification shall be used as a basis for the second AP 242ed2 DIS ballot:

1. Agreement to retain ISO/TS 10303-4442 in the standard
2. Agreement to continue to provide an EXPRESS representation for the Integrated Resources, AIM, ARM and Domain Model
3. Agreement to retain EXPRESS



N4118

2026 – 05 – 15

4. Agreement to use SysML as the modelling language for AP 242ed2 domain model in ISO/TS 10303-4442
5. Agreement to re-ballot the standing documents N3210, N3211, and N3212 with a revision to focus the documents only on the specific materials required to support AP 242ed2 and ISO/TS 10303-4442. All materials that speak to the future state of the extended architecture will be moved to a technical report and addressed by an ad hoc group on the strategy for extended architecture with Allison Barnard Feeney as chief editor. SC 4 invites members from each NSB to join and contribute to the refinement of the content.
6. Agreement to create SMRL V8 with the required updates that support AP 242ed2.
7. Agreement to create one or more NPs that address the implementation of SysML as a modelling solution into the internal procedures of SC 4. This work will primarily impact WG 11 and coordination has taken place at the Chicago meeting to define the work required by WG 11.
8. Acknowledgement that the strategy for extended architecture contained in the technical report will require further action by the ad hoc group as there exists a strong dependency between the strategy and the implementation of follow on capabilities for AP 239ed3 and AP 243.
9. Acknowledgement that future releases of extended architecture capabilities will require update and revision of the standing documents noted. These standing documents shall be balloted incrementally as changes are declared.
10. Acknowledgement that the current priority is to resolve documented comments against the ballot content for AP 242ed2. Any discovery or comments not formally declared will be noted as a bug in the bug tracking system against a future edition.

Resolution 1000: (Chicago – Nov 2018)

Transfer ISO 22745 and ISO 29002 to WG 13

SC 4 requests its Secretariat to coordinate with ISO/CS and transfer ISO 22745 and ISO 29002 and all their parts from WG 2 to WG 13. SC 4 welcomes the interest of IEC SC3D in the joint development of the new edition of ISO 29002, and requests its Secretariat to reactivate its existing liaison to TC 3 to include SC3D.

Resolution 1001: (Chicago – Nov 2018)

Disband WG 02 and allocate the work programme to active WGs

SC 4 assigns ISO 13584 to JWG 24 and SC 4 disbands WG 2, thanking its experts for

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

their contribution. SC 4 requests the IEC SC 3D to concur.

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

SC 4 expresses its appreciation to our SC 4 Liaisons and the presentations that were provided:

- TC 68/SC 8 (LEI) and ECCMA, Peter Benson
- TC 171/SC 2, Kenny Swope
- JTC 1/SC 24 and Web3D Consortium, Christophe Mouton
- Energistics, Jay Hollingsworth
- TC 37/SC 3, Written Report Submitted
- TC 59/SC 13, Written Report Submitted
- TC 67, Written Report Submitted
- ISO/TC 172/SC 1, Written Report Submitted
- TC 251, Written Report Submitted

ACCLAMATION

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the 76th ISO/TC 184/SC 4 Open Technical Forum on in Chicago, IL USA 2018-11-05:

- ISO 8000-2, Tim King
- Extended Architecture, Jean Brange
- ISO 15926, Lillian Hella
- PLCS Library, Jonas Rosen
- ISO 10303 & StepMod, Keith Hunten

ACCLAMATION

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the 76th ISO/TC 184/SC 4 Industry Day on in Chicago, IL USA 2018-11-05:

- Jiwan Hayre, Boeing
- Kenny Swope, Boeing
- Bob Deragish, Parker Aerospace
- Martin Hardwick, StepTools

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Jochen Haenisch, Jotne
- Moneer Helu, NIST
- Curtis Brown, DMSC/QIF
- Chris Johnson, Lockheed Martin
- Peter Benson, ECCMA
- Tim King, Babcock International
- Ashutosh Agarwal, Uptake
- Dan Carnahan
- Peter Eales, MRO Insyte
- Van Bicknell
- Daryl Crockett, ValidDatum

ACCLAMATION

SC 4 expresses its appreciation to Peter Benson, Brandi Fisher, and the rest of ECCMA's staff for hosting the 76th ISO/TC 184/SC 4 Plenary in Chicago, IL USA during 2018-11-05/09. Special acknowledgement is extended to UI Labs and its staff for their seamless and attentive support throughout the week.

Resolution 1002: (Toulouse – May 2019)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 approves the following Project Leaders and Project Editors for the associated Projects:

Project	Working Group	Project Leader/Editor
ISO 8000-65	WG 13	Leader: Jean Brange
ISO TR 15926-14	WG 3	Leader: Nils Sandsmark
ISO 8000-63	WG 13	Leader: Peter Hoehn
ISO 10303-113	WG 12	Leader: Tom Thurman
ISO 15926-4	WG 3	Leader: Paul van Exel Editor: David Leal
ISO 23247-1	WG 15	Leader: Jin Young Lee
ISO 23247-2	WG 15	Leader: Sangkeun Yoo

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ISO 23247-3	WG 15	Leader: Hyunjeong Lee
ISO 23247-4	WG 15	Leader: Sung Hei Kim
ISO 10303-235	WG 12	Leader: Norman Swindells
ISO 15926-6	WG 3	Leader: Paul van Exel

Resolution 1003: (Toulouse – May 2019)

Cancellation of ISO 8000-63

SC 4 requests its Committee Manager to cancel ISO 8000-63 and, after receiving the appropriate documentation from WG 13, launch an NP Ballot of ISO 8000-63 to proceed to stage 40.00.

Resolution 1003: (Toulouse – May 2019)

Conditional Cancellation and relaunch of ISO 10303-242 ed2

In the event the project team is unable to achieve the final WG 21 Change Request Validation Ballot start date of 2019-06-12, SC 4 requests its Committee Manager to cancel ISO 10303-242 ed2. SC 4 requests its Committee manager to initiate an 8-week NP ballot to restart the project at stage 40.00. SC 4 further instructs the ISO 10303-242 ed2 project team to retain all project materials for use in the future new work item proposal.

Resolution 1004: (Toulouse – May 2019)

Cancellation of ISO 10303-239 ed3

Noting the schedule for ISO 10303-239 ed3, the status from the project team, and dependencies on other projects in the committee, SC 4 requests its Committee Manager cancel project ISO 10303-239 ed3. SC 4 instructs the project team to preserve existing project materials for future use.

Resolution 1005: (Toulouse – May 2019)

Establish PWI for ISO 10303-239ed3

Noting the contribution of previous work on ISO 10303-239 ed3 and the gaps that



N4118

2026 – 05 – 15

remain to be filled in the standard, SC 4 requests its Committee Manager to establish a PWI for ISO 10303-239ed3. SC 4 encourages its member bodies to nominate experts to the project.

Resolution 1006: (Toulouse – May 2019)

Creation of a Committee Innovation

Noting the project results from the Committee Innovation project to evaluate Jira and GIT and the positive outcome, SC 4 requests its Committee Manager submit a Committee Innovation proposal to the Living Lab task force requesting a committee innovation to evaluate continuous integration methods.

Resolution 1007: (Toulouse – May 2019)

Establish Category A Liaison with ASM International

SC 4 welcomes the application of ASM International as a category A Liaison. SC 4 requests its Committee Manager to submit their application to ISO CS for processing.

Resolution 1008: (Toulouse – May 2019)

Move ISO 10303-243 to 48-month timeline

SC 4 requests its Committee Manager to apply the 48-month development track to the ISO 10303-243 project.

Resolution 1009: (Toulouse – May 2019)

Request 9-month extension for ISO 10303-210ed4

SC 4 requests the ISO TMB to grant a 9-month extension to ISO 10303-210 ed4 in accordance with the justification provided.

Resolution 1010: (Toulouse – May 2019)

Establish PWI for Digital Validation

SC 4 establishes a PWI to create a Technical Report on Digital Validation by effective use of simulation. SC 4 encourages its member bodies to nominate experts to this

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

preliminary work.

Resolution 1011: (Toulouse – May 2019)

Establish PWI on Visualization elements of digital twins

SC 4 establishes a PWI to create a Technical Report on Visualization elements of digital twins. SC 4 encourages its member bodies to nominate experts to this preliminary work.

Resolution 1012: (Toulouse – May 2019)

Convert ISO/PAS 17506 (COLLADA) to IS

Noting the document as a fundamental technology of AutomationML (IEC 62714 in IEC/TC 65/SC 65E), SC 4 resolves to convert the ISO/PAS 17506 to an IS with no change in scope with target dates: DIS 2020-06-01, IS 2021-06-01. SC 4 assigns the work to JWG 16 and requests ISO CS to preserve the project and prevent cancellation while JWG 16 confirms the project leader and the Committee Manager reports the leader to ISO CS. The organization originally responsible for the format is invited to collaborate with the development project to formulate the development strategy including how to cope with new versions of the standard.

Resolution 1013: (Toulouse – May 2019)

Nomination of SC 4 Liaison to JTC 1 Digital Twin AG

SC 4 acknowledges the report of JTC 1 N14262 and Resolution 20 and nominates Sangkeun Yoo to be a liaison to the digital twin advisory group in accordance with membership category 2. SC 4 recommends that TC 184 also nominate a representative to the advisory group.

Resolution 1014: (Toulouse – May 2019)

Correct Title of ISO 8000-81

SC 4 requests its Committee Manager to inform ISO CS that the new title of ISO 8000-81 is "Data quality -- Part 81: Data quality assessment: Profiling".

Resolution 1015: (Toulouse – May 2019)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Establish Ad Hoc Group on Core industrial data set of terms

SC 4 establishes an Ad Hoc Group on Core industrial data set of terms and appoints Nils Sandmark as Convenor to lead the team of Tor Arne Irgens, David Leal, Hiroshi Murayama, Allison Barnard Feeney, Phil Spiby, Jeong Sam Yang, and Matthew West. SC 4 invites other experts to participate in the group.

Resolution 1016: (Toulouse – May 2019)

Nomination of SC 4 Liaison to TC 68/SC 8 and TC 68/SC 9

SC 4 requests liaison to ISO/TC 68/SC8 and SC9 and appoints Peter Benson as liaison officer from ISO/TC 184/SC 4. SC 4 invites ISO/TC 68 SC 8 and SC 9 to appoint liaisons to ISO/TC 184/SC 4 in order for them to participate in the work in WG13.

Resolution 1017: (Toulouse – May 2019)

Place JWG 8 on Standby

Recognizing that all work is complete in JWG 8, SC 4 requests its Committee Manager to notify SC 5 and ISO CS of the same and place JWG 8 on standby. SC 4 expresses thanks to Anne-Françoise Cutting-Decelle for her leadership and to the experts that contributed to the success of the work programme.

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

SC 4 expresses its appreciation to our SC 4 Liaisons and the presentations that were provided:

- JTC 1/SC 24 and Web3D Consortium, Christophe Mouton
- TC 37/SC 3, Written Report
- TC 59/SC 13, Written Report
- TC 59/SC 13/JWG 12, Written Report
- TC 171/SC 2, Written Report
- TC 172/SC 1, Written Report

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ACCLAMATION

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the Open Technical Forum:

- Brief of JETI (JTC 1 Emerging Technology and Innovation) TTR (technology trend report) on Digital Twin - Sangkeun YOO (KATS/ETRI)
- Towards Construction of Innovative Digital Validation Environment - Masatomo INUI, Prof. Ibaraki Univ
- Digital Twin for Smart Facility: Variables for Integrated Asset Management - Youngsoo Jung
- Digital Thread / Digital Twin concepts in France - Jean Brange
- Digital Twin for Manufacturing framework - Martin Hardwick
- Visualization elements of digital twins – Soonhung Han
- Digital Transformation at AIA – Howard Mason

ACCLAMATION

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the Industry Day event:

Opening:

- Thierry COTELLE, Région Occitanie / Pyrénées-Méditerranée
- Bertrand SERP, Toulouse Métropole
- Patrick LAMBOLEY, ISO/TC 184, Schneider Electric
- Kenneth SWOPE, ISO/TC 184/SC 4, Boeing

Session 1:

- Karl DEVILLIERES, Airbus
- Philippe BOERI, AFIS, Naval Group
- Yves BAUDIER, AFNET

Session 2:

- Kenneth SWOPE, Boeing
- Jean-Pierre SOUZY, Airbus

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

- Jean-Yves DELAUNAY, Airbus

Session 3:

- Convenor : Yves KERARON, ISADEUS
- Jean-François HAMELIN, EDF
- Jean-Charles LECLERC, TOTAL
- Alain BAUDRY, EGIS

Round table 1:

- Chair: Timothy KING, Babcock International
- Rebecca TOTH TOBLER, EDF
- Christophe BROSSE, Airbus
- Christophe CASTAING, EGIS

Round table 2:

- Chair: Kenneth SWOPE, Boeing
- Olivier MARCHAND, ISO/TC 85, EDF
- Laurent GUISE, IEC, Schneider Electric
- Jean-Yves DELAUNAY, Airbus
- Etienne CAILLEAU, AFNOR
- Christophe CASTAING, EGIS

Concluding remarks:

- Alban SCHMUTZ, OVH

ACCLAMATION

SC 4 expresses its appreciation to the following individuals for organizing the 77th ISO/TC 184/SC 4 Plenary in Toulouse, France during 2010-05-13/17:

- Alain Costes, Isabelle Rimbert, Stéphane Moulière, Caroline de Condé, Martine Degardin, Melissa Jean, and Roxana Turcanu – AFNOR Normalisation
- Christophe Mouton – Chair of French Mirror Committee to ISO/TC 184, SC 4 and SC 5

Special acknowledgement is extended to Hotel Mercure Toulouse Centre Compans and its staff for their seamless and attentive support throughout the week.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ACCLAMATION

SC 4 expresses its sincere thanks to the following organizations for their support and sponsorship of the meeting and weekly events:

- EDF
- Dassault Systèmes
- Région Occitanie / Pyrénées-Méditerranée

ACCLAMATION

SC 4 expresses its thanks to Airbus for the excellent Airbus A380 assembly line tour and dinner afterwards.

ACCLAMATION

SC 4 expresses its sincere appreciation to Howard Mason for 35 years of dedicated service and consummate leadership of ISO/TC 184/SC 4. During this time, Howard spent 17 years as Chairman of SC 4, leading numerous accomplishments and innovations in industrial data standardization. This included the 2007 ISO Lawrence D. Eicher Award recognizing unmatched excellence in standards development. SC 4 wishes all the best to Howard in his next journey and future endeavours.

Resolution 1018: (Marina del Rey – Nov 2019)

Approval of Project Leaders for SC 4 Projects

SC 4 approves the following Project Leaders and Project Editors for the associated Projects:

Project	Working Group	Project Leader/Editor
ISO/NP TS 15926-11 ed 2	WG 3	Leader: Paul van Exel
ISO/NP 8000-51	WG 13	Leader: Peter Benson
ISO/NP 8000-82	WG 13	Leader: Sungjoon Lim
ISO/PRF TS 10303-1021	WG 12	Leader: Jean Brangé
ISO/AWI TS 10303-4000	WG 12	Leader: Jean Brangé
ISO/PWI TR 24463	WG 12	Leader: Masatomo Inui

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ISO/CD TR 24464	JWG 16	Leader: Soonhung Han
ISO/NP 3151	JWG 16	Leader: Soonhung Han

Resolution 1019: (Marina del Rey – Nov 2019)

Appointment of Quality Committee Convenor

SC 4 appoints Véronique Dubillot as the Convenor for the Quality Committee for a three- year term, with appreciation for her offer to serve in that capacity. SC 4 also wishes to express thanks to Véronique Dubillot for her prior service as the secretary for the Quality Committee.

Resolution 1020: (Marina del Rey – Nov 2019)

Appointment of Quality Committee Secretary

SC 4 appoints Melissa Harvey as the Secretary of the Quality Committee for a three-year term, with appreciation for her offer to serve in that capacity.

Resolution 1021: (Marina del Rey – Nov 2019)

Reappointment of WG 12/21 Deputy Convenor

Noting that continuity of operations and management of work is an important consideration to the successful operation of large working groups, SC 4 appoints Jochen Haenisch as Deputy Convenor for the WG 12 “STEP Modeling Product and Resources” and WG 21 “SMRL validation team” for a further three-year term, with appreciation for his offer to serve in that capacity.

Resolution 1022: (Marina del Rey – Nov 2019)

Appointment of Liaison Officer to NAFEMS

Noting that coordination with external liaisons is an important consideration to the successful development and implementation of standards, SC 4 appoints David Leal for a three-year term, with appreciation for his offer to serve in that capacity.

Resolution 1022: (Marina del Rey – Nov 2019)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Establish Category A Liaison with MTConnect Institute.

Having been informed by ISO CS that MTConnect Institute meets the requirements of a Category A Liaison, SC 4 welcomes the application of MTConnect as a Category A Liaison and requests its Committee Manager to submit their application to ISO CS for final processing.

SC 4 appoints Dr. Moneer Helu as Liaison Officer for a three-year term, with appreciation for his offer to serve in that capacity.

Resolution 1023: (Marina del Rey – Nov 2019)

Move ISO15926-14 to 24-month timeline

Upon reviewing the work plan of ISO 15926-14, SC 4 recognizes the progress made to date and requests its Committee Manager apply the 24-month development track to the ISO 15926-14 project in order to complete required deliverables for the next stage of work.

Resolution 1024: (Marina del Rey – Nov 2019)

Extend the term of the Ad Hoc Group onCore Industrial Data Set of Terms

Noting the progress made by AHG 1 “Core Industrial Data Set of Terms” and the additional work needed to develop recommendations for inclusion in their final report, SC 4 extends the term for AHG 1 to the next meeting where a final report will be provided.

Resolution 1025: (Marina del Rey – Nov 2019)

Create a Task Force for SMRL

Noting the workshop held 29 – 30 October, 2019 between ISO CS and SC 4, a significant opportunity to enhance efficiency and improve quality has emerged regarding the publication of the ISO 10303 STEP Modular Reference Library. SC 4 creates a Task Force titled “ISO 10303 SMRL Architecture innovation” with the following term of reference:

1. Taking into account the workshop report out and phasing plan proposal, develop

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- a detailed plan with recommendations for implementation;
- 2. Coordinate with other affected groups in SC4 on plan elements;
- 3. Provide periodic status reports to SC 4/PPC;
- 4. Deliver a final report to SC 4 on a schedule agreeable between the Task Force and SC4.

SC 4 nominates Brandon Sapp to lead the Task Force and requests nominations from member bodies and relevant liaisons to participate in the Task Force.

Resolution 1026: (Marina del Rey – Nov 2019)

Establish PWI on ontology for geometry and topology

SC 4 establishes a PWI to create a Technical Report on Ontology for geometry and topology.

The report is to be delivered in May 2020 for review at the next SC 4 meeting.

SC4 nominates Ben Urick (US) as project leader with appreciation to serve in this capacity and requests member bodies and liaisons to nominate experts to this preliminary work.

Resolution 1027: (Marina del Rey – Nov 2019)

Cancellation of ISO 8000-64

Noting that the existing 48-month project may no longer receive a 9-month extension from ISO TMB, SC 4 requests its Committee Manager to cancel ISO 8000-64 and, after receiving the appropriate documentation from WG 13, launch an NP Ballot of ISO 8000-64 with the document attached to the ISO Form 4 to be registered as a Committee Draft (stage 30.00) and to apply the 8-week ballot duration to that ballot.

In addition, SC 4 requests the WG 13 to deliver the NP/CD ballot to the committee manager by 2020-01-15.

Resolution 1028: (Marina del Rey – Nov 2019)

Cancellation of ISO 8000-110 Edition 2 project

SC4 requests WG13 to prepare a technical report outlining the steps necessary to

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

combine existing (and any future) standards in the ISO 8000 series of parts relating to master data and names Peter Benson as project leader. This technical report will allow the national bodies to prepare their own publication strategies for future editions of these parts.

SC4 requests member bodies and liaisons to nominate experts to this work.

Resolution 1029: (Marina del Rey – Nov 2019)

Review of ISO/DIS 29002

SC 4 requests its Committee Manager to ask member National Bodies and SC 4 Working Group Convenors to identify experts to contact Peter Eales, project leader of ISO 29002 Edition 2, and Tim King, Convenor of WG 13, to volunteer their availability to participate in review of the draft of ISO/DIS 29002 from 2020-03-02 to 2020-05-01.

Resolution 1030: (Marina del Rey – Nov 2019)

Formal documentation of the mapping between STEP and QIF to harmonize terms and definitions

SC 4 requests its Committee Manager launch a PWI/TR to develop a Technical Report formally documenting the correspondence between the entities in ISO 10303 and ISO 23952 using a mapping language. The following persons did volunteer as members of the project team:

Mapping expert: David Loffredo
Mapping expert: Kevin Le Tutour
Geometry Expert: Ben Urick
Geometry Expert: Victor Mikushin
QIF expert: Curtis Brown
QIF expert: Mark Nielsen
STEP expert: Pierre Duchier
STEP expert: Allison Barnard Feeney
End user expert: Atsuto Soma
End user expert: Martin Hardwick

SC 4 nominates Dr. Martin Hardwick as project leader and Mr. Atsuto Soma as editor

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

with appreciation to support in this capacity and requests member bodies and liaisons to nominate experts to this preliminary work.

Resolution 1031: (Marina del Rey – Nov 2019)

Initiate CIB for SC 4 Committee Specific Procedure

SC 4 requests the Committee Manager to circulate a Committee Internal Ballot with a draft of the ISO Committee Specific Procedure (CSP) for comment and invite relevant ISO committees to join in liaison to JWG24 for development of the process and management of the CDD content.

SC 4 invites IEC to comment through its membership to JWG24.

Resolution 1032: (Marina del Rey – Nov 2019)

Reestablish Liaison Officers to IEC SC 3D

SC 4 nominates Yoshiaki Sonoda to serve as liaison officer to IEC SC 3D for a three-year term, with appreciation for his offer to serve in that capacity.

Resolution 1033: (Marina del Rey – Nov 2019)

Leadership Changes for SC 4 Implementation Forum

SC 4 appoints Mario Tucci as Deputy Convenor of SC 4/AG 02 “Implementation Forum” for a three-year term, with appreciation for his offer to serve in that capacity. SC 4 also wishes to express thanks to Anna Moreno for her prior service on the Implementation Forum.

Resolution 1034: (Marina del Rey – Nov 2019)

Ad Hoc Group to draft NWI Nuclear Digital Ecosystem Specification

Noting the interest in developing a TR assessing the nuclear industry needs for information standards, SC 4 request its Committee Manager launch an Ad hoc team with following participants:

- Christophe Mouton, EDF, France

Dana Tripp

Secretary, ISO/TC 184/SC 4

trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Yves Keraron, Isadeus, France
- Hiroshi Seki, Hitachi, Japan
- Yoshiaki Sonoda, Mitsubishi Heavy Industries, Japan'
- Soonhung Han, KAIST, South Korea
- Youngsoo Jung, Myongji University South Korea
- Kevin Zhang, Plant Resource Technology, China
- Resources from other TC's as required.

SC 4 appoints Christophe Mouton as project leader with appreciation to serve in that capacity, and requests that Christophe coordinate with ISO/TC 85 with the opportunity to contribute to that report. The Ad hoc team will have regular teleconferences and report bi-monthly to the IF Convenor and at the TC 184/SC 4 meeting in Bergen, Norway in May 2020.

Resolution 1035: (Marina del Rey – Nov 2019)

Call for requirements for data visualization in the context of industrial artefacts

SC 4 calls SC 4 WG conveners and liaison organizations to deliver to JWG 16, their use cases, examples, and requirements about data visualization in the context of industrial artefacts before end January 2020.

JWG 16 will consolidate them for end of February into a DTR that will be submitted through a ballot so that the comments will be discussed at the next SC 4 plenary in Bergen and be published in 2020.

Resolution 1036: (Marina del Rey – Nov 2019)

CD Ballot duration of ISO 10303-59 ed3

SC4 requests its Committee Manager to make the CD ballot duration of ISO 10303-59 ed3 to eight weeks.

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

SC 4 expresses its appreciation to our SC 4 Liaisons and the presentations that were provided:

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

 Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

- | | |
|-------------------------------|-------------------|
| • ECCMA | Peter Eales |
| • ECCMA | Peter Benson |
| • ISO/IEC JTC 1/SC 24 & Web3D | Christophe Mouton |
| • TC 37/SC 3 | Written |
| • TC 59/SC 13 | Written |
| • TC 172/SC 1 | Written |
| • TC 171/SC 2 | Written |

ACCLAMATION

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the Open Technical Forum:

- SMRL Workshop Report Out: Kenny Swope
- Modelling Language of the SC: Jean Brange'
- Ad Hoc Group on Ontology: David Leal
- Common Ontology for Industrial Data Standards: Yves Keraron
- UML for 15926: Paul van Exel

ACCLAMATION

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the 78th SC 4 Industry Day event:

Opening:

- Kenneth Swope, The Boeing Company

Sponsor:

- Jack Harris, PDES, Inc.

Presenters:

- Dr. Micheal Grieves, Florida Institute of Technology
- Dr. John Dong, The Boeing Company
- Dr. Martin Hardwick, STEP Tools
- Marion Toussaint, NIST

Panel Discussion:

- Dr. Sangkeun Yoo, ETRI

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

- Paul Nelson, Northrop Grumman Corporation
- Alan T. Johnston, MIMOSA

Facilitators:

- Melissa Harvey, The Boeing Company
- Brandon Sapp, The Boeing Company

ACCLAMATION

SC 4 expresses its appreciation to Peter Benson, Brandi Fisher, and the rest of ECCMA's staff for organizing the 78th SC 4 Plenary in Marina del Rey, CA, USA during 2010-11-03/08:

ACCLAMATION

Special acknowledgement is extended to Hotel MdR and its staff for their seamless and attentive support throughout the week.

Resolution 1037: (Virtual Meeting – May 2020)

Approval of Project Leaders for SC 4 Projects

SC 4 approves the following Project Leaders and Project Editors for the associated Projects:

Project		Working Group	Project Leader/Editor
ISO/AWI 17506		JWG16	Project leader: Neil TREVETT ntrevett@nvidia.com Project Editor: Hwanyong LEE hwanyong.lee@gmail.com

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Resolution 1038: (Virtual Meeting – May 2020)

To establish reduced ballot period for New Proposal.

SC 4 requests its Committee Manager to apply the 8-week ballot duration to the following NP ballots:

Project	Approval
NP ISO 10303-18	☒
NP ISO 10303-2	☒
NP ISO 8000-117	☒

Resolution 1039: (Virtual Meeting – May 2020)

Appointment of Deputy Convenor to WG 3

SC 4 appoints Nils Sandmark as the Deputy Convenor for WG 3 for a three-year term, with appreciation for his offer to serve in that capacity.

Resolution 1040: (Virtual Meeting – May 2020)

Appointment of Deputy Convenor to WG 13

SC 4 appoints Anne Dourgnon as the Deputy Convenor for WG 13 for a three-year term, with appreciation for her offer to serve in that capacity.

Resolution 1041: (Virtual Meeting – May 2020)

Appointment of Convenor to WG 11

SC 4 appoints Dave Loffredo as Convenor for WG 11 for a three-year term, with appreciation for his offer to serve in that capacity.

Resolution 1042: (Virtual Meeting – May 2020)

To extend the term of AHG 1.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Noting the progress made by AHG 1 “Core industrial data set of terms” and the further work needed, SC 4 extends the term of the AHG 1 and requests its final report at the 80th Plenary meeting of SC 4. AHG 01 is requested to provide a plan for the further work itemized in the introduction to this resolution.

Resolution 1043: (Virtual Meeting – May 2020)

To extend the term for AHG 2 “Nuclear Digital Ecosystem Specification” to allow the group to complete this work

Noting the progress made by AHG 2 “Nuclear Digital Ecosystem Specification” and the additional work needed to develop recommendations for inclusion in their final report, SC 4 extends the term for AHG 2 to the next meeting where a final report will be provided.

Resolution 1044: (Virtual Meeting – May 2020)

Deprecate SC4N0535 and SC4N1290 as standing documents.

Noting the work performed by ISO/TC 184/SC 4/PPC during its February meeting to perform a comprehensive review of existing SC 4 standing documents for purpose, relevance, and alignment to SC 4 operations, SC 4 concurs with the recommendation to deprecate SC4N0535 and SC4N1290 as standing documents of SC and instructs the PPC to update the SC 4 Handbook accordingly. Further, SC 4 instructs its Committee Manager to remove the named documents from the standing documents folder in eCommittees.

Resolution 1045: (Virtual Meeting – May 2020)

Define and publish a reference model for SC4 that builds upon the work of IDIOM, STEP extended architecture and the on-going activities on ontologies.

In order to define and publish an SC4 reference model, SC 4 creates a Task Force titled “**SC 4 Reference Model for Industrial Data**” with the following term of reference:



N4118

2026 – 05 – 15

1. Build upon the work of IDIOM, STEP extended architecture and the on-going activities on ontologies, such as ISO 15926, ISO 13584, ISO 22745, ISO 29002 and others,
2. Coordinate with other affected groups in SC4 on plan elements;
3. Provide periodic status reports to SC 4/PPC;
4. Deliver a reference model using model based computer interpretable technologies ;
5. Publish human readable guide (technical report) to the use of the reference model.

The task force is requested to liaise with the "Core terms" work being carried out by AHG 01 and with the work on an ontology for geometry and topology being carried out by WG 12/T 1.

SC 4 nominates Jean Brangé and David Leal to lead the Task Force and appoints the following experts to the Task Force:

- Nils Sandmark
- Tor Arne Irgens
- Allison Barnard Feeney
- Paul van Exel
- Mario Tucci
- Ben Urick
- Ronald Tse
- Tom Thurman
- Yves Keraron

In addition, SC4 requests nominations from member bodies and relevant liaisons to participate in the Task Force.

Resolution 1046: (Virtual Meeting – May 2020)

To organize a meeting with ISO/TC 184, ISO/TC 184/SC 4, ISO/TC 184/SC 4/AHG 2, ISO/TC 85/SC 6, and ISO/TC 184/WG 6 for identifying a collaboration model and recommendation for the appropriate TC for “digital handover of engineering data for nuclear power plant “.

Noting the comments and recommendations from AHG 2 on the content and how to

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

collaborate on developing the project “digital handover of engineering data for nuclear power plant”, SC 4 requests SC 4 Chair and Committee Manager to organize a meeting with ISO/TC 184, ISO/TC 184/SC 4, ISO/TC 184/SC 4/AG 2, ISO/TC 184/SC 4/AHG 2, ISO/TC 85/SC 6, and ISO/TC 184/WG 6 chairs, committee and workgroup conveners and AHG 2 experts to discuss potential collaboration between the parties. SC4 instructs its Committee Manager to schedule the meeting before the 80th Plenary.

Resolution 1047: (Virtual Meeting – May 2020)

The objective of the resolution is to deliver ISO 15926-4ed.3 building on the experience as gathered in the POSC Caesar Association (PCA) data base with improvements on the following points:

Consistent naming and definitions

1. Reference to other RDIs in text definition
2. Reference to source
3. Improve specialization and classification according results of the Top-Class structure task force.
4. Rules for use of entity types `inanimate_physical_object` and `functional_object`

The ISO 15926-4 ed.3 document will reflect the changes made and will be in line with the electronic inserts update of ISO 15926-4e2. It will contain a time plan in line with the incremental steps that will be taken.

Noting that new updates to ISO 15926-4 is needed and there is no change in scope to the standard, SC4 instructs its Committee Manager to launch edition 3 using the SDT 24 track and names Onno Paap as Project Leader and allocates the work to WG 3 with the following terms of reference.

- Update ISO 15926-4 ed.3 as presented to the Change Board during the 79th Plenary of SC4 on 05 May 2020
- Update the electronic inserts to ISO/NP 15926-4 ed.3 per SC4N2591, SC4N2984, and SC4N2986 standing documents,
- Request national bodies to nominate their representatives and deputies to the Validation Team (VT) as noted in SC4N2541, and
- To keep a register of the VT members for the practice of sending out notifications to vote and other administrative items.

Resolution 1048: (Virtual Meeting – May 2020)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Changing the project to a 24-month schedule will give the team sufficient time to finish ballot preparation, circulate the CD/TS ballot, resolve comments and publish.

Noting the progress of the project team and the updated project plan to circulate a CD/TS ballot and resolve comments, SC 4 requests its Committee Manager to apply the 24-month development track to the ISO 10303-17 project.

Resolution 1049: (Virtual Meeting – May 2020)

Request ISO TMB grant a nine month extension to 10303-243

SC4 requests the ISO TMB to grant a 9 month extension to ISO 10303-243 allowing the project team additional time to complete necessary coordination using virtual working arrangements dictated by restrictions in meeting due to COVID-19.

Resolution 1050: (Virtual Meeting – May 2020)

Fix errors in 103030-242:2020

SC 4 notes the release of ISO 10303-242 ed2 with errors and instructs the Convenor of WG12 to coordinate a correction to the AP242 edition 2 with the project leader. Further, SC 4 instructs its Committee Manager to notify ISO CS of the errors and seek guidance per the directives on next steps.

Resolution 1051: (Virtual Meeting – May 2020)

Request 9 month extension for ISO 10303-238 ed2

SC 4 notes the successful DIS ballot and the comments from members bodies that indicate an improvement can be achieved before publication and requests the ISO TMB to grant a nine month extension for Edition 2 of AP238 to enable complete industry testing and validation of the model produced by the DIS ballot comment resolution during the COVID-19 crisis.

ACKNOWLEDGEMENTS and APPRECIATIONS

ACCLAMATION

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC4 expresses its appreciation to our SC 4 Liaisons and reports provided to the 79th Plenary

- | | |
|----------------------------|------------------------------|
| • ISO/TC 10 | Richard Lindqvist (written) |
| • ISO/TC 37 | written |
| • ISO/TC 37/SC 3 | Dr. Detlef Reineke (written) |
| • ISO/TC 171/SC 2 | Phil Sprier (to JWG 16) |
| • ISO/TC 171/SC 2 | Leonard Rosenthal (written) |
| • ISO/TC 172/SC 1 | Jochen Haenisch (written) |
| • ISO/IEC/JTC 1/SC 24 | Christophe Mouton |
| • ISO/IEC JTC 1/AG 11 | Sangkeun Yoo |
| • IEC/SC3D | Yoshiaki Sonda |
| • Khronos Group | Neil Trevett |
| • Small Business Standards | Jochen Haenisch |
| • Web3D Consortium | Vince Marchetti |

ACCLAMATION

SC4 expresses its appreciation to Melissa Harvey and the ISO Web team for standing up the ISO/TC 184/SC 4 website and populating the site with initial content.

ACCLAMATION

SC 4 expresses its appreciation to Soonhung Han, Christophe Mouton and JWG 16 experts for experimenting with Microsoft Teams and providing valuable feedback on the utility of the application for virtual standards work.

ACCLAMATION

SC 4 expresses its appreciation to Tim King and WG 13/23 for utilizing a full complement of digital tools for standards development and providing valuable feedback to SC 4 on their experiences.

ACCLAMATION

SC 4 expresses its profound thanks to Brian Stanton for 15 years of service as ISO/TC 184/SC 4's Editorial Program Manager and his countless contributions to the quality output of SC 4 standards and innovation in supporting our industry requirements. SC 4

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

wishes Brian the best in his continuing role as ISO Editorial Team Leader.

ACCLAMATION

SC 4 expresses its sincere appreciation to all the experts, delegates, and leaders for executing a completely virtual plenary meeting successfully while undergoing numerous challenges during this extraordinary time for stakeholders worldwide.

Resolution 1051: (Virtual Meeting – Nov 2020)

Appointment of Convenor to AG 00 Change Management

SC 4 appoints Kenny Swope as the Convenor for AG 00 – Change Management for a three-year term, with appreciation for his offer to serve in that capacity.

Resolution 1052: (Virtual Meeting – Nov 2020)

Appointment of Deputy Convenor to WG 23

Noting that continuity of operations and management of work is an important consideration to the successful operation of large working groups, SC 4 appoints Hayley Thompson as the Deputy Convenor for WG 23 for a three-year term, with appreciation for her offer to serve in that capacity.

Resolution 1053: (Virtual Meeting – Nov 2020)

Additional liaison representative from SC 4 to ISO/IEC JTC 1/SC 7 (Software and systems engineering)

SC 4 appoints Tim King as an additional liaison representative from SC 4 to ISO/IEC/JTC 1/SC 7 (Software and systems engineering) for a three-year term, with appreciation for his offer to serve in that capacity.

Resolution 1054: (Virtual Meeting – Nov 2020)

Extend term of Ad Hoc Group 1 on Core Industrial Data Set of Terms

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Noting the progress made by AHG 1 “Core industrial data set of terms” and the further work needed, SC 4 extends the term of the AHG 1 and requests its final report at the 81st Plenary meeting of SC 4. AHG 01 is requested to provide a plan for the further work itemized in the introduction to this resolution. Further, France nominates Yves Keraron and Yves Baudier as experts to this group.

Resolution 1055: (Virtual Meeting – Nov 2020)

Extend term of the Ad Hoc Group 2 on Nuclear Digital Ecosystem

Noting the progress made by AHG 2 “Nuclear Digital Ecosystem Specification”, the ongoing national survey and the additional work needed to develop recommendations for inclusion in their final report, SC 4 extends the term for AHG 2 to the 81st plenary where a final report will be provided.

SC 4 encourages its members to join Ad Hoc Group 2 and take part to the ongoing national survey to collect national needs and interoperability requirements in terms of specifications for a digital ecosystem for the nuclear industry.

Resolution 1057: (Virtual Meeting – Nov 2020)

Liaison appointment review

Recognizing that Liaisons are an important component to a healthy technical competency for SC 4 and that Liaison representative play a critical role in establishing and maintaining effective lines of communication between organizations, SC 4 regularly publishes the current Liaisons through the SC 4 Newsletter twice yearly. Noting the cursory review of SC 4’s published liaisons through the SC 4 Newsletter and the significant number of errors, SC 4 requests the Committee Manager initiate a review of the Liaisons and validate Liaisons through interrogation of the ISO Global Directory for membership and/or correspondence with each identified Liaison representative. Further, if no response is received within four weeks of initiation, the liaison assignment will be considered inactive.

Resolution 1058: (Virtual Meeting – Nov 2020)

Approve ISO/AWI 17506 to move to DIS ballot

Noting the progress of ISO/AWI 17506 and the use of ISO/PAS 17506 in industry, SC 4 resolves to promote ISO/AWI 17506 to a DIS ballot and requests the Committee

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Manager to launch a DIS ballot.

Resolution 1059: (Virtual Meeting – Nov 2020)

Move ISO 10303-17 from SDT 24 to SDT 36

Noting the progress of the project team and the updated project plan to circulate a CD/TS ballot on 15 January, 2021 and resolve comments, SC 4 requests its Committee Manager to apply the 36-month development track to the ISO 10303-17 project.

Resolution 1060: (Virtual Meeting – Nov 2020)

Request 9-month extension for ISO 10303-209 ed3

SC4 requests the ISO TMB to grant a nine months extension to ISO 10303-209 edition 3 allowing the project team additional time to complete CD by 15 January 2021, and DIS by 15 July 2021 through necessary coordination using virtual working arrangements dictated by restrictions in meeting due to COVID-19.

Resolution 1061: (Virtual Meeting – Nov 2020)

Approval to use Jira/Git procedure for select projects

Noting the successful conclusion of the Committee Innovation Project (Living Lab) for the use of Jira/Git in conjunction with Agile development methods and the desire of SC 4 to continuously improve internal operations while balancing risk of adoption resolve to:

Approve the Jira/Git procedure as an interim process for the purpose of managing change using the Jira/Git platform,

Grant an exception to select projects in SC 4 from using Bugzilla as the change management solution noted in the SC 4 Handbook,

Concur with the recommendation of the PPC to allow ISO/AWI TS 15926-4, ISO/AWI TS 23301, ISO/PWI TR 4758 and TF 02 work products to use the process,

Request the SC 4 Committee Manager to maintain a record of projects that use the process,

Direct the PPC to periodically review the performance of the pilot activity and report

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

status at the next Plenary meeting of SC 4.

Resolution 1062: (Virtual Meeting – Nov 2020)

Approve draft for CIB ballot of JWG 24 procedure

Noting the mandate from the ISO/TMB through Resolution 81/2017 and development by experts from ISO and IEC in ISO/TC 184/SC 4/JWG 24 (IEC SC 3D/JWG 1), a draft procedure has been drafted for committee use in managing standards in database form. SC 4 acknowledges the contribution of the draft procedure and requests its Committee Manager distribute a six-week Committee Internal Ballot for comments in preparation of submitting a Committee Specific Procedure to ISO/TMB.

ACCLAMATION

SC4 expresses its appreciation to our SC 4 Liaisons and reports provided to the 80th Plenary

- ISO/TC 171/SC 2 (Written)
- IEC/SC 3D
- ISO/IEC/JTC 1/SC 24 and Web 3D Consortium

ACCLAMATION

SC 4 expresses its sincere appreciation to all the experts, delegates, and leaders for executing a completely virtual plenary meeting successfully while undergoing numerous challenges during this extraordinary time for stakeholders worldwide.

Resolution 1063: (Virtual Meeting – May 2021)

To establish a reduced ballot period for New Work Item Proposals.

SC 4 requests its Committee Manager to apply the 8 week ballot duration to the following NP ballots.

Project	Approval
ISO NP 8000-114	☒

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ISO NP eXtended Master Connection File	☒
ISO NP/CD 10303-1 ed 3	☒

Resolution 1064: (Virtual Meeting – May 2021)

Appointment of Convenors and Deputy Convenors

Noting that continuity of operations and management of work is an important consideration to the successful operation of advisory groups, SC 4 appoints the following Convenors and Deputy Convenors for a three-year term, with appreciation to serve in this capacity.

WG	Role	Appointment
AG 2	Convenor	Martin Hardwick
AG 2	Deputy Convenor	Paul van Exel
AG 2	Deputy Convenor	Cheong-Hyuk Yoon
AG 2	Deputy Convenor	Changsoo Lee
QC	Deputy Convenor	Changsoo Lee
WG 12	Convenor	Keith Hunten
WG 13	Convenor	Tim King
WG 13	Deputy Convenor	Anne-Marie Mougeolle
JWG 16	Convenor	Christophe Mouton
JWG 16	Deputy Convenor	Soonhung Han
JWG 24	Convenor	Hiroshi Murayama

Resolution 1065: (Virtual Meeting – May 2021)

Place ISO 10303-209 ed3 on hold

Noting the progress to date of ISO 10303-209 ed3 (project 75060) and the challenges due to COVID-19, SC 4 requests the TMB to place the project on hold for six months in order that the project team can address challenges with the project staffing.

Resolution 1066: (Virtual Meeting – May 2021)

Change to project duration of ISO 8000-110 edition 2 – Master data: Exchange of characteristic data: Syntax, semantic encoding, and conformance to data specification



N4118

2026 – 05 – 15

SC 4 resolves to adjust the standards development track of ISO 8000-110 edition 2 (project id = 78501) to SDT 36.

Resolution 1067: (Virtual Meeting – May 2021)

Schedule Change for ISO/TS 23301

Noting the progress on ISO/AWI TS 23301 and the proximity to limit dates, SC4 requests the ISO TMB grant a nine-month extension to the schedule of ISO/AWI TS 23301 (project 75189) allowing the project team additional time to complete CD/TS by 2021-12-09 through necessary coordination using virtual working arrangements dictated by restrictions in meeting due to COVID-19. The project team has completed the necessary documents and plans to release a CD/TS ballot on 2021-05-25 with a publication targeted for October 2021.

Resolution 1068: (Virtual Meeting – May 2021)

Create an Advisory Group on Core Terminology for industrial data

Noting the report from Ad Hoc Group 1 “Core industrial data set of terms” (AhG01/N34), ISO/TC 184/SC 4:

- Accepts the recommendations identified in the report,
- Thanks the Ad Hoc Group members for their work,
- Requests ISO/TC 184/SC 4 circulate AhG01/N34 “Core terminology for industrial data” in a CIB for an ISO Technical Report,
- Launches an advisory group entitled “Core terminology for industrial data” to:
 - Interpret/map the core terminology to the different Top Level Ontologies (TLOs),
 - Align with relevant activities outside ISO/TC 184/SC 4,
 - Extend the content of the “Core terminology for industrial data” to a higher degree of specialization,
 - Recommend interfaces to ontologies and reference data,
 - Develop examples to be used for pilots with the different TLOs,
 - Complete the presentation intended for domain engineers and business decision makers,
- Nominates Nils Sandmark as convener of the advisory group, and
- Retains existing members and invites member bodies to nominate additional experts to the advisory group.



N4118

2026 – 05 – 15

Resolution 1069: (Virtual Meeting – May 2021)

Extend term of the Ad Hoc Group 2 on Nuclear Digital Ecosystem Specification

Noting the progress made by AHG 2 “Nuclear Digital Ecosystem Specification”, the ongoing additional work needed to develop their final report to plan a meeting with ISO/TC 85 representatives, SC 4 extends the term for AHG 2 to the 82nd plenary where a final report will be provided.

SC 4 encourages its members to join Ad Hoc Group 2 and take part to the continuous effort to collect needs and interoperability requirements in terms of specifications for a digital ecosystem for the nuclear industry.

Resolution 1070: (Virtual Meeting – May 2021)

Conduct a review of ISO/TC 184/SC 4 N1167 “SC4 Industrial Data Framework”

Noting that SC4N1167 “Industrial Data Framework” is out of date, but contains fundamental principles of the approach to standardizing industrial data, ISO/TC 184/SC 4 requests its CM to initiate a Committee Internal Ballot (CIB) similar to an ISO/SR inviting member bodies to note their support for the document continuing as a standing document and comment on the document’s contents.

Resolution 1071: (Virtual Meeting – May 2021)

Launch combined NP/CD for ISO 10303-239 ed3

Recalling Resolution 1004 and 1005 that launched edition 3 of ISO 10303-239 as a PWI and the development work of the project team to draft a CD ready revision to part 239 noting that there is no change in the scope from edition 2, SC 4 requests its Committee Manager to launch edition 3 using the SDT 24 track, naming Yves Baudier (yves.baudier@afnet.fr) and Patrick Walsh (patrick.l.walsh@boeing.com) as Project Leaders, and allocating the work to WG 12 with the following target dates:

- DIS Submission: 2021-11-30
- Publication: 2022-04-30

Resolution 1072: (Virtual Meeting – May 2021)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Launch minor revisions to ISO 10303-210 ed 4, ISO 10303-238 ed 2, and ISO 10303-242 ed 2

Recalling resolution 1050 from the May 2020 Plenary meeting on the AP 242 EWIS, geometry and topology open points from 2020, and noting two WG21 Change Requests CR_Part42 and CR_PMI_4 published as SMRL v9, SC4 requests its Committee Manager to notify ISO/CS via completion of Form 8B for ISO 10303-210 ed4, ISO 10303-238 ed2, and ISO 10303-242 ed2 of the decision to create a minor revision and submit FDIS documents for ISO 10303-238 ed2, ISO 10303-242 ed2 and ISO 10303-210 ed4 noting the schedule for FDIS ballot as:

- ISO 10303-242: 2021-07-16
- ISO 10303-210: 2021-08-16
- ISO 10303-238: 2021-08-16

Resolution 1073: (Virtual Meeting – May 2021)

Establish an Ad Hoc Group on UUID management for industrial data

Noting the technical discussions among experts from WG 3, WG 12, WG 15 and others on the opportunity to align concepts on the use of UUID in industrial data, SC 4 establishes an Ad Hoc Group “UUID management for industrial data” with the following terms of reference:

- Define the opportunity space for UUID in context of SC 4 standards,
- Identify use cases for UUID in industrial data,
- Provide recommendations to SC 4 on the scope and applicability of UUIDs to SC 4 standards,
- Provide a report to SC 4 upon completion.

SC 4 nominates Martin Hardwick as Convenor with appreciation to serve in that capacity, appoints Allison Barnard Feeney

Peter Benson

Keith Hunten

Alan Johnston

David Odendahl

Thomas Thurman

Jean Brange

Mikael Hedlind

Bengt Olsson

Max Ungerer

Dana Tripp

Secretary, ISO/TC 184/SC 4

trippdsc4@gmail.com



N4118

2026 – 05 – 15

Hiroshi Murayama
Hyunjeong Lee
Nils Sandsmark

to the group and invites National Bodies and Liaisons to nominate experts.

Resolution 1074: (Virtual Meeting – May 2021)

Update the SC 4 Handbook to incorporate standing documents

Noting the procedures for ISO/TC 184/SC 4 and recent approvals of SC4 relative to standing documents, SC4 requests the Committee Manager launch an eight week CIB to modify the SC4 Handbook to account for the following changes in approved standing documents.

- Deprecate SC4 N2412 “Supplementary directives — Rules for the structure and drafting of SC 4 standards for industrial data”
- Add SC 4 N3567 “SC 4 Supplementary directives — Rules for the structure and drafting of SC 4 standards for industrial data”
- Deprecate SC4 N3422 “STEP extended architecture high-level description”
- Add SC4 N3501 “STEP extended architecture high-level description”
- Deprecate SC4 N3433 “STEP extended architecture high-level requirements”
- Add SC4 N3502 “STEP extended architecture high-level requirements”
- Deprecate SC4 N3424 “STEP extended architecture detailed STEPlib specifications”
- Add SC4 N3503 “STEP extended architecture detailed STEPlib specifications”
- Deprecate SC4 N3425 “STEP extended architecture publication design”
- Add SC4 N3504 “STEP extended architecture publication design”

Note: The Handbook has not been updated to account for recent deviations to ISO Operations as a result of COVID-19. Subsequent to TMB and ISO guidance, procedures for managing ISO TCs and SCs are covered by continuing resolutions from the TMB and supersede guidance provided by this Handbook on specific points covered.

Resolution 1075: (Virtual Meeting – May 2021)

ISO 8000-2 Vocabulary for the ISO 8000 series Living Lab

Recalling the original proposal for a Living Lab on managing vocabulary using

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

standards in a database format and the final report delivered by WG 13, SC 4:

- Thanks WG 13/23 and all participants that contributed to the innovation,
- Accepts the final report,
- Endorses the recommendations noted to ISO/TMB,
- Requests ISO/CS continue to support the existing process for ISO 8000-2, and
- Concludes this Living Lab.

Resolution 1076: (Virtual Meeting – May 2021)

ISO 15926-4 Uniform Resource Identifiers Living Lab

Recalling Resolution 979 taken during the May 2018 Plenary in Beijing, the work on the Living Lab for ISO 15926-4 Uniform Resource Locators and the final report, SC 4:

- Thanks WG 3 and all participants that contributed to the innovation,
- Accepts the final report,
- Endorses the recommendations noted to ISO/TMB,
- Requests ISO CS continue to support the existing process for ISO 15926-4, and
- Concludes this Living Lab.

ACCLAMATION

SC 4 expresses its appreciation to Anne-Françoise Cutting-Decelle for her role as Convenor of JWG 8 and extends best wishes in her next assignment supporting the development of international standards.

ACCLAMATION

SC 4 expresses its appreciation to Anne Dourgnon for her role as Deputy Convenor of WG 13 and extends best wishes in her next assignment supporting the development of international standards.

ACCLAMATION

SC 4 expresses its appreciation to ISO CS and the ISO Documents project team for a trouble free transition from eCommittees to ISO Documents and for all of the technical support provided to SC 4 during the transition.

ACCLAMATION

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 expresses its appreciation to ISO IT for supporting multiple Committee Innovations with Living Lab projects and for their responsiveness to requests from SC 4 to modify and provide support to multiple activities. SC 4 sincerely appreciates the sponsorship and enthusiasm of ISO staff to experiment with new methods and ways of working in promotion of quality standards.

ACCLAMATION

SC 4 expresses its appreciation to our SC 4 Liaisons and reports provided to the 81st Plenary.

- ISO/TC 171/SC 2
- IEC/SC 3D
- ISO/IEC/JTC 1/SC 24 and Web 3D Consortium
- Khronos Group

ACCLAMATION

SC 4 expresses its appreciation to Nils Sandsmark for organizing and facilitating the multi-group coordination meeting held on 2021-05-04 and to the participants who provided technical content to the betterment of SC 4 level harmonization.

ACCLAMATION

SC 4 expresses its appreciation to Soonhung Han and Christophe Mouton for proposing to exchange places as Convenor and Deputy Convenor of JWG 16, thus maturing the concept of the Deputy position in SC 4 and promoting new ways of working together on international standards.

ACCLAMATION

SC 4 expresses its appreciation to Allison Barnard Feeney and Melissa Harvey for their work on SC 4's submission to ISO CS for consideration in the competition for the LD Eicher award.

ACCLAMATION

SC 4 expresses its sincere appreciation to all the experts, delegates, and leaders for executing a completely virtual plenary meeting successfully while undergoing numerous

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

challenges during this extraordinary time for stakeholders worldwide.

Resolution 1077: (Virtual Meeting – November 2021)

Establish ballot duration of new work item proposals

SC 4 requests its Committee Manager to apply the 8-week ballot duration to the following NP ballots:

Project	Approval
ISO/NP 10303-242 ed3	☒
ISO/NP 15926-6 ed1	☒
ISO/NP 8000-114	☒
ISO/NP 8000-210	☐
ISO/NP 3151-2	☒

Resolution 1078: (Virtual Meeting – November 2021)

Appointment of Convenors and Deputy Convenors

Noting that continuity of operations and management of work is an important consideration to the successful operation of advisory groups, SC 4 appoints the following Convenors and Deputy Convenors for a three-year term [2022 – 2024], with appreciation to serve in this capacity. Note the Convenor for the PPC is appointed to a two-year term to align with the Change Management AG.

WG	Role	Appointment
*PPC	Convenor	Kenny Swope
WG 3	Convenor	Paul van Exel
WG 15	Convenor	Martin Hardwick
WG 15	Deputy Convenor	Olsson, Bengt
WG 21	Convenor	Hunten, Keith
WG 21	Deputy Convenor	Haenisch, Jochen

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

WG 22	Convenor	Sandsmark, Nils
WG 22	Deputy Convenor	Paul van Exel
WG 23	Convenor	Tim King

Resolution 1079: (Virtual Meeting – November 2021)

Liaison arrangements for ISO/TC 184/SC 4

Noting the importance of Liaison Representatives to effective collaboration with SC 4 in the development of standards, SC 4 appoints and recognizes the following individuals. The appointments are for a three-year term [2022 – 2024]. SC 4 appreciates the service of all the identified individuals. Where indicated, SC 4 requests members to volunteer to fill vacancies.

Organization	Appointed from SC 4	Recognized to SC 4
MIMOSA (category A)	Nils Sandsmark	Alan Johnston
ECCMA (category A)	Vacant	Peter Benson

Resolution 1080: (Virtual Meeting – November 2021)

Extend the term of Ad Hoc Group 2 on Nuclear Digital Ecosystem Specification

Noting that ISO/TC 85 is reviewing the DTR “Nuclear Digital Ecosystem Specification” provided by AHG 2 and that their comments and feedback need to be accommodated, AHG 2 requires additional time to complete the consensus processes and therefore requests SC 4 extend the term for AHG 2 to the 83rd SC 4 Plenary. AHG 2 plans to report on all developments to the PPC and at the 83rd SC 4 plenary.

Further, SC 4 encourages its members to join Ad Hoc Group 2 and take part to the continuous effort to collect needs and interoperability requirements in terms of specifications for a digital ecosystem for the nuclear industry.

Resolution 1081: (Virtual Meeting – November 2021)

Request One-year Extension for ISO 15926-4 ed3

SC4 requests its Committee Manager coordinate with ISO CS and move ISO/TS 15926-

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

4 ed3 from SDT 24 to SDT 36 allowing the project team additional time to complete the project by 05 Jun 2023.

Resolution 1082: (Virtual Meeting – November 2021)

Committee Specific Procedure for JWG 24 to manage database-based standards of ISO data dictionaries and ontologies.

Noting the action assigned by ISO/TMB through resolution 81/2017 to form an ISO/IEC Joint Working Group and approve in principle the exception for ISO/TC 184 and other interested committees to publish and maintain data in the IEC CDD database through a Committee Specific Procedure (CSP) approved by the TMB, SC 4:

- thanks the experts from both ISO and IEC in developing a CSP that meets the needs of both organizations toward the management of standards in database format,
- concurs with the work of ISO/IEC JWG 24 in developing the proposed CSP,
- recommends that ISO/TMB approve the proposed CSP,
- requests the SC 4 Committee Manager work with ISO/CS and deliver to the ISO/TMB secretariat this CSP as per the ISO/TMB resolution 81/2017 Part E, and
- upon approval by the TMB, invites member bodies nominate experts to the ISO VT-CDD.

Resolution 1083: (Virtual Meeting – November 2021)

Change to project duration of ISO 10303-2 – STEP Vocabulary

Noting the progress of the project team to complete a CD draft of ISO 10303-2 and the actions to address the resource, technology and data quality issues surfaced, SC 4 requests its Committee Manager coordinate with ISO/CS to move ISO 10303-2 (project id = 81196) from SDT 24 to SDT 36 with the following dates:

Stage 30 – Committee Draft Registered: 2021-12-01 (Original Target 2021-06-14)

Stage 40 – DIS Registered: 2022-05-01 (Original Target 2021-10-11)

Stage 60 – IS Published: 2022-10-01 (Original Target 2022-05-15)

Resolution 1084: (Virtual Meeting – November 2021)

Target Date change to ISO 10303-239 ed3

Noting the progress of the project team that is developing ISO/CD 10303-239 ed3 (Project ID 78832) and the balance to go in delivering the project as scheduled on 2023-

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

06-15, SC4 requests its Committee Manager modify the target date for Stage 40.00 (DIS Registered) to 2022-04-15.

Resolution 1085: (Virtual Meeting – November 2021)

Modify WGs for SMRL items from WG 21 to WG 12

Noting the procedures laid out in the ISO/TC 184/SC 4 Handbook, standing document SC4N2591, and the results of a project audit conducted in 2021, SC4 requests its Committee Manager coordinate with ISO CS and reallocate the following projects from ISO/TC 184/SC 4WG 21 to ISO/TC 184/SC 4/WG 12:

ISO/CD TS 10303-4443 <https://sd.iso.org/projects/project/83491/overview>
ISO/AWI TS 10303-1847 <https://sd.iso.org/projects/project/81278/overview>
ISO/AWI TS 10303-1852 <https://sd.iso.org/projects/project/82033/overview>
ISO/AWI TS 10303-1853 <https://sd.iso.org/projects/project/82522/overview>
ISO/CD TS 10303-4439 <https://sd.iso.org/projects/project/83492/overview>

Resolution 1086: (Virtual Meeting – November 2021)

Launch ISO 23301 ed2 by resolution

Noting that ISO 23301 edition pilots in 2020-2021 have demonstrated that the AP242 ed2 fully shattered product structure is an efficient way to manage complexity and diversity of geometry datasets implementations and that updates are needed to address certain needs for meta data, and specifications with no change in scope to edition 1, SC 4 requests its Committee Manager coordinate with ISO CS and launch ISO 23301 ed2 at Stage 20 with SDT 18 and the following schedule dates:

Circulation of 1st Working Draft to experts: 2022-04-15

Committee Draft ballot (if any): 2022-07-30

DIS submission*: 2022-12-01

Publication*: 2022-12-01

And nominates Jacques Heinisch as project leader with an invitation to member bodies to nominate experts to the project team.

Resolution 1087: (Virtual Meeting – November 2021)

Retirement of Bugzilla and Cutover to JIRA

Dana Tripp

Secretary, ISO/TC 184/SC 4

trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

Noting the progress made in developing an agile environment for SC 4 CR management and the detailed plan for cutover to the new environment, SC 4 supports the recommendation of WG 12 and agrees to retire Bugzilla as the system of record for CR management and authorizes WG12 to start working in JIRA per the following schedule:

Stop using Bugzilla on	December 15, 2021
Archive Bugzilla to ISO WG12 Member Area by	December 20, 2021
Migrate Bugzilla projects to JIRA by	December 26, 2021
Start using JIRA on	January 1, 2022

Further, SC 4 concurs with the recommendation of the Living Lab to disposition the Bugzilla projects as follows:

Topic	Description	Bug Count	Proposed Action
10303 example files	Example files used by implementors to understand the standard.	0	Recommend ARCHIVAL
10303 Recommended Practices	Includes new materials as well as old AP210 materials. Changed product name because we are adding rec prac for other ISO 10303 parts. The old documents are closed for new bug entries so there is no conflict	114	Recommend ARCHIVAL
AP210 IS1	ISO 10303-210:2001	1105	Recommend MIGRATION: TC184SC4-10303
Express Engine	EXPRESS based purpose built software for facilitating EXPRESS model development and integration. This is input as a separate product in order to avoid issues with version numbers with the XSLT environment.	83	Recommend MIGRATION: TC184SC4-Utilities

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

 Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

<u>ISO 10303</u>	all issues on ISO 10303 standards	<u>6644</u>	Recommend MIGRATION: TC184SC4-10303
<u>ISO 13584</u>	Parts library	<u>16</u>	Recommend MIGRATION: TC184SC4-13584
<u>ISO 14306</u>	Open JT	<u>327</u>	Recommend MIGRATION: TC184SC4-14306
<u>ISO 29002</u>	Exchange of characteristic data	<u>0</u>	Recommend ARCHIVAL
<u>SC4 Documents</u>	SC4 Documents that are not standing documents	<u>2</u>	Recommend MIGRATION: TC184SC4- StandingDocs
<u>SC4 Standing documents</u>	SC4 Standing documents	<u>51</u>	Recommend MIGRATION: TC184SC4- StandingDocs
<u>STEPIlib environment and utilities</u>	same	<u>3</u>	Recommend MIGRATION: TC184SC4-Utilities
<u>stepmod environment and utilities</u>	stepmod environment and utilities	<u>548</u>	Recommend MIGRATION: TC184SC4-Utilities
<u>Technical Reports</u>	Technical reports authored by SC4	<u>0</u>	Recommend ARCHIVAL
<u>test</u>	A dummy product used to test user interaction with permissions subsystem.	<u>1</u>	Recommend ARCHIVAL
<u>WG12 policy</u>	Record of WG12 policy items.	<u>5</u>	Recommend MIGRATION: TC184SC4- StandingDocs

WG 12 recommends that SC 4 PPC take the action to revise the SC 4 Handbook at its next revision noting the new system of record, and thanks PDES, Inc. for providing outstanding custodianship of the Bugzilla environment.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Resolution 1088: (Virtual Meeting – November 2021)

Retirement of CVS and Cutover to Bitbucket GIT

Noting the progress made in developing a new repository for the SC 4 code base utilizing git technology and demonstrated performance by projects on the STEP Extended Architecture, STEP Utilities and prototyping the new SRL build processes, SC 4 approves the retirement of CVS and migrate content to ISO Bitbucket per the schedule below:

Stop using CVS on December 15, 2021
Archive CVS STEPMod to ISO WG12 Member Area by December 20, 2021
Migrate CVS STEPMod to Bitbucket repositories by December 26, 2021
Migrate GitLab STEPLib to Bitbucket repositories by December 26, 2021
Start using Bitbucket on January 1, 2022

SC 4 invites member bodies to identify key users that require training and to notify the project team.

SC 4 acknowledges and thanks AFNeT Services (formally Boost Conseil) for providing outstanding custodianship of the development environment.

Acclamations and Acknowledgements

Acclamation

SC 4 recognizes the outstanding performance and contribution of Nigel Shaw to the development of ISO 10303 and to the standards community with the ISO Excellence award. The award recognizes a lifetime of achievement and technical excellence from the first edition of ISO 10303 through the development and extension of the standard through the full product lifecycle.

Acclamation

SC 4 recognizes and thanks Laura Mathew and the staff at ISO/CS for their expediency in supporting the processing of the ISO Excellence Award for Nigel Shaw and for their attention to detail during the process.

Acclamation

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 expresses its appreciation to our SC 4 Liaisons and reports provided to the 82nd Plenary.

- ISO/TC 171/SC 2 through written report
- ISO/TC 307 through Patrick Curry
- IEC/SC 3D through Gernot Rossi
- ISO/IEC/JTC 1/SC 24 and Web 3D Consortium through Christophe Mouton
- ASM International through Afina Lupulescu
- ECCMA through Peter Benson

Acclamation

SC 4 expresses its appreciation to Nils Sandsmark for organizing and facilitating the multi-group coordination meeting held on 2021-11-02 and to the participants who provided technical content to the betterment of SC 4 level harmonization.

Acclamation

SC 4 expresses its appreciation to Tim King, Frans Nilsén, and Nils Sandsmark for their leadership and participation in the CEN/CENELAC/ETSI Industrial Data Steering Group and for promoting the scope of SC 4 Industrial Data and the demonstrated history of international standards that exist today.

Acclamation

SC 4 expresses its appreciation to Barnaby Lewis and the ISO communications team for their excellent work in developing the L.D. Eicher Award introductory video for SC 4 that will be used in promotional materials for the Committee.

Acclamation

SC 4 expresses its sincere appreciation to all the experts, delegates, and leaders for executing a completely virtual plenary meeting successfully while undergoing numerous challenges during this extraordinary time for stakeholders worldwide.

Resolution 1089: (Virtual Meeting – May 2022)

Approval of Project Leaders and Project Editors for SC 4 Projects



N4118

2026 – 05 – 15

SC 4 Approves the following Project Leaders and Project Editors for the listed Projects noted:

Project	Working Group	Project Leader/Editor
ISO 10303-2 (ID 81196)	WG 12	Stuart Galt (Leader)
ISO/PAS 8329 (ID 83119)	WG 12	Matthias Weinert (Leader)

SC 4 encourages the above-named individual(s) to approach their respective National Body to request they are nominated as a participating expert in ISO/TC 184/SC 4/QC (Quality Committee) giving them access to the latest advice on ensuring SC 4 projects deliver products that satisfy ISO and SC 4 expectations for quality.

Resolution 1090: (Virtual Meeting – May 2022)

Establish ballot duration of new work item proposals

Noting the report of ISO/TC 184/SC 4 Change Management Advisory Group, SC 4 requests its Committee Manager to apply the 8-week ballot duration to the following NP ballots:

Project	Approval
Framework for Nuclear Digital Ecosystem (NDE)	☒
ISO 8000-1 ed3, Data quality – Part 1: Overview	☒
ISO 8000-200, Data quality – Part 200: Transaction data: Quality of transaction data	☒

Resolution 1091: (Virtual Meeting – May 2022)

Appointment of Convenors and Deputy Convenors

Noting that continuity of operations and management of work is an important consideration to the successful operation of advisory groups, SC 4 appoints the following Convenors and Deputy Convenors for a three-year term, with appreciation to serve in this capacity.

WG	Role	Appointment
TF1 ISO 10303 SMRL architecture innovation	Convenor	Stuart Galt

Resolution 1092: (Virtual Meeting – May 2022)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Establish liaison from ISO/TC 184/SC 4 to ISO/TC 85

Noting that AHG 2 is currently reviewing and managing the comments received from ISO/TC 85 and that a comments resolution meeting will be organized with ISO/TC 85, SC 4 welcomes this fruitful discussion and proposes to ISO/TC 85 to prepare a future coordination on development and implementation of standards in establishing a liaison from ISO/TC 184/SC 4 to ISO/TC 85.

SC 4 welcomes the proposal to establish a liaison from ISO/TC 184/SC 4 to ISO/TC 85 and appoints Christophe MOUTON for a three-year term, with appreciation for his offer to serve in that capacity.

Resolution 1093: (Virtual Meeting – May 2022)

Identify experts in support of ISO 16792 and ISO/TC 10 Liaison

Recalling the liaison relationship between ISO/TC 184/SC 4 and ISO/TC 10 and the request from ISO/TC 10 for support in the development of ISO 16792 ed4 with expertise from SC 4, SC 4 nominates Atsuto Soma, Tom Thurman, Keith Hunten, Leo van Ruijven, Mikael Hedlind, Pierre Duchier and Allison Barnard Feeney as experts from ISO/TC 184/SC 4 and requests its Committee Manager coordinate with ISO/CS to add the experts noted to the project in the ISO systems as required for access to necessary documents and participation in the project's development.

Resolution 1094: (Virtual Meeting – May 2022)

Update to the ISO/TC 184/SC 4 Handbook

Noting the work accomplished by ISO/TC 184/SC 4/TF 1 in establishing the Jira/Git repositories for SC 4 use, the updates from the Quality Committee, and a review by ISO/TC 184/SC 4/PPC, SC 4 requests its Committee Manager submit ISO/TC 184/SC 4/PPC N201, ISO/TC 184/SC 4/N 3492, and ISO/TC 184/SC 4/N 3619 for a Committee Internal Ballot of eight weeks duration for approval as standing documents of SC 4.

Resolution 1095: (Virtual Meeting – May 2022)

Incorporation of ISO/TS 10303-4000 into the SMRL

Noting that ISO 10303-4000 ed1 is a technical specification approved by ballot in 2021

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

and released in early 2022 and requires update to the STEP Extended Architecture and modular architecture APs consistent with the procedures of SC4N2591 and the SC 4 Handbook, SC 4 requests that ISO/CS incorporate ISO 10303-4000 into the SMRL covered by the procedures for managing standards in database format and the standing document SC4N2591 and launch edition 2 noting no change in scope and supporting SDG 9. SC 4 appoints Jean Brange as project leader and notes the following schedule:

Change Request ballot: 2022-05-30

SMRL integration: 2022-07-30

Submitted for publication as part of SMRLv10: 2022-09

Resolution 1096: (Virtual Meeting – May 2022)

Modification of ISO 10303-1819 from IS to TS

Noting that

- ISO 10303 SMRL is a body of work collectively managed as a set of Technical Specification and subject to the procedures noted in SC4N2591 for managing standards in database format,
- ISO 10303-1819 is managed in SC 4 as a required element of the SMRL and has been since its inception,
- ISO/CS has found that ISO 10303-1819 is currently listed as an IS in the publication process for SMRL V9, and
- having found no root cause for the discrepancy,

SC4 requests its Committee Manager coordinate with ISO/CS and modify ISO 10303-1819 from IS to TS in the ISO Project Portal and continue with its incorporation into SMRL V9.

Resolution 1097: (Virtual Meeting – May 2022)

Change to project duration of ISO/CD 10303 – 239 ed3

Noting the progress of the project team to complete a DIS quality ISO 10303 – 239 ed3 and the actions taken to address the resources and technical mapping issues, SC 4 requests its Committee Manager coordinate with ISO/CS to move ISO 10303 – 239 ed3 (project id = 78832) from SDT 24 to SDT 36 with the following target dates:



N4118

2026 – 05 – 15

Stage 40 – DIS Registered: 2022-10-15 (Original Target: 2022-04-14)
Stage 50 – FDIS Registered: 2023-06-15 (Original Target: not set)
Stage 60 – IS Published: 2023-11-28 (Original Target: 2023-04-14)

Resolution 1098: (Virtual Meeting – May 2022)

Call for Experts to AG 2 - Implementation Forum

Noting the extensive work underway in industry, ISO, and IEC to formalize digital twin capabilities and the opportunity to share more broadly the value of SC 4 standards to the maturity of digital twin activities, SC 4 recognizes the value of the Implementation Forum and the rich discussion regarding digital twins and requests all interested experts to coordinate with their respective organizational representative and join the AG02 Implementation Forum.

Resolution 1099: (Virtual Meeting – May 2022)

Establish a TR for Reference model for industrial data

Recalling Resolution 1045 and the progress of Task Force 2 in accomplishing their terms of reference, notably the creation of a human readable guide complementary to the reference model, SC 4 establishes a TR for the reference model for industrial data, nominates Jean Brange as project leader and David Leal as project editor and encourages its P-members and A-liaisons to review their current participation in Task Force 2, appointing additional Experts as appropriate to support the development of the Technical Report.

Resolution 1100: (Virtual Meeting – May 2022)

Establish liaison from ISO/TC 184/SC 4 to ISO/IEC JTC 1/SC 41

Recalling the decision by ISO/JTC 1 to revise the scope of ISO/IEC JTC 1/SC 41 and incorporate digital twin with a mandate to serve as the focus and proponent for JTC 1's standardization programme and the creation of ISO/IEC JTC 1/SC 41/WG 6 – Digital Twin, and noting that ISO/TC 184/SC 4 possesses numerous requirements in the space of industrial data for digital twin, ISO/TC 184/SC 4 establishes a liaison to ISO/IEC JTC 1/SC 41 and nominates the following liaison representatives for a three-year term with appreciation to serve in that capacity:

- Soonhung Han

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Martin Hardwick
- Sangkeun Yoo
- Duan Haibo

Acclamations and Acknowledgements

Acclamation

SC 4 expresses its appreciation to our SC 4 Liaisons and reports provided to the 83rd Plenary.

Acclamation

SC 4 expresses its appreciation to participants and presenters of the Open Technical Forum, Implementation Forum, and Change Management Committee during the pre-plenary week of activities.

Acclamation

SC 4 expresses its appreciation to Meinolf Gröpper for his many years of service to ISO/TC 184/SC 4 and to the international standardization community. Meinolf entered into retirement between Plenary meetings and SC 4 wishes him all the best in this next phase of his journey.

Acclamation

SC 4 expresses its appreciation to Brandon Sapp for his leadership of ISO/TC 184/SC 4/TF 1, ISO/CD 10303-2, and his extensive contribution to the development of SC 4's Jira / GIT infrastructure. SC 4 extends its best wishes to Brandon in his new career endeavours.

Acclamation

SC 4 expresses its appreciation to Melissa Harvey for her service in the role of Quality Committee Secretary and her valuable contributions to its success. SC 4 extends its best wishes to Melissa in all her future career endeavours.

Acclamation

SC 4 expresses its sincere appreciation to all the experts, delegates, and leaders for

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

executing a completely virtual plenary meeting successfully while undergoing numerous challenges during this extraordinary time for stakeholders worldwide.

Resolution 1101: (Hamamatsu – Nov 2022)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 Approves the following Project Leaders and Project Editors for the listed Projects noted:

Project	Working Group	Project Leader/Editor
ISO/DTR 3151-1 (ID: 84703)	JWG 16	Soonhung Han

Resolution 1102: (Hamamatsu – Nov 2022)

Establish ballot duration of new work item proposals

SC 4 requests its Committee Manager to apply the 8-week ballot duration to the following NP ballots:

Project	Approval
ISO/NP 10303-238 ed 4	☒
ISO/NP 10303-62 ed2	☒
ISO/NP 8000-220	☒
ISO/NP 14306-1	☐
ISO/NP 14306-4	☐

Resolution 1103: (Hamamatsu – Nov 2022)

Appointment of Convenors and Deputy Convenors

Noting that continuity of operations and management of work is an important

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

consideration to the successful operation of advisory groups, SC 4 appoints the following Convenors and Deputy Convenors for a three-year term, with appreciation to serve in this capacity.

WG	Role	Appointment
QC	Convenor	Véronique Dubillot
WG 11	Convenor	David Loffredo

Resolution 1104: (Hamamatsu – Nov 2022)

Target Dates change to ISO/TS 23301 ed2

Noting the progress of the project team that is developing ISO/TS 23301 edition 2 and the balance to go in delivering the project as scheduled on 2022-12-01, SC4 requests its Committee Manager modify the target date for
Stage 20.20 (Working Draft study initiated) to 2022-11-05.
Stage 30.00 (Committee draft registered to 2023-03-05
Stage 60.60 (International Standard published) to 2023-09-01

Resolution 1105: (Hamamatsu – Nov 2022)

Moving ISO 14306 to a multipart standard

Noting the work conducted by JWG 16 and the ISO/PWI 14306 (project id. 85802) project team resolving industry requirements to address new capabilities in the ISO 14306 standard, SC 4 accepts the recommendation of JWG 16 and requests it's Committee Manager coordinate with ISO/CS and designate project id. 85802 as ISO/PWI 14306-3 and recognize ISO 14306 as a multi-part standard.

Resolution 1106: (Hamamatsu – Nov 2022)

AHG 2 Disbandment and NDES TR publication ballot launch

Noting the progress made by AHG 2 “Nuclear Digital Ecosystem Specification” and the draft technical report that has been delivered, SC 4 thanks AHG 2 members for their work and the fruitful collaboration with ISO/TC 85. SC 4 accepts the draft technical report and requests it's Committee manager to launch a DTR ballot and disbands AHG 2.

Resolution 1107: (Hamamatsu – Nov 2022)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Creation of WG 25 with the title “ISO CDD Validation Team”

SC 4 approves the creation of the WG 25 with the title “ISO CDD Validation Team” under the structure of ISO/TC 184/SC 4 and nominates Gernot ROSSI as convenor for a 3-year term (until the end of 2025) and Hiroshi MURAYAMA as deputy convenor for a 3-year term (until the end of 2025). ISO/TC 184/SC 4 Committee Manager is asked to take the necessary steps for the establishment of the WG 25 and requests national bodies appoint the validation team members to WG 25 through the ISO global directory.

Resolution 1108: (Hamamatsu – Nov 2022)

Cancellation of ISO 15926-14 DTR

Noting the ISO 15926-14 DTR (project id. 75949) no longer is considered useful in the light of the new developments on ISO 15926-14, SC 4 will request ISO CS to cancel the ISO 15926-14 DTR project.

Noting the desire to start a NP for an ISO 15926-14 (IS) and the current lack of full consensus on the NP in WG3, SC 4 instructs WG 3 to do a study with the top SC 4 experts on ISO 15926 and OWL and the project experts on the position of the proposed part in ISO 15925 and to examine whether ISO 15926-12 could have been used for reasoning on data sets with the same results. The study results should be reported in the PPC meeting of Jan 2023.

Resolution 1109: (Hamamatsu – Nov 2022)

Change to project duration of ISO/TS 10303-1848/1849/1850/1851/1852/1853

Noting the development work of SMRL v9 and the forecast for SMRL v10 and SMRL v11, SC 4 resolves to adjust the standards development track of the following ISO 10303 modules to SDT 36:

- ISO/AWI TS 10303-1848 (ed.1- id.81490), “Annotated 3d model data quality criteria”
- ISO/AWI TS 10303-1849 (ed.1- id.81491), “Annotated 3d model data quality inspection result”
- ISO/AWI TS 10303-1850 (ed.1- id.81492), “Triangulated shape data quality”
- ISO/AWI TS 10303-1851 (ed.1- id.81493), “PMI data quality”
- ISO/AWI TS 10303-1852 (ed.1- id.82033), “Retention period”

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- ISO/AWI TS 10303-1853 (ed.1- id.82522), “Datum and datum systems”

The modules will be published according to the following schedules:

- ISO 10303-1852 in SMRL v10, publication on 2023-05-31;
ISO 10303-1848, 1849, 1850, 1851 and 1853 in SMRL v11 publication on 2023-10-15.

Resolution 1110: (Hamamatsu – Nov 2022)

Conduct of a CIB for ISO 8000-3

SC 4 resolves to initiate a CIB that asks the following questions to the members of SC 4.

- 1) Do you approve adding a preliminary work item “ISO/PWI 8000-3, Data quality – Part 3: Information and data management system: Requirements” with Peter Eales as the project leader to the work programme of WG 13 as per the contents of ISO/TC 184/SC 4/WG 13/N0982? (responses: yes | no | abstain + allow comments via dialog box)
- 2) Please provide the name, affiliation and contact details of any of your experts who wish to work on this project. (allow comments via dialog box)
- 3) Please provide the name and type of any organizations that are based in your country and have expressed an interest in the benefits of this proposed MSS. (allow comments via dialog box)

SC 4 instructs WG 13 to present the proposal for review by the SC 4 Change Management Advisory Group before the CIB is initiated by the SC 4 Committee Manager.

Resolution 1111: (Hamamatsu – Nov 2022)

ISO 29002

SC 4 resolves to initiate a project to develop ISO 29002 (Industrial automation systems and integration — Exchange of characteristic data). This project has no change in scope from the project 78517.

The project is to begin at Stage 40 and use the SDT18 development track, with the target dates: circulation of the enquiry draft on 2023-04-30; publication of the International Standard on 2024-04-30. SC 4 appoints Peter Eales as project leader and invites its P-members and A-liaisons to nominate experts to participate in the project.

Resolution 1112: (Hamamatsu – Nov 2022)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Liaison Appointments

Noting the importance of Liaison Representatives to effective collaboration with SC 4 in the development of standards, SC 4 appoints and recognizes the following individuals. The appointments are for a three-year term [2022 – 2025]. SC 4 appreciates the service of all the identified individuals. Where indicated, SC 4 requests members to volunteer to fill vacancies.

Liaison Organization	Appointment
ISO/IEC JTC 1	Christophe Mouton
ISO/IEC JTC 1/SC 24	Soonhung Han
ISO/IEC JTC 1/SC 32	Sungjoon Lim
ISO/TC 29	Bengt Olsson
ISO/TC 59	Hiroshi Murayama
ISO/TC 67	Nils Sandmark
ISO/TC 154	Hiroshi Murayama
ISO/TC 211	Yunsub Lee
ISO/TC 213	Atsuto Soma

Resolution 1113: (Hamamatsu – Nov 2022)

Termination of Liaisons

Noting the analysis of liaisons, and the vacancies in nominated representatives from SC 4 to the committees noted below, SC 4 requests its Committee Manager to coordinate with ISO/CS and terminate the following liaison relationships.

REFERENCE	TITLE
ISO/IEC JTC 1/SC 31	Automatic identification and data capture techniques
ISO/TC 10/SC 1	Basic conventions
ISO/TC 37	Language and terminology
ISO/TC 118	Compressors and pneumatic tools, machines and equipment
ISO/TC 118/SC 1	Process compressors
ISO/TC 171	Document management applications
ISO/TC 172	Optics and photonics
ISO/TC 172/SC 1	Fundamental standards

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

[ISO/TC 184/SC 5](#) Interoperability, integration, and architectures for
enterprise systems and automation applications

[ISO/TC 207/SC 5](#) Life cycle assessment

[ISO/TC 229](#) Nanotechnologies

Acclamations and Acknowledgements

Acclamation

SC 4 expresses its appreciation to the following individuals for organizing the 84th ISO/TC 184/SC 4 Plenary in Hamamatsu, Japan from 2022-10-30 to 2022-11-04:

(JNC)

- Akihiko Ohtaka
- Atsuto Soma

(MSTC)

- Satoshi Abe
- Kenji Kitayama
- Toshiyuki Shirabe
- Yasunari Fujinuki

(Elysium)

- Mayumi Harakawa
- Kotoe Onoda
- Mark Melendy
- Yushi Shiozaki

Special acknowledgement is extended to the staff at Act City Hamamatsu Congress Center, the Crown Palais Hotel, and Okura Act City Hotel for their seamless and attentive support throughout the week.

Acclamation

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the SC 4 Open Technical Forum

- Paul van Exel, Convenor, WG 3

Dana Tripp

Secretary, ISO/TC 184/SC 4

trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Patrick Walsh, Project Co-leader, AP 239
- Stuart Galt, Convenor, TF 1

Acclamation

SC 4 expresses its appreciation to the following individuals and organizations for their participation in the SC 4 Plenary Industry Day event:

Akihiko Ohtako and Atsuto Soma, Moderators

Session I:

Presenters:

- Jean Brange, AFNet Services
- Takao Senko, Honda Motor Co., Ltd.

Session II:

Presenters:

- Atsuto Soma, JEITA 3D ISTEK (Elysium Co., Ltd.)
- Hiroyuki Hiraoka, Chuo University
- Mikitaka Hayashi, JGC Corporation, DAMA Japan
- Leo van Ruijven, Adesys
- Tim King, Babcock International

Session III:

Presenters:

- Sun Lim, Korea Electronics Technology Institute
- Hiroshi Nishi, LIGHTz Inc.
- Max Ungerer, PROSTEP AG
- Kenny Swope, The Boeing Company

Acclamation

SC 4 expresses its appreciation to our SC 4 Liaisons and their reports provided to the 84th Plenary.

- ISO/IEC Joint Task Force on Risk
- ISO/IEC JTC 1/SC 7
- ISO/TC 171/SC 2 written report
- PDES Inc.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- NAFEMS
- Small Business Standards
- Khronos
- Web3D Consortium
- ISO/IEC JTC 1/SC 24

Acclamation

SC 4 expresses its appreciation to Nils Sandsmark for organizing and facilitating the multi-group coordination meeting held on 2022-11-01 and to the participants who provided technical content to the betterment of SC 4 level harmonization.

Acclamation

SC 4 expresses its sincere appreciation to all the experts, delegates, and leaders for executing our first hybrid plenary meeting successfully while undergoing numerous challenges.

Resolution 1114: (Paris – June 2023)

Establish ballot duration of new work item proposals.

SC 4 requests its Committee Manager to apply the 8-week ballot duration to the following NP ballots:

Project	Approval
ISO/NP 23247-5	☒
ISO/NP 23247-6	☒
ISO/PWI 14306-2	☒

Resolution 1115: (Paris – June 2023)

Appointment of Convenors and Deputy Convenors

Noting that continuity of operations and management of work is an important consideration to the successful operation of advisory groups, SC 4 appoints the following Convenors and Deputy Convenors for a three-year term, with appreciation to serve in this capacity.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

WG	Role	Appointment
WG 3	Deputy Convenor	Nils Sandmark

Resolution 1116: (Paris – June 2023)

Create a Task Force for the SC 4 Common Change Process

Following the workshop held on 19 April 2023 by members of the PPC and the recommendations developed as a result of reviewing the procedures for managing standards in database format for WG 21, WG 22, WG 23, and JWG 24, SC 4 requests its Committee Manager establish a Task Force on the SC 4 Common Change Process with the following terms of reference:

- Create a Committee Specific Procedure for WG 21, WG 22, WG 23, and JWG 24 that harmonizes the process for maintaining standards as a database and propose the procedure to SC 4 for consideration as a standing document,
- Revise the existing procedures for WG 21 (SC4N2617, SC4N2591), WG 22 (SC4N2984), WG 23 (SC4N2612), and JWG 24 (SC4N3708) to incorporate the proposed Common Change Process and propose to SC 4 for consideration as a standing document,
- Draft/revise user level documents for implementing the Common Change Process at the technology application level for each WG, and
- Update SC4 at each meeting on progress until expected completion in Fall 2024. SC 4 nominates the following to the Task Force with appreciation for serving on this important task:
 - Nils Sandmark
 - Jochen Haenisch
 - Paul van Exel
 - Dr. Tim King (Convenor)
 - Hiroshi Murayama

And invite National Bodies to nominate members to the task force.

Resolution 1117: (Paris – June 2023)

Continuation of ISO/TC 184/SC 4/TF 2 following up on ISO/TR 17999

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Noting the progress made on ISO/TR 17999 - SC 4 Reference Model for Industrial Data and the need to make available the acquired knowledge to the various potential users of it, SC 4 concurs with the recommendation of ISO/TC 184/SC 4/TF 2 to extend the mandate of TF 2 and add to its terms of reference the following:

- Build upon the ISO/TR 17999 SC 4 Reference Architecture;
- Share with other committees, subcommittees and industrial standardisation groups the opportunity proposed in ISO/TR 17999 to their standards;
- Develop a guide (technical report) on how ISO/TR 17999 could assist the implementation of computer processable ontologies, terminologies, and the activities necessary to manage them;
- And invite National Bodies to nominate members.

Resolution 1118: (Paris – June 2023)

Create a liaison relationship from ISO/TC 184/SC 4 to ISO/TC 164/SC 1

Noting the need to collaborate with ISO/TC 164/SC 1, SC 4 requests its Committee Manager establish a liaison to ISO/TC 164/SC 1 and appoints David Leal to serve as liaison representative for a three-year term [2023 – 2026], with appreciation for his offer to serve in that capacity.

In addition, SC 4 appoints Tom Thurman as a liaison representative from ISO/TC 184/SC 4 to ISO/TC 213 for a three-year term [2023 – 2026], with appreciation for his offer to serve in that capacity.

Resolution 1119: (Paris – June 2023)

ISO/DIS 29002 change to SDT 36 months

SC 4 resolves to assign ISO 29002 (Industrial automation systems and integration — Exchange of characteristic data) to the SDT 36 development track, with the target dates: circulation of the enquiry draft on 2023-07-31; publication of the International Standard on 2024-07-31.

Resolution 1120: (Paris – June 2023)

Launch ISO/NP Automation systems and integration – Ontology-based interoperability

Noting the proposal from Standard Norge (SN) for a new multi part standard, ISO/TC



N4118

2026 – 05 – 15

184/SC 4:

- Requests its Committee Manager launch ISO/NP Automation systems and integration – Ontology-based interoperability with an eight-week ballot duration;
- Create a Work Group entitled WG 26 “Ontology-based interoperability” to:
 - Develop and maintain ISO xxxxx: “Automation systems and integration - Ontology-based interoperability,”
 - Develop and maintain guidance for the implementation and use of the standard series,
 - Develop and maintain detailed specifications for integration and re-use of existing reference data libraries and ontologies using the standard series,
- Requests WG 26 to work in close cooperation with WG 3, WG 12, WG 15, WG 21, and WG 22 to align with, reuse and complement their existing standards, and avoid overlaps and conflicts with ISO 10303, ISO 15926, and ISO 23247 and other SC 4 standards,
- Nominates Pål Rylandsholm as convener of the work group, and
- Invites member bodies to nominate additional members to the work group.

Resolution 1121: (Paris – June 2023)

Launch ISO/TR – Automation systems and integration - Core terminology for industrial data

Noting the report from ISO/TC 184/SC 4/AhG 1 - Core industrial data set of terms, and the decision taken in Resolution 1068, ISO/TC 184/SC 4 requests its Committee Manager to ballot AG 3/N2 “Automation systems and integration - Core terminology for industrial data” as a DTR. SC 4 nominates David Leal as project leader.

Resolution 1122: (Paris – June 2023)

Change ISO/TS 23301 ed2 to SDT 24

Noting the progress of the project team developing ISO/TS 23301 edition 2 and the submission of the document to ISO for FDIS ballot on 2023-06-12, and the concern that a second FDIS may be required, SC4 requests its Committee Manager to set the SDT at 24 and adjust the target date for Stage 60.60 (International Standard published) to 2024-03-01.

Resolution 1123: (Paris – June 2023)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Start of a PWI for an amendment of ISO 13584-501

Noting that:

- WG 25 embodies the ISO Validation Team for Change Requests that uses the CSP that is already installed, and;
- WG 25 already started processing of the first CR on ICS (International Classification for Standards) submitted by Japan National Committee for ISO TC 184/SC 4, on June 13, 2023 and;
- There is a strong request from ISO/CS to formally and substantially restore the lost content of the ISO 13584-501 RA; ISO TC 184/SC 4 supports the start of preliminary work item for publishing an amendment on ISO 13584-501 with an aim to replace the maintenance procedure described in the standard based on RA with the CSP for CDD. SC 4 nominates Hiroshi Murayama as project leader.

Resolution 1124: (Paris – June 2023)

Skip CD and proceed to DIS ballot for ISO 14306-1

Recalling Annex SS of the ISO/IEC Directives Part 1, and noting the consensus found in JWG 16 during the comments resolution process of ISO/WD 14306-1 and that the resulting draft is estimated to have a DIS maturity, SC4 resolves to skip the CD ballot (stage 30.0) and move directly to a DIS ballot (Stage 40.0).

Resolution 1125: (Paris – June 2023)

Create a liaison relationship from ISO/TC 184/SC 4 to ISO/TC 268

Noting the need to collaborate with ISO/TC 268, SC 4 requests its Committee Manager establish a liaison to ISO/TC 268 and appoints Yunsub Lee to serve as liaison representative for a three-year term [2023 – 2026], with appreciation for his offer to serve in that capacity.

Acclamations and Acknowledgements

Acclamation

SC 4 expresses its appreciation to the following individuals for organizing the 85th ISO/TC 184/SC 4 Plenary in Paris, France from 2023-06-11 to 2023-06-17:

AFNOR

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Franck LEBEUGLE,
- Stéphane MOULIERE
- Mélissa DE MEDEIROS
- Nathalie DA SILVA
- Arthur RIBEMONT

Special acknowledgement is extended to all the staff at AFNOR for their seamless and attentive support throughout the week.

Acclamation

The French organization team representing the French ISO/TC 184/SC 4 mirror committee, AFNOR/CP IDMI:

Patrick LAMBOLEY, Schneider Electric
Christophe MOUTON, EDF

Acclamation

The Digital transformation ISO/TC 184 symposium host at EDF Lab in Saclay:

- Pascal TERRIEN, EDF

Acclamation

SC 4 expresses its appreciation to the following individuals for organizing and facilitating three informative workshops for this super-meeting;

- Martin Hardwick - Industrial Implementation Forum
- Véronique Dubillot - Quality and Standards Development
- Kenny Swope - Digital Twin Roadmap

Acclamation

SC 4 expresses its appreciation to our SC 4 Liaisons and their reports provided to the 85th Plenary.

Acclamation

SC 4 expresses its appreciation to Nils Sandsmark for organizing and facilitating the

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

multi-group coordination meeting held on 2023-06-13 and to the participants who provided technical content to the betterment of SC 4 level harmonization.

Acclamation

SC 4 expresses its sincere appreciation to all the experts, delegates, and leaders for executing our hybrid plenary meeting successfully while undergoing numerous challenges.

Resolution 1126: (Saratoga Springs, USA – October 2023)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 Approves the following Project Leaders and Project Editors for the listed Projects noted:

Project	Working Group	Project Leader/Editor
ISO 23247-5	WG 15	Robert Rencher (Co-leader)
ISO 23247-6	WG 15	Guodong (Gordon) Shao (Co-leader)
ISO/PWI 14306-3	JWG 16	Trevor Leeson

Resolution 1127: (Saratoga Springs, USA – October 2023)

Establish ballot duration of new work item proposals.

SC 4 requests its Committee Manager to apply the 8-week ballot duration to the following NP ballots and coordinate with the appropriate committees in joint work:

Project	Approval
ISO/NP 8000-119 Data quality — Part 119: Application of ISO 8000-115 to transport unit identifiers	☒
ISO/NP 14306-4 Industrial automation systems and integration — JT file format specification for 3D visualization — Part 4: Version 3 (In coordination with ISO/TC 171/SC 2 and ISO/IEC JTC 1/SC 24)	☒



N4118

2026 – 05 – 15

Resolution 1128: (Saratoga Springs, USA – October 2023)

Appointment of Convenors and Deputy Convenors

Noting that continuity of operations and management of work is an important consideration to the successful operation of advisory groups, SC 4 appoints the following Convenors and Deputy Convenors for a three-year term, with appreciation to serve in this capacity.

WG	Role	Appointment
AG 0	Convenor	Kenny Swope
TF 2	Convenor	Jean Brange
PPC	Convenor	Kenny Swope
WG 23	Deputy Convenor	Hayley Thompson
WG 26	Deputy Convenor	Maja Miličić Brandt

Resolution 1129: (Saratoga Springs, USA – October 2023)

Launch DTR ballot of ISO TR 24464 ed2

Noting that

- A revision to ISO/TR 24464 ed1 started in 2021-11 and was reviewed at the change management meeting,
- A working group consultation was held in 2021-11, and
- A CIB was launched in 2022-10,

SC 4 appreciates the recommendation of ISO/TC 184/AG 0 and requests its Committee Manager coordinate with ISO/CS and launch a DTR ballot of Visualization elements of digital twin - Visualization fidelity as edition 2.

Resolution 1130: (Saratoga Springs, USA – October 2023)

Revise the title of ISO/DIS 3151-2

Noting that resolution of comments for ISO/CD 3151-2 has revised the title to be: Visualization elements of PLM-MES interface - 3D error feedback in heavy industry, and the project id (84702), project portal states the title as: Visualization elements of PLM-MES interface — Part 2: 3D error feedback in the plant industry, SC4 requests its Committee Manager coordinate with ISO/CS to change the title of project 84702 to be Visualization elements of PLM-MES interface - 3D error feedback in heavy industry and



N4118

2026 – 05 – 15

proceed with the DIS ballot.

Resolution 1131: (Saratoga Springs, USA – October 2023)

Restart ISO/DTR 23164 as ISO/DTS 23164

Noting the report from ISO/TC 184/SC 4/AG 3, the recommendation of the ISO Editor and the consensus of the PPC ISO/TC 184/SC 4, SC 4:

- Requests its Committee Manager to coordinate with ISO/CS to proceed with balloting ISO/CD TR 23164 ed1 (project id. 87378),
- Requests its Committee Manager to launch ISO/NP 23164 – Automation systems and integration – Core terminology for industrial data as a TS with an eight-week ballot duration at stage 40.00,
- Nominates David Leal as project leader, and
- Invites member bodies to nominate additional experts to this project.

Resolution 1132: (Saratoga Springs, USA – October 2023)

Restart SMRL modules in support of SMRL V11

Recalling the creation of ISO/AWI TS 10303-1848, -1849, -1850, -1851, and -1853 as required modules of the ISO 10303 SMRL subject to the special procedures for managing standards in a database format, and noting the progress of the project team to complete a CD quality ISO TS 1848, 1849, 1850, 1851, and 1853, SC 4 notes the pending limit date for this set of projects and appreciates the actions taken to address the integration issues. Further, SC 4 requests its Committee Manager coordinate with ISO/CS to restart ISO/AWI TS 1848 (project id = 81490), 1849 (project id = 81491), 1850 (project id = 81492), 1851 (project id = 81493) and 1853 (project id = 82522) on track with SDT 18 at stage 10.99 with the following target dates:

Stage 30 – CD registered: 2023-12-15

Stage 50 – Final text received: 2024-3-15

Stage 60 – IS published: 2024-6-15

And nominates Hiroyuki Hiraoka as project leader for -1848, -1849, -1850, and -1851, and nominates Tom Thurman as project leader for -1853.

Resolution 1133: (Saratoga Springs, USA – October 2023)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Launch an Amendment to ISO 17506 ed2

Noting the recommendation of ISO/TC 184/SC 4/AG 0, SC4 requests its Committee Manager to coordinate with ISO/CS to amend ISO 17506:2022 with the following schedule:

- DIS (40.00): 2024-02-28
 - Publication (60.00): 2024-04-30,
- And nominates Hwanyong Lee, hwan@ajou.ac.kr as project leader.

Resolution 1134: (Saratoga Springs, USA – October 2023)

Launch ISO/DTR 8000-320

SC 4 requests its Committee Manager to coordinate with ISO/CS to initiate the development of ISO/TR 8000-320 with the title “Data quality – Part 320: AI training data quality for smart manufacturing” with Giljo Kim (KOR) as project leader and the target dates:

- circulation of draft technical report by 2023-12-31,
- publication of the technical report on 2024-06-30.

Resolution 1135: (Saratoga Springs, USA – October 2023)

Launch ISO 8000-115 – Minor revision

SC 4 resolves to initiate the development of a minor revision of ISO 8000-115 (Master data: Exchange of quality identifiers: Syntactic, semantic and resolution requirements) with Hayley Thompson (USA) as project leader and the target dates:

- submission of revised document by 2023-12-31,
- publication of the minor revision by 2024-06-30.

SC 4 requests its Committee Manager to prepare and submit to ISO/CS the necessary ISO Form 8B for this minor revision.

Acclamations and Acknowledgements

Acclamation

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 expresses its appreciation to the following individuals at STEP Tools, Inc. for organizing the 86th ISO/TC 184/SC 4 Plenary in Saratoga Springs, NY from 2023-10-22 to 2023-10-27.

- Lesli Garnett
- Martin Hardwick
- Joe Fritz

Special acknowledgement is extended to all the staff at the Gideon Putnam Hotel, and the Holiday Inn in Saratoga Springs for their seamless and attentive support throughout the week.

Special acknowledgement is extended to all the staff of SUNY Empire State University, including:

- Josh Fleming
- Renee Lienau
- Jeffrey Kurto

Acclamation

SC 4 expresses its appreciation to the following individuals for contributing and presenting at the Open Technical Forum discussion on Monday evening.

- Hiroshi Murayama
- Paul van Exel
- Stuart Galt

Acclamation

SC 4 expresses its appreciation to Nils Sandsmark for organizing and facilitating the multi-group cross-coordination meeting held on 2023-06-24 and to the participants who provided technical content to the betterment of SC 4 level harmonization.

Acclamation

SC 4 expresses its appreciation to the following individuals for contributing and presenting at the Implementation Forum on Tuesday evening,

- Martin Hardwick
- David Loffredo
- Soonhung Han

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Acclamation

SC 4 expresses its appreciation to the following individuals for contributing and presenting at Industry Day on Wednesday morning.

- Sangkeun Yoo
- Sung Hei Kim
- Chi Ho Noh
- Brad S. Carlberg

Acclamation

SC 4 expresses its appreciation all the staff of the Lake George Steamboat Company for a fantastic cruise of Lake George on the Mohican.

Acclamation

SC 4 expresses its appreciation to the staff at Star Trek Original Series Set Tour for an amazing tour.

- James Cawley
- Patrick Cawley
- Anthony Willet
- Myleigh Willet
- Ashley Brand
- Sue Cross
- Marybeth Ritkouski
- Dave Loffredo

Acclamation

SC 4 expresses its appreciation to the staff at Burleigh's Luncheonette for a great dining experience.

- Debbie, Don, and Mark Barber
- Dylan Menke
- Matthew Michalak

Acclamation

SC 4 expresses its appreciation to all our SC 4 Liaisons who provided reports at the 86th Plenary.

- ISO/IEC JTC1/SC 32 - Liaison Mr. Sungjoon Lim (F2F)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- ISO/IEC JTC 1/SC 24 Liaison Mr. Christophe Mouton (V)
- ISO/IEC JTC 1/SC 7 Liaison Dr. Timothy King (V)
- ISO/TC 37/SC 3 Liaison Mrs. Sabine Mahr (V)
- ISO/TC 85 Liaison Mr. Christophe Mouton (V)
- ISO/TC 171/SC 2 Liaison Mr. Stuart Galt (F2F)
- DMSC A-Liaison Ms. Jennifer Herron and Ms. Rosemary Astheimer (F2F)
- ECCMA A-Liaison Mr. Peter Benson (F2F)
- IACS A-Liaison Mr. He Ci (V)
- NAFEMS A-Liaison Mr. David Leal (F2F)
- OMG A-Liaison Ms. Allison Barnard Feeney (F2F)
- PDES A-Liaison Mr. Patrick Walsh (F2F)
- SBS A-Liaison Mr. Jochen Haenisch (V)
- Web 3D Consortium A-Liaison Mr. Vincent Marchetti (V)

Acclamation

SC 4 expresses its sincere appreciation to all the experts, delegates, and leaders for executing our hybrid plenary meeting successfully while undergoing numerous challenges.

Resolution 1136: (Seoul, Korea – May 2024)

Approval of Project Leaders and Project Editors for SC 4 Projects.

SC 4 Approves the following Project Leaders and Project Editors for the listed Projects noted:

Project	Working Group	Project Leader/Editor
ISO 14306 series	JWG 16	Trevor Coulson (Editor)

Resolution 1137: (Seoul, Korea – May 2024)

Establish ballot duration of new work item proposals.

SC 4 requests its Committee Manager to apply the 8-week ballot duration to the following NP ballots and coordinate with the appropriate committees in joint work:

Project	Approval
ISO/NP 15926-100 Automation systems and integration – Integration of life-cycle data for process plants including oil and gas production facilities – Part 100: Vocabulary	☒
ISO/NP 8000-230 Data quality – Part 230: Sensor Data: Guidelines for	☒

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

data cleansing	
ISO/NP Automation systems and integration - Core terminology for simulation data management	☒
ISO/NP Automation systems and integration – Industrial digital twin interface architecture	☒

Resolution 1138: (Seoul, Korea – May 2024)

Appointment of Convenors and Deputy Convenors.

Noting that continuity of operations and management of work is an important consideration to the successful operation of advisory groups, SC 4 appoints the following Convenors and Deputy Convenors for a three-year term, with appreciation to serve in this capacity.

WG	Role	Appointment
AG 02 IF	Convenor	Van Exel, Paul
AG 02 IF	Co-Deputy	Yoon, Jeong-Hyuk
AG 02 IF	Co-Deputy	Lee, Changsoo
AG 02 IF	Co-Deputy	Hardwick, Martin
AG 02 IF	Co-Deputy	ZHANG Wei Qun, Kevin
AG 03	Convenor	Sandsmark, Nils
Quality	Deputy	Lee, Changsoo
TF 1	Convenor	Galt, Stuart
WG 12	Convenor	Hunten, Keith
WG 13	Convenor	King, Tim M.
WG 13	Deputy	Mougeolle, Anne-Marie
JWG 16	Co-Convenor	Mouton, Christophe
JWG 16	Co-Convenor	Han, Soonhung
JWG 16	Secretary	Lee, Jaein
WG 21	Convenor	Hunten, Keith
WG 23	Convenor	King, Tim
JWG 24	Co-Convenor	Murayama, Hiroshi
JWG 24	Deputy	Sonoda, Yoshiaki

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Resolution 1139: (Seoul, Korea – May 2024)

Move ISO 18136 to a multi-part standard.

Noting that WG 3 and the ISO/CD 18136 project team foresee the development of new capabilities of the ISO 18136 standard to meet further nuclear industry requirements, SC 4 accepts the recommendation of WG 3 and requests its Committee Manager coordinate with ISO/CS and designate project id 85184 as ISO/CD 18136-1 and recognize ISO 18136 as a multi-part standard and update the title as: Automation systems and integration — Nuclear digital ecosystem — Part 1: Overview and framework noting no change in scope to the project or its schedule.

Resolution 1140: (Seoul, Korea – May 2024)

Update to the ISO/TC 184/SC 4 Handbook.

Noting the work accomplished by ISO/TC 184/SC 4/PPC, SC 4 requests its Committee Manager submit ISO/TC 184/SC 4/PPC N308 for a Committee Internal Ballot of eight weeks duration for approval as a standing document of SC 4 and replace the current SC 4 Handbook (SC4N3713).

Resolution 1141: (Seoul, Korea – May 2024)

Start a PWI for a TR on Industrial requirements for product data visualization and perspectives.

Noting the proposal from JWG 16 for creating a technical report supporting future developments in the scope of visualization for industrial data and especially industrial visualization requirements, SC 4 requests its Committee Manager to coordinate with ISO/CS to initiate the development of a technical report with the title “Automation systems and integration — Industrial data — Industrial requirements for product data visualization and perspectives”. SC4 appoints Christophe MOUTON and Atsuto SOMA as project leaders, Soonhung HAN as editor, and requests SDT 18 with the following target dates:

- Circulation of 1st Draft Technical Report to experts: for the Fall 2024 SC 4 plenary
- Draft Technical Report ballot: beginning 2025
- Publication: Fall 2025

And allocates the project to JWG 16

SC 4 requests its Committee Manager to activate the OSD once the PWI is initiated.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Acclimation

SC4 expresses its appreciation to the following individuals at KSTEP for organizing the 87th ISO/TC 184/SC 4 Plenary in Seoul, Korea from 2024-05-12 to 2024-05-17.

- Soonhung Han
- Chiho Noh
- Seongjae Kim
- Myeongjun Kim
- Jaein Lee
- Jungbin Im

Special acknowledgement is extended to all the staff at the K-Hotel for their seamless and attentive support throughout the week.

Acclimation

SC4 expresses its appreciation to the following individuals for contributing and presenting at the Open Technical Forum discussion on Monday evening.

- Soonhung Han, Kevin Zhang, Christophe Mouton, Atsuto Soma, Hiroshi Murayama
- Stuart Galt, Allison Barnard Feeney, Pal Rylandsholm, Max Ungerer
- Jean Brange, Martin Hardwick, Keith Hunten
- Paul van Exel, Nils Sandmark, David Loffredo
- Kenny Swope, Veronique Dubillot, Dana Tripp

Acclimation

SC4 expresses its appreciation to the following individuals for contributing and presenting at Industry Day on Tuesday.

Track 1	Track 2
Soonjo Kwon (Moderator)	Sukkeun Cha (Moderator)
Kenny Swope	Kyungik An
Sungyeon Park	Gunwoo Kwon
Christophe Mouton	Byungjoo Kim
Hyunsik Lee	Haegon Lee
Kevin Zhang	Chunsik Lee

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

Kyoungjae Sun (Moderator)	
Kangjae Lee	Sukkeun Cha
Jiyeon Son	Chongwook Chin
Max Ungerer	Taehwan Kim
Jaehyun Lee	Peter Eales
Hiroshi Murayama	Liu Dan
SooJin Park	Yoshiaki Sonoda
Jean Brange	Sangkeun Yoo
Yuichi Sagawa	

Special acknowledgement is extended to all the staff at COEX and all the sponsors or Industry Day; KATS, ETRI, TUKOREA, Grand ICT, WeDataLab, SEABURY Solutions, YUIL, KOIIA, and KSTEP.

Acclimation

SC 4 expresses its appreciation to the staff of Asunje for an enjoyable Korean dining experience at our social event on Tuesday evening.

Acclimation

SC 4 expresses its appreciation to all our SC 4 Liaisons who provided reports at the 87th Plenary.

- ISO/IEC JTC 1/SC 24 Liaison Mr. Christophe Mouton (F2F)
- ISO/TC 85 Liaison Mr. Christophe Mouton (F2F)
- ISO/TC 171/SC 2 Liaison Stuart Galt (F2F)
- PDES A-Liaison Mr. Patrick Walsh (Written)
- Web 3D Consortium A-Liaison Mr. Vincent Marchetti (V)
- The Khronos Group Liaison Hwanyong Lee (F2F)

Acclimation

SC 4 expresses its sincere appreciation to all the experts, delegates, and leaders for successfully executing our Spring plenary meeting in a hybrid format.

Resolution 1142: (Stavanger, Norway – October 2024)

Establish ballot duration of new work item proposals

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 requests its Committee Manager to apply the 8-week ballot duration to the following NP ballots and coordinate with the appropriate committees in joint work:

Project	Approval
ISO/NP 23726-2 Industrial automation systems and integration – Ontology based interoperability – Part 2: Vocabulary	☒
ISO/NP 23726-100 Industrial automation systems and integration – Ontology based interoperability - Part 100: Schedule data ontology	☒
ISO/NP 8000-200 Data quality — Part 200: Transaction data: Quality of transaction data	☒

Resolution 1143: (Stavanger, Norway – October 2024)

Appointment of Convenors and Deputy Convenors

Noting that continuity of operations and management of work is an important consideration to the successful operation of advisory groups, SC 4 appoints the following Convenors and Deputy Convenors for a three-year term, with appreciation to serve in this capacity.

WG	Role	Appointment
WG 3	Convenor	Paul van Exel
WG 12	Deputy Convenor	Melissa Harvey
WG 15	Convenor	Martin Hardwick
WG 15	Deputy Convenor	Bengt Olsson
WG 21	Deputy Convenor	Melissa Harvey
WG 22	Convenor	Nils Sandmark
WG 22	Deputy Convenor	Paul van Exel

Resolution 1144: (Stavanger, Norway – October 2024)

Reconfirm Deputy Convenor special responsibilities for WG 13/23

Noting that continuity of operations and management of work is an important consideration to the successful operation of working groups, ISO/TC 184/SC 4 recognizes the role of a Deputy Convenor to support the Convenor in the completion of

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

his/her tasks when the Convenor is temporarily unavailable to do so. Though not necessary to have a resolution from a procedural perspective to enact this function of the Deputy Convenor, ISO/TC 184/SC 4 nonetheless reconfirms this Deputy Convenor role and approves the following Deputy Convenors to serve in this capacity until further notice.

Working Group	Role	Name
ISO/TC 184/SC 4/WG 13	Deputy Convenor	Mougeolle, Anne-Marie
ISO/TC 184/SC 4/WG 23	Deputy Convenor	Thompson, Hayley

Resolution 1145: (Stavanger, Norway – October 2024)

Establish a liaison from ISO/TC 184/SC 4 to ISO/TC 258

Recognizing that the development of ontology standards is best served by access to domain experts for the terminology and vocabulary of the subject domain and the proposal to develop schedule data ontology is before SC4, and necessary domain expertise exists within ISO/TC 258: Project, program and portfolio management, SC 4 requests its Committee Manager establish a liaison from SC4 to ISO/TC 258 and appoints Lillian Hella to serve as liaison representative for a three-year term [2024 – 2027], with appreciation for her offer to serve in that capacity.

Resolution 1146: (Stavanger, Norway – October 2024)

Update liaison from ISO/TC 184/SC 4 to ISO/TC 10 and IEC/TC3

Noting the need to collaborate with ISO/TC 10, and with IEC/TC 3, SC 4 requests its Committee Manager to establish a liaison from SC4 to ISO/TC 10 and to IEC/TC 3 and appoints Christophe Mouton, Yves Keraron, Hiroshi Murayama, Arild Waaler, Hayley Thompson, and Yoshiaki Sonoda as liaison representatives for a three-year term [2024 – 2027], with appreciation for their offer to serve in that capacity.

Resolution 1147: (Stavanger, Norway – October 2024)

Launch NP for ISO/TS 15926-4 ed4 by resolution

Noting that a new update to ISO/TS 15926-4 is needed and there is no change in scope to the standard, SC4 requests its Committee Manager to coordinate with ISO/CS and launch edition 4 using the SDT 18 track, names Onno Paap as Project Leader, and

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

allocates the work to WG 3 with the following terms of reference:

- Update ISO/TS 15926-4 to ed.4 as presented to the Change Board during the 88th Plenary of SC4 on 22 Oct 2024 with the following schedule milestones:
 - Proposed date for first meeting: 2024-11-21
 - Circulation of 1st Working Draft (if any) to experts: 2024-11-07
 - Committee Draft ballot (if any): 2025-05-07
 - DIS submission: 2025-11-07
 - Publication: 2026-05-07
- Update the electronic inserts to ISO/TS 15926-4 ed.4 per SC4N2591, SC4N2984, and SC4N2986 standing documents,
- Request national bodies to nominate their representatives and deputies to the Validation Team (VT) as noted in SC4N2541, and
- To keep a register of the VT members for the practice of sending out notifications to vote and other administrative items.

Resolution 1148: (Stavanger, Norway – October 2024)

Launch PWI for ISO 10303-242 ed5 and nominate experts

Noting the proposal from WG 12 for creating edition 5 of ISO 10303 – 242 to incorporate additional capabilities from industry requirements, SC 4 requests its Committee Manager to coordinate with ISO/CS and initiate the revision of ISO 10303 – 242 ed4 as a Preliminary Work Item with an expected submission of NP ballot on or before 2025-10-15. SC4 allocates the project to WG 12 and appoints Jean Brangé as project leader with nominations of the following experts to participate:

- AFNOR: Frédéric Darré, Véronique Dubillot, Pierre Duchier, Jacques Heinisch,
- ANSI: Rosemary Astheimer, Allison Barnard Feeney, Keith Hunten, Ben Urick, Stuart Galt, Melissa Harvey, Michelle Drummond, Brian Bentrim, Benjamin Standfield, John Van Horn, Pat Walsh, Tom Thurman,
- BSI: Robert Swindells,
- DIN: Bernd Feldvoss, Guillaume Hirel, Lothar Klein, Max Ungerer,
- JISC: Hiroyuki Hiraoka, Fumiki Tanaka, Atsuto Soma,
- SAC: Hyman Duan,
- SIS: Mikael Hedlind, Bengt Olsson, Magnus Lundgren, Torbjörn Holm,
- SN: Nils Sandmark,

Further, due to the special nature of ISO 10303 – 242 SC 4 requests ISO to grant a waiver on the mandatory use of the ISO Online Standards Development platform for this project.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Resolution 1149: (Stavanger, Norway – October 2024)

Launch a PWI for ISO 23726-1

Noting the proposal from WG 26, ISO/TC 184/SC 4 decides to initiate a project which will develop a New Work Item Proposal (NP) by October 2025 with the following terms of reference:

- Register a Preliminary Work Item ISO/PWI 23726-1 Overview and fundamental principles,
- Allocate the work to ISO/TC 184/SC 4/WG 26,
- Nominate Melinda Hodkiewicz as the project leader, and
- Invites member bodies to nominate additional experts to the project.

Resolution 1150: (Stavanger, Norway – October 2024)

Revise schedule for ISO/AWI TR 8000-320

Noting the progress on ISO/AWI TR 8000 – 320 ed1 and the commitment of the project development team to resolve the current resource constraints, SC 4 requests its Committee Manager to coordinate with ISO/CS and move the ISO/AWI TR 8000-320 ed.1, ISO.PID:88287

project from SDT 18 to SDT 36 with the following target dates for each stage:

- Committee Draft: 2025-03-31
- Draft Technical Report: 2025-08-31
- Publication date: 2025-12-31.

SC 4 requests national bodies to nominate additional experts to the project.

Resolution 1151: (Stavanger, Norway – October 2024)

Revise schedule for ISO/CD 10303 – 238 ed4

Noting the action taken at the 87th Plenary meeting of ISO/TC 184 SC/4 to place ISO/CD 10303 – 238 ed4 on hold to execute round robin testing of capabilities prior to incorporation, SC 4 resolves to modify the target dates for the project as follows:

- Enquiry stage (DIS): 2025-12-30,
- Publication stage: 2026-12-30.



N4118

2026 – 05 – 15

Acclamation

SC 4 expresses its sincere appreciation to Jochen Haenisch for many years of dedicated service and leadership as the deputy convenor of ISO/TC 184/SC 4/WG 12. SC 4 wishes all the best to Jochen in his next journey and future endeavors.

Acclamation

SC 4 expresses its appreciation to the following organizations for their contributions in sponsoring 88th ISO/TC 184/SC 4 Plenary in Stavanger Norway. Those contributions include meeting room facilities for our working groups, daily refreshments in the business center, the welcome reception on Sunday, and the social event on Wednesday.

- Equinor
- Aker BP
- DNV
- Standards Norway

Acclamation

Special acknowledgement is extended to the following individuals for organizing and hosting the 88th ISO/TC 184/SC 4 Plenary in Stavanger, Norway from 2024-10-21 to 2024-10-25.

- Inghild Kaarstad, Standards Norway
- Knut Aune, Standards Norway
- Nils Sandsmark, Standards Norway
- Heljar Ballo, Equinor

Acclamation

SC 4 expresses its appreciation to all the staff at the Scandic Stavanger Forus Hotel, for their seamless and attentive support throughout the week while hosting the PPC meeting, the QC Workshop, the Welcome Reception and the Open Technical Forum.



N4118

2026 – 05 – 15

Acclamation

SC 4 expresses its appreciation to Véronique Dubillot for leading the Quality Committee Workshop on Friday and engaging our members with quizzes and activities all week. Her dedication contributes to the improvement of standards development within SC 4.

Acclamation

SC 4 expresses its appreciation to Jean Brangé for leading and presenting at the Open Technical Forum on Monday evening to facilitate the discussion on the ISO 10303 Roadmap.

Acclamation

SC 4 expresses its appreciation to Nils Sandsmark for organizing and facilitating the multi-group cross-coordination meeting held on 2024-10-22 and to the participants who provided technical content to the betterment of SC 4 level harmonization.

Acclamation

SC 4 expresses its appreciation to the following individuals for contributing and presenting at Industry Day on Wednesday. The delegation from Sweden is very grateful for the well managed Industry Day.

- Heljar Ballo, Equinor
- Kenneth Swope, Boeing
- Per Erik Bøe Hansen, Equinor
- Åse Lunde, SN
- Roberta Gerasymchuk, ISO
- Anders Gierver Aibel
- Jerker Delsing, LTU Univ Luleå
- Heiner Temmen, DEXPI
- Jean-Charles Leclerc, Total Energies
- Heljar Ballo, Equinor, Chair ILAP
- Thor Kvamsdal, Sintef
- Marius Slagsvold, Step Solutions
- Maja Brandt, Siemens
- Helge Schiøtt, AkerBP
- Espen Schulze, Cobuilder
- Svein Harald Olsen, Statnett

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Karl Johan Pedersen, DNV
- Yong-Kwan Lee, KSTEP

Acclamation

SC 4 expresses its appreciation to the staff at Norsk Oljemuseum for the informative tour and the staff at Bølgen & Moi for a great dining experience during our Social Event on Wednesday evening.

Acclamation

SC 4 expresses its appreciation to all our SC 4 Liaisons who provided reports at the 88th Plenary.

- ISO/IEC JTC 1/SC 24
- ISO/TC 85
- ISO/TC 171/SC 2
- Web 3D Consortium
- The Khronos Group
- ECCMA

Acclamation

SC 4 expresses its sincere appreciation to all the experts, delegates, and leaders for successfully executing our hybrid plenary meeting in Stavanger, Norway.

Resolution 1152: (Little Rock, Arkansas, USA – May 2025)

Approval of Project Leaders and Project Editors for SC 4 Projects

SC 4 Approves the following Project Leaders and Project Editors for the listed Projects noted:

Project	Working Group	Project Leader/Editor
ISO 15926-100 Industrial automation systems and integration — Integration of life-cycle data for process plants including oil and gas production	WG 3	XU Kai Cheng (replacing Wang Zhi Qiang as Co-Project Leader)

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

facilities — Part 100: Vocabulary

Resolution 1153: (Little Rock, Arkansas, USA – May 2025)

Establish ballot duration of new work item proposals

SC 4 requests its Committee Manager to apply the 8-week ballot duration to the following NP ballots and coordinate with the appropriate committees in joint work:

Project	Approval
ISO/NP 10303-1 Industrial automation systems and integration — Product data representation and exchange — Part 1: Overview and fundamental principles	☒

Resolution 1154: (Little Rock, Arkansas, USA – May 2025)

Appointment of Convenors and Deputy Convenors

Noting that continuity of operations and management of work is an important consideration to the successful operation of advisory groups, SC 4 appoints the following Convenors and Deputy Convenors for a three-year term, with appreciation to serve in this capacity.

WG	Role	Appointment
TF 3	Deputy Convenor	Jean Brangé
TF 3	Deputy Convenor	Véronique Dubillot

Resolution 1155: (Little Rock, Arkansas, USA – May 2025)

Appoint Convenors for WG 13 and WG 23

Noting that the role of Convenor in a Working Group is critical to continuity of operations and management of work and requires sustained engagement with experts, project leaders, and the committee, SC 4 recognizes the extended absence of the Convenor for WG 13 and WG 23 and appoints the current Deputy Convenor (Anne-Marie Mougeolle for WG 13 and Hayley Thompson for WG 23) from each Working Group to the role of Convenor to serve out the remainder of the term, currently through November 2027. SC 4 welcomes nominations for the next four weeks from member bodies for the role of Deputy Convenor for both WG 13 and WG 23.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Further, SC 4 recognizes the significant contributions of Tim King for his leadership and contribution over the years to the development of SC 4 standards and for setting the strong foundation for the ISO 8000 series of standards.

Resolution 1156: (Little Rock, Arkansas, USA – May 2025)

AhG 3 - UUID management for industrial data final report and disbandment

Noting the final report of AhG 3 – UUID management for industrial data and the recommendation to follow current experiments through to their conclusion, SC 4 accepts the report and its recommendations, thanks the AhG for its efforts and disbands AhG 3.

Resolution 1157: (Little Rock, Arkansas, USA – May 2025)

Launch Study Group 1 - Industrial data and digital twin

Noting the proposal from JWG 16 regarding the situation ISO/TC 184/SC 4 finds with the current implementation of digital twin as a technical concept into the scope of industrial data, and the existing activities of ISO/TC 184/AG 2 and IEC/ISO JTC 1/SC 41 – Internet of Things and Digital Twin, SC 4 establishes Study Group 1 – Industrial data and digital twin with the following terms of reference:

- Examine the current implementations of digital twin in SC4 standards and related organizations inside and outside ISO, notably ISO/TC 184, ISO/TC 184/SC 5, IEC/TC 65,
- Develop pros and cons to each proposal put forward from JWG 16 and any developed internal to the study group,
- Recommend one of the proposals to SC4 with a draft implementation plan including timing,
- Report to SC4 at each plenary meeting the status of the study group until the group is disbanded, and
- Provide its final report to SC4 on or before June 2026.

SC4 names Kenneth Swope, Convenor, Soonhung Han, Deputy Convenor and Martin Hardwick, Deputy Convenor, with appreciation to serve in that capacity, recognizes the following experts

- Hiroshi Murayama,
- Hyman Duan,
- Sangkeun Yoo,

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Gordon Shao,
- Bengt Olsson,
- Paul van Exel,
- Mario Tucci,
- Benjamin Martinez,
- Richard Merkel,
- Anne-Francoise Cutting-Decelle, and
- Christophe Mouton,

who have volunteered to participate and invites member bodies to nominate experts to participate in the group's work.

Resolution 1158: (Little Rock, Arkansas, USA – May 2025)

Establish a Category C Liaison from ALICIA Project to WG 15

Noting the application of the ALICIA project, a EU funded Horizon Europe activity, to ISO in support of Category C Liaison status to ISO/TC 184/SC 4/WG 15 and recognizing the opportunity to collaborate with the technical experts in the project regarding digital twin for manufacturing, SC4 concurs with the request and welcomes the ALICIA project as a Category C Liaison to WG 15.

Resolution 1159: (Little Rock, Arkansas, USA – May 2025)

Establish a Category C Liaison from ALICIA Project to WG 26

Noting the application of the ALICIA project, a EU funded Horizon Europe project, to ISO in support of Category C Liaison status to ISO/TC 184/SC 4/WG 26 and recognizing the opportunity to collaborate with the technical experts in the project regarding ontology-based interoperability, SC4 concurs with the request and welcomes the ALICIA project as a Category C Liaison to WG 26.

Resolution 1160: (Little Rock, Arkansas, USA – May 2025)

Establishment of Category A Liaison with POSC Caesar Association

Recognizing the contribution POSC Caesar Association is making to SC4, the committee accepts the request from the POSC Caesar Association to establish a two-way Category A liaison, request its Committee Manager to confirm their eligibility with ISO CS, and appoints Lars D. Berthinussen to serve as liaison representative for a three-year term [2025 – 2028], with appreciation for his offer to serve in that capacity.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Resolution 1161: (Little Rock, Arkansas, USA – May 2025)

Establish a Maintenance Agency for ISO 23726 Ontology based interoperability

Noting the preparatory work conducted by WG 26, SC4:

- Recommends establishing a maintenance agency (MA) for ISO 23726 – 3 and ISO 23726 – 100 ontologies with scope, duties, and operations defined in the terms of reference documented in WG 26/N 78,
- Appoints POSC Caesar Association represented by managing director Lars D. Berthinussen to serve as the secretariate for the MA, and
- Requests its Committee Manager send the terms of reference for the MA and the proposed assignment to ISO/CS for consideration by the TMB for approval of the MA.

Resolution 1162: (Little Rock, Arkansas, USA – May 2025)

Revise Schedule for ISO 23726-3 IDO for an exceptional 9-month extension

Noting the progress on ISO 23726-3, the unplanned work of establishing a Maintenance Agency for the same, and the commitment of the project development team, SC4 requests its Committee Manager submit the project limit date extension request to ISO/CS with the following target and limit dates:

- Stage 40 Target date: 2025-09-13
- Stage 40 Limit date: 2026-07-13
- Stage 60 Target date: 2026-09-13
- Stage 60 Limit date: 2027-07-13.

Resolution 1163: (Little Rock, Arkansas, USA – May 2025)

Revise schedule for ISO/AWI 23726-2 and change from SDT 24 to SDT 36

Noting the progress on ISO/AWI 23726-2 and the commitment of the project development team to resolve all the received comments, SC4 requests its Committee Manager to coordinate with ISO/CS and move the ISO/AWI 23726-2 project from SDT 24 to SDT 36 with the following target dates for each stage:

- 30.00 Committee draft (CD) registered: 2026-01-20
- 40.00 DIS registered: 2026-06-20
- 60.60 International standard published 2027-06-20.



N4118

2026 – 05 – 15

Resolution 1164: (Little Rock, Arkansas, USA – May 2025)

Revise schedule for ISO/AWI 15926-4 ed 4 and change SDT 18 to SDT 24

Noting the progress on ISO/AWI TS 15926-4 ed4 and the commitment of the project development team to resolve the current resource constraints, SC 4 requests its Committee Manager to coordinate with ISO/CS and move the ISO/AWI TS 15926-4 ed4, ISO PID:90901

project from SDT 18 to SDT 24 with the following target dates for each stage:

- Committee Draft ballot: 2025-09-01
- DTS submission: 2026-04-01
- Publication: 2026-07-01

Resolution 1165: (Little Rock, Arkansas, USA – May 2025)

Launch revision to ISO 8000-2 ed 6

Noting that a new update to ISO 8000-2 ed 6 is needed and there is no change in scope to the standard, SC4 requests its Committee Manager coordinate with ISO/CS and launch edition 6 using the SDT 18 track, nominates Hayley Thompson as Project Leader, and allocates the work to WG 23 with the following terms of reference:

Update ISO 8000-2 ed 6 as presented to the Change Board during the 89th Plenary of SC4 on 20 May 2025 with the following schedule milestones:

- DIS submission: 2025-08-22
- Publication: 2026-05-22

Nominate the following experts and welcome additional experts to participate in the project

- Changsoo Lee (KATS),
- ZHANG Wei Qun, Kevin (SAC),
- Julian Schwarzenbach (BSI),
- Anne-Marie Mougeolle (AFNOR), and
- Peter Benson (ANSI).



N4118

2026 – 05 – 15

Resolution 1166: (Little Rock, Arkansas, USA – May 2025)

Convert ISO/CD TR 17999 to a PAS

Noting the report from SC 4/TF 2, the current balloted and approved DTR 17999, the recommendation of the ISO Technical Editor and the consensus of the PPC, SC4:

- Requests its Committee Manager to launch ISO/PAS 17999 – *Automation systems and integration — Reference model for industrial data standards* with SDT 18 track and at stage 50.00 utilizing the existing ISO/DTR 17999 as content, and
- Nominates David Leal as project leader with appreciation to serve in that capacity.

Resolution 1167: (Little Rock, Arkansas, USA – May 2025)

Call for contributions for ISO/PWI TR 25272

Noting the need to collect contributions from ISO/TC 184/SC 4, ISO/IEC JTC 1/SC 24 and ISO/TC 171/SC 2 for supporting future developments in the scope of visualization for industrial data, SC4 requests its Committee Manager launch a CIB to share this call to the committee and committees in liaison. SC4 also encourages convenors, team leaders, and liaisons representatives to consider submitting contributions (requirements, use-cases, and examples) to JWG 16 co-convenors and secretary before 1 October 2025.

Resolution 1168: (Little Rock, Arkansas, USA – May 2025)

Launch a PWI for ISO 13584-26 ed2

Noting the proposal from JWG 24, SC4 resolves to initiate a project to update ISO 13584-26 ed2 to develop a New Work Item Proposal (NP) at the next plenary in October 2025 with the following terms of reference:

- Register a Preliminary Work Item ISO/PWI 13584-26 Industrial automation systems and integration — Parts library — Part 26: Logical resource: Information supplier identification
- Allocates the work to ISO/TC 184/SC 4/JWG 24,
- Nominates Hiroshi Murayama as the project leader,
- Nominates Jean-Charles LECLERC, Yves KERARON experts and Yoshiaki Sonoda, and
- Invites member bodies to nominate additional experts to the project.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Acclamations and Acknowledgements

Acclamation

SC 4 expresses its sincere appreciation to the staff of University of Arkansas, Little Rock for their contributions in hosting the 89th ISO/TC 184/SC 4 Plenary in Little Rock, Arkansas, USA.

- Professor John Talburt, and Department of Computer and Information Sciences, University of Arkansas at Little Rock, for the initial offer to host our plenary and to coordinate locally,
- Professor Jeffrey B. Connelly, Interim Dean of the Donaghey College of Science, Technology, Engineering, and Mathematics for the welcoming message on Monday,
- Professor Tansel Karabacak, Incoming Dean of the Donaghey College of Science, Technology, Engineering, and Mathematics for the welcoming message for Industry Day, and,
- Javon Fells, Dean's office staff member who helped behind the scenes with the room reservations, floor plans, auditorium keys, and other details.

Acclamation

SC 4 expresses its sincere appreciation to the following Information Quality Graduate Students of University of Arkansas, Little Rock for their assistance and IT expertise:

- Rudra Pratap Mukherjee
- Hendrika Maclean
- Harshitha Kancheti
- Fumiko Kobayashi
- Serajum Munira
- Samuel Sackey

Acclamation

SC 4 expresses its sincere appreciation to all presenters at the Open Technical Forum on Monday evening.

- Paul van Exel
- Jean Brange

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Anne-Marie Mougeolle
- Hayley Thompson
- Soonhung Han
- Pål Rylandsholm

Acclamation

SC 4 expresses its sincere appreciation to Benjamin Martinez and all members of TC 184/AhG 4 for their participation and collaboration during the Smart Manufacturing Next Steps meeting.

- Baba Takenori
- Fumihiko Kimura
- Youichi Nonaka
- Ulf Carlsson
- Yiqin Wang
- Richard Martin

Acclamation

SC 4 expresses its sincere appreciation to the following Industry Day Organizers and Presenters.

- Peter Benson
- Hayley Thompson
- Kenny Swope
- Professor John Talburt

Resolution 1169: (Nagasaki, Japan – Oct 2025)

SC 4 requests its Committee Manager to apply the 8-week ballot duration to the following NP ballots and coordinate with the appropriate committees in joint work:

Project	Approval
ISO/PWI 23726 – 1 Industrial automation systems and integration – Ontology based interoperability – Part 1: Overview and fundamental principles	☒
ISO/NP 8000 – 260: Data quality – Part 260: Sensor data life cycle management: Concept and framework	☒

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

ISO/NP XXXX: Industrial automation systems and integration – Framework for visualization of product data using AI models	☒
ISO/NP XXXX: Industrial automation systems and integration – 3D shape descriptors for fidelity metrics of digital twin shapes	☒
ISO/NP 15926 – 200: Industrial automation systems and integration — Integration of life-cycle data for process plants including oil and gas production facilities — Part 200: RDFS Implementation of the data model	☒

Resolution 1170: (Nagasaki, Japan – Oct 2025)

Noting that continuity of operations and management of work is an important consideration to the successful operation of advisory groups, SC 4 appoints the following Convenors and Deputy Convenors for a three-year term, with appreciation to serve in this capacity.

WG	Role	Appointment
QC	Convenor	Ms. Véronique Dubillot
WG 3	Deputy Convenor	Mr. Yoshiaki Sonoda
WG 11	Convenor	Dr. David Loffredo
WG 12	Convenor	Mr. Stuart Galt
WG 21	Convenor	Mr. Jean Brangé
WG 13	Convenor	Mr. Christophe Mouton
JWG 16	Deputy Convenor	Mr. Atsuto Soma
WG 25	Convenor	Mr. Yoshiaki Sonoda
WG 25	Deputy Convenor	Dr. Klaus Dickmann

Resolution 1171: (Nagasaki, Japan – Oct 2025)

Noting that a new update to ISO 10303-59:2022 is needed and there is no change in scope to the standard, SC4 requests its Committee Manager to coordinate with ISO/CS and launch the new edition using the SDT 24 track, nominating Atsuto Soma as Project Leader, and allocating the work to WG 12 with the following terms of reference:

Update ISO 10303-59:2022 as presented to the Change Board during the 90th Plenary of SC4 on 28 Oct 2025 with the following schedule milestones:

- CD consultation: 2026-03-31
- DIS submission: 2026-07-31
- Publication: 27-01-31

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Nominate the following experts to participate in the project

- Fumiki Tanaka (JISC)
- Akihiko Ohtaka (JISC)
- Hiroyuki Hiraoka (JISC)
- Yoshihito Kikuchi (JISC),
- Kazuya Kobayashi (JISC).

Further, the SC 4 Committee Manager is requested to launch a call for experts from National Bodies and Liaisons to participate in the project's work.

Finally, due to the special nature of ISO 10303 – 59 SC 4 requests ISO to grant a waiver on the mandatory use of the ISO Online Standards Development platform for this project.

Resolution 1172: (Nagasaki, Japan – Oct 2025)

Noting that a new update to ISO 10303-62:2025 is needed and there is no change in scope to the standard, SC4 requests its Committee Manager to coordinate with ISO/CS and launch the new edition using the SDT 24 track, nominating Atsuto Soma as Project Leader, and allocating the work to WG 12 with the following terms of reference:

Update ISO 10303 - 62:2025 as presented to the Change Board during the 90th Plenary of SC4 on 28 Oct 2025 with the following schedule milestones:

- CD consultation: 2026-03-31
- DIS submission: 2026-07-31
- Publication: 2027-01-31

Nominate the following experts to participate in the project

- Fumiki Tanaka (JISC)
- Akihiko Ohtaka (JISC)
- Hiroyuki Hiraoka(JISC)
- Yoshihito Kikuchi (JISC),
- Kazuya Kobayashi(JISC).

Further, the SC 4 Committee Manager is requested to launch a call for experts from National Bodies and Liaisons to participate in the project's work.

Finally, due to the special nature of ISO 10303 – 62 SC 4 requests ISO to grant a



N4118

2026 – 05 – 15

waiver on the mandatory use of the ISO Online Standards Development platform for this project.

Resolution 1173: (Nagasaki, Japan – Oct 2025)

Noting the proposal from WG 12 for creating edition 5 of ISO 10303 – 242 to incorporate additional capabilities from industry requirements, SC 4 requests its Committee Manager to coordinate with ISO/CS and initiate the revision of ISO 10303 – 242 as a new project using the SDT 36 track with the following dates:

Proposed date for first meeting: 2025-11-20

Dates for key milestones:

- Circulation of 1st Working Draft (if any) to experts: 2025-11-15
- Committee Draft Consultation (if any): 2025-12-15
- DIS submission: 2026-12-15
- Publication: 2027-09-15

SC4 allocates the project to WG 12 and appoints Jean Brangé as project leader and nominates the following experts to participate in the project

- AFNOR: Frédéric Darré, Véronique Dubillot, Pierre Duchier, Jacques Heinisch, Erwan Livolant
- ANSI: Brian Bentrin, Allison Barnard, Michelle Drummond, Stuart Galt, Melissa Harvey, John Van Horn, Benjamin Standfield, Tom Thurman, Asa Trainer, Ben Urick, Pat Walsh
- BSI: Robert Swindells,
- DIN: Bernd Feldvoss, Guillaume Hirel, Lothar Klein, Max Ungerer,
- JISC: Hiroyuki Hiraoka, Fumiki Tanaka, Atsuto Soma,
- SAC: Hyman Duan,
- SIS: Mikael Hedlind, Magnus Lundgren, Torbjörn Holm
- SN: Nils Sandmark, Pal Rylandsholm

Further, the SC 4 Committee Manager is requested to launch a call for experts from National Bodies and Liaisons to participate in the project's work.

Finally, due to the special nature of ISO 10303 – 242 SC 4 requests ISO to grant a waiver on the mandatory use of the ISO Online Standards Development platform for this project.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Resolution 1174: (Nagasaki, Japan – Oct 2025)

Noting the contribution of China to WG 15 of a use case for robotic multilayer and multipass gas-shielded metal arc welding, and the expansion of use cases utilizing digital twin for manufacturing framework, SC4 requests its Committee Manager to coordinate with ISO/CS to ballot a Technical Report with the title “Automation Systems and Integration – Industrial Data – Digital Twin framework for manufacturing: – Part 101: Use case on management of robotic multilayer and multipass gas-shielded metal arc welding processes” as Part 101 in the ISO 23247 series.

Resolution 1175: (Nagasaki, Japan – Oct 2025)

Noting that an update to ISO 15926-2 is required and that there is no change in scope to the standard, SC4 requests its Committee Manager launch ISO 15926-2 edition 2 using the SDT 36 track and names Onno Paap as Project Leader and Jonathan Eyre as Editor for the project. SC 4 allocates the work to WG 3 with the following terms of reference:

- Update ISO 15926-2 to ed.2 as presented to the Change Board during the 90th Plenary of SC4 on 28 Oct 2025
- Implement the target schedule as shown:
 - Circulation of 1st Working Draft to experts: 2026-05-01
 - Committee Draft consultation: 2026-09-01
 - DIS submission*: 2027-09-01
 - Publication*: 2028-11-01
- Nominates the following experts to the project
 - Leo van Ruijven, NL
 - Jonathan Eyre, GB
 - Kevin Zhang, CN
 - Mr. Chandrakant Bhardwaj, IN
 - Salomon de Jager, ZA
 - Youngsoo Jung, KO

Further, the SC 4 Committee Manager is requested to invite experts from National Bodies and Liaisons to participate in the project’s work.

Resolution 1176: (Nagasaki, Japan – Oct 2025)

Noting resolution 1161 taken during the 89th Plenary meeting and subsequent approval from ISO/TMB (TMB Resolution 63/2025) on the application to establish a Maintenance



N4118

2026 – 05 – 15

Agency for ISO 23726 ontologies, SC 4 requests its Committee Manager to launch a call for experts to participate in the Standards as Database team for the Maintenance Agency on ISO 23736 part 3 and part 100 with gratitude to ISO CS, and SC4 support in establishing the MA, TMB Resolution 63/2025.

Resolution 1177: (Nagasaki, Japan – Oct 2025)

Noting the proposal from WG 12 for creating ISO/NP 10303 – 63 to define representation structures for isogeometric analysis (IGA), SC4 requests its secretary launch a New Work Item for 10303-63 with an 8 weeks duration, due to the special nature of ISO 10303 – 63, SC 4 requests ISO to grant a waiver on the mandatory use of the ISO Online Standards Development platform for this project.

Resolution 1178: (Nagasaki, Japan – Oct 2025)

Noting the proposal from WG 12 for creating ISO/NP 10303 – 63 to define representation structures for isogeometric analysis (IGA), SC4 requests its secretary launch a New Work Item for 10303-63 with an 8 weeks duration, due to the special nature of ISO 10303 – 63, SC 4 requests ISO to grant a waiver on the mandatory use of the ISO Online Standards Development platform for this project.

Resolution 1179: (Nagasaki, Japan – Oct 2025)

In light of the shared interest in 3D printing, scanning addressed by ISO/IEC JTC 1/WG 12, and industrial data, particularly in terms of formats for visualization and other forms of product data addressed by JWG 16, SC 4 requests its Committee Manager establish a liaison from SC4 to ISO/IEC/JTC 1/WG 12 and appoints JWG 16 co-convenors, Soonhung Han and Christophe Mouton, to serve as liaison representatives for a three-year term [2025 – 2028], with appreciation for their offer to serve in that capacity.

Acclamation

SC 4 expresses its sincere appreciation to the organizers and sponsors of the 90th Plenary meeting held in Nagasaki, Japan:

- Manufacturing Science and Technology Center (MSTC)
- Engineering Advancement Association of Japan (ENAA)
- ELYSIUM Co., Ltd.
- OWLS Consulting Group,
- System Five, Corporation, Inc.
- Yokogawa Electric Corporation

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

Acclamation

SC 4 expresses its sincere appreciation to the on-site Nagasaki meeting secretariat and supporting staff to the 90th Plenary meeting of ISO/TC 184/SC 4.

- Mr. Kenji Kitayama, MSTC
- Mr. Yasunari Fujinuki, MSTC
- Mr. Kazunori Yamaguchi, ENAA
- Mr. Tetsuya Fujino
- Ms. Satomi Miyata
- Ms. Miku Hashimoto, Univ. of Nagasaki
- Mr. Haruto Igawa, Univ. of Nagasaki
- Mr. Agito Kajiwara, Univ. of Nagasaki,
- Mr. Kensuke Nishimura, Univ. of Nagasaki
- Mr. Koji Matsumoto

Acclamation

SC 4 expresses its sincere appreciation to the organizations and staff of the hosting facilities to the 90th Plenary meeting of ISO/TC 184/SC 4:

- Nagasaki City
- Nagasaki International Tourism and Convention Association (長崎県国際観光コンベンション協会)
- Nagasaki Diamon Staff
- Nagasaki Industrial Federation (長崎県工業連合会)
- Nagasaki IT Solution Industry Association (長崎県情報産業協会)
- DEJIMA MESSE Nagasaki
- Hilton Nagasaki
- KAGETSU

Acclamation

SC 4 expresses its sincere appreciation to the Daitaku Kiln for their excellent work and attention to detail creating the souvenir for delegates to the 90th Plenary meeting of ISO/TC 184/SC 4:

- Mr. Takuo Ogushi, Daitaku Kiln
- Ms. Youko Ogushi, Daitaku Kiln
- Mr. Takahiro Ogushi, Daitaku Kiln

Acclamation

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

SC 4 expresses its sincere appreciation to all presenters at the Open Technical Forum on Monday evening.

- Mr. Peter Eales
- Dr. Martin Hardwick
- Mr. Hiroshi Murayama
- Mr. Jumyung Um'
- Mr. Mikkel Brynildsen
- Ms. Caitlin Woods
- Mr. Jean Brangé
- Ms. Hayley Thompson

Acclamation

SC 4 expresses its sincere appreciation to the following Industry Day Guests and Presenters.

- Special Guests
 - o Mr. Nakano Shingo, Director, International Standardization Division, Innovation and Environment Policy Bureau, Ministry of Economy, Trade and Technology
 - o Mr. Takeyuki Ohara, International Standardization Division, Innovation and Environment Policy Bureau, Ministry of Economy, Trade and Technology
 - o Mr. Shiro Suzuki, Mayor of Nagasaki City
 - o Dr. Ikuo Yamamoto, Vice President, Nagasaki University
 - o Mr. Satoshi Abe, Chief Executive Director, Manufacturing Science and Technology Center (MSTC)
 - o Mr. Kiyotaka Terajima, President, Engineering Advancement Association of Japan (ENAA)
 - o Mr. Kazunori Yamaguchi, Senior Manager, Planning Division, Engineering Advancement Association of Japan (ENAA)
- Presenters
 - o Mr. Kenneth Swope, The Boeing Company
 - o Mr. Jean Brangé, AFNeT Services
 - o Dr. Timothy King, BAE Systems plc
 - o Dr. Shinichi Urakawa, Ouranos Ecosystem Promotion Center (OEPC)
 - o Mr. Hiroshi Nagakura, Mitsubishi Heavy Industries, Ltd.
 - o Dr. Martin Hardwick, STEP Tools, Inc.
 - o Mr. Fumiki Ohtani, JAMA taskforce on Data interoperability in digital engineering
 - o Dr. Max Ungerer, Independent Engineering Consultant



N4118

2026 – 05 – 15

- Mr. Kazuya Furuichi, Chiyoda Corporation
- Dr. Leo van Ruijven, Croonwolder&dros B.V.
- Mr. Mikitaka Hayashi, MODS Management Limited
- Dr. Melinda Hodkiewicz, The University of Western Australia
- Dr. Dan Liu, China Railway Construction Heavy Industry Corporation Limited
- Dr. Youngsoo Jung, Myongji University
- Mr. Atsuto Soma, ELYSIUM Co., Ltd.

Acclamation

SC 4 expresses its sincere appreciation to Gernot Rossi and Hiroshi Murayma and for their leadership as Convenor and Deputy Convenor of WG 25 and wishes them the very best in their future endeavours.

Acclamation

SC 4 expresses its sincere appreciation to Keith Hunten, Convenor of WG 12 since 2003, for his passion, technical leadership, in-depth knowledge of the ISO 10303 standard and recognizes him with the ISO Award of Excellence and wishes him the very best in his future endeavours.

Resolution 1180: (VIRTUAL – May 2026)

SC4 requests its Committee Manager to apply the 8-week ballot duration to the following NP ballots and coordinate with the appropriate committees in joint work:

Project	Approval
ISO/NP 18136 – 2: Automation systems and integration — Nuclear digital ecosystem — Part 2: Vocabulary	☒

Resolution 1181: (VIRTUAL – May 2026)

Noting the utilization of quantities and units in industrial data and the increasing involvement of SC4 in developing standards for industrial data with recognition that ISO/TC 12 – Quantities and units has expertise in this domain, SC4 appoints Mr. Robert Swindells, Mr. Yoshiaki Sonoda, and Mr. Ronald Tse as liaison representatives from ISO/TC 184/SC 4 to ISO/TC 12 for a three-year term (2026–2029), with appreciation for their willingness to serve in this capacity.



N4118

2026 – 05 – 15

Resolution 1182: (VIRTUAL – May 2026)

Noting the increasing involvement of SC4 in vocabularies including ISO 15926 – 100, ISO 23726 – 2, and ISO/NP 18136 – 2 with recognition that ISO/TC 37 – Language and terminology has expertise in this domain, SC4 appoints Dr. LIU Peng, Mr. Hiroshi Murayama, Ms Hayley Thompson, Mr. Yves Keraron, Mr. Christophe Mouton, and Mr. Peter Eales as liaison representatives from ISO/TC 184/SC 4 to ISO/TC 37 for a three-year term (2026–2029), with appreciation for their willingness to serve in this capacity.

Resolution 1183: (VIRTUAL – May 2026)

Noting the formation of ISO/IEC JTC 5 – Digital product passport and the relevance of the topic to standards developed in SC4, SC4 requests its Committee Manager to establish the liaison from ISO/TC 184/SC 4 to ISO/IEC JTC 5 and appoints Mr. Christophe Mouton, Mr. Peter Benson, Ms Hayley Thompson, Mr. Yoshiaki Sonoda, Mr. Jean Brangé, Mr. Hiroshi Murayama, Mr. Erik Molin, Mr. Peter Eales, and Mr. Zhang Wei Qun (Kevin) as new liaison representatives for a three-year term [2026–2029], with appreciation for their willingness to serve in this capacity.

Resolution 1184: (VIRTUAL – May 2026)

Recognizing:

- that ISO/TC 154 is developing ISO/NP 26087 Processes and data in carbon market monitoring reporting verification (MRV) — Data quality assessment, which introduces structured methods for evaluating completeness, timeliness, traceability, accuracy, and comparability of MRV data, as described in ISO/TC 154 N 1641 and N 1642;
- that these concepts align closely with the data quality principles, vocabularies, and industrial data frameworks maintained by ISO/TC 184/SC 4;
- that WG 13 has expressed strong interest in contributing expertise to ensure consistency and interoperability between SC 4 data quality standards and the MRV data quality methodologies under development;

SC 4 requests its Committee Manager to extend the liaison from ISO/TC 184/SC 4 to ISO/TC 154, and appoints Mr. Christophe Mouton, Mr. Peter Benson, Ms Hayley Thompson, Mr. Yoshiaki Sonoda, Mr. Yves Keraron, and Mr. Zhang Wei Qun, (Kevin) as new liaison representatives for a three-year term [2026–2029], with appreciation for their willingness to serve in this capacity.

Resolution 1185: (VIRTUAL – May 2026)

Noting that ISO 18136-11 is in current development and benefits from a robust relationship to ISO/TC 85, SC4 requests its Committee Manager to appoint Mr.

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



**Technical Committee 184: Automation
Systems and Integration
Subcommittee 4: Industrial data**

 Winner
2007 & 2021
Lawrence D. Eicher Leadership Award

N4118

2026 – 05 – 15

Youngsoo, Jung, Mr. Yves Keraron, and Mr. ZHANG Wei Qun, Kevin as an additional Liaison Representative from ISO/TC 184/SC 4 to ISO/TC 85 (Nuclear energy, nuclear technologies, and radiological protection) for a three year term (2026 – 2029) with appreciation to serve in this capacity.



N4118

2026 – 05 – 15

Resolution 1186: (VIRTUAL – May 2026)

Recalling Annex SS of the ISO/IEC Directives Part 1 and noting the consensus found in WG 12 regarding the maturity of ISO/AWI 10303 – 59 ed5 to be at DIS level, SC4 resolves to skip the CD ballot (stage 30.0) and move directly to a DIS ballot (stage 40.0), with the DIS ballot to be initiated no later than 2026-07-31. SC4 instructs WG 12 to prepare the necessary documents for submission and requests its Committee Manager to make the submission through the normal procedures.

Resolution 1187: (VIRTUAL – May 2026)

Recalling Annex SS of the ISO/IEC Directives Part 1 and noting the consensus found in WG 12 regarding the maturity of ISO/AWI 10303 – 62 ed5 to be at DIS level, SC4 resolves to skip the CD ballot (stage 30.0) and move directly to a DIS ballot (stage 40.0), with the DIS ballot to be initiated no later than 2026-07-31. SC4 instructs WG 12 to prepare the necessary documents for submission and requests its Committee Manager to make the submission through the normal procedures.

Resolution 1188: (VIRTUAL – May 2026)

Noting results of the NP ballot for ISO/NP 26379 taken on 2026-01-29 where a 2/3 majority of voting P-members approved the proposal with no disapprove votes and two P-members nominated experts combined with the subsequent Committee Internal Ballot taken on 2026-03-24 that garnished an additional two P-members to nominate experts, SC4;

- recognizes that the mandatory requirement of five expert nominations has not been satisfied; and
- notes the reconsideration by DIN to nominate an expert and confirm the inclusion of the NP into their work program.

Resolves to launch ISO/NP 26379, 3D Shape Descriptor for Digital Twin with the following terms of reference:

- Prof. Soonhung Han as Project Leader; and
- Mr. Tarique Hasan Khan as Project Editor,

with appreciation for their willingness to serve in these capacities.

SC4 further recognizes the following experts who have volunteered to participate in the project:

- Mr. Trevor Coulson
- Mr. Jayendra Ganguli
- Dr. Matthias Grau
- Mr. Christophe Mouton

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- Mr. Astuto Soma
- Mr. Hyman Duan
- Mr. Chiho Noh
- Mr. Tahir Abbas Jauhar
- Mr. Jaein Lee,

SC4 invites additional member bodies to nominate experts to participate in the projects.

Resolution 1189: (VIRTUAL – May 2026)

Noting that a revision to ISO 15926-6 ed2 is required by industry, that the resources are available, and that there is ample support from NSB's, SC4 requests its Committee Manager to launch ISO 15926 – 6 ed2 as a Preliminary Work Item in the ISO OSD with the following terms of reference:

- Project Leader: Yoshiaki Sonoda
- Project Editor: Hiroshi Seki
- Quality Committee Representative: Hiroshi Murayama as QC role
- Nominated experts
 - JP, Mr. Yoshiaki Sonoda,
 - JP, Mr. Hiroshi Seki,
 - JP, Mr. Yoshinari Hori,
 - JP, Mr. Kazuo Sueyoshi,
 - JP, Mr. Masahiko Tomi,
 - NL, Mr. Paul van Exel,
 - NL, Mr. Onno Paap,
 - CN, Mr. Zhang, Wei Qun,
 - CN, Mr. Sun, Yi,
 - GB, Mr. Jonathan Eyre,
 - FR: Mr. Christophe Mouton,
 - FR: Mr. Yves Keraron,
 - US, Mr. Ronald Tse, and

Invites member bodies not listed to nominate experts to the project.



N4118

2026 – 05 – 15

Resolution 1190: (VIRTUAL – May 2026)

Noting that ISO 18136 – 11 is required by industry to establish the standards for configuration management of nuclear ecosystem data, that the resources are available, and that there is ample support from NSB's, SC4 requests its Committee Manager to launch ISO 18136 – 11 in the ISO OSD as a Preliminary Work Item with the following terms of reference:

- Project Leader: Mr. Youngsoo Jung
- Project Editor: Mr. Yunsub Lee
- Nominated experts
 - TC 85: (Mr. Jim Herrold)
 - AFNOR: France (Mr. Christophe Mouton, Mr. Yves Keraron)
 - ANSI: US (Mr. Alan Johnston)
 - BIS: India (Mr. Amit Thakur, Mr. Kundan Giri)
 - JISC: Japan (Mr. Yoshiaki Sonoda, Mr. Hiroshi Seki)
 - KATS: Korea (Mr. Youngsoo Jung, Mr. Yunsub Lee)
 - NEN: Netherlands (Mr. Paul van Exel, Mr. Leo van Ruijven)
 - SAC: China (Mr. ZHANG, Wei Qun, Mr. LIU, Xu Jia, Mr. SUN, Yi , Mr. QIN, Qiang)

Invites member bodies not listed to nominate experts to the project.

Resolution 1191: (VIRTUAL – May 2026)

Noting the opportunity to manage vocabulary in a common way within the Working Group 3 body of experts for ISO 15926 and ISO 18136, SC4 requests its Committee Manager circulate an 8-week Committee Internal Ballot (CIB) the draft SC4 standing document:

“Procedure for the alignment and maintenance of terms in the ISO 15926 and ISO 18136 series” for comment.

SC4 further requests WG 3 to review the comments received and to submit a revised draft, together with proposed dispositions, to SC4 for further consideration as a standing document of SC4.



N4118

2026 – 05 – 15

Resolution 1192: (VIRTUAL – May 2026)

SC4 notes that the Systematic Review ballot for ISO/TS 10303-17:2022 indicated that a simple majority of the P-members of the committee propose confirmation. Furthermore, the results of the ballot indicates adoption/use in Denmark, Japan, Finland and Norway, which together with ongoing ISO 10303-242 development activity in France and Germany demonstrate that the document is used internationally and has been adopted with or without change or is used in at least five countries.

SC4 concludes that this document meets the criteria for retention after systematic review and shall be made available in its existing form. SC4 requests its Committee Manager disposition the SRR results accordingly and notify ISO/CS of the Committee's decision.

Resolution 1193: (VIRTUAL – May 2026)

Noting the proposal from WG 11 for creating edition 3 of ISO 10303 – 15 to incorporate capability enhancements from industry requirements identified in ISO JIRA [TCSC410303-2756](#), SC4 requests its Committee Manager to coordinate with ISO/CS and initiate the revision of ISO 10303 – 15, as a New Project on SDT 18 track with the following terms of reference:

Proposed date for first meeting: 2026-05-21

Dates for key milestones:

Circulation of 1st Working Draft (if any) to experts:	2026-05-07
Committee Draft Consultation (if any):	2026-06-15
DTS submission:	2026-09-15
Publication:	2027-01-15

Identifies SDG Goal 9 with the following rationale:

- The ISO 10303 standard in total is identified against Goal 9 and this part is included in that series, and
- To enable the interoperability between industry and IT architectures in industrial data applications.

SC4 allocates the project to WG 11 and appoints Mr. Kevin Le Tutour as Project Leader with nominations of the following experts to participate and welcome additional experts to participate in the project:

- AFNOR: Mr. Frédéric Darré, Mr. Pierre Duchier, Mr. Erwan Livolant, Mr. Kevin Le Tutour
- DIN: Mr. Guillaume Hirel, Mr. Max Ungerer
- BSI: Mr. Phil Spiby, Mr. Robert Swindells
- ANSI: Mr. Patrick Walsh, Mr. Stuart Galt, Mr. Tom Thurman

Dana Tripp
Secretary, ISO/TC 184/SC 4
trippdsc4@gmail.com



N4118

2026 – 05 – 15

- SIS: Mr. Mikael Hedlind

Utilize the ISO OSD with a document provided by WG 11 that will replace the edition currently in publication.

Resolution 1194: (VIRTUAL – May 2026)

Noting the ballot results of the Systematic Review for ISO/SRR TS 8000 – 60 taken on 2025 – 12 – 02 and the recommendation by WG 13, SC4 requests its Committee Manager to Register with ISO/CS ISO/AWI TS 8000-60 edition 2 with the following terms of reference:

- Incorporate the new scope to include parts 65 and 66 from the ISO 8000 series,
- Development stage 20 with a standard development track (SDT) of 36 months,
- Target dates:
 - Circulation of 1st Working Draft: 2026-12-14
 - Committee Draft consultation: 2027-03-01
 - DIS submission: 2027-12-13
 - Publication: 2029-05-31
- Allocates the work to WG 13 with Christophe Mouton as Project Leader, Prof. Changsoo Lee as Project Co-Leader, and Ms Hayley Thompson as Project Editor
- ISO/TS 8000-60 supports the UN Sustainable Development Goal 9 (Industry, Innovation and Infrastructure) by providing concepts for implementing, assessing and improving data quality management in various industrial sectors.
- Register the following list of NSBs with their nominated representatives as active participants in the revision:
 - USA: Mr. Peter Benson
 - Norway: Mr. Tor Arne Irgens,
 - Korea: Mr. Sungjoon LIM
 - South Africa: Dr. Salomon De Jager
- Provide OSD access to enable WG 13 to prepare the Working Draft using the previous edition as a basis.



N4118

2026 – 05 – 15

Resolution 1195: (VIRTUAL – May 2026)

Noting the work of JWG 24 in the refinement of ISO/PWI 13584 – 26 ed2 by the project team, SC4 resolves to advance ISO/PWI 13584 – 26 ed2 to stage 20 with the following terms of reference:

- Development Stage 20 (NP) with a standard development track (SDT) of 24 months,
- Target dates:
 - Circulation of 1st Working Draft: 2026-10-09
 - Committee Draft consultation: 2026-12-12
 - DIS submission: 2027-05-31
 - Publication: 2027-12-11
- ISO 13584 – 26 supports the UN Sustainable Development Goal 9 (Industry, Innovation and Infrastructure) by ensuring efficient use of data in industrial settings in concert with the larger part of the ISO 13584 series.
- ISO 13584 – 26 supports the UN Sustainable Development Goal 12 (Responsible Consumption and Production) by properly identifying data, its meaning and ensuring the responsible use of that data in industrial settings.
- Allocates the work to JWG 24 with Mr. Hiroshi Murayama as Project Leader,
- Nominate the following experts:
 - AFNOR: Mr. Christophe Mouton, Mr. Yves Keraron,
 - ANSI: Mr. Ronald Tse,
 - DIN: Mr. Gernot Rossi,
 - JISC: Mr. Yoshiaki Sonoda
 - KATS: Prof. Soonhung Han
 - SABS: Dr. Salomon De Jager
- Provide OSD access using the existing document to enable JWG 24 to prepare the Working Draft, and
- Requests member bodies nominate experts to the project's work.



N4118

2026 – 05 – 15

Acclamation

SC4 recognizes Dana Tripp for her role as Committee Manager for the past several years and for the significant contribution to the management of the Committee and wishes her the best in her next endeavours.

Acclamation

SC4 welcomes Leslie Garnett and Step Tools, Inc. to the anticipated role of Committee Manager and for the enthusiastic engagement in the Committee's work while the approval process concludes with ANSI.

Acclamation

P Member and Liaison reports are an important part of progressing the standardization of industrial data. SC4 recognizes and appreciates the contributions of the following:

- Mr. Yiqun Yang, China
- Mr. Christoph Mouton, France, Web3D, ISO/JTC 1/SC 24, ISO/TC 10, IEC/TC 3, ISO/TC 85
- Ms. Hayley Thompson, ECCMA
- Mr. Paul van Exel, USPI
- Mr. Stuart Galt, ISO/TC 171/SC 2

Acclamation

SC4 recognizes the members of the Committee for their flexibility and patience in the move from a hybrid in-person meeting to fully virtual. SC4 recognizes the Convenors, Project Leaders, and experts for adapting to the new format and timeline with meetings held ahead of the Plenary to support the Plenary's goals.